

RECOMMENDED OPERATING PROCEDURES FOR ANALYSIS IN THE VERIFICATION OF CHEMICAL DISARMAMENT

2011 Edition

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* Chapter is not available in the 2011 Edition.

Chapter I. Introduction

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1. Introduction

The Blue Book, a collection of the Recommended Operating Procedures for Sampling and Analysis in the Verification of Chemical Disarmament, was last published in 1994. Starting in 2009, the Recommended Operating Procedures (ROPs) for Analysis in the Verification of Chemical Disarmament have been updated via international collaboration with expert laboratories working in the field of Chemical Weapons Convention (CWC)-related analysis. These methods are the basis for accreditation and the guidelines of the designated laboratories of the Organisation for the Prohibition of Chemical Weapons (OPCW) or the laboratories applying for designation. The ROPs are also used to train personnel working in the field of chemical warfare agents.

These new recommendations will be announced in the International Workshop on December 8 and 9, 2011 in Helsinki, Finland. The Blue Book, 2011 Edition, will be available as a hardcopy from VERIFIN (kira.sivula@helsinki.fi) or as a free-of-charge electronic copy from the VERIFIN web-site (<http://www.helsinki.fi/verifin/bluebook>).

The work to update these guidelines will continue on the intranet (ROPSite). User rights and instructions are available from Martin Söderström (martin.soderstrom@helsinki.fi). The aim is to make regular updates on an annual basis. The December 2011 workshop in Helsinki will serve as the first occasion to discuss these recommendations, followed by discussion at the OPCW in connection with the validation group and proficiency test meetings in The Hague. The first author, who is the responsible author, should be contacted if there are any proposals regarding a particular procedure. Editor, Paula Vanninen (paula.vanninen@helsinki.fi), should be kept informed about all planned activities. New and emerging methods/procedures are welcomed after they have been tested with the proficiency tests organised by the OPCW or by other international collaboration.

VERIFIN will send a regular newsletter to those interested. Please, follow-up on the VERIFIN website. The current 2011 Edition is the first attempt to update the recommendations. It should be noted that some issues have been left out or are not yet included in this 2011 Edition. Input from other institutes is required to make these recommendations more comprehensive to users or target audience. It is the wish of the

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authors of this 2011 Edition that input from other laboratories is received to update future editions of the Blue Book.

The first monograph of this kind was published in 1993 followed by an updated version published in 1994. This, so-called Blue Book, was not considered complete at the time, and should never be, as new and better procedures will always emerge for sampling and analysis. In 1994, it was seen that continuous development of the Recommended Operating Procedures throughout the lifetime of the Convention will be needed. This can also be stated now; continuous efforts are needed to keep these ROPs up-to-date, even after the 2011 Edition. The increased pace of scientific and technological development further emphasises this need. Additionally, there are gaps in the coverage of current procedures. Additional chapters for trace analysis, analysis of biomedical samples, scheduled chemicals like ricin and saxitoxin and possibly other toxins and riot control agents might be added in future editions.

All authors and reviewers of the Blue Book, 2011 Edition, and their corresponding organisations as well the financial contributions from the Ministry for Foreign Affairs of Finland are highly acknowledged.

2. Authors and reviewers

Scientists from the fourteen laboratories listed below, including the OPCW Laboratory, have collaborated in updating the Recommended Operating Procedures (ROPs). More than forty authors and reviewers have taken part in this project. The Blue Book, 2011 Edition, has been edited by the Finnish Institute for Verification of the Chemical Weapons Convention (VERIFIN), Department of Chemistry, University of Helsinki (P.O. Box 55, FI-00014 University of Helsinki, Finland; telephone: +358-9-191 50444, fax: +358-9-19150437).

The following institutes have contributed to the preparation of the Blue Book, 2011 Edition:

1. Finnish Institute for Verification of the Chemical Weapons Convention (VERIFIN), Finland
2. Laboratory of Toxicant Analysis, Academy of Military Medical Sciences (AMMSLTA), China
3. Danish Emergency Management Agency (DEMA), Denmark
4. DGA CBRN Defence (DGA), France
5. Vertox Laboratory, Defence Research and Development Establishment, Gwalior, India
6. Norwegian Defence Research Establishment (FFI), Norway
7. DSO National Laboratories, Singapore
8. Laboratorio de Verificación (LAVEMA), Unidad de Defensa Química, Área NBQ y Materiales, Instituto Tecnológico La Marañosa (ITM), Spain
9. Swedish Defence Research Agency (FOI), Sweden
10. Spiez Laboratory, Swiss NBC Defence Establishment, Analytical Chemistry Branch, Switzerland
11. Defence Science and Technology Laboratory (Dstl), UK
12. Forensic Analytical Center, Edgewood Chemical Biological Center (ECBC), USA

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13. Teal Consulting, USA
14. OPCW Laboratory, The Organisation for Prohibition of the Chemical Weapons (OPCW)

Chapter III. Chemicals

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1. Scope

This chapter provides an overview of chemicals relevant to verification activities under the Chemical Weapons Convention (CWC). The definition of what constitutes a relevant chemical is not rigid but depends on the context of the activity. The Convention defines a chemical weapon as meaning toxic chemicals, munitions and devices as well as any equipment related to the previous. Toxic chemical are defined mainly by their effects:

any chemical which through its actions on life processes can cause death, temporary incapacitation or permanent harm to humans and animals. This includes all such chemicals regardless of their origin or of their method of production, and regardless of whether they are produced in facilities, munitions or elsewhere.

This definition encompasses virtually any toxic chemical when used for purposes not permitted by the Convention.

For the purpose of implementing the Convention, the text is much more specific in identifying those toxic chemicals and their precursors that are subject to declaration and/or verification measures. Chemicals are listed in three schedules in the Annex on Chemicals, according to the risk they are considered to pose to the object and purposes of the Convention. Verification activities associated with routine and challenge inspections of research, production, storage and other relevant facilities are restricted to these scheduled chemicals, plus their well-established degradation products.

The situation is somewhat different in cases of alleged use of chemical weapons, in which riot control agents, non-scheduled toxic industrial chemicals or novel agents may also be relevant, depending on the scenario.

OPCW Proficiency Tests are generally run along the lines expected of a challenge inspection. They normally include scheduled chemicals (agents, precursors and other compounds captured by generic structures specified in the schedules), their degradation products and sometimes products that are formed upon decontamination of a scheduled chemical.

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This chapter reviews the chemicals covered by Schedules 1, 2 and 3 of the Convention as well as their well-established degradation products, some of which are also scheduled as precursor chemicals. The chapter also discusses some chemicals relevant to old and abandoned chemical weapons, which were considered obsolete when the text of the Convention was concluded and therefore excluded from the schedules.

2. The schedules of chemicals

2.1 Background to the schedules

The schedules of chemicals that are subject to verification measures were compiled during the lengthy negotiations that culminated in the agreed text of the Convention. These schedules were based mainly on chemicals known to have been previously weaponised, although a number of agents used in WWI and considered obsolete were excluded. In compiling the schedules, generic structures were devised, which were intended to encompass analogues that were known or predicted to have properties similar to the agents, and could therefore pose a risk to the Convention. These generic structures define well over one million compounds. In reality, the generic structures err on the side of caution, and many of the analogues captured by them have little or no CW potential. No additions to the Schedules have been formally proposed or made since the Convention entered into force in 1997.

The majority of toxic chemicals and precursors included in the schedules relate to the three structural classes of nerve agents and three structural classes of vesicants shown in Figure 1. With the exception of tabun and nitrogen mustards, these agents accounted for most of the stockpile of toxic chemicals subsequently declared under the terms of the Convention. Other toxic chemicals included in the schedules are the toxins saxitoxin and ricin, the incapacitant quinuclidinyl benzilate (BZ), the obsolete pesticide amiton, plus several toxic industrial chemicals such as phosgene, some of which were used as warfare agents in WWI. The precursors included in the schedules are based on the main known production routes to Schedule 1 or Schedule 2 chemicals. They do not necessarily cover all possible routes that could be used.

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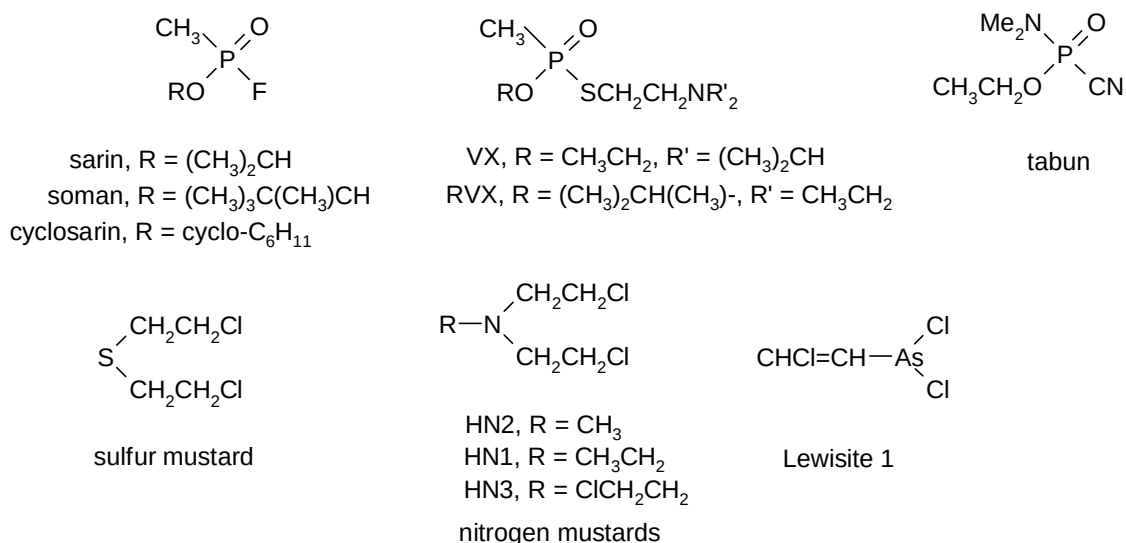


Figure 1. The CW agents on which most (but not all) of the scheduled chemicals were based.

2.2 Guidelines adopted for inclusion in the schedules

2.2.1 Schedule 1

The following criteria shall be taken into account in considering whether a toxic chemical or precursor should be included in Schedule 1:

- (a) It has been developed, produced, stockpiled or used as a chemical weapon as defined in Article II;
- (b) It poses otherwise a high risk to the object and purpose of this Convention by virtue of its high potential for use in activities prohibited under this Convention because one or more of the following conditions are met:
 - (i) It possesses a chemical structure closely related to that of other toxic chemicals listed in Schedule 1, and has, or can be expected to have, comparable properties;
 - (ii) It possesses such lethal or incapacitating toxicity as well as other properties that would enable it to be used as a chemical weapon;
 - (iii) It may be used as a precursor in the final single technological stage of production of a toxic chemical listed in Schedule 1, regardless of whether this stage takes place in facilities, in munitions or elsewhere;
- (c) It has little or no use for purposes not prohibited under this Convention.

2.2.2 Schedule 2

The following criteria shall be taken into account in considering whether a toxic chemical not listed in Schedule 1 or a precursor to a Schedule 1 chemical or to a chemical listed in Schedule 2, part A, should be included in Schedule 2:

- (a) It poses a significant risk to the object and purpose of this Convention because it possesses such lethal or incapacitating toxicity as well as other properties that could enable it to be used as a chemical weapon;

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- (b) It may be used as a precursor in one of the chemical reactions at the final stage of formation of a chemical listed in Schedule 1 or Schedule 2, part A;
- (c) It poses a significant risk to the object and purpose of this Convention by virtue of its importance in the production of a chemical listed in Schedule 1 or Schedule 2, part A;
- (d) It is not produced in large commercial quantities for purposes not prohibited under this Convention.

2.2.3 Schedule 3

The following criteria shall be taken into account in considering whether a toxic chemical or precursor not listed in other Schedules should be included in Schedule 3:

- (a) It has been produced, stockpiled or used as a chemical weapon;
- (b) It poses otherwise a risk to the object and purpose of this Convention because it possesses such lethal or incapacitating toxicity as well as other properties that might enable it to be used as a chemical weapon;
- (c) It poses a risk to the object and purpose of this Convention by virtue of its importance in the production of one or more chemicals listed in Schedule 1 or Schedule 2, part B;
- (d) It may be produced in large commercial quantities for purposes not prohibited under this Convention.

2.3 Lists of chemicals

These lists are extracted directly from the text of the Convention but with added structures, and added comments shown in *italic*.

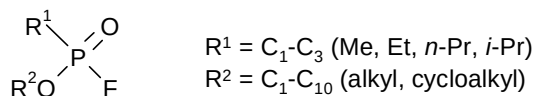
2.3.1 Schedule 1

A. Toxic chemicals: (CAS registry number)

Nerve agents

- (1) O-Alkyl ($\leq C_{10}$, incl. cycloalkyl) alkyl (Me, Et, n-Pr or i-Pr)
-phosphonofluoridates

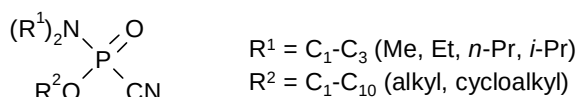
Commonly known as 'G agents' (but not GA)



e.g. Sarin: O-Isopropyl methylphosphonofluoridate (107-44-8)
Soman: O-Pinacolyl methylphosphonofluoridate (96-64-0)

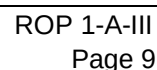
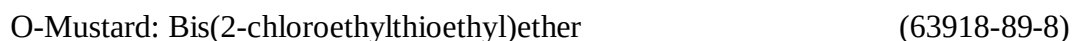
- (2) O-Alkyl ($\leq C_{10}$, incl. cycloalkyl) N,N-dialkyl (Me, Et, n-Pr or i-Pr)
phosphoramidocyanidates

Tabun (GA) and analogues



Chapter III. Chemicals

and corresponding alkylated or protonated salts



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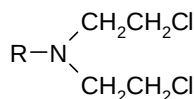
(6) Nitrogen mustards:

HN1: Bis(2-chloroethyl)ethylamine (538-07-8)

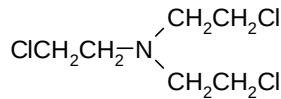
HN2: Bis(2-chloroethyl)methylamine (51-75-2)

HN3: Tris(2-chloroethyl)amine (555-77-1)

Salts of nitrogen mustards are not included



HN1, R = Et; HN2, R = Me

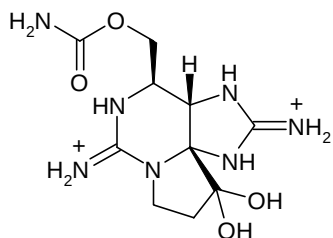


HN3

Toxins

(7) Saxitoxin (35523-89-8)

A marine toxin; analogues are not included



(8) Ricin (9009-86-3)

A heterogeneous glycoprotein extracted from the castor bean, approximate molecular mass 65 kDa.

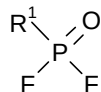
The Scientific Advisory Board to the Director-General of the OPCW has recommended the following definition be adopted for verification purposes:

All forms of ricin originating from Ricinus communis, including any variations in the structure of the molecule arising from natural processes, or man made modification designed to maintain or enhance toxicity, are to be considered ricin as long as they conform to the basic 'native' bipartite molecular structure of ricin that is required for mammalian toxicity, i.e. A and B chains linked only by a disulfide bond (A-S-S-B). Once the inter-chain S-S bond is broken or the protein denatured, it is no longer ricin.

B. Precursors:

(9) Alkyl (Me, Et, n-Pr or i-Pr) phosphonyldifluorides

Precursors for Schedule 1A(1) chemicals, G agents



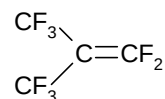
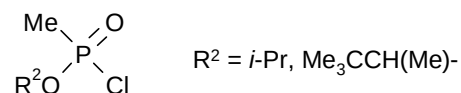
R¹ = C₁-C₃ (Me, Et, n-Pr, i-Pr)

e.g. DF: Methylphosphonyldifluoride (676-99-3)

Precursor for sarin, soman, cyclosarin

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- and corresponding alkylated or protonated salts



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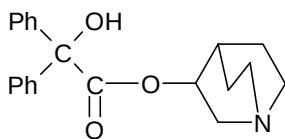
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- (3) BZ: 3-Quinuclidinyl benzilate (6581-06-2)

A previously weaponised centrally & peripherally acting incapacitant



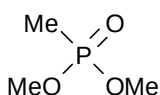
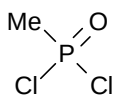
B. Precursors:

- (4) Chemicals, except for those listed in Schedule 1, containing a phosphorus atom to which is bonded one methyl, ethyl or propyl (normal or iso) group but not further carbon atoms.

e.g. Methylphosphonyl dichloride (676-97-1)

Dimethyl methylphosphonate (756-79-6)

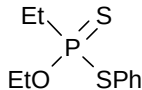
Precursors for Schedule 1A(1) chemicals, G agents



Exemption: Fonofos:

O-Ethyl S-phenyl ethylphosphonothiolothionate (944-22-9)

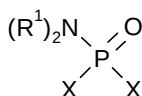
Commercial pesticide



N.B. This schedule defines only one substituent (alkyl) on phosphorus and includes P(III) and P(V) chemicals; it is therefore an unlimited set of compounds.

- (5) N,N-Dialkyl (Me, Et, n-Pr or i-Pr) phosphoramidic dihalides

Precursors for Schedule 1A(2) chemicals, tabun & analogues

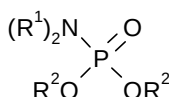


R¹ = C₁-C₃ (Me, Et, n-Pr, i-Pr)

X = halide

- (6) Dialkyl (Me, Et, n-Pr or i-Pr) N,N-dialkyl (Me, Et, n-Pr or i-Pr) -phosphoramidates

Possible precursors for, or impurities in, Schedule 1A(2) chemicals



R¹ = C₁-C₃ (Me, Et, n-Pr, i-Pr)

R² = C₁-C₁₀ (alkyl, cycloalkyl)

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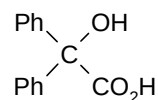
- (7) Arsenic trichloride (7784-34-1)

Precursor for Schedule 1A(5) chemicals, Lewisites



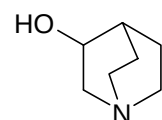
- (8) 2,2-Diphenyl-2-hydroxyacetic acid (benzilic acid) (76-93-7)

Precursor for BZ



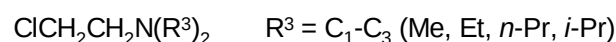
- (9) Quinuclidin-3-ol (1619-34-7)

Precursor for BZ



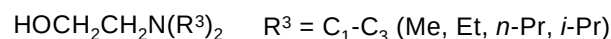
- (10) N,N-Dialkyl (Me, Et, n-Pr or i-Pr) aminoethyl-2-chlorides
and corresponding protonated salts

Precursors for Schedule 1A(3) chemicals, V agents



- (11) N,N-Dialkyl (Me, Et, n-Pr or i-Pr) aminoethane-2-ols
and corresponding protonated salts

Precursors for Schedule 1A(3) chemicals, V agents



Exemptions:

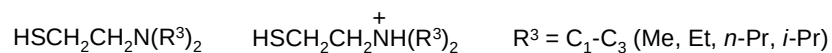
N,N-Dimethylaminoethanol and corresponding protonated salts (108-01-0)

N,N-Diethylaminoethanol and corresponding protonated salts (100-37-8)

These both have widespread industrial uses

- (12) N,N-Dialkyl (Me, Et, n-Pr or i-Pr) aminoethane-2-thiols
and corresponding protonated salts

Precursors for Schedule 1A3 chemicals, V agents



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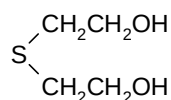
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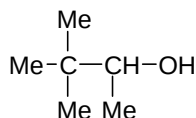
- (13) Thiodiglycol: Bis(2-hydroxyethyl)sulfide (111-48-8)

Precursor for sulfur mustard



- (14) Pinacolyl alcohol: 3,3-Dimethylbutan-2-ol (464-07-3)

Precursor for soman



2.3.3 Schedule 3

A. Toxic (*industrial*) chemicals:

- (1) Phosgene: Carbonyl dichloride, COCl_2 (75-44-5)
(2) Cyanogen chloride, CNCl (506-77-4)
(3) Hydrogen cyanide, HCN (74-90-8)
(4) Chloropicrin: Trichloronitromethane, CCl_3NO_2 (76-06-2)

B. Precursors:

Nerve agent precursors

- (5) Phosphorus oxychloride, $\text{P}(\text{O})\text{Cl}_3$ (10025-87-3)
(6) Phosphorus trichloride PCl_3 (7719-12-2)
(7) Phosphorus pentachloride PCl_5 (10026-13-8)
(8) Trimethyl phosphite $(\text{MeO})_3\text{P}$ (121-45-9)
(9) Triethyl phosphite $(\text{EtO})_3\text{P}$ (122-52-1)
(10) Dimethyl phosphite $(\text{MeO})_2\text{P}(\text{O})\text{H}$ (868-85-9)
(11) Diethyl phosphite $(\text{EtO})_2\text{P}(\text{O})\text{H}$ (762-04-9)

Vesicant precursors

- (12) Sulfur monochloride S_2Cl_2 (10025-67-9)
(13) Sulfur dichloride SCl_2 (10545-99-0)
(14) Thionyl chloride SOCl_2 (7719-09-7)
(15) Ethyldiethanolamine $\text{EtN}(\text{CH}_2\text{CH}_2\text{OH})_2$ (139-87-7)
(16) Methyldiethanolamine $\text{MeN}(\text{CH}_2\text{CH}_2\text{OH})_2$ (105-59-9)
(17) Triethanolamine $\text{N}(\text{CH}_2\text{CH}_2\text{OH})_3$ (102-71-6)

2.4 Number of compounds encompassed by generic structures

2.4.1 Schedule 1

The organophosphorus nerve agents covered by Schedules 1.A.1, 1.A.2 and 1.A.3 are defined by generic structures that allow any combination of substituents as defined by alkyl or cycloalkyl substituents R^1 , R^2 and R^3 . In the case of the P-alkyl substituent R^1 and the N substituent R^3 , there are only four possibilities, Me, Et, n-Pr and i-Pr. Where two R^3 groups are present on a nitrogen atom, they can be symmetrical or unsymmetrical, i.e. any combination of Me, Et, n-Pr and i-Pr. In the case of R^2 , which may contain 1–10 carbon atoms, it has been calculated that there are 25,524 possible carbon chains when all isomers (alkyl and cycloalkyl) are considered. For example, there are 512 possible linear and branched chain alkyl isomers, 8 simple cycloalkyl groups, plus additional possibilities for alkyl-substituted cycloalkyl analogues. When all possible permutations and combinations are considered, Schedule 1.A alone captures some 1,324,296 possible chemicals. These do not include protonated salts and unspecified alkyl quaternary salts of V agents (in the latter case the number of carbon atoms on the additional alkyl substituent on the nitrogen is not defined). Nerve agents can also exist as two or more enantiomers or diastereoisomers, but the Convention does not distinguish among these (and it would be impractical to resolve them in the context of a chemical weapon). Similar considerations apply to the precursors defined by Schedules 1.B. The structures of vesicants and saxitoxin are more specifically defined. Schedules 1.A.4, 1.A.5 and 1.A.6 include only 15 chemicals. No isomers are possible within the definitions, although Lewisites can exist as *cis* and *trans* isomers in relation to the double bond. A definition of what constitutes the heterogeneous proteinaceous toxin ricin has been proposed by the Scientific Advisory Board (SAB) of the OPCW.

The reality of which analogues are likely to be seriously considered as CW agents by a potential aggressor is of course somewhat different. It is very unlikely that obscure and expensive branched O-alkyl or alkylated cycloalkyl side chains would be considered for CW agents. Many analogues, e.g. P-propyl and O-*n*-alkyl (C_4 – C_{10}) G agents, have much lower toxicities than sarin and soman. Unsymmetrical N-alkyl substituents on GA or V agents are also unlikely because the starting materials are generally more expensive and difficult to obtain, and the agent is unlikely to offer any advantage.

2.4.2 Schedule 2

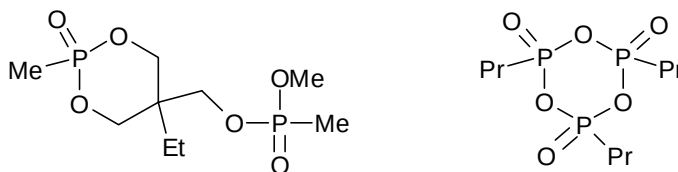
Similar arguments apply to the undefined quaternary salts of amiton in Schedule 2.A.1 and to the generic structures for nerve agent precursors defined in Schedules 2.B.4, 2.B.5, 2.B.6, 2.B.10, 2.B.11 and 2.B.12. Furthermore, Schedule 2.B.4 includes all chemicals other than those listed in Schedule 1 that contain a phosphorus atom to which is bonded one Me, Et, n-Pr or i-Pr group but not further carbon atoms. This constitutes an infinite set of compounds, as only one substituent on phosphorus is defined, and phosphorus can be in oxidation states +3 or +5 [P(III) or P(V)]. Some industrial chemicals and intermediates that fall within this category are used as flame retardants or their precursors. Two examples are shown below.

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2.4.3 Schedule 3

Schedule 3 refers to compounds in widespread industrial use, and all are specifically defined.

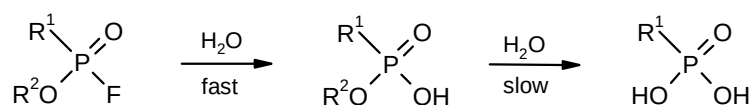
3. Degradation products of scheduled chemicals

Most CW agents are reactive electrophiles; i.e. they react predominantly with nucleophiles, which constitute a component of their biochemical mechanism of toxicity. In the environment (and in the body), they react with water to form more polar hydrolysis products, most of which have relatively low toxicity (in the case of mustards the primary degradation products are also precursors). These are important analytes with regard to verification, as they may provide more persistent indicators than the agents of prior use or production. In the case of sulfur mustard, oxidation of the S atom, in combination with hydrolysis, is an additional degradation pathway.

3.1 Nerve agents

3.1.1 Alkyl alkylphosphonofluoridates, Schedule 1A(1)

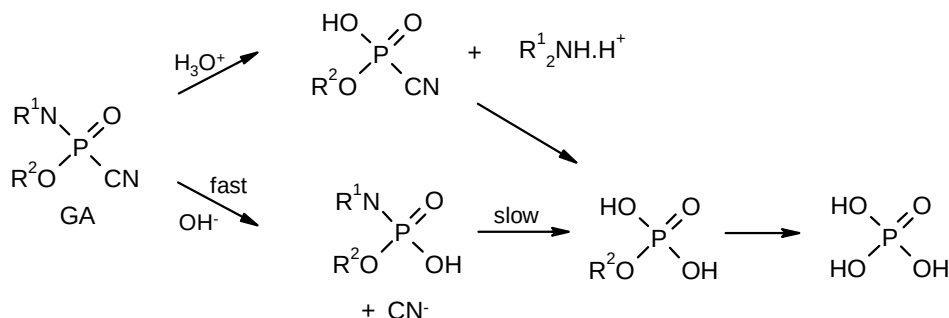
The main degradation products of alkyl alkylphosphonofluoridates are the primary hydrolysis products, alkyl alkylphosphonates (alternative name, alkyl alkylphosphonic acids), formed by displacement of fluoride. Further slow hydrolysis may occur to alkylphosphonic acids and the alcohol R_2OH , as shown below:



Alkyl alkylphosphonates and alkylphosphonic acids retain the P-alkyl substituent and are captured by the generic definition for Schedule 2.B.4 chemicals. The alcohols, which are also precursors, are not scheduled and in most scenarios have little value for verification purposes because of their widespread use for legitimate purposes. An exception is pinacolyl alcohol, the precursor alcohol for soman, which has very limited industrial use and is included in Schedule 2.

3.1.2 O-Alkyl N,N-dialkylphosphoramidocyanidates, Schedule 1A(2)

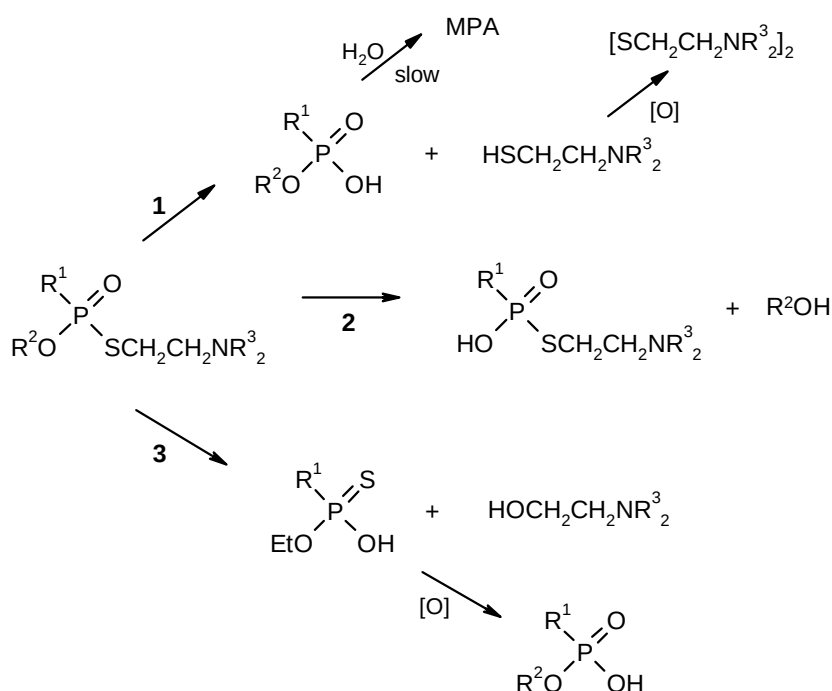
These nerve agents react with water via two different pathways, displacement of CN^- or $\text{R}^1_2\text{NH}_2^+$, depending on the pH. The initial hydrolysis products, O-alkyl N,N-dialkylphosphoramidates or alkyl phosphorocyanidates, hydrolyse further to alkyl phosphates.



Because this class of nerve agents does not contain a P-alkyl substituent, the degradation products are not scheduled chemicals. Some alkyl phosphoric acids, e.g. ethyl phosphoric acid, are ubiquitous environmental contaminants derived from pesticides and plasticizers.

3.1.3 O-Alkyl S-2-dialkylaminoethyl alkylphosphonothiolates, Schedule 1.A.3

In the case of V agents, hydrolysis may occur by three pathways (P-S, P-O and S-C cleavage), depending on pH, temperature and concentration, as shown in the scheme below. Pathways 1 and 2 predominate under basic conditions; pathway 3 is a minor pathway for VX under weakly basic conditions.



Pathway 1 produces alkyl alkylphosphonates, which are also hydrolysis products of alkyl alkylphosphonofluoridates. The hydrolysis product $\text{R}^1\text{P}(\text{O})(\text{OH})\text{SCH}_2\text{CH}_2\text{NR}^3_2$ derived

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from pathway 2 is still a Schedule 1.A.3 chemical ($R_2 = H$) because, at least in the case of VX, it retains high toxicity. The organophosphorus hydrolysis products from pathways 1 and 3 are captured by Schedule 2.B.4.

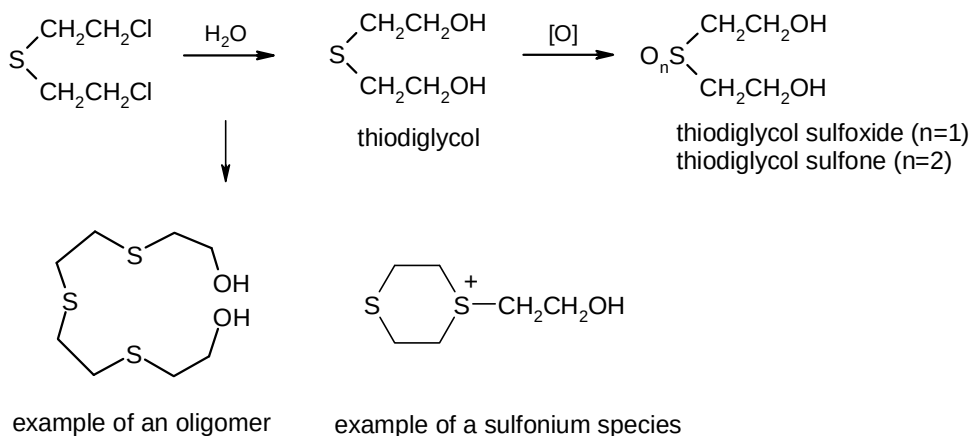
The hydrolysis products derived from the N,N-dialkylaminoethylthio side chain, $HOCH_2CH_2NR_2^3$ and $HSCH_2CH_2NR_2^3$, are included in Schedule 2.B as precursors, with the exceptions of $HOCH_2CH_2NMe_2$ and $HOCH_2CH_2NEt_2$, which are exempted because of widespread industrial use.

N,N-Dialkylaminoethylthiols, formed in pathway 1, oxidise to the non-scheduled disulfides in the environment.

3.2 Vesicants

3.2.1 Sulfur mustards, Schedule 1.A.4

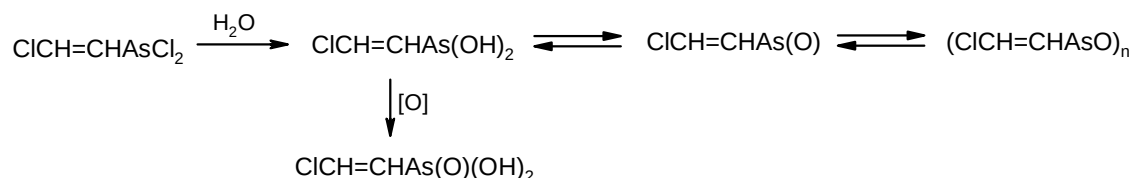
Under most conditions, sulfur mustard degrades predominantly by hydrolysis to thiodiglycol (Schedule 2 precursor). Thiodiglycol may be oxidised to the non-scheduled thiodiglycol sulfoxide, and more slowly to the non-scheduled sulfone. In the environment oxidation tends to stop at the sulfoxide, but if oxidative decontamination has been undertaken, further oxidation to the sulfone occurs. Depending on conditions and concentration, some more complex oligomeric products may be produced in the environment via sulfonium intermediates formed from intermolecular reactions between sulfur mustard and thiodiglycol.



Homologues/analogues of sulfur mustard degrade by analogous pathways. Where two sulfur atoms are present, mixed sulfide/sulfoxides/sulfones may be formed.

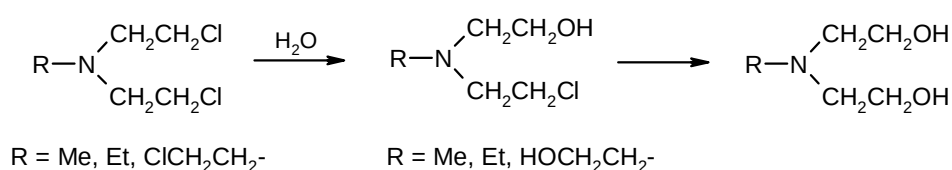
3.2.2 Lewisites, Schedule 1.A.5

Lewisites 1 and 2 degrade initially by simple hydrolysis of the As–Cl bond to give the corresponding non-scheduled As(III) acids, but the initial products may reversibly polymerise. In the environment As(III) degradation products are oxidised to non-scheduled As(V) compounds. The scheme for Lewisite 1 is shown below.



3.2.3 Nitrogen mustards, Schedule 1A(6)

Nitrogen mustards hydrolyse predominantly to the corresponding N-alkyldiethanolamines or triethanolamine (HN3), plus their intermediate partial hydrolysis products, depending on conditions. The ethanolamines are included in Schedule 3 as precursors. Triethanolamine in particular has widespread industrial use.



3.3 BZ, Schedule 2A(3)

BZ hydrolyses to benzoic acid and 3-quinuclidinol, both of which are Schedule 2.B precursors.

3.4 Some well-established degradation products

Some of the well-established degradation products of scheduled chemicals are not covered by the schedules, either as precursors or generic structures such as those defined in Schedule 2.B.4. Some of the more common non-scheduled degradation products are shown in Table 1. Some of these have been included in OPCW Proficiency Tests, although in a number of cases their spectra (as derivatives) are not in the OCAD. In addition to the chemicals shown in the table, many degradation products or impurities in weapon-grade sulfur mustard and its environmental residues have been identified. Examples are 2-chloroethyl vinyl sulfide, 2-chloroethyl vinyl sulfoxide, bis(2-chloroethyl) disulfide, 1,4-dithiane and its sulfoxides, 1,4-thioxane etc. Decontamination products of scheduled chemicals fall outside the scope of this chapter, but products derived from reactions of nerve agents with alkoxyalkanols, e.g. butoxyethanol and ethoxyethanol, under alkaline conditions (as used in several caustic decontaminants) have been used as spiking chemicals in proficiency tests.

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Table 1. Some well established but non-scheduled degradation products.

Name	Structure	Source
O-Alkyl N,N-dialkyl-phosphoramidates	$\begin{array}{c} \text{R}^1\text{N} \\ \diagdown \\ \text{P}=\text{O} \\ \diagup \\ \text{R}^2\text{O} \quad \text{OH} \end{array}$	Hydrolysis of Schedule 1.A.2 chemicals; tend to hydrolyse further.
O-alkyl phosphorocyanidates	$\begin{array}{c} \text{HO} \\ \diagdown \\ \text{P}=\text{O} \\ \diagup \\ \text{R}^2\text{O} \quad \text{CN} \end{array}$	As above; tend to hydrolyse further.
O-Alkyl phosphates	$\begin{array}{c} \text{HO} \\ \diagdown \\ \text{P}=\text{O} \\ \diagup \\ \text{R}^2\text{O} \quad \text{OH} \end{array}$	As above; also ubiquitous contaminants from pesticides & plasticizers.
Bis(N,N-dialkylaminoethyl) disulfides	$\text{R}^3_2\text{NCH}_2\text{CH}_2\text{S-SCH}_2\text{CH}_2\text{NR}^3_2$	Hydrolysis/oxidation of Schedule 1.A.3 chemicals; can be important environmental markers; corresponding tri- and tetra-sulfides may be formed.
N,N-Dialkylaminoethanesulfonic acids	$\text{R}^3_2\text{NCH}_2\text{CH}_2\text{S(O)}_2\text{OH}$	Hydrolysis/oxidation (e.g. from decontamination) of Schedule 1.A. 3 chemicals.
Bis(2-chloroethyl) sulfoxide; Bis(2-chloroethyl) sulfone	$\begin{array}{c} \text{CH}_2\text{CH}_2\text{Cl} \\ \diagdown \\ \text{O}_n\text{S} \\ \diagup \\ \text{CH}_2\text{CH}_2\text{Cl} \end{array} \quad n = 1, 2$	Oxidation of sulfur mustard (or minor impurities).
1,2-Bis(2-hydroxyethylthio)-ethane and homologues	$\begin{array}{c} \text{SCH}_2\text{CH}_2\text{OH} \\ \diagdown \\ (\text{CH}_2)_n \\ \diagup \\ \text{SCH}_2\text{CH}_2\text{OH} \end{array} \quad n = 1-5$	Hydrolysis products and precursors of sulfur mustard homologues.
1,2-Bis(2-hydroxyethylsulfinyl)-ethane; 1,2-bis(2-hydroxyethylsulfonyl)ethane; and homologues	$\begin{array}{c} \text{S(O)}_n\text{CH}_2\text{CH}_2\text{OH} \\ \diagdown \\ (\text{CH}_2)_m \\ \diagup \\ \text{S(O)}_n\text{CH}_2\text{CH}_2\text{OH} \end{array} \quad m = 1-5, n = 1, 2$	Hydrolysis/oxidation of sulfur mustard homologues; mixed sulfide / sulfoxide / sulfones are also possible.
Bis(2-hydroxyethylthiomethyl) ether	$\begin{array}{c} \text{CH}_2\text{SCH}_2\text{CH}_2\text{OH} \\ \diagdown \\ \text{O} \\ \diagup \\ \text{CH}_2\text{SCH}_2\text{CH}_2\text{OH} \end{array}$	Hydrolysis product & precursor of bis(2-chloroethylthiomethyl)-ether (oxidation products have not been reported).
Bis(2-hydroxyethylthioethyl)-ether	$\begin{array}{c} \text{CH}_2\text{CH}_2\text{SCH}_2\text{CH}_2\text{OH} \\ \diagdown \\ \text{O} \\ \diagup \\ \text{CH}_2\text{CH}_2\text{SCH}_2\text{CH}_2\text{OH} \end{array}$	Hydrolysis product & precursor of bis(2-chloroethylthioethyl)-ether

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Name	Structure	Source
Bis(2-hydroxyethylsulfinylethyl)-ether; Bis(2-hydroxyethyl-sulfonylethyl)ether	$\begin{array}{c} \text{CH}_2\text{CH}_2\text{S(O)}_n\text{CH}_2\text{CH}_2\text{OH} \\ \\ \text{O} \\ \\ \text{CH}_2\text{CH}_2\text{S(O)}_n\text{CH}_2\text{CH}_2\text{OH} \end{array}$ $n = 1, 2$	Hydrolysis/oxidation of bis(2-chloroethylthioethyl)ether
N-2-Chloroethyl-N-alkyl-ethanolamine	$\begin{array}{c} \text{CH}_2\text{CH}_2\text{OH} \\ \\ \text{R}-\text{N} \\ \\ \text{CH}_2\text{CH}_2\text{Cl} \end{array} \quad \text{R} = \text{Me, Et}$	Partial hydrolysis of HN1, HN2
N-2-Chloroethyl-N,N-diethanolamine	$\begin{array}{c} \text{CH}_2\text{CH}_2\text{OH} \\ \\ \text{HOCH}_2\text{CH}_2-\text{N} \\ \\ \text{CH}_2\text{CH}_2\text{Cl} \end{array}$	Partial hydrolysis of HN3
2-Chlorovinylarsonous acid	$\text{ClCH}=\text{CHAs(OH)}_2$	Hydrolysis of Lewisite 1
Bis(2-chlorovinyl)arsinous acid	$(\text{ClCH}=\text{CH})_2\text{AsOH}$	Hydrolysis of Lewisite 2
As(V) compounds	$\begin{array}{l} \text{CHCl}=\text{CHAs(O)(OH)}_2 \\ (\text{CHCl}=\text{CH})_2\text{As(O)(OH)} \\ (\text{CHCl}=\text{CH})_3\text{As(O)} \end{array}$	Hydrolysis / oxidation of Lewisites; oxidation products not well characterised
Benzophenone	PhCOPh	Degradation product of BZ

4. Old and abandoned chemical weapons

4.1. Scope

This section reviews old and abandoned chemical weapons, riot control agents and so-called non-lethal chemical weapons.

4.2. History of old and abandoned chemical weapons (OACW)

World War I was the first war in which chemical weapons (CW) were used on a large scale. Almost 66 million artillery shells containing CW were fired during WWI.

In WWII the usage of CW was less than in WWI. The Allied forces chose not to use CW to avoid retaliation and escalation of chemical warfare.¹ The only known occurrences of extensive usage of CWs during WWII were in China by Japanese forces and in Abessinia by Italy.

The development of nerve agents was in progress in Germany during WWII, but for some reason these new agents were not used. The German 'Blitzkrieg' tactic, i.e. quickly moving troops and less static frontlines, was possibly the reason why CW was not used. Also, the Geneva Protocol (1925) limited the usage of CW. Many of the unused munitions were left behind on battlefields and in storage facilities. Hundreds of shells are recovered every year from former WWI and WWII battlefields.²

4.3. Chemical weapons dumped at sea

After WWII the German army left behind large numbers of different types of chemical munitions, most of them in special storage areas. The subsequent destruction of these

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munitions was performed by burning or dumping them at sea. Many types of CW were disposed of by sinking them to the bottom of the Baltic Sea. Bornholm Basin and Little Belt were the main disposal areas. Sulfur mustard, Clark I, Clark II, adamsite, chloroacetophenone, phosgene, nitrogen mustards and tabun (Figure 2) were dumped in different types of bombs, drums and canisters. Finally, most of the contaminated boats were sunk with the last cargo.

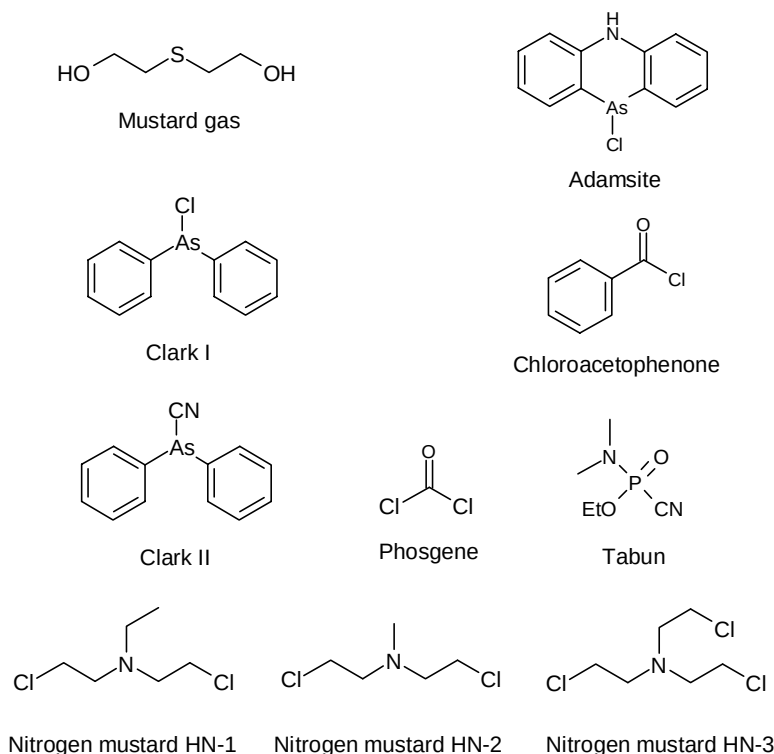


Figure 2. Chemical structures of some of the dumped chemical warfare agents.

Other areas were also used for dumping. In the Adriatic Sea, the Island of Ischia, near Bari, was used for dumping phosgene, hydrogen cyanide and cyanogen chloride bombs. Australia dumped their own and also Japanese munitions of mustard, tear gases, lewisite and solid adamsite in sea areas around Australia. The US dumped sulfur mustard, phosgene, hydrogen cyanide and cyanogen chloride into the basins near O'ahu Island.

The dumping areas were often marked in the nautical charts as 'forbidden areas' for anchoring and fishing. Over the decades, the restrictions lost their original importance, and many fishermen started to fish in the surrounding areas or even within the areas. Often, nets trawling the sea bed caught grenades of mustard, or leaked droplets (sometimes the size of 'footballs') containing polymerized mustard on their surface and liquid mustard inside the droplet. When trawl nets were handled the fishermen suffered skin and respiratory injuries. These accidents were frequent, and later the fishing boats were equipped with protective equipment such as gas masks. The frequency of these accidents is now decreasing, possibly due to better knowledge of the hazard and better protective equipment. More than 65 years later, most of the agent that has been in contact with water is assumed to have hydrolysed (although a polymeric skin may protect sulfur mustard). The risk, however, remains; unbroken canisters and bombs may still contain active agent, particularly sulfur mustard.

Russia and Germany are constructing a gas pipeline between their countries across the sea bed of the Baltic Sea. As the pipeline was planned to cross the Little Belt Basin, the environmental inspection of dumped chemical weapons was obligatory. Several hundreds of sediment samples were analyzed, but no traces of CW were found.³ Some of the degradation products of arsenical warfare agents were found at trace level.

4.4. Old chemical weapons in ground areas

In China, sulfur mustard and phosgene munitions were abandoned when Japanese troops departed. There has been a significant amount of discussion of the destruction of these OACW and remediation of contaminated land areas. Due to the high cost of these operations the issue remains open.

Belgium and The Netherlands still recover phosgene, mustard and chlorine gas munitions from WWI and WWII battlefields. During WWI the ground areas turned into mud and many of the grenades and bombs were buried. These are still being uncovered during agricultural and construction activities.

5. Riot control agents and their use

While the classical riot control agents (RCAs) such as CR and CS are used for riot control purposes, their use for war purposes is forbidden by the CWC.⁴ Newer riot control agents, such as pepper spray, which contains capsaicin as an active agent, are under continuous development.

6. New developments of 'non-lethal agents'

The increased threat of terrorist attacks, and particularly incidents in which hostages may be held, has encouraged the development of non-lethal weapons, in most cases from known RCAs or pharmaceutical and veterinary drugs. The CWC states that any toxic chemical, if it has been used in military operations or developed for military use, is a chemical weapon (the so-called 'General Purpose Criterion' in CWC). The verification of production and usage of such chemicals may be problematic because verification procedures have not yet been established (and because verification is complicated by their allowance for 'law enforcement purposes').

Plant breeding is being used to increase the concentration of capsaicin in pepper plants in order to develop more effective pepper sprays. Also, synthetic capsaicin or an analogue can be used for more effective pepper sprays and easier large-scale production. The technology for the method of delivery is also under development.

In 2002, 50 armed Chechen terrorists took 850 hostages in a Moscow theatre and demanded the withdrawal of Russian forces from Chechnya and an end to the Second Chechen War. After a two-and-a-half day siege, Russian Spetsnaz forces pumped an unknown chemical agent (later identified by a Russian Minister as derived from the opioid drug fentanyl) into the building's ventilation system. In the ensuing rescue operation, approximately 170 people died, although most of the civilians were saved.

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Fentanyl is commonly used in anaesthesia and is typically approximately 100 times more potent than morphine. Derivatives of fentanyl can be up to 10,000 times more potent than morphine. Dissemination can be achieved using aerosol technology.

Malodorants have been developed for non-lethal warfare. According to the Israeli police, the new crowd-control method, which they called a 'skunk bomb', was used for the first time in the village of Naalin. Israeli police showered Palestinian demonstrators with liquid containing a malodorant using a water-spraying device, forcing most to rush off to change their clothes.

Malodorants can be based on skunk spray, which contains 38–44% of (E)-2-buten-1-thiol, 18–26% of 3-methyl-1-butanethiol and 3–12% of 2-quinolinemethanethiol and corresponding thioacetates. These chemicals can be produced synthetically on a large scale.

In the future the biotechnology and development of new drugs may be a source of new 'non-lethal' warfare agents.

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Chapter IV. Quality control

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1. Introduction

This chapter describes the testing laboratory quality assurance system in general. The quality requirements and aspects of individual methods and techniques are discussed in detail in the corresponding chapters.

To become an OPCW (Organisation of the Prohibition of Chemical Weapons) designated laboratory, a laboratory must successfully participate in the official Proficiency Tests organised by the OPCW annually and must have an accredited quality assurance system certified by an internationally recognised accreditation body.¹

2. Standard ISO/IEC 17025:2005

ISO/IEC 17025 is an international standard being commonly used by testing laboratories. The latest version is ISO/IEC 17025:2005 – general requirements for the competence of testing and calibration laboratories. The standard can be purchased directly from the International Organization for Standardization (ISO)² or through the national accreditation body.

The standard ISO/IEC 17025:2005 contains all the requirements that testing and calibration laboratories must meet in order to demonstrate that they follow a quality system, are technically competent, and are able to produce technically valid results. The standard has two parts: management requirements and technical requirements. Management requirements are primarily related to the operation and effectiveness of the quality management system. Technical requirements include factors that determine the correctness and reliability of the tests and calibrations performed.

Another important document is CITAC/Eurachem Guide to Quality in Analytical Chemistry.³ This guidance is intended to help laboratories implement the quality assurance program. It concentrates on technical issues, with an emphasis on areas in which a particular interpretation is required for chemical testing and analysis.

3. Quality Assurance and Accreditation

The value of chemical analysis depends on the level of confidence that can be placed in the results. Appropriate quality assurance means that a laboratory is able to prove that they have adequate facilities and equipment for the chemical analysis and that the work is carried out by trained and competent staff members in a controlled way.

As mentioned above, internationally recognized accreditation is a prerequisite to obtaining the status of an OPCW-designated laboratory. Accreditation is a procedure by which an authoritative body gives formal recognition that a body or person is competent to carry out specific tasks. It is an instrument that provides confidence in technical competence, but the accreditation itself cannot guarantee that 100% of the individual results will be reliable. However, commercial, industry, and public authorities recognize accreditation as significant evidence of competence in carrying out testing work.

Accreditation is granted to a laboratory after the assessment conducted by the accreditation body. The assessment normally includes an examination of the analytical procedures (e.g., official OPCW Proficiency Tests), the quality system, and the quality documentation. The assessment can also include a “performance audit,” in which the laboratory is required to analyse samples supplied by the accreditation body.

4. Analysis Process

The analysis process has several steps, beginning with sampling and ending with the delivery of the analysis report to the customer (or the evaluator). A laboratory should have a written analysis strategy or process that includes at least the following sub-tasks: sampling (if appropriate), receiving the samples, sample preparation, preliminary screening analysis, technical analysis, identification of the target chemicals, and reporting. The procedures for testing the analytical equipment, including calibrating the pipettes and balances as well as controlling the freezers, should also be described. The necessary planning and follow-up meetings and reviews can also be included in the written analysis process. The analysis process is described in Section I, Part B.

5. Training and Competence of Personnel

The laboratory must ensure that qualified, competent, and trained personnel perform the sample analysis. Only analysts who can demonstrate the necessary competence or who are adequately supervised may perform tests on samples. The continuous training and retraining of analytical skills are also important elements of personnel competence. The laboratory shall maintain an up-to-date record of the competence and training of each employee involved in sample analysis. The purpose of this is to prove that every member of the staff has been adequately trained and is competent to perform sample analysis. In order to ensure that the laboratory is capable of performing analysis at any time, it should also define and train deputies or backup staff for the key areas of analysis work.

6. Chain of Custody

If the laboratory wishes to prove that it has reliable results, it must ensure that the chain of custody procedure has been strictly followed. Chain of custody is used in most chemical sampling and analysis situations to maintain the integrity of the results of sample analysis

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by providing documentation of the coding, control, transfer, and analysis of samples. The chain of custody procedure starts with sample collection and continues through the destruction of the sample. Chain of custody is described in Section 4, Chapter III.

7. Summary

If a laboratory wishes to ensure outside bodies, e.g., customers, that it can produce valid and reliable results, it must follow a written quality system procedure and must have documented data from every step of the analysis process. The laboratory must show that the sample analysis is done with functioning and tested equipment and by trained and qualified personnel. Chain of custody is one of the key elements of the sample analysis. The laboratory must be able to demonstrate that the principles of the chain of custody have been followed throughout the analysis process and that there is no possibility of cross-contamination or a mix-up of the samples.

8. References

1. V. Dubey, S. Velikeloth, M. Sliwakowski, G. Mallard, "Official Proficiency Tests of the Organisation of the Prohibition of Chemical Weapons: Current Status and Future Direction", *Accred Qual Assur* **14** (2009) 431-437
2. <http://www.iso.org/iso/home.htm>
3. CITAC/EURACHEM GUIDE: Guide to Quality in Analytical Chemistry, An Aid to Accreditation, Edition 2002
(www.eurachem.org/guides/pdf/CITAC%20EURACHEM%20GUIDE.pdf)

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Chapter I Screening

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1. Scope

This chapter describes the screening step in the analysis of chemicals related to the Chemical Weapons Convention (CWC). In this step, various characteristics of the CWC-related chemicals are used to detect them among the irrelevant background chemicals. The next step is the actual identification, which is discussed in the following chapters. Generally, the methods aim to detect around 1 mg/kg concentration levels. In practice, many of the screening methods are able to detect traces of relevant chemicals at the µg/kg level.

This chapter discusses the screening of only CWC-related small molecule weight chemicals. The analysis of toxin molecules is discussed separately in Section 3, Part F.

2. Summary of the methods

A wide range of different methods is available for the screening step in the analysis of CWC-related chemicals. The gas chromatography (GC)-based techniques, GC using selective detectors and gas chromatography–electron ionisation mass spectrometry (GC–EI/MS) are the most used and most sensitive screening techniques. It should, however, be remembered that other techniques, such as liquid chromatography–mass spectrometry (LC–MS) and nuclear magnetic resonance spectroscopy (NMR), can be used very efficiently in screening. One major advantage these two methods have over GC-based

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techniques is the ability to screen aqueous samples with a minimal amount of sample preparation.

The most important feature of a screening technique is to be able to detect a CWC-related chemical among other chemicals in the samples. This ability is based on various chemical properties, but the screening methods are divided into two distinct groups: targeted screening methods and non-targeted screening methods. In the former group, only a defined set of chemicals is being searched for, typically using mass spectrometry (either GC or LC-based). In the latter group, a more universal property, such as an element (typically phosphorus, sulphur or nitrogen) or a common part of the molecule is being searched for.

Typically, the instruments and methods used for screening (especially spectrometric methods) are the same as those used for actual identification. Only the way of looking at the analysis results is different.

The screening must be based on a pre-existing screening strategy, which contains lists of all ions to be searched for. The screening strategy must be carefully planned and evaluated in advance. It is important to understand what the advantages and limitations of each screening technique are and, especially, the screening step as a whole.

This chapter gives an overview of different screening methods and their possibilities. It is recommended that the following related chapters are consulted for more details on the methods discussed in this chapter:

- 2-A-III Derivatisation for gas chromatography
- 3-B-I Gas Chromatography (GC)
- 3-B-II Gas Chromatography–Electron Ionisation/Mass Spectrometry (GC–EI/MS)
- 3-C-II Liquid chromatography–API Mass Spectrometry
- 3-C-III Liquid chromatography–Tandem Mass Spectrometry (LC–MS/MS)
- 3-E-II Basic ^{31}P NMR techniques
- 3-E-III Basic ^{19}F NMR techniques
- 3-E-IV Nonselective 1D ^1H – ^{31}P HSQC technique
- 3-F-I Analysis of saxitoxin: LC–MS/MS
- 3-F-VII Analysis of ricin: LC–MS/MS

3. Screening strategies

To enable a reliable and comprehensive analysis of CWC-related chemicals, it must be assured that all relevant chemicals can be detected and identified. One of the tool sets applied to this task is screening samples using various techniques.

Each screening technique relies on one or more properties of the CWC-related chemicals, which separate them from irrelevant chemicals. Table 1 shows examples of selected CWC-related chemicals and some typical screening methods for each chemical.

It should be remembered that the screening step has to be a combination of various techniques. The nature of the sample impacts the selection of screening methods. Paragraph 7 will discuss the selection of screening methods and paragraph 8 gives two examples of screening strategies applied by a designated laboratory for analysis of CWC-related chemicals.

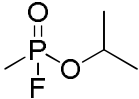
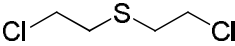
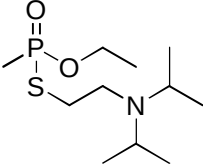
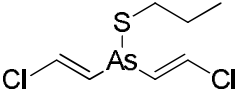
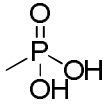
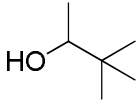
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Table 1. Examples of selected CWC-related chemicals and typical screening methods.

Chemical	Structure	Property	Screening method
Sarin		Phosphorus	Selective GC detector NPD, FPD(P), PFPD(P), AED ³¹ P NMR
		Protons with phosphorus coupling	1D ¹ H- ³¹ P HSQC
		Fluorine	¹⁹ F NMR
		Group ions	GC-EI/MS
Mustard		Sulphur	Selective GC detector FPD(S), PFPD(S), AED
		Chlorine	Selective GC detector AED, ECD
		Group ions	GC-EI/MS
VX		Phosphorus	Selective GC detector NPD, FPD(P), PFPD(P), AED ³¹ P NMR
		Protons with phosphorus coupling	1D ¹ H- ³¹ P HSQC
		Sulphur	Selective GC detector FPD(S), PFPD(S), AED
		Nitrogen	Selective GC detector NPD, PFD(P(N)), AED
		Group ions	GC-EI/MS LC-MS
Lewisite I <i>derivatised using propane-1-thiol*</i>		Sulphur	Selective GC detector FPD(S), PFPD(S), AED
		Arsenic	Selective GC detector PFPD(As), AED
		Group ions	GC-EI/MS
Methylphosphonic acid		Phosphorus	Selective GC detector NPD, FPD(P), PFPD(P), AED <i>requires derivatisation*</i> ³¹ P NMR
		Protons with phosphorus coupling	1D ¹ H- ³¹ P HSQC
		Group ions	LC-MS
Pinacolyl alcohol		Fragment ions & retention time	GC-EI/MS

* See the chapter "Derivatisation for gas chromatography" (2-A-III) for more information

3.1 Targeted and non-targeted screening

There are different ways of approaching the screening task: one possibility is to use the most general method as possible (*non-targeted screening*) and the other is to try to look for individual chemicals (*targeted screening*). In reality, many of the screening methods are somewhere in between these two extremes.

The advantage of non-targeted screening is that the method searches for a common property of a chemical or a chemical group and, therefore, one does not have to know each of the chemicals to be screened. On the other hand, the screening result is very tentative and requires confirmation by other methods.

Very typical non-targeted screening methods are mass screening methods such as selective GC detectors (paragraph 4.1). While still quite general, ^{31}P NMR (paragraph 6) gives already more information on the actual structure of the target molecule.

Mass spectrometric methods are typically more or less targeted screening methods. In the least targeted modes, MS methods are looking for chemical groups based on characteristic ions (paragraph 5.1) such as m/z 99 for sarin-type chemicals, m/z 97 for dialkyl alkylphosphonates or m/z 109 and 63 for mustard-type agents (see Figure 1 for structures). When LC-MS is used in a non-targeted mode (using in-source CID), the characteristic group ions can be screened in a similar way to the GC-EI/MS screening.

In the most targeted modes, single chemicals are being searched for. This method, for example, can be used in GC-EI/MS to find chemicals that are not detectable using selective GC detectors (paragraph 5.1.1). The most selective MS screening method is the multiple reaction monitoring (MRM) of CWC-related chemicals in LC-MS (paragraph 5.2.2).

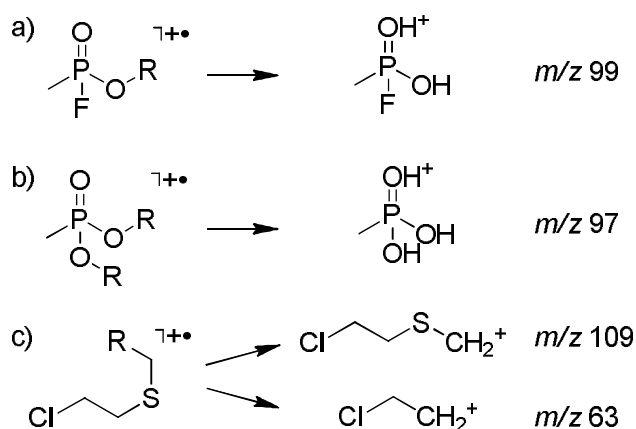


Figure 1. Typical group ions used for GC-EI/MS screening of CWC-related chemicals: a) sarin-type chemicals, b) dialkyl alkylphosphonates and c) mustard-type chemicals.

4. Elemental Screening

The most comprehensive coverage in screening could be gained by screening for the heteroatoms in the CWC-related chemicals. Almost all of the relevant chemicals contain at least one of the following elements: phosphorus, sulphur or nitrogen. Additionally many of the intact agents also contain halogen atoms (either chlorine or fluorine).

Naturally, a huge amount of irrelevant chemicals will also contain one of the above elements, but the number of possible target chemicals is normally reduced to a much more manageable number. Figure 2 shows an example of the removal of the hydrocarbon background from a sample and the detection of mustard gas using a flame photometric detector (FPD) in sulphur mode.

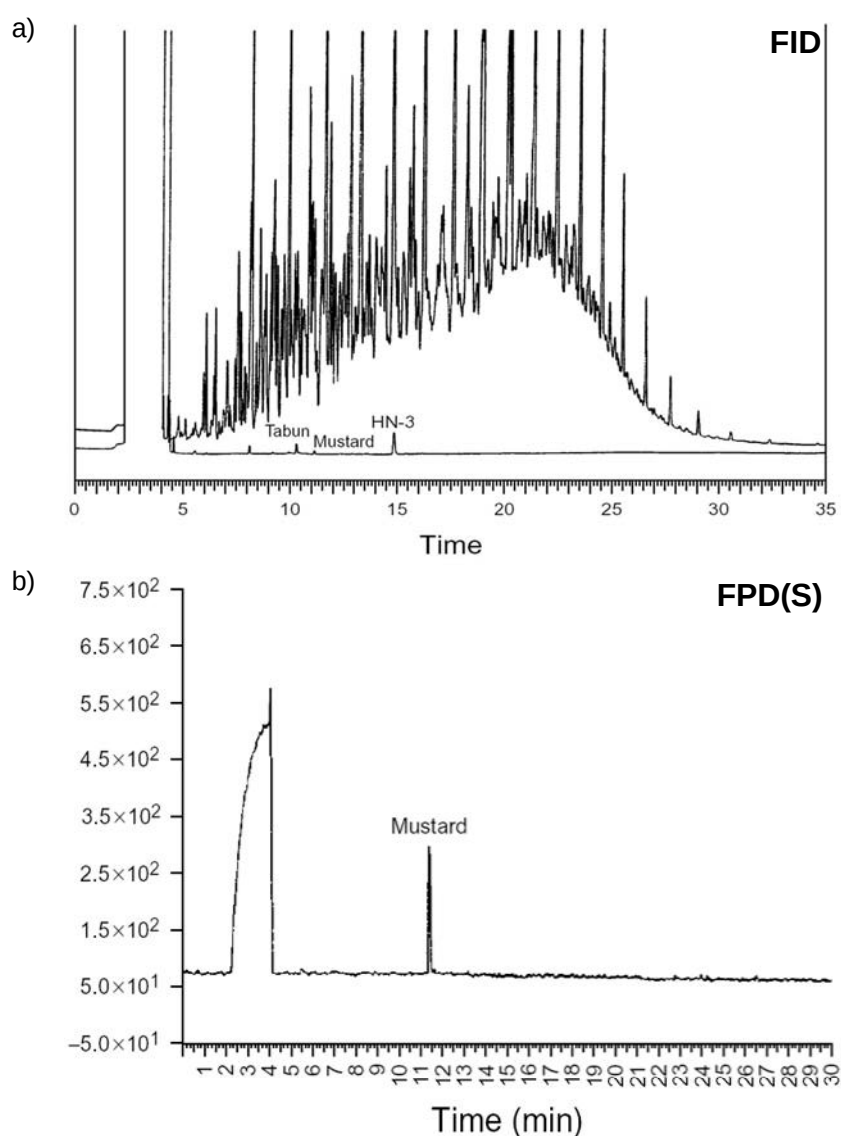


Figure 2. Chromatograms of a spiked sample analysed using a) flame ionisation detector (FID) and b) flame photometric detector (FPD) in sulphur mode. The chromatogram a) also shows the spiked sample without the hydrocarbon background.

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4.1 GC-based screening techniques

A single gas chromatograph instrument can be configured with different options to provide multiple simultaneous screening results. Typically, this requires splitting the column flows to the different detectors by using a press-fit connector (recommended only for GC use) or a Deans switch (recommended for GC–MS use).

There are several practical limitations in managing GC techniques using multiple channels. The main limitation is that it is not advisable to use more than two columns per injector or to split the column flow between more than two detectors. Examples of possible GC detectors or their combinations are presented in Table 2.

Table 2. Examples of practical detector combinations for a single GC instrument.

Combinations	Elements	Comments
NPD + FPD(S)	N+P, S	Gives detection of P, S and N, but is not able to differentiate P and N.
GC–EI/MS + dual FPD	MS, P, S	This instrument is compact, but reserves both mass spectrometer and selective detectors for all analysis. Suitable for mobile laboratory units. Does not give detection of N.
dual FPD	P, S	Does not require splitting of the GC flow and, therefore, gives maximum sensitivity for both channels. Does not give detection of N.
pulsed FPD	P, S, As, N	Gives detection of P, S, N and As, but the differentiation between the elements is a compromise with the detection sensitivity. Not able to detect all four elements simultaneously with both best selectivity and sensitivity.
AED	all	Gives the detection of most elements present in the CWC-related chemicals: C, H, As, N, O, P, S, F, Cl. May require several runs to cover all the elements as the plasma conditions may have to be varied. More complex instrument than other GC detector and more expensive to use due to the high purity required and consumption of gases.

4.1.1 Flame photometric detector (FPD)

A flame photometric detector (FPD) uses an optical band-pass filter¹ for the selection of the detected element: phosphorus, FPD(P), or sulphur, FPD(S). To convert the detector from one element to the other requires a change of filter. The detector is more sensitive to phosphorus than sulphur.²

***Note!** The photomultiplier tube is very sensitive to ambient light. Therefore, it has to be switched off before opening the detector. Also, the open end of the photomultiplier tube should always be covered (e.g., by using aluminium foil) when it is not connected to the detector body.*

In some detector systems, two photomultiplier tubes are fitted into one detector body. This way both phosphorus and sulphur can be monitored simultaneously without splitting the column flow.

On some occasions, a **very strong** phosphorus peak can create a small ghost peak on the sulphur channel. This is due to an increased width of the phosphorus emission peak so that the signal is also increased in the sulphur wavelength. To avoid the wrong interpretation of these signals, one has to pay attention to the relative signal intensities. Comparison to a known sample, for example, a column performance test, helps to evaluate the correct signal levels. For sulphur, however, the detector is not linear and suffers from quenching by co-eluting hydrocarbons.

4.1.2 Pulsed FPD (PFPD)

The pulsed flame photometric detector (PFPD) is able to detect phosphorus, sulphur, arsenic and nitrogen-containing compounds. The ability to detect arsenic or nitrogen-containing chemicals makes the PFPD very useful for screening CWC-related chemicals.

PFPD exploits the time-dependence of the photon emission process during combustion of the compounds in the flame, which is not continuous, but turned on and off every 300 ms. The flame pulse profile over time is dependent on the elements in the chemicals, and thus the signal may be selectively associated to an element. The PFPD does not suffer the quenching of co-eluting hydrocarbon chemicals. Unfortunately, the emission spectra of the elements frequently overlap with others.

***Note!** Avoid the use of chlorinated solvents with the PFPD detector due to the risk of photomultiplier degradation.*

The sensitivity of the PFPD is better as compared to the FPD detector for a single element in conditions optimised specifically for the detection of that element.

The main optimisable parameters affecting the selectivity and sensitivity of the PFPD are the gate width and the gate delay. Careless maximisation of the sensitivity towards a single

¹ Filter wavelengths: 394 nm for sulphur, 526 nm for phosphorus.

² Agilent specifications for 7890A gas chromatograph: <60 fg of phosphorus/s; <3.6 pg of sulphur/s.

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element may result in false positive detections, due to a decrease of selectivity. Conversely, the optimisation of selectivity is as detrimental to sensitivity. It is important to realise that subtle variations in these parameters of the detector have a strong influence on signal emission, acquisition and processing.

The simultaneous detection of all four elements is, in practice, impossible due to the different parameter values depending of each element. Table 3 lists optimised parameters for a typical PFPD detector.

Table 3. Optimised values for the different elements for a typical PFPD detector (GC Varian 3800).

Element	Gate Delay	Gate Width
Sulphur	6	20
Phosphorus	4	10
Nitrogen	3.5	8
Arsenic	6	20

4.1.3 Nitrogen-Phosphorus Detector (NPD)

The nitrogen-phosphorus detector (NPD) gives signals for both phosphorus and nitrogen simultaneously, which means that it is not possible to say whether a chemical contains only phosphorus, only nitrogen or both.

An NPD is very selective and more sensitive towards phosphorus than towards nitrogen. The sensitivity of the detector depends strongly on the temperature of the NPD bead and the flow rate of the detector gases. In addition, some models are not compatible with chlorinated solvents and/or silylation reagents and, in this case, the analysis of silylated samples (i.e., after derivatisation using BSTFA) results in the reduced performance of the detector. This is due the silicon-containing residue depositing on the collector surface of the detector. Due to this, it is recommended that the silylated samples be run only after the other samples.

4.1.4 Atomic Emission Detector (AED)

The atomic emission detector (AED) is particularly well-suited to screening complex samples for CWC-related chemicals containing heteroatoms such as phosphorous, sulphur, nitrogen, chlorine and arsenic.

This detector uses microwave-induced plasma in which molecules eluting from the chromatographic system are atomised in excited states. The relaxation of excited atoms results in the emission of photons at wavelengths characteristic to various elements, which are, in turn, detected by means of a spectrometer.

The AED is a sensitive technique for the detection of carbon, hydrogen, sulphur, phosphorus, chlorine, nitrogen, arsenic and silicon (for silylated chemicals). Fluorine and oxygen can also be detected but with a much poorer limit of detection.

During the acquisition, specific reagent gases are required for optimal sensitivity for different elements. The elements form groups where the reagent gas mixture is the same and the used wavelengths are close to each other, as shown in Table 4. Elements within

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one group can be analysed in the same chromatographic run. If it is necessary to analyse elements from more than one group, several runs have to be made.

Table 4. Most common emission lines used for screening CWC-related chemicals. Elements in the same group can be run during the same injection.

Group	Element	Wavelength (nm)	Reagent gas(es)
1	Carbon	193	O ₂ and H ₂
	Sulphur	181	
	Nitrogen	174	
2	Carbon	496	O ₂
	Chlorine	479	
	Bromine	478	
	Hydrogen	486	
3	Phosphorous	186	H ₂ and He high flow
	Arsenic	189	
4	Silicon	252	O ₂ , H ₂ and He high flow
5	Oxygen	171	H ₂ and 10% CH ₄ /90% N ₂
6	Fluorine	690	H ₂

The selectivity between the elements is usually very high, although some interference can arise because of the proximity of characteristic wavelengths (e.g., chlorine and bromine; nitrogen and carbon).

For optimal performance, periodic maintenance operations such as discharge tube and ash filter replacement are required.

Four GC–AED chromatograms from a spiked sample are shown in Figure 3. The four elements can be detected by running three runs: first for carbon and sulphur (*group 1*), second for chlorine (*group 2*) and third for phosphorus (*group 3*). It should be noted that the carbon chromatogram is quite close to what a GC–FID chromatogram would be.

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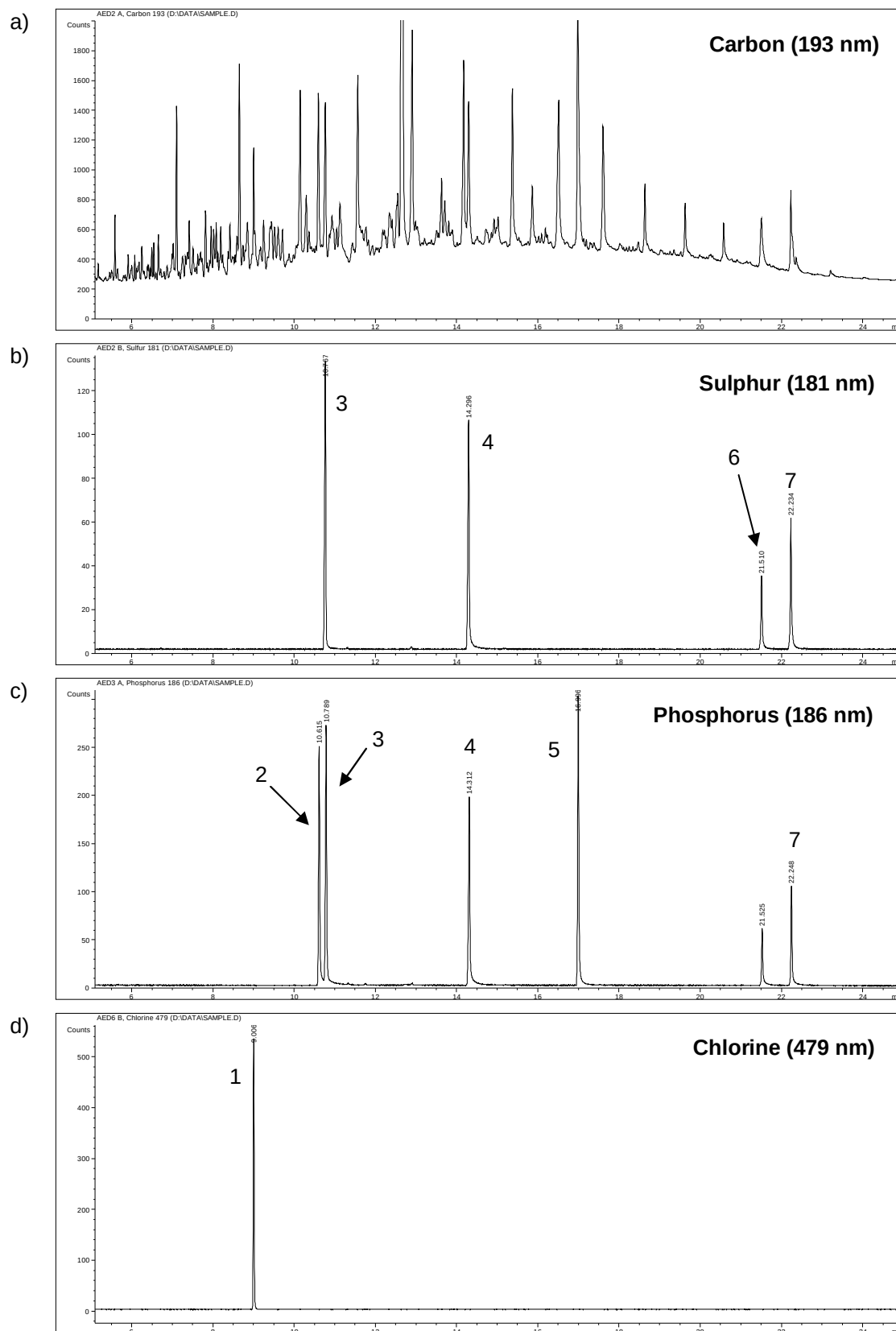


Figure 3. Chromatograms of a spiked sample analysed by GC-AED on a) carbon line at 193 nm, b) sulphur line at 181 nm, c) phosphorous line at 186 nm and d) chlorine line at 479 nm. The comparison of the retention times for different peaks shows the presence in the sample of one chlorine-containing chemical (1), two phosphorus-containing chemicals (2 and 5) and four chemicals with both sulphur and phosphorus (3, 4, 6 and 7).

4.1.5 Electron Capture Detector (ECD)

Some CWC-related compounds containing chlorine atoms, e.g., vesicants, are detectable with an ECD, but the sensitivity for some compounds is rather low depending on the electron affinity of the compounds.

The sensitivity of the ECD towards fluorine is quite poor. Therefore, NPD or FPD(P) would give better sensitivity for most CWC-related fluorine-containing chemicals.

The ECD can be considered as moderately sensitive towards vesicants, but, in most cases, not selective enough. This is especially true for environmental and biological samples, which often contain irrelevant ECD-active chemicals, which will interfere with the detection of the CWC-related compounds.

4.2 Other methods

4.2.1 Inductively Coupled Plasma (ICP) Spectroscopy and ICP–MS

Some CWC-related compounds containing special elements like arsenic can be detected by ICP–MS. The sensitivity is quite good for arsenic, which can be detected in the low ppb-range even in diluted samples.

The technique is quite simple to use. Water samples can be run directly or after dilution with diluted nitric acid. Organic samples or solids can be analysed after digestion in nitric acid followed by oxidation with hydrogen peroxide. In this case, the samples are normally diluted with water to a nitric acid concentration in the range of 0.65% to 6.5%.

The ICP–MS analysis can be done either as a single element analysis or as a multi-element analysis. In the latter case, it will often be possible to verify the presence of other heteroatoms such as phosphorous and sulphur.

5. Mass spectrometric screening

Mass spectrometric screening has to rely on searching for ions typical of the target chemicals. These ions can be a molecular ion (typically in LC–MS), a fragment ion or an MRM transition. Various levels of screening methods can be selected from relatively non-targeted to specifically targeted methods.

5.1 GC–EI/MS-based screening

As electron ionisation almost always produces fragment ions, the screening methods for GC–EI/MS typically rely on more or less characteristic fragments.

5.1.1 Screening phosphorus-, nitrogen- and sulphur-deficient chemicals

Even though selective detectors for phosphorus, nitrogen and sulphur are often used for screening CWC-related samples, all relevant chemicals are not detected with these detectors. The reason for this is that there are some relevant chemicals that do not contain any of these three elements. The number of these chemicals is so small that they can be manually screened using their typical ions. The list of these chemicals and the specific ions for screening are shown in the Table 5. Some of the chemicals can be found with GC only as derivatives. Figure 4 shows an example of manual screening of 3,3-dimethylbutan-2-ol using its characteristic ions.

Note! *HCN, phosgene and PFIB elute before normal organic solvents. Either headspace or split analysis should be used when these chemicals are suspected.*

Note! *Chloropicrin and 3,3-dimethylbutan-2-ol (i.e., pinacolyl alcohol) elute very close to the solvent peak. These two chemicals will not be seen if the solvent delay time is too long.*

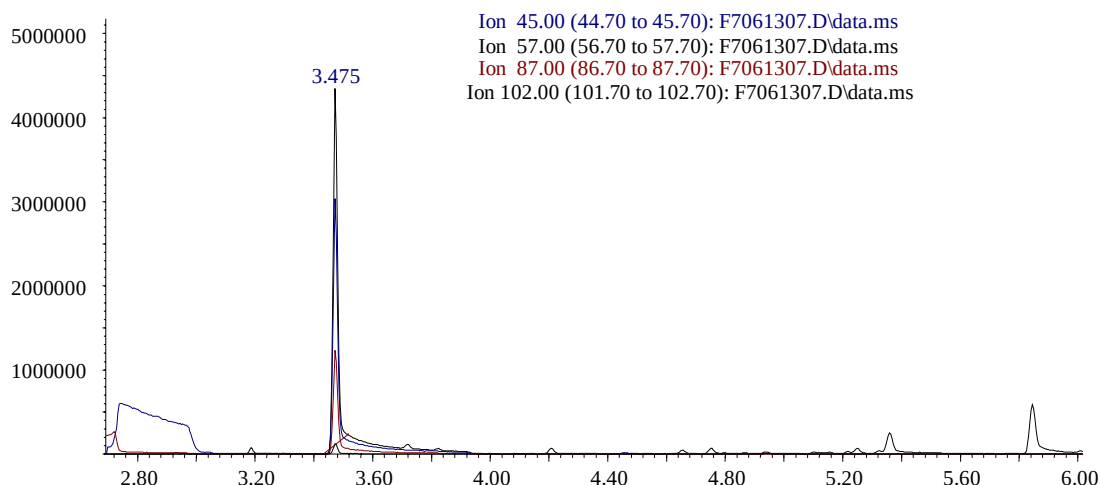


Figure 4. Screening of 3,3-dimethylbutan-2-ol with characteristic ions m/z 45, 57, 87 and 102. The retention time (here 3.475 min) is one of the major identification criteria as there are several other alcohols almost producing this spectrum. The solvent delay should be kept as short as possible to allow the detection of this early eluting chemical. [GC program: 40 °C (1 min) – 10 °C/min – 280 °C (5 min); splitless injection (1 min splitless time)].

RECOMMENDED OPERATING PROCEDURES FOR CWC-RELATED ANALYSIS**Section 1.** General**Part B.** Analytical Strategy**Chapter I.** Screening**Table 5.** Specific ions for chemicals that are not detected with phosphorus, nitrogen and sulphur selective detectors.

Chemical	Derivative*	Screening ions (<i>m/z</i>)	Notes
HCN	<i>none</i>	15, 26, 27	Elutes before solvent; use either headspace or split analysis <i>See Note 1</i>
Phosgene	<i>none</i>	35, 63, 65, 98	Elutes before solvent; use either headspace or split analysis <i>See Note 1</i>
PFIB	<i>none</i>	93, 181, 200	Elutes before solvent; use either headspace or split analysis
Chloropicrin	<i>none</i>	82, 117, 119, 121	NO ₂ group does not show with NPD Check the Rt carefully; same ions are found in CCl ₄
3,3-Dimethylbutan-2-ol (pinacolyl alcohol)	<i>none</i>	45, 57, 87, 102	Check the Rt carefully; other C ₆ alcohols produce very similar spectra TMS derivative would be covered by the large reagent peak
BZ	Silylation (TMS)	110, 255	Often also gives mono-TMS derivative in addition to the expected di-TMS
Benzilic acid	Methylation	77, 105, 183	May also be formed in methanol-containing samples
Lewisite I	Silylation (TMS)	255, 329	
	DMT	229, 290	Same derivative as for Lewisite II <i>See Note 2</i>
	1,3-butyldithiol	107, 176, 286	
	Propane-1-thiol	150, 176, 286	<i>See Note 2</i>
Lewisite II	Butane-1-thiol	164, 204, 314	<i>See Note 2</i>
	DMT	229, 290	Same derivative as for Lewisite I <i>See Note 2</i>
	1,3-butyldithiol	145, 203, 272	
	Propane-1-thiol	145, 272, 274	<i>See Note 2</i>
Arsenic trichloride	Butane-1-thiol	203, 229, 286	<i>See Note 2</i>
	DMT	229, 107, 121	<i>See Note 2</i>
	Propane-1-thiol	109, 151, 193	<i>See Note 2</i>
	Butane-1-thiol	197, 253, 342	<i>See Note 2</i>

* See the chapter “Derivatisation for gas chromatography” (2-A-III) for more information

Note 1: The mass range has to start lower than normal

Note 2: Detected with FPD(S), but normally screened using GC–EI/MS

5.1.2 AMDIS-based screening

It is recommended that AMDIS screening also be used for all samples analysed with GC–EI/MS. AMDIS is a software designed for the deconvolution of GC–EI/MS data followed by automated spectral cleaning (i.e., removal of background) and subsequent searching for target chemicals in a library. Originally, this tool was intended for the use by OPCW inspectors for on-site GC–MS analysis, but it has also proven very successful for off-site laboratory use.

The major advantage of the software is the combination of automatic deconvolution and library search of each of the components in the GC run. The system can often find relevant low-concentration components that would be missed by a human analyst.

The automatic library search is also the weakest point of the system. First, if a spectrum of a chemical is not present in the library, it is not possible to detect the chemical. Second, the library search – as normal – may result in several close matches, which might lead to false positive identification without careful manual checking of the results.

AMDIS is also able to perform automatic calculations of retention indices for all components of the run. For this calculation, it is necessary to calibrate the retention indices using an alkane mixture [typically from typically octane (C₈) to tetracosane (C₂₄)] which is included in the normal quality control mixture for GC and GC–MS [see discussion in Chapter “*Gas Chromatography–Electron Ionisation/Mass Spectrometry (GC–EI/MS)*” (3-B-II)]. The combination of mass spectral search and retention index search can solve most of the problems with the false positive arising from closely related chemicals.

The normal retention index calculation is always limited to values between the first and last component of the retention index standard series – typically from C₈ to C₂₄. AMDIS offers an easy – but unofficial – way to extrapolate for very early eluting chemicals such as 3,3-dimethylbutan-2-ol (example shown in Figure 5 below). This helps in differentiating it from other similar alcohols.

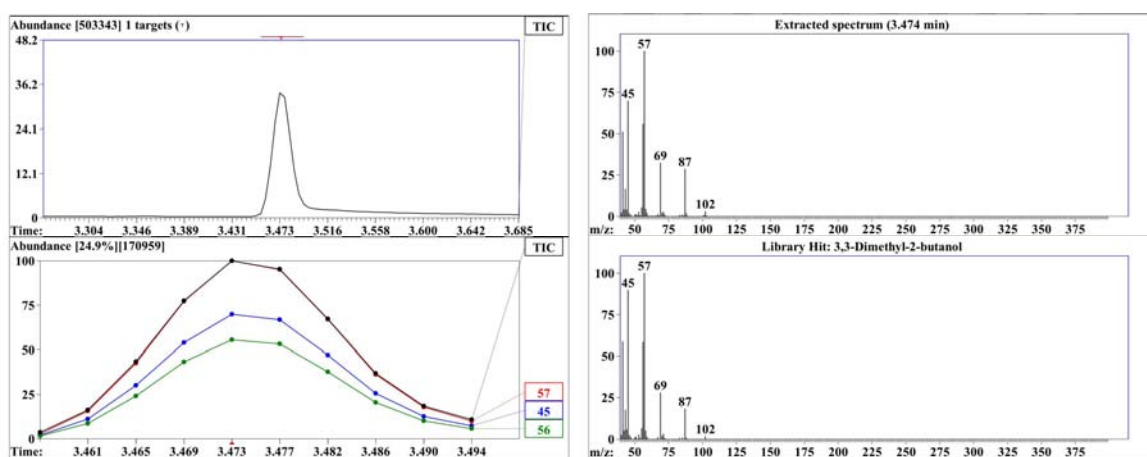


Figure 5. Screening result for 3,3-dimethylbutan-2-ol with AMDIS software. *Top left:* total ion chromatogram, *bottom left:* ion chromatograms for the 3,3-dimethylbutan-2-ol peak, *top right:* extracted spectrum (i.e., spectrum cleaned up by AMDIS) from the sample and *bottom left:* library spectrum from OCAD. The match factor for these spectra is 94 (maximum 100) and the extrapolated retention index is 756.6.

5.1.3 Group ion-based screening

In addition to the target screening discussed in the paragraph 5.1.1, the mass spectrometer can be used to screen groups of chemicals based on characteristic ions for each group as shown in Figure 2, above.

These ions can be used to find chemicals in complex samples if their presence is suspected for any reason, based on GC or NMR screening, for example. Sometimes, this may be necessary to help locate a low concentration target peak in a high background.

Appendix 1 of the chapter “*Gas chromatography–electron ionisation/mass spectrometry (GC–EI/MS)*” (3-B-II) contains the specific ions of the most important groups of phosphorus-containing CWC-related chemicals.

Note! *Group ions are not always very selective: ion m/z 99 used for sarin-type chemicals is also found in phosphates, which are irrelevant chemicals.*

5.2 LC–MS-based screening

In contrast to GC–MS, LC–MS cannot normally benefit from element selective detectors as discussed above. Therefore, LC–MS-based screening has to rely more on the mass spectral behaviour of the target chemicals. With modern LC–MS/MS systems, a variety of scan modes are available for the screening of discrete chemicals with known structures (‘knowns’) as well as chemicals or groups of chemicals with a common basic structure with unknown structures and molecular weight (‘unknowns’) in samples. Generic strategies for the application of LC–MS equipment for screening of CWC-relevant chemicals will be given in this chapter. For further information on LC–MS analysis and the ions used for screening, refer to the chapter “*Liquid chromatography-atmospheric pressure ionisation/mass spectrometry*” (3-C-II).

Atmospheric pressure ionisation (API) techniques, including electrospray ionisation (ESI) and atmospheric pressure chemical ionisation (APCI), are the most commonly used ionisation techniques. As ESI and APCI are very soft ionisation techniques, deprotonated or protonated molecules or adduct ions with a low degree of fragmentation are generated depending on the analysed chemical and its polarity. MS/MS techniques such as single reaction monitoring (SRM), multiple reaction monitoring (MRM) or *in-source* collision induced dissociation (CID) can be used for fragmenting and detecting known, discrete chemicals or unknown chemicals in combination with a corresponding in-house spectral database. However, the availability of a database is not a prerequisite for a successful screening and identification procedure.

For the latest generation of mass spectrometers, which provide high resolution mass spectrometry (HRMS), such as the hybrid quadrupole time-of-flight mass spectrometer with MS/MS capability (Q–TOF) or the Orbitrap, screening will make use of the high mass accuracy these instruments provide. These instruments deliver resolutions $R(\text{FWHM}) > 10\,000$ and thus allow for the determining of the m/z values of ions with very high accuracy of lower than 5 ppm. Specifically, Q–TOFs can be operated at higher scan speeds. These properties allow one to:

- measure in full scan and MS/MS mode simultaneously in one LC run (e.g., Q–TOFs)
- provide empirical formula of unknown chemicals while screening for knowns and unknowns using m/z values with extremely narrow mass windows, typically in the mDa-range

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- identify CWC-related chemicals based on CWA-specific elements in their empirical elemental composition, such as S, P, N, Cl, As and F

To apply a successful LC–MS screening strategy for CWC-related chemicals in environmental samples, one has to find which specific chemical properties of CWC-related chemicals and which technical properties of the LC and the mass spectrometer can be used for screening.

5.2.1 Chemicals

To define appropriate screening methods, the nature and specific properties of the different chemicals must be taken into account in the first place. Common chemical properties have to be identified and searched for by a screening method. For a simplified discussion of different screening strategies, CWC-related chemicals can be roughly divided into ‘knowns’ and ‘unknowns’.

At first glance, the ‘known’ chemicals seem to pose the least problems for screening and detection, because their structure and molecular weight are known and, therefore, the retention time (t_R) and the MS/MS spectra can be determined. For this type of chemical, an MRM-based method can be established, with transitions using the molecular ion as well as one or two fragment ions.

The ‘unknowns’ pose an inherently difficult problem: how to screen for chemicals which have unknown structures and molecular weights? This situation arises in all labs where environmental samples or biomedical samples are analysed. Many approaches are therefore published. Most of the proposed solutions to this so called ‘general unknown screening’ (GUS) problem make use of at least known common substructure elements, or of characteristic isotopic distributions and other known, common physicochemical properties of the ‘unknowns’. The same holds true for CWC-related chemicals: substructures and fragment ions typical for distinct substance classes can be used as markers for the screening process. It is important to notice that this group contains an unknown high number of chemicals. These chemical class specific fragment ions are discussed in the chapter “*Liquid chromatography (LC) separation for mass spectrometry*” (3-C-I).

5.2.2 Mass spectrometric techniques for CWC-related screening

Besides the chemical properties of the CWC-related chemicals, the technique of the mass spectrometer also defines the screening strategy. While for ‘conventional’ instruments, several LC runs are necessary to screen a sample in different scan modes, the HRMS instruments need only one run in positive mode and one run in negative mode to gather all necessary data for screening. These instruments permit measuring in the full scan mode and acquiring MS/MS data at the same time. Such data sets contain all information for the screening of known and unknowns as well as for a retrospective analysis. Additional MS/MS experiments are often not necessary.

5.2.2.1 ‘Conventional’ LC–MS screening

When screening is based on the known molecular weight, the known retention time and the known MS/MS spectra, it is referred to as **target screening**. The screening of an

‘unknown’ must be based on properties common to the respective chemical family and it is referred to as **non-target screening**.

Target screening using SRM/MRM

The method of choice for the detection of known chemicals in complex matrices is the SRM/MRM-method. It is widely used for the detection and quantification of toxins, metabolites, drugs, dopes and environmental pollutants.

The procedure to develop and optimise an SRM/MRM method entails the following steps and can be quite labour intensive (Q1 and Q3 refer to quadrupoles of a triple quadrupole mass spectrometer):

1. Measure the MS/MS spectrum of the reference chemical preferably using the quasi-molecular ion (e.g. $[M+H]^+$) as the precursor ion (Q1).
2. Based on the MS/MS spectrum, choose specific fragment masses for Q3. Normally, the most intense fragment ion is chosen (‘quantifier’). Preferentially, more than one transition (Q1→Q3) per chemical should be used for unambiguous identification (‘quantifier’ and ‘qualifier(s)’). Fragments formed by the loss of water are not recommended for MRM transitions.
3. Use split-flow injection (same flow as in the LC–MS run) to optimise MRM-parameters such as declustering potential, collision energy, collision gas pressure and quadrupole and ion trap potentials. The influence of the parameters on the MRM signal is dependent on the type and manufacturer of the mass spectrometer used.
4. Check if the optimised method is still specific for the chemical of interest (a matrix injection without the chemical should yield no signal).
5. Determine the LOD with a serial dilution.

If the number of MRM transitions, i.e., the number of screened chemicals is high, a scheduled MRM-method should be used. But be careful not to exclude transitions that can occur at several retention times during an LC run when setting the retention time regions.

Non-target screening using LC–MS with in-source CID

In conventional MS/MS experiments, the Q1 filter is set to the precursor ion mass of the known target chemical. As the molecular mass of ‘unknowns’ is not known, the MS/MS mode cannot be applied. For general unknown screening, in-source CID in the electrospray source can be used. For that purpose, the ions are accelerated sufficiently in the ‘high-pressure region’ of the mass spectrometer to produce fragments when they collide with background gas molecules. This technique allows for the screening of fragment ions without knowing the molecular mass of the parent ion. Even a single quadrupole mass spectrometer would be sufficient. An example of in-source CID fragmentation for alkyl phosphonates is shown in Figure 6.

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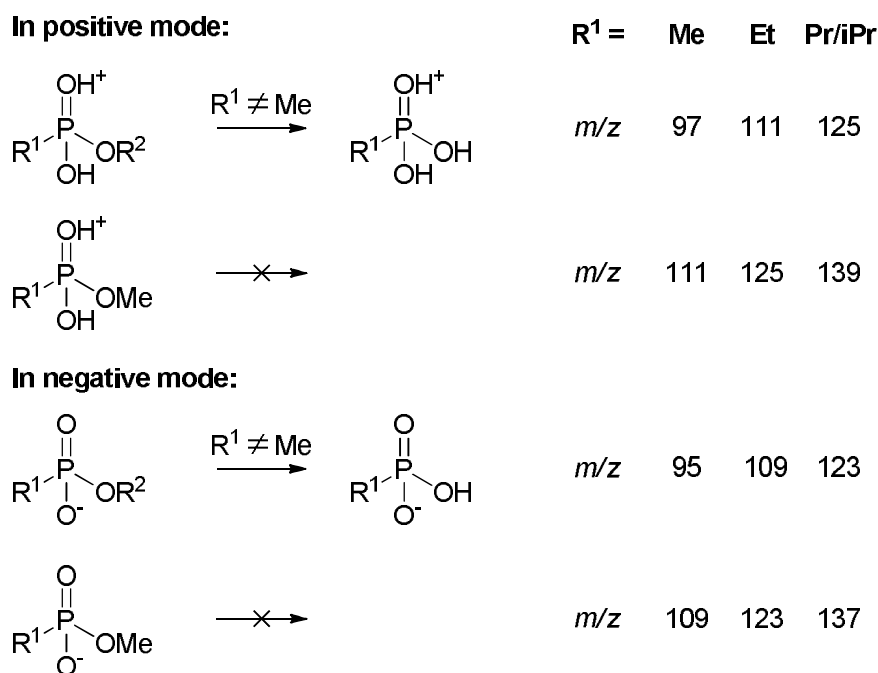


Figure 6. An example of in-source fragmentation of alkyl alkylphosphonates in both positive and negative ion modes. For methyl esters, the protonated molecules are observed instead of fragment ions. (For scheduled chemicals: R^1 = methyl, ethyl, propyl or isopropyl; R^2 = C_1 – C_{10} alkyl (including cycloalkyl)).

In-source CID occurs between orifice and skimmer. The degree of fragmentation can be controlled by adjusting the orifice-skimmer potential on electrospray-based instruments. This type of CID is ideal for non-target screening. The potential has to be set to a compromise value at which most of the expected classes of chemicals produce sufficient in-source CID fragments.

The screening ions used for LC–MS screening are discussed in the chapter “*Liquid chromatography–atmospheric pressure ionisation/mass spectrometry*” (3-C-II).

Possible screening methods using quadrupole or ion trap instruments

Taking into account the capabilities of today’s LC–MS systems, several approaches could, in fact, be used. Some of these methods are summarised in Table 6 together with their advantages and disadvantages.

Considering the advantages and disadvantages of all of the listed screening methods, the full scan/extracted ion chromatogram (EIC) method is a favourable choice for non-target screening due to its non-selective property (all ions are detected and can be used for screening – this in contradiction to MRM, where only selected ions are detected). For the EIC method, the ion source parameters which control the extent of the in-source CID must be optimised and checked for all chemical classes that are to be screened. Compromise values should then be used, under which all chemicals undergo in-source CID fragmentation to a certain degree. In principle, a single quadrupole instrument would be sufficient for this type of screening. If HRMS instruments are used, the screening becomes even more specific.

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SRM/MRM for non-target screening

The boundary between targeted and non-targeted screening is not sharp. If a class of chemicals produces identical in-source CID fragments, their m/z values can be used in an MRM method to detect all chemicals of this class.

As an example, when fragmenting alkyl methylphosphonates (hydrolysed sarin-type nerve agents) under appropriate CID conditions, fragment m/z 95 $[\text{MeP}(\text{O})(\text{OH})\text{O}]^-$ in the negative and m/z 97 $[\text{MeP}(\text{OH})(\text{OH})\text{OH}]^+$ in the positive ion mode is generated. Under normal CID conditions, these chemicals further decay into fragment m/z 79 $[\text{PO}_3]^-$ in the negative mode or m/z 79 $[\text{MeP}(\text{O})\text{OH}]^+$ in the positive mode. Thus, a SRM method using the m/z 95 $\rightarrow$ 79 (in negative mode) transition will detect all alkyl methylphosphonates that fragment into m/z 95/97 and 79, respectively, provided the ion source parameters are set to generate these in-source CID fragments. A subsequent precursor ion scan of m/z 95 or m/z 97, respectively, can reveal the molecular weight of the intact chemical and structural elements can be determined. As the retention time of the chemicals is not known, this type of transition should not be included in a scheduled MRM method, unless the respective transition is monitored in each time segment.

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Table 6. Screening strategies for chemicals using a 'conventional' LC–MS instrument.

Principle	Description	Procedure	Screening	Pros	Cons
Extracted ion chromatogram (EIC)	Pseudo-molecular ions and specific in-source CID fragments	Full scan with compromise in-source CID parameters to detect common fragment ions of the chemical classes involved as well as the quasi-molecular ions	Non-target Target	Reliable principle Can be automated Transparent method Fast: only 2 measurements necessary (pos. and neg.) Low false-positive, false-negative rate Applicable on single/triple quad and ion trap instruments	Limited sensitivity and selectivity Background Co-elution
SRM/MRM-experiments	Q1: quasi-molecular ion Q3: fragment	SRM- or MRM-scan, respectively	Target Non-target (chemical class-specific in-source CID fragments)	Sensitivity Selectivity Automatable Up to 800 chemicals can be scanned	Individual parameter set for each chemical required Data limited to scanned ions
SRM-MS/MS and MS/MS database	SRM: see above MS/MS measurement triggered by information-dependent acquisition (IDA)	Survey scan: SRM/MRM Dependent scan: MS/MS	Target	Sensitivity Selectivity Automatable	Database needed Difficult to find universal IDA-parameters Data limited to scanned ions
Precursor scan-MS/MS	Precursor scan of typical in-source CID fragments Subsequent MS/MS scan of detected parent ion	Survey scan: Precursor scan with optimised fragmentation Dependent scan: MS/MS	Target Non-target	Automatable Applicable to chemical classes M _w is obtained	Collision energy- for precursor scan is chemistry dependent False-negative results possible
SRM/MRM-precursor scan	SRM transition of typical in-source CID fragments triggers a precursor ion scan	Survey scan: SRM-transition of typical in-source CID fragments Dependent scan: Precursor scan	Non-target (compounds with known, specific in-source CID fragments)	Sensitivity Selectivity Chemical classes	Only for chemical classes with identical in-source CID fragments Only as an addition to an existing non-target screening method
In-source CID-precursor scan	Scan for typical in-source CID-fragments Higher ion signals trigger precursor scans	Survey scan: Full scan with optimised fragmentation Dependent scan: Precursor scan	Non-target Target	Automatable For general unknown screening No database necessary Fragment rate can be high, detection of quasi-molecular ion not necessary	Signal intensity dependent on degree of CID fragmentation Background Co-elution
Full scan MS and data mining software	Full scan spectra are processed by numerical algorithms to extract ions from background noise	Full scan spectra are numerically processed	Target Non-target	All ions are detected Data can be analysed retrospectively	False negative results possible Eventually increased number of false-positive results
Full scan MS, HRMS, data library	Full scan spectra are measured. High mass accuracy and isotopic distribution reveal the molecular formula of unknowns	Full scan spectra at high mass accuracy and resolution	Target Non-target	All ions are detected, data can be analysed retrospectively Measurement can lead directly to identification. Search in high resolution data library	HRMS instrument necessary High mass accuracy data library

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Example of a screening strategy for a hybrid LC–MS system

The strategy presented in Figure 7 was developed for a hybrid triple quadrupole/linear ion trap system (AB MDS SCIEX 3200QTRAP) and has been successfully applied in OPCW Proficiency Testing in Spiez Laboratory in Switzerland.

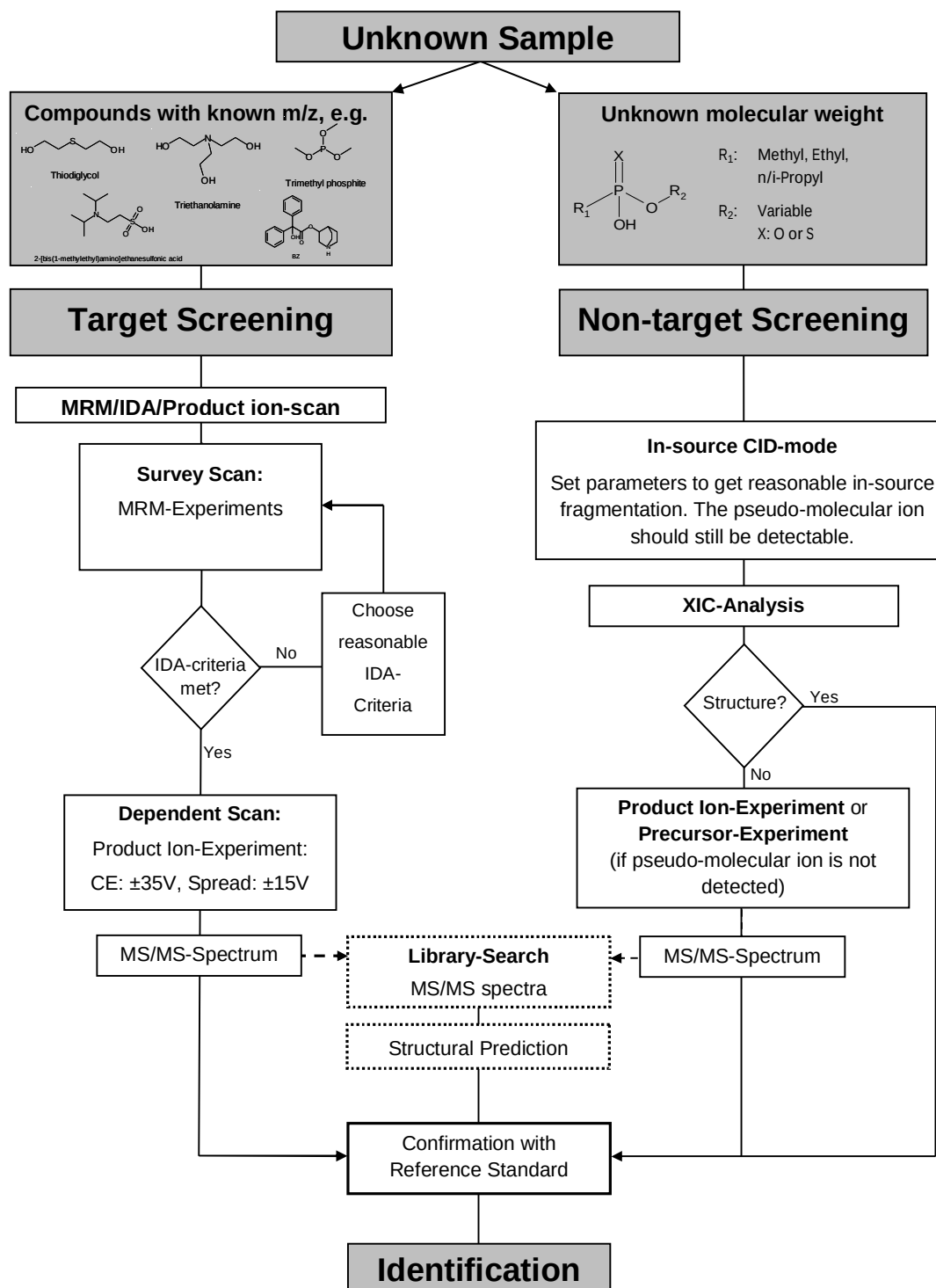


Figure 7. Screening strategies and scan modes for 'conventional' LC–MS systems. All measurements have to be done in the positive as well as in the negative ion mode. Dashed lines indicate the implementation of a data library in the screening process.

5.2.2.2 Screening using HRMS instruments

The instruments with high mass resolution and mass accuracy allow a different approach in screening. In Q-TOF instruments, due to their high scan speed, it is possible to acquire full scan MS and MS/MS data simultaneously.

The respective methods are dependent on the mass spectrometer manufacturer and can be used for both target and non-target screening.

The high resolution and mass accuracy allows for setting the mass window for extracted ion chromatograms so narrow (typically < 10 mDa), that even in a full scan spectrum, co-eluting chemicals do not impair the signal, and almost no background is seen in the EICs. In addition to this, the HRMS instruments produce exact masses (< 5 ppm), which can be used to calculate the elemental composition of the ions. Additionally, isotopic distribution for typical common fragment ions, specific fragment ions and quasi-molecular ions can be obtained.

It should be remembered that a product ion spectrum will not show any isotopic peaks if the resolution of the first MS stage (e.g., Q1 in a Q-TOF instrument) is unit resolution or better. It is normally possible to lower the resolution of the first stage to also allow the isotopic ions through. This will increase the background noise in full scan data, but the effect can be compensated by using the narrow window EICs.

Depending on the type of instrument, the two screening methods – target and non-target – can be carried out simultaneously. For these instruments, databases containing exact masses of chemicals, exact fragment ion masses and isotopic distributions should be used.

To confirm the identity of chemicals, a reference chemical has to be measured and the data compared with that of the unknown sample. In this respect, there is no difference between the ‘conventional’ and the HRMS screening approach.

5.2.2.3 Screening problems

Besides the chemicals of interest, the very sensitive mass spectrometers also detect matrix components. As an example, an often occurring interference should be addressed briefly. Reoccurring sources of problems are the polyethers, the almost ubiquitous polyethylene (PEG) and polypropylene glycol (PPG). They are especially problematic, because they can be easily detected by LC-MS in positive ion mode. The oligomers can be protonated but also easily form adducts with cations such as NH_4^+ , Na^+ or K^+ and in-source CID fragments are often present. All these ions produce a considerable set of m/z values that can be detected almost everywhere in the total ion chromatogram (TIC).

Further, matrix components can suppress the signals of the target chemicals. The presence of matrix components in environmental samples can thus impair the screening process, either leading to false-positive or false-negative results. This possibility has to always be taken into account, but a comprehensive discussion of the role of the matrix and its components in LC-MS analyses are beyond the scope of this chapter.

6. NMR-based screening techniques

Screening for the heteroatoms like phosphorus and fluorine in the CWC-related chemicals is also feasible using NMR spectroscopy.

Direct observation of phosphorus and fluorine with $^{31}\text{P}\{^1\text{H}\}$ NMR and ^{19}F NMR, respectively, provides a selective overview of the compounds containing these heteroatoms. The resonance multiplicity in ^{19}F NMR also provides information about the structure of the alkyl chain bonded to phosphorus in the Schedule 1.A.1 chemicals.

It is also possible to use the characteristic $^1\text{J}_{\text{HP}}$ couplings found in organophosphorus compounds that are related to the CWC, and screen the samples with 1D ^1H - ^{31}P HSQC. The main advantage of the 1D ^1H - ^{31}P HSQC experiment is its selectivity: signals from protons with a scalar coupling to ^{31}P nuclei are observed exclusively while the signals from the sample background of obscuring signals from organophosphorus compounds in ^1H NMR spectra are efficiently removed. Compared to the $^{31}\text{P}\{^1\text{H}\}$ NMR experiment, the sensitivity is considerably enhanced.

Examples of ^1H NMR and 1D ^1H - ^{31}P HSQC spectra measured from the same sample are shown in Figure 8 below. Further details can be found in the chapter “Nonselective 1D ^1H - ^{31}P HSQC technique” (3-E-IV).

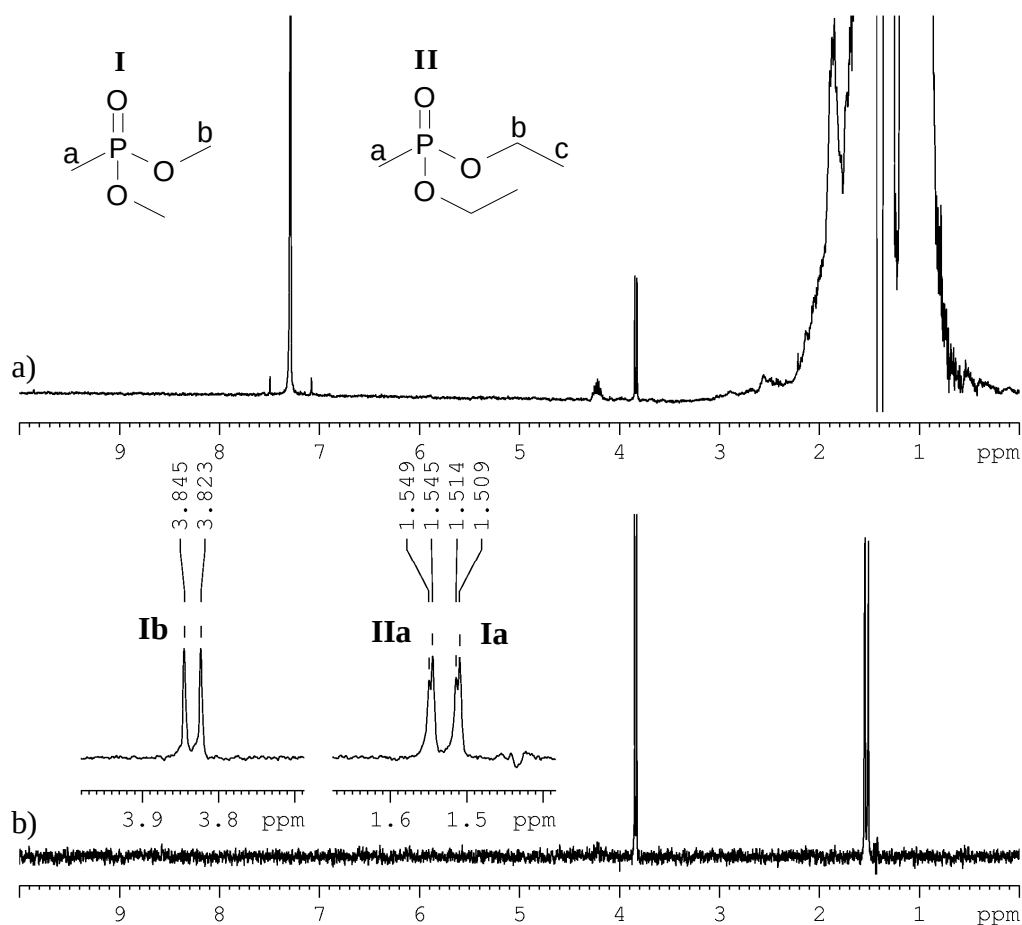


Figure 8. Sample OX from the 23rd Proficiency Test (concentration 10 $\mu\text{g}/\text{ml}$, solvent *n*-decane, 100 μl chloroform- d_1 added to 500 μl OX): a) 500 MHz ^1H NMR spectrum, b) 500 MHz 1D ^1H - ^{31}P HSQC spectrum.

7. Selection of screening methods

The actual selection of the screening methods depends on the available instrumentation. Most common methods for screening are element selective GC detectors and different mass spectrometric methods (both GC–MS- and LC–MS-based methods). Additionally, other methods such as NMR can be successfully applied.

When the actual analysis is being carried out, it is important to get the screening information as early as possible to allow the location of the peaks of interest in more complex data, such as GC–EI/MS chromatograms. This requirement puts emphasis on the speed and simplicity of the screening analysis. It would, for example, be preferable to obtain the screening results with minimal sample preparation. A good example of this is the screening of water samples using both NMR and LC–MS techniques, which do not require the time consuming evaporation of water that is required for GC-based techniques. On the other hand, samples in organic solvents can very easily be screened using GC-based techniques, but may require more complicated sample preparation before LC-based analysis.

Many laboratories use element detective screening techniques for screening, for example, in the OPCW Proficiency Tests, but do not actually report them. The emphasis in reporting is normally put on spectrometric techniques. Additionally, co-eluting chemicals may prevent the reporting of a GC-based technique. In mass spectrometry, by carefully selecting the ions, the co-eluting peaks can be removed from the extracted ion chromatograms (EICs).

8. Usage of screening in various laboratories

The choice of the methods used for screening depends on the available instrumentation. In a typical OPCW designated laboratory, GC, GC–EI/MS, LC–MS and NMR are always available for screening. In the following, examples of screening strategies used in two OPCW designated laboratories are discussed.

8.1 AMMS

The samples are screened for the target chemicals by GC, GC–MS, LC–MS and NMR in the Laboratory of Toxicant Analysis, Academy of Military Medical Sciences (see Figure 9).

GC, together with selective detectors such as FPD, NPD and AED, are used to monitor chemicals containing phosphorus, sulphur, nitrogen and arsenic. The retention indices (RI) of these chemicals are obtained at the same time, and are used to pinpoint the location of the chemicals in total ion chromatograms of GC–EI/MS and GC–CI/MS.

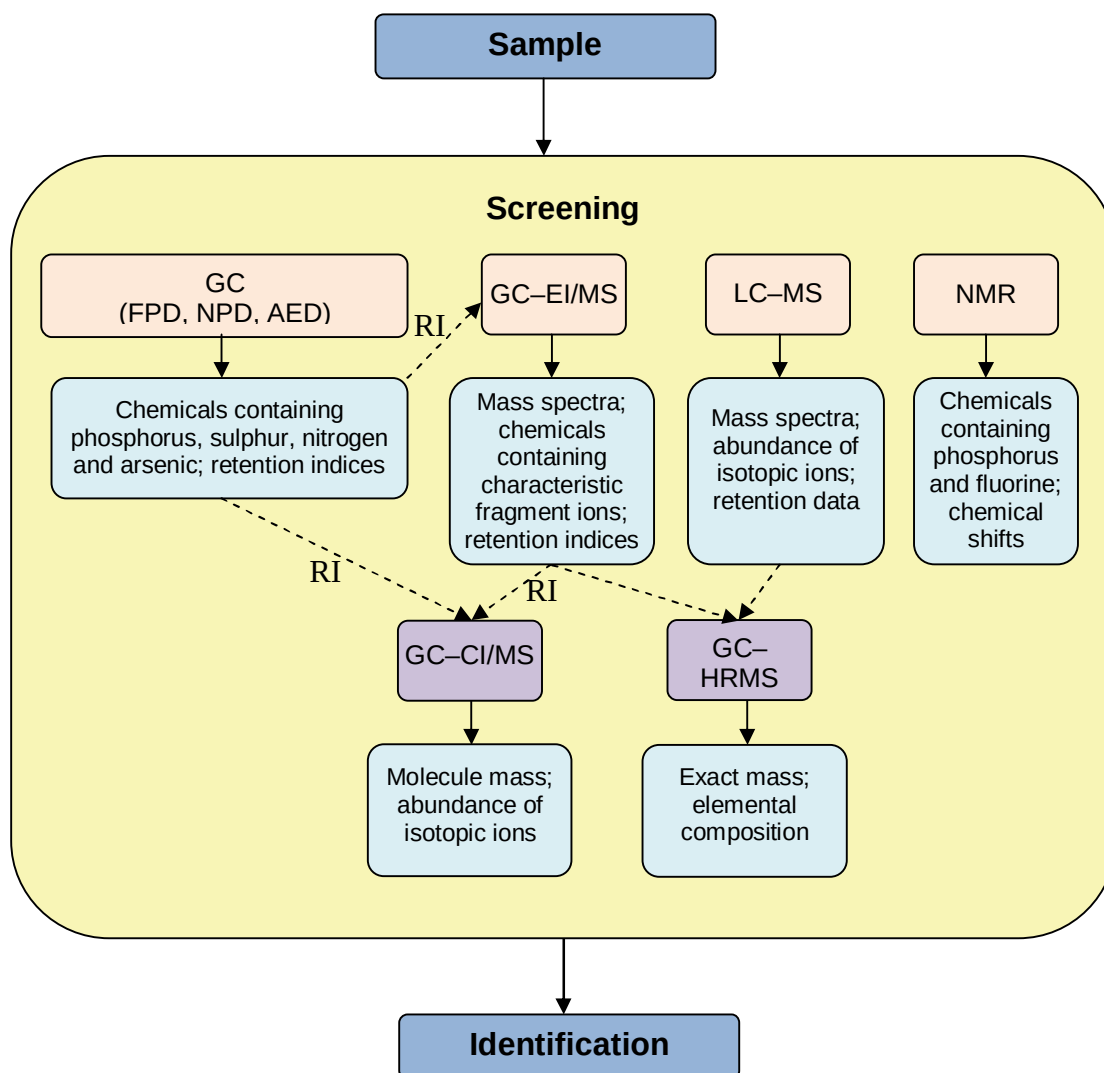


Figure 9. Screening techniques employed at AMMS in testing CWC-related chemicals.

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GC–EI/MS in scan mode is used to screen for all chemicals of interest. By using AMDIS, combined with a proper target library (OCAD, VGWD or other target libraries), the data acquired can be analysed automatically while the chemicals of interest are extracted. Here, the search control factors of AMDIS are always set leniently to get more hits, and some analogues not in the library can also be extracted. In addition, typical fragment ions produced by various groups of CWC-related chemicals are monitored. Special attention is paid to CWC-related chemicals not containing phosphorus, sulphur, nitrogen and arsenic by examining the presence of their characteristic fragment ions.

GC–CI/MS analyses always follow GC and GC–EI/MS analyses to give information on molecular weight.

LC–ESI/MS analyses are carried out in both positive ion mode and negative ion mode, which provide screening for chemicals of interest, especially a wide range of hydrolysis products of CW agents, on the basis of molecular mass, fragment ions present and abundance of isotopic ions. In negative ion mode analyses, N,N-dialkylaminoethane-2-sulfonic acids with C₁–C₃ alkyl substituents are monitored in particular since they cannot be detected by GC–MS after silylation with BSTFA.

If necessary, the exact mass, determined by LC–HRMS or GC–HRMS, can be used to calculate the elemental composition of the molecular or the fragment ions.

³¹P and ¹⁹F NMR analyses are performed to reveal phosphorus- and fluorine-containing chemicals.

8.2 VERIFIN

VERIFIN routinely uses GC–NPD/FPD(S), GC–EI/MS, LC–APCI/MS and NMR for screening CWC-related samples. The purpose of the screening is to cover all CWC-related chemicals in the samples. All positive chemicals will be identified, if possible, and their relevance will be determined.

Several screening-related sample preparation steps are always done. Typically, separate sample aliquots are prepared to screen for Lewisites, arsenic trichloride and inorganic chlorides from Schedule 3. To allow rapid screening of the received samples using LC–MS and NMR, sample type-dependant sample preparation, such as adding a suitable solvent, is often carried out. For example, a non-polar organic solvent sample can be analysed using LC–MS after the addition of more polar solvent (e.g., methanol).

8.2.1 GC-based screening

The first instrument is fitted with one column which is connected to a Y-shaped press-fit connector. Short capillaries are connected from this connector to both NPD and FPD detectors. The FPD detector is used with the sulphur filter. This way one instrument detects for phosphorus, nitrogen and sulphur. The major disadvantages of this system are:

- it is much more sensitive to phosphorus than to nitrogen or sulphur
- real differentiation of nitrogen and phosphorus signals is not possible (if necessary, this problem can be solved by switching the FPD to phosphorus mode)
- the analysis of several silylated samples starts to affect the sensitivity of the NPD

The GC–EI/MS instrument is fitted with exactly the same type of column (normally SE-54-type) – preferably also batch – as well as column dimensions (typically 30 m, 0.25 mm

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i.d., 0.25 µm film thickness) as the GC–NPD/FPD instrument. The flow of the GC–NPD/FPD instrument is adjusted so that the retention times match on both instruments. This way, the comparison of the data is much easier. The same quality control sample is run with both instruments and can be used to check the retention times.

The GC–EI/MS data is routinely screened using two methods: 1) manual searching of ions – as discussed in paragraph 6.1 – and 2) automatic analysis using AMDIS software and the OCAD database. If necessary, the characteristic ions for various chemicals can also be used based on other screening information.

The retention indices are automatically calculated for all detected chemicals by AMDIS. These are often used as a tool for identification of chemicals, but are normally not reported.

These first two screening procedures are repeated for each sample fraction suitable for GC analysis.

8.2.2 LC–MS-based screening

All possible sample aliquots suitable for LC–MS analysis are screened using both positive and negative LC–APCI/MS using non-targeted in-source fragmentation method (see paragraph 5.2.2.1). Both acidic and basic elution conditions are used to modify the elution and ionisation of the acidic and basic target chemicals.³ The chromatography method is based on C18 separation (150 mm x 4.6 mm, 5 µm particles) with a flow of 1 ml/min, which is suitable for the APCI source.

Saxitoxin is screened separately using an MRM-based targeted screening method using LC–electrospray (ESI)/MS/MS. The method is based on hydrophilic interaction chromatography (HILIC).

8.2.3 NMR-based screening

All liquid state samples are screened as such using NMR spectroscopy; the only sample preparation that is needed is an addition of a small amount of a compatible deuterated solvent for lock reference. The water and dichloromethane extracts of soil samples are also screened as such after the addition of the deuterated solvent (D₂O and dichloromethane-*d*₂, respectively). The screening is performed with ¹H NMR, 1D ¹H-³¹P HSQC and 2D ¹H-³¹P HMQC techniques. Additionally, ³¹P{¹H} NMR is used for confirmation of the phosphorus chemical shifts of the found organophosphorus compounds. If the presence of fluorine-containing CWC-related chemicals (i.e., phosphonofluoridates and phosphonic difluorides) is detected with these techniques, the presence of fluorine is further confirmed with ¹⁹F NMR. While the organophosphorus chemicals can be fairly easily screened with the above techniques, the screening of other CWC-related chemicals that have protons (i.e., Lewisites, sulphur mustards and nitrogen mustards as well as related degradation products) is also feasible, provided that the sample matrix does not obscure the peaks. The assignment of the relevant peaks can be assisted using proton-proton correlation experiments (COSY and TOCSY).

³ *Basic conditions:* Methanol : water system with ammonium acetate buffer. The pH is adjusted to 8.6 using NH₄OH. *Acidic conditions:* Methanol : water system with formic acid as an additive. The pH is around 2.6.

9. Evaluation of methods

9.1 Area of applicability

The screening methods should be applied to each sample. Each specific technique has its own limitations, but when a screening strategy has been carefully designed, it should detect most of the scheduled chemicals in any type of sample. The more individual screening techniques a laboratory has, the more flexibility it has.

9.2 Advantages/disadvantages

The clear advantage of the screening step is that the relevant chemicals are easily picked up, even from quite difficult samples.

However, the limitations of a screening method should always be remembered. Thus, the gaps in the overall screening method can be identified and, finally, closed.

Although not discussed in detail in this chapter, it is essential that there is always a clear connection between sample preparation and screening as well as the actual identification steps. This way, it can be assured that the analysed sample is really suitable for the analysis methods.

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Change history of the ROP

1.0 08 Dec 2011 First Edition

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Chapter IV. Reference chemicals

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1. Scope

The verification analysis is based on the identification of chemicals that results from the comparison between the measured data and the spectra libraries or structure elucidation when similar chemical spectra are available. The best and most reliable verification method is comparing the analytical data to the analytical data of authentic reference chemicals.

This chapter provides guidelines for purchasing and preparing reference chemicals for verification analysis, quality criteria for reference chemicals and chain of custody.

2. Commercially available reference chemicals

2.1 Purchase of chemicals

The treaty-related chemicals are often difficult to purchase. Most of the chemical suppliers do not have any Schedule 1 chemicals available and there are few laboratories that can produce these chemicals for analytical purposes. Also, the Chemical Weapons Convention (CWC) has regulations for a transfer from one State Party to another. The transfer of Schedule 1 chemicals must be announced 120 days before the transfer. Some of the degradation products of Schedule 1 chemicals are commercially available. Hazardous chemicals may have prolonged distribution time due to the transport regulations.

3. Own synthesis of reference chemicals

The own synthesis of reference chemicals on a miniature scale is quite simple and the safety measures are not too difficult. Good equipment and knowledge provides a good starting point for the synthesis.

3.1 Starting materials

The starting materials for synthesis of treaty-related chemicals and their degradation products are sold by many chemical suppliers. Different kinds of alcohols and phosphorus chemicals are available for synthetic purposes. Also, various sulphur and nitrogen containing compounds are available for the preparation of mustards and their degradation products. The starting materials can be purchased in advance and stored properly. Thus, when a reference chemical is needed, quick start of synthesis can be possible. End-use certificate of CWC is obligatory for the purchase of certain reference chemicals. Also, the commercially available chemicals can be used for preparation of starting materials that are not sold.

3.2 Synthetic methods

The political sensitivity of the synthesis of chemical warfare agents (CWAs) is possibly the main reason why there are only few research papers in the literature describing synthetic methods. Some scientific books contain some methods for synthesising degradation products of CWAs. The OPCW also plans to arrange courses for synthesis of reference chemicals in the future. Furthermore, the collaboration with OPCW-designated laboratories can be useful.

3.3 Equipment

Micro-scale synthesis does not have special requirements for glassware. The amounts of the used chemicals and solvents are minimal, as well as the heat energy formed in the reaction. Usual vials (typically 1.5–2 ml) can be used for most of the synthesis. Stirring of the reaction can be done by miniature magnetic stirring bars and the measurement of liquid volumes by digital pipette. Analysis of synthesis products can be done by GC–MS, LC–MS or NMR (the deuterated solvent can have a dual use as a reaction and analysis solvent). For example, the concentration of the reagents can be set so that the dilution of the reaction mixture (1:100) leads to a suitable concentration for the GC–MS analysis.

4. Identification of reference chemical

4.1 Analytical methods

At least two separate analytical methods are needed for identification and purity check of the synthesised chemical. It is preferable to have the same analytical method for identification that was used for the analysis. Often no exact purity of the chemical is needed.

5. Synthesis of reference chemicals as a part of quality system

Good laboratory procedures are important in the synthesis of reference chemicals. The synthetic route, starting materials and identification of the chemical must be described. Book-keeping of the prepared reference chemicals must be accurate. The origin of the chemical must be clearly identified.

The Quality systems standard often defines, in detail, how the information of the synthesis of the reference chemical has to be written. The guidelines for the synthetic work can be described in the quality system working instructions.

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Chapter V. False positive identification of Chemical Weapons Convention-related compounds

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1. Scope

This chapter describes the issues related to the possible false positive identifications during analysis of chemicals related to the Chemical Weapons Convention (CWC). The reporting of a false positive for a chemical – i.e., a chemical that is in actuality not present in the sample – in CWC-related analysis might have serious political consequences.

This discussion concentrates on gas chromatography–electron ionisation mass spectrometry (GC–EI/MS), which is used both during on-site analysis carried out by OPCW inspection teams and in, for example, designated laboratories for off-site analysis. See chapter *Gas chromatography–electron ionisation/mass spectrometry (GC–EI/MS)*(3-B-II).

Issues related to analytical data searched against both mass spectral and retention index databases are also discussed.

2. Summary

This chapter focuses on the understanding and correct evaluation of the results of mass spectral search results. Data are presented for false positive matches between non-relevant chemicals and CWC-related chemicals.

The OPCW Official Analytical Database (OCAD)^{*} is an excellent database for CWC-related chemicals, but a number of problems are inherent in these types of limited spectral

^{*} OCAD contains analytical data evaluated and validated by an international group of experts (the Validation Group). Current version (v.13, November 2010) contains 4608 electron ionisation mass spectra for 3606 chemicals as well as 3273 retention indices (3649 in PDF format), 936 FTIR spectra and 1391 NMR spectra.

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databases. The advantages are clear for using a larger database, such as the NIST database, in addition to the OCAD.

Different types of false positive hits are discussed and several possible ways of recognising and avoiding these false positives are introduced. One of the main methods for circumventing false positives is by using a combination of mass spectral and retention index searches for identification of the chemicals.

Other analytical methods can be used to help rule out possible false positive matches. A typical example is the use of a non-relevant chemical that produces good mass spectral and retention index matches with a phosphorus-containing relevant chemical, but that does not contain phosphorus.

Finally, the misidentification of one scheduled chemical for another different scheduled chemical is discussed. These types of false positive matches arise due to the very similar spectra of some groups of scheduled chemicals.

The behaviour of some of the scheduled chemicals that produce false positive hits can be used in a productive manner. Matches with similar chemicals can be used in classification of scheduled chemicals, while keeping in mind that the match is not correct.

3. Introduction

While misidentification of a compound in an analysis is to be avoided in all cases, the consequences of false positives in the area of chemical weapons are particularly more serious. False positives in civil or military defence-related analysis could give rise to false alarms and consequent extra costs. In the context of the Chemical Weapons Convention (CWC), the occurrence of a false positive identification during an inspection or an investigation of alleged use could have adverse diplomatic consequences.

Two broad areas are covered in this discussion:

1. Identification of a chemical unrelated to chemical weapons as a CWC-related chemical (*false positive identification*)
2. Identification of one CWC-related chemical as another (*misidentification*)

The terms in parenthesis are often viewed as the same thing. In this discussion, the stronger term “false positive” will be used to focus on the greater risk associated with this type of identification. The identification of one Schedule 1.A.1 chemical as another is also a false positive identification, but typically the consequences are less serious.

Emphasis in this discussion will be placed on the analysis using gas chromatography–electron ionisation mass spectrometry (GC–EI/MS) instruments, since these are common to both typical field analysis and laboratory analysis. Understanding identification using a GC–EI/MS requires understanding the nature of the components of the identification. The GC–EI/MS analysis produces two independent measures of the component: the spectrum and the retention index. Most of the details of the spectra of CWC related chemicals are beyond the scope of this chapter, so the focus here is only on those details of the spectra that result in false positive identification and upon the underlying algorithms that are used for identification.

The specific issue of false positive identification during on-site inspections is derived in part from the fact that these analyses rely on the OPCW Central Analytical Database (OCAD) as the sole resource for primary identification. This means that a spectrum of an

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unknown may have a good match with something in the OCAD and therefore would be identified as a CW-related chemical, whereas it would actually have a far better match to a non-CWC-related chemical if such a spectrum was available. However, this is not readily evident since the data for a broader set of chemicals are unavailable due to confidentiality requirements in the inspection.

Note! OCAD contains only CWC-related chemicals, with the exception of the chemicals included in the QC test for GC and GC–MS. No so-called general purpose database, such as NIST or Wiley, is allowed on OPCW inspections.

4. Identification of chemicals by mass spectral searching

The identification of chemicals by mass spectra requires that the experimental spectrum of the unknown to be identified can be compared to a library of spectra that have been measured for known chemicals. The criterion for identification is that the unknown spectrum matches one of the library spectra to a preset degree and does not match any other spectrum to a similar degree. This identification is usually expressed in algorithm terms that are minor variations of formula (1):

$$\text{Match Factor (MF)} = \frac{(\sum m_i^q (I_l I_u)^{p/2})^2}{\sum m_i^q I_l^p + \sum m_i^q I_u^p} \quad (1)$$

where m_i is the range of m/z values in the library and unknown spectrum, I_l and I_u are the intensity of the library (l) and unknown (u) spectra at each m_i , p and q are variables in the various algorithms, typically $p \leq 1$ and $q \geq 0$, and the sums run over the range of m/z values for the library and the unknown.

The powers p and q reduce the impact of intensity variation on the match factor (p) and increase the relative importance of the higher masses (q). As shown above, the MF will range from 0 to 1. This value is multiplied by 100 or 1000 to give the MF normally reported. Identification is defined as some minimum match factor (e.g., the OPCW criterion for identification in an OPCW Proficiency Test is 80 or 800).

In addition, a separate match factor is often used to deal with cases where the target compound is masked by another chemical – that is, the spectrum that is seen is the result of a mixture two compounds and not a pure spectrum. In these cases, the reverse search is described by formula (2):

$$\text{Reverse Match Factor (RMF)} = \frac{(\sum m_i^q (I_l I_u)^{p/2})^2}{(\sum m_i^q I_l^p)^2} \quad (2)$$

The only difference between the MF and RMF is that the denominator of the RMF does not have any information about the search spectrum. As a result, extra peaks that are in the search spectrum, but not in the library spectrum, do not *increase* the denominator and thus do not *decrease* the match factor. The reverse match factor is more likely to produce false positives, but it is also more likely to be able to reveal true positives in very complex matrices.

Note 1! Some mass spectrum search programs present only one value, which is normally the MF.

Note 2! Care should be taken when comparing two quite different spectra that have a high RMF value, as this could mean either that at least two co-eluting compounds are present or that the match is totally incorrect.

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A combined Net Match Factor (NMF) that combines both MF and RMF is defined by formula (3):

$$\text{Net Match Factor (NMF)} = 0.75MF + 0.25RMF \quad (3)$$

As indicated above, the spectrum is tentatively identified if the net match factor is greater than some preset value – typically 80 (out of 100). Note that the relative proportion of the MF and RMF in the NMF is somewhat arbitrary. The 3:1 ratio used above has been found to give good results – that is, relatively few additional false positives are added by the use of the RMF and some increase is seen in the ability to detect CWC-related compounds in practical situations.

From the point of view of the analysis of chemical weapons related compounds, two distinct problems could occur – first, the spectrum is that of a compound that is related to the chemical weapons convention (CWC), but the data are not in the database; and second, the spectrum is of a compound that is not related to the CWC but the spectrum is similar to one of the spectra in the database. The first case, as will be discussed in detail subsequently, is quite unlikely to occur for the most important CWC-related chemicals, given the current state of the OCAD. The second case, which is the false positive case, does occur in practice. While it is impossible to look at every case where a false positive might occur, determining to what extent the spectra in the OCAD might lead to false positives is useful. To assess this risk, the spectra in the OCAD can be compared to a large database representing a significant fraction of the possible chemicals.

This comparison, shown in Table 1, has been conducted by searching the spectra of the OCAD against the NIST database. The search criteria were set up so that no scheduled compounds (of which there are many in the NIST database) are found. The result is a set of match factors for each compound in the OCAD against non-scheduled chemicals in the NIST database.

Table 1. False Positives for data from OCAD matched to NIST Database (*NMF = Net Match Factor*)

Schedule	Total Spectra in OCAD	OCAD Chemicals matching non-CW Spectra		Total number of matched unscheduled compounds
		NMF ≥ 80	80 > NMF ≥ 75	
1.A.1	569	17	49	33
1.A.2	329	0	3	1
1.A.3	460	140	261	20
1.A.4	22	0	0	0
1.A.5	6	0	0	0
1.A.6	8	0	0	0

The forward (MF) and the reverse (RMF) search values were calculated and a net match factor (NMF) was calculated using formula (3). A NMF of 80 is the default criterion for identification, but it is important to realise that compounds for which NMF > 75 when matched vs. library spectra may have NMF > 80 in field experiments and thus they show up as false positives. The table illustrates the major problem with false positives from data in the OCAD – most of the VX family (1.A.3) of chemicals have some risk of false positives, while minimal risk of false positives occurs for the tabun family chemicals

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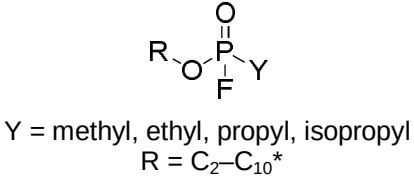
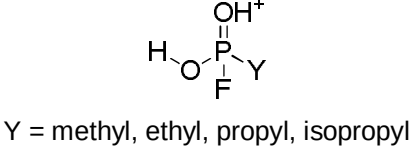
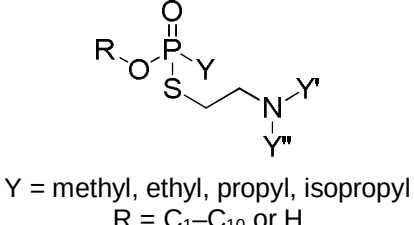
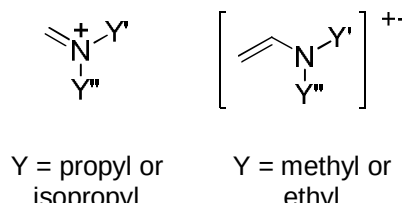
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(1.A.2) and no risk at all is seen for the sulphur or nitrogen mustards (1.A.4 or 1.A.6) or for the Lewisites (1.A.5).

This problem arises because, for both the Schedule 1.A.1 and 1.A.3 compounds, the spectra are dominated by one or two peaks due to the much greater thermodynamic stability of these particular ions over other possible ions. This means that any other compound that happens to have the same ion – even if is not from the same source – may have a high risk of generating a false positive when compared to an OCAD spectrum. The formation of these ions is shown in Figure 1.

Schedule	Structure of scheduled chemical	Dominant ion(s)
1.A.1		
1.A.3		

* The fragmentation is different when R = methyl

Figure 1. Formation of dominant ions in mass spectra of Schedule 1.A.1 and 1.A.3 chemicals (see Table 2 for the actual ions).

Table 2 below lists the dominant mass spectral ions for the various subgroups of the Schedule 1.A.1 and 1.A.3 chemicals. Note that the structure of the dominant ions for Schedule 1.A.3 is slightly dependent on the R group.

Table 2. Dominant ions for Schedule 1.A.1 and 1.A.3 chemicals

Schedule 1.A.1		Schedule 1.A.3	
Y alkyl	Dominant Ion (<i>m/z</i>)	Y' / Y'' alkyl	Dominant Ion(s) (<i>m/z</i>)
Methyl	99	Methyl	58 (71)
Ethyl	113	Ethyl	86 (99)
Propyl	127	Propyl	114 (127, 98)
Isopropyl	127	Isopropyl	114 (127)

For both the 1.A.1 and the 1.A.3 chemicals, the information about the R group is not present in the most dominant ion. In addition, for the 1.A.3 chemicals, the dominant ions do not contain phosphorus and thus might have arisen from a non-phosphorus source and thus represent non-chemical weapons related chemicals. However, note that despite the large number of the Schedule 1.A.3 compounds that give false positives, only a very small number of unscheduled chemicals are involved (see Table 1). This is because a great deal of similarity exists between the spectra of the 1.A.3 compounds and so the same few

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unscheduled chemicals will be involved in giving false positives for a large number of scheduled chemicals.

One type of false positive occurs in a few cases where the spectrum of the 1.A.1 scheduled chemical is dominated by more than just a few major ions. In such cases, the O-alkyl group yields a set of ions from the molecular ion, in addition to the dominant ion, as indicated in Figure 1. When this occurs, the fragmentation pattern is very similar to the corresponding alkene (RH-H₂) or the corresponding alcohol (ROH), with the addition of the normal dominant ion. This type of fragmentation becomes more common as the stability of the ions associated with the O-alkyl fragmentation grows. In most cases, these O-alkyl groups are either ring systems (as in the example shown in Figure 2) or longer chain branched O-alkyl groups.

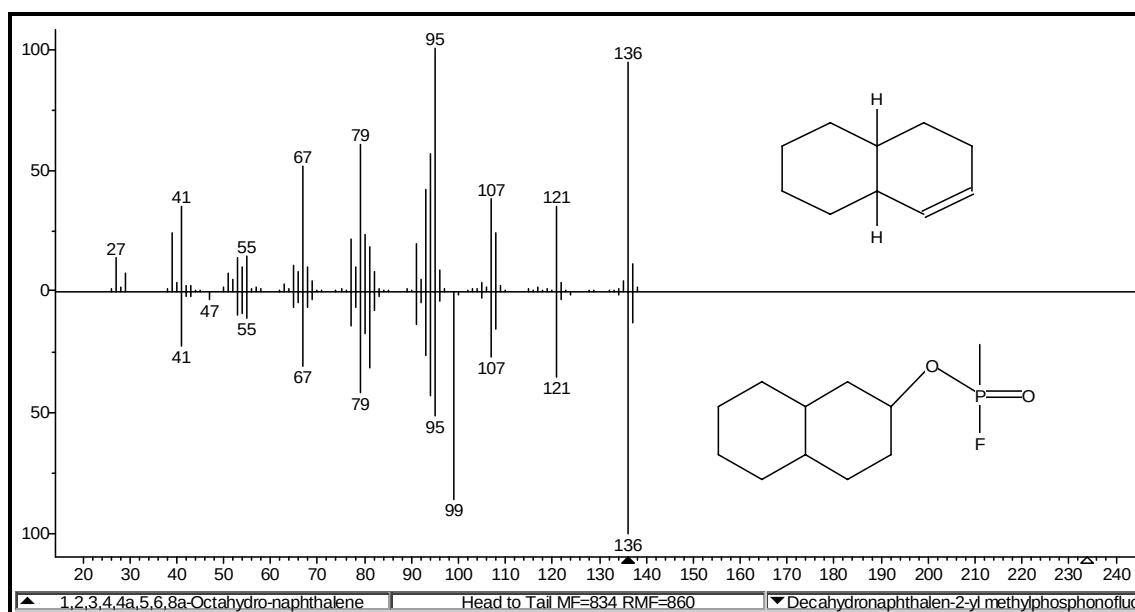


Figure 2. False positive due to high match in non-phosphorus-containing fragments [Top: false positive match; bottom: spectrum of decahydronaphthalen-2-yl methylphosphonofluoridate (Schedule 1.A.1); MF=834 & RMF=860 → NMF=841].

Evaluation of the spectra: The ion m/z 99 (typical for alkyl methylphosphonofluoridates) in the bottom spectrum is missing in the top spectrum. Most of the ions in the spectra are at least partially from the cyclic ester group.

In the above case, the normally dominant ion (i.e., m/z 99) is only one of many ions. Here, the thermodynamics do not as strongly favour the major ion noted above. The false positive occurs because the spectrum is combination of fragments typical for a methylphosphonofluoridate spectrum (dominated by m/z 99) and for a spectrum of an alkene arising from the breaking of the O-alkyl bond – in this case the bicyclic ten-carbon ring.

As shown in Figure 3, the non-scheduled chemicals have spectra that are coincidentally very similar to those for the schedule 1.A.1 chemicals. The non-scheduled chemicals (top) have simple spectra that match both the dominant ions corresponding to those of the scheduled chemical as well as the smaller mass ions. Since each of these non-scheduled compounds has such a simple spectrum, they have a net match factor that is greater than 750 with several scheduled chemicals (the upper one with five matching Schedule 1.A.1 spectra, the lower one with four). These spectral similarities are more or less random and are caused by the matching (low resolution) ion masses. In the upper pair in Figure 3, the strong m/z 127 ion originates from the molecular ion for the unscheduled chemical with a

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formula of $C_4H_5N_3O_2$, and from the dominant ion for the scheduled chemical with a formula of $C_3H_9FO_2P$. In low resolution spectra, the mass of both ions is 127 u. For the lower pair in Figure 3, the mass at 113 corresponds to a formula of $C_5H_7NO_2$ for the unscheduled chemical and $C_2H_7FO_2P$ for the scheduled chemical.

As these types of matches are completely random, it is fortunate that they occur rarely. Based on the spectrum, it is difficult to determine if the spectrum is of a scheduled chemical or not.

Note! In high resolution data ($R > 10,000$), the masses of these would be 127.0382 and 127.0324 u for the upper pair and 113.0477 and 113.0168 u for the lower pair.

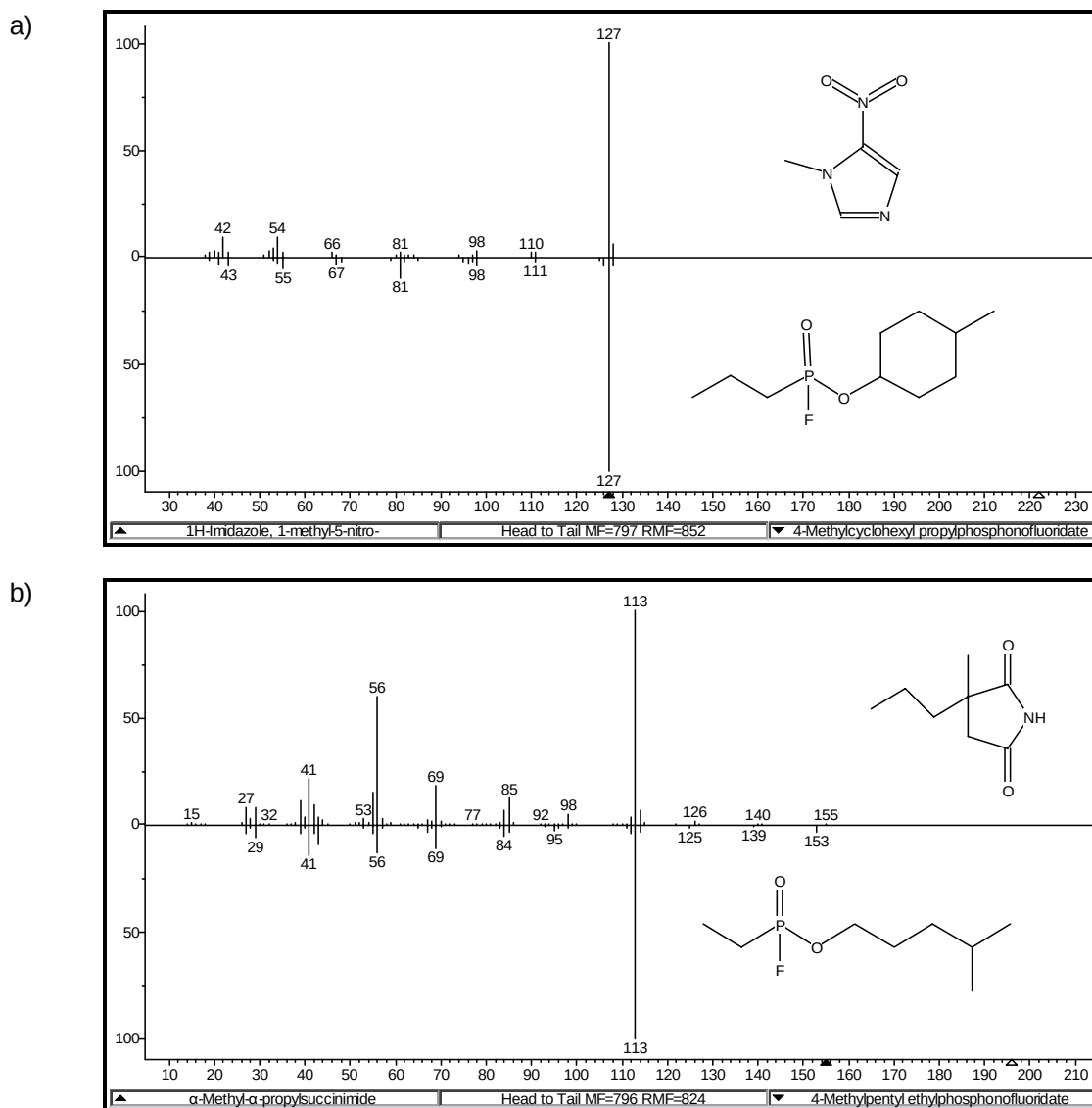


Figure 3. False positives due to coincidental spectra matches: a) *Top*: false positive match; *bottom*: spectrum of 4-methylcyclohexyl propylphosphonofluoridate (Schedule 1.A.1); MF=797 & RMF=852 → NMF=811, b) *Top*: false positive match; *bottom*: spectrum of 4-methylpentyl ethylphosphonofluoridate (Schedule 1.A.1); MF=796 & RMF=824 → NMF=803.

Evaluation of the spectra: The dominant ion [a) m/z 127; b) m/z 113] (typical for alkyl propylphosphonofluoridates and alkyl ethylphosphonofluoridates, respectively) are incidentally matched with isobaric masses. The masses with lower m/z value also match quite well.

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From a practical point of view, the case illustrated in Figure 2 is easier to resolve than the cases shown in Figure 3. While the Figure 2 false positive has the highest net match factor, an examination of the spectra will quickly show that the false positive *does not* have an m/z 99 peak, whereas in the spectrum of the scheduled chemical the m/z 99 peak is almost 90% of the base peak. The high match factor occurs because of the match of m/z 136, as well as a fairly good match of a large number of fairly strong peaks below m/z 99. The fact that this is a clear false positive is proven by the absence of the diagnostic m/z 99 ion. In contrast, no major peak is missing and no real extra peaks are evident in either of the cases illustrated in Figure 3. In contrast to the false positive in Figure 2, which only matches a single spectrum in the OCAD – closely tied to the structure of the O-alkyl group – the spectra in Figure 3 have multiple matches due to the general similarity to the corresponding alkyl alkylphosphonofluoridates. The lack of any definitive features (in addition to the m/z 113 or 127 ions) in these spectra makes them more difficult to rule out as false positives.

For the Schedule 1.A.3 chemicals, the situation is different: the actual ions from both the scheduled chemicals and the false positives are the same, since the compounds have the same structural part, which then forms a very stable ion. The spectra in Figure 4 show two examples of the kinds of problems that can occur for these chemicals.

As noted above, in the case of the spectra of Schedule 1.A.3 chemicals, the phosphorus-containing portion of the molecule does not produce any intense ions. In both of the examples shown in Figure 4, the ions that are dominant in both the false positive and the scheduled chemical are from the same structural element in molecule: the S-2-dialkylaminoethyl part. Since the same structural element is present in many other unscheduled chemicals, they match a large number of scheduled spectra – in the case of the first spectrum shown above, 56 scheduled spectra exist with NMF > 750, in the case of the second, there are 64.

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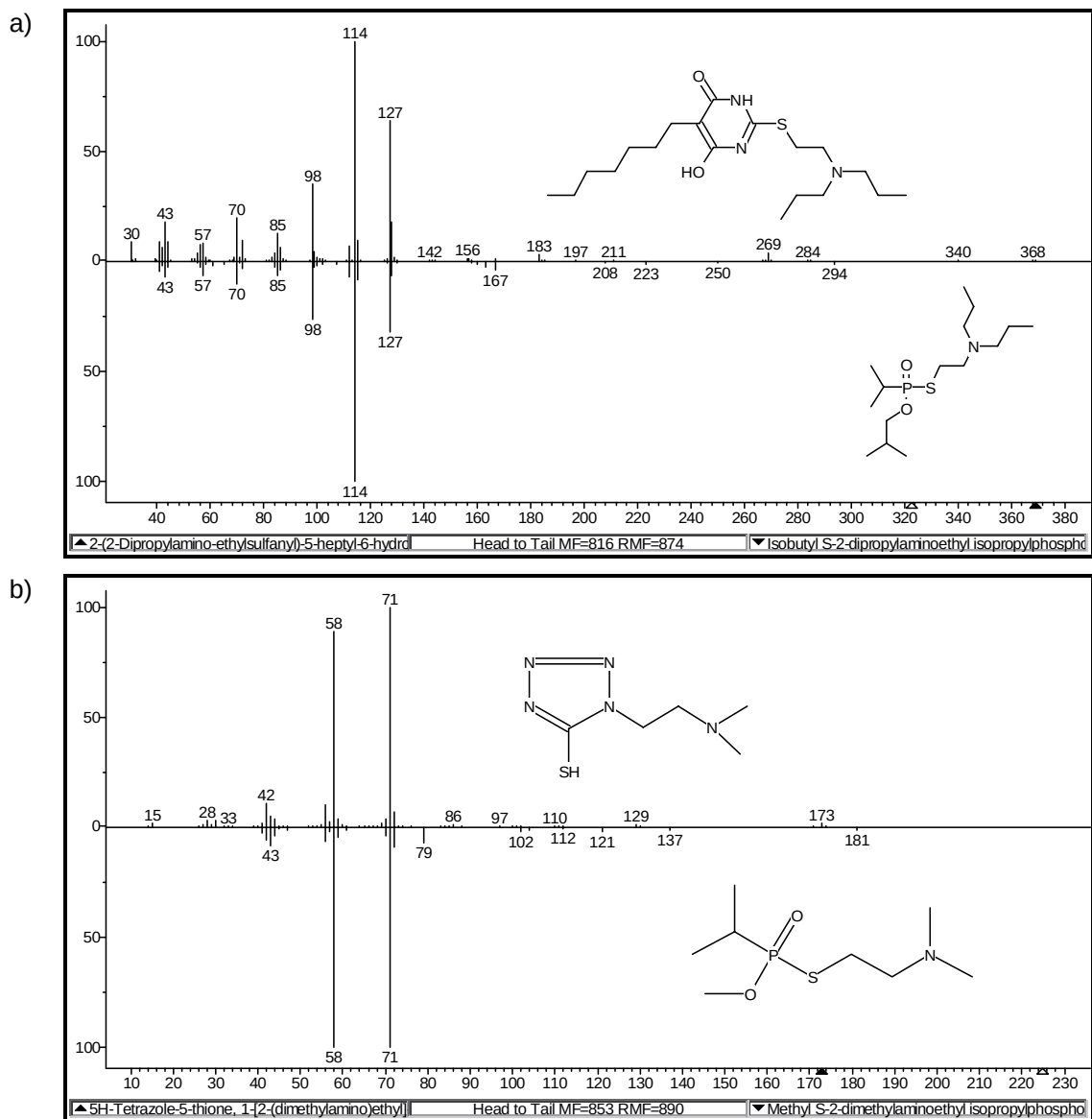


Figure 4. False positive matches with Schedule 1.A.3 chemicals due common major ions:
a) *Top*: false positive match; *bottom*: spectrum of isobutyl S-2-dipropylaminoethyl propylphosphonothiolate (Schedule 1.A.3); MF=816 & RMF=874 → NMF=830,
b) *Top*: false positive match; *bottom*: spectrum of methyl S-2-dimethylaminoethyl isopropylphosphonothiolate (Schedule 1.A.3); MF=853 & RMF=890 → NMF=862.

Evaluation of the spectra: The major ions from the dialkylaminoethyl part [a) m/z 114 and 127; b) m/z 58 and 71] as well as the lower m/z ions match well. The small ions at higher m/z values do not match well. In the spectra of the 1.A.3 chemicals, these are the ions that still contain the phosphorus part.

Matches to trichloronitromethane (chloropicrin; schedule 3.A.4) are quite unique cases of false positive identification. The electron ionisation spectrum for trichloronitromethane does not show a molecular ion (M^+). The highest m/z ion in the spectrum is fragment ion m/z 117 ($[CCl_3]^+$ with its clear chlorine isotope ions at m/z 119 and 121). As the spectral data are related to this ion only, any chemical producing the same ion in a high abundance is a possible false positive match. The most probable false positive match for this chemical is carbon tetrachloride (MF > 850), which, like trichloronitromethane, does not show a

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molecular ion. In addition, chemicals in the NIST database, such as trichloroacetyl chloride, dichloro[(chloromethyl)sulphonyl]methanesulphonyl chloride and 2,2,2-trifluoroethyl trichloromethanesulphonate will produce a match with $MF > 700$. These three chemicals will not be a problem when present at high concentrations, but at low concentrations, with the increasing noise, the spectral comparison may become more difficult. A comparison of GC-EI/MS spectra for trichloronitromethane, carbon tetrachloride and trichloroacetyl chloride is presented in Figure 5.

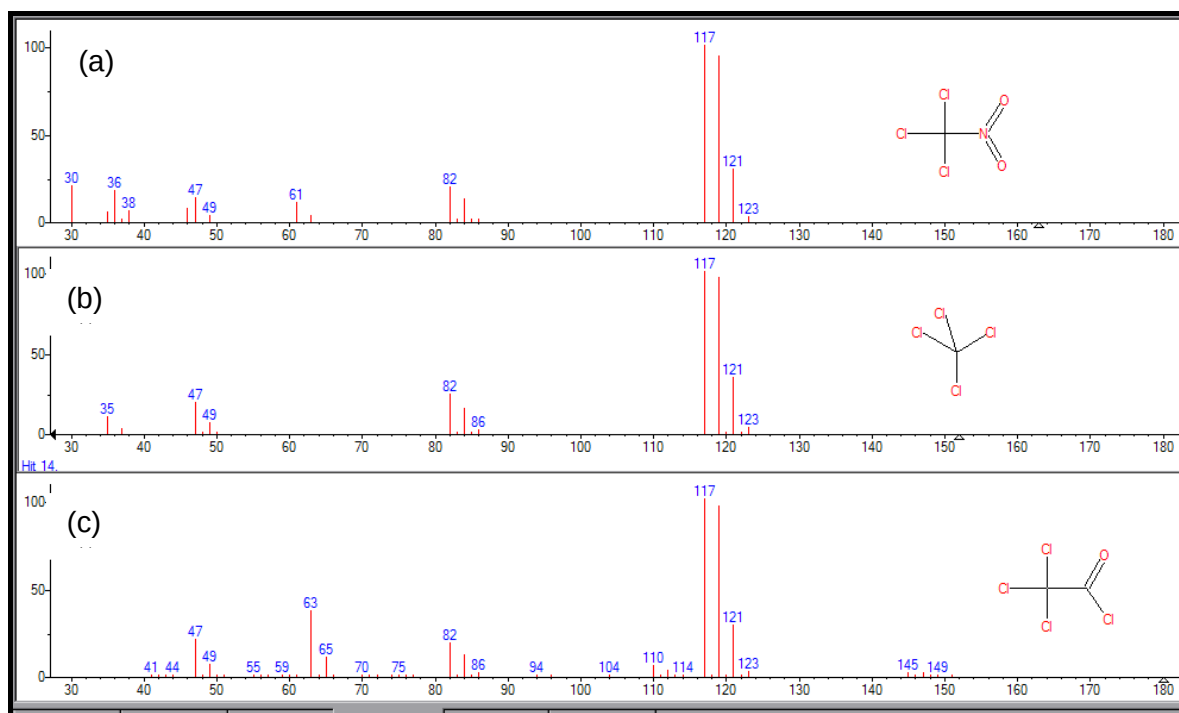


Figure 5. Comparison of the GC-EI/MS spectra for (a) trichloronitromethane (schedule 3.A.4), (b) carbon tetrachloride and (c) trichloroacetyl chloride (b and c are irrelevant chemicals).

Evaluation of the spectra: All of these spectra show an intense m/z 117 ion related to the $[CCl_3]^+$ ion and no molecular ion M^+ .

Note! Because trichloronitromethane has such a low RI (ca. 760), it will not normally be seen by on-site analysis (lowest retention index standard is octane with RI of 800). However, laboratories are likely to find any trichloronitromethane present in off-site analysis of samples.

5. Role of the limited database

In all of the above cases, we have discussed the similarity of the spectra of the scheduled chemical to that of an unscheduled chemical. Were we to compare the spectrum to a large database, we would, in all cases, find these chemicals correctly identified. However, as noted earlier, the analysis of on-site samples under the Chemical Weapons Convention is undertaken with a limited database. In these analyses, only those compounds present in the OCAD are to be reported, in order to ensure confidentiality to the inspected party.

The major drawback of a limited database, such as OCAD, is that a spectrum may have a MF > 80 and thus be reported. Since it is an OCAD chemical, it will be a CWC-related compound, even though if the same spectrum were searched against a larger database, a different – and non-CWC-related – chemical might be found. A search of the data against a larger database, such as the NIST or Wiley databases, will almost always yield a better match (higher MF) for the correct non-scheduled compound. However, the possibility still exists that a compound may match a spectrum in the OCAD with MF > 80 and not be among the spectra present in the larger database.

***Note!** In an off-site laboratory (e.g. an OPCW designated laboratory), no limitation is imposed for the usage of different databases. Use of as wide a database as possible is advisable to prevent the false positive identification, by actually identifying the chemicals in question.*

6. Use of retention index to reduce false positive risk

Even in the absence of a larger database (as is the case for on-site inspections), other information may be used to resolve false positives. The discussion above has focused on one of the two parameters that are available in a GC–EI/MS experiment, the mass spectrum. In addition to the mass spectrum, the GC–EI/MS data gives a retention time – the time that a compound takes to elute from the column. This can be converted to a retention index (RI) by comparing the retention time to the time for a set of standards – typically straight chain hydrocarbons. The times of elution for the hydrocarbons are assigned an RI equal to 100 times their carbon number (octane = 800, dodecane = 1200, etc.) and the experimental retention time is converted to a retention index by interpolation between adjacent standard times.

Retention index is an independent parameter that is uncorrelated with the mass spectra. As such, it can be used to resolve the multiple identifications that are based solely on the mass spectral match factor. As will be discussed subsequently, the RI is especially useful in resolving the multiple matches that occur within the various schedules, but the initial focus is on reducing the risk of false positive identifications of unscheduled chemicals.

For practical identification, both the difference in the spectral pattern of the library chemical and the unknown (the match factor) and the difference between the RI of a library hit and the experimental RI of the unknown are used. Typically, the RI difference [RI – RI(lib)] is either used as a cut-off filter – that is, if the retention index of the unknown is more than some predefined distance from the library value, the identification is rejected – or as variable filter, where the MF is reduced more and more as the difference

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between the experimental RI and the library RI increases, but without an absolute cut-off. The latter mode is used in the AMDIS software[†] for on-site analysis with the OCAD.

In AMDIS, if the retention index of the unknown is within a user definable window (i.e., ± 20 retention index units) of the library match, no penalty is given. If the absolute difference is outside the window, an increasing penalty will be incurred, typically increasing by 10 match factor points (out of 100) per window – again, a user-definable parameter. Thus, if the experimental value of RI is 1825, the spectrum of a chemical from the library has MF = 87 and the RI of that chemical in the library was 1789 (i.e., a difference of 36), using the example values for the window and penalty level, the NMF would be 79 – the RI difference minus the window ($36 - 20 = 16$), to allow for the no penalty region, divided by the window (20) times the penalty per window (10) would produce the penalty:

$$\frac{36 - 20}{20} * 10 = 8$$

Therefore, the final match factor would be below the identification threshold of 80.

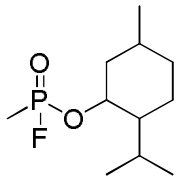
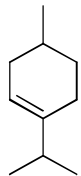
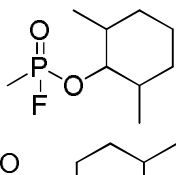
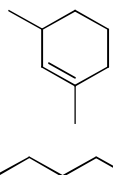
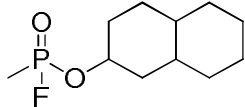
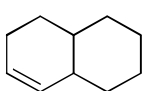
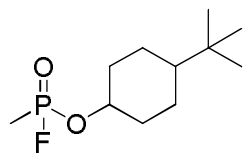
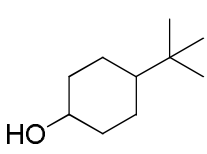
Note! In OPCW Proficiency tests, the identification criteria have been defined as follows:

- mass spectrometric match ≥ 80
- absolute retention index difference to OCAD value (only this OCAD is acceptable) ≤ 20 index units

Figure 2 above illustrated a type of false positive that occurs for the schedule 1.A.1 chemicals with large ring-containing side chains. For a few of these compounds, the ions originating from the side chain are far more dominant than the normal dominant ions typical of 1.A.1 chemicals. Several examples are listed in Table 3. The spectral match factors for these pairs vary from just over 700 to over 800. The RI differences, however, are very large, with the smallest (between the alcohol and the corresponding scheduled chemical) being over 300 RI units. These cases, which are on the edge of being false positives when using the spectral information alone, are of no concern when the RI data is taken into account. Using a normal RI window of 20 or 30 and a penalty level of either 5 or 10 per excess window, the closest of these matches would be 10 to 15 window-widths away from the RI of the library spectral match and would thus be penalised a minimum of 50 out of 100 match factor points.

[†] The Automated Mass Spectral Deconvolution and Identification System (AMDIS) is a program that extracts spectra for individual components in a GC–MS data file and identifies target compounds by matching these spectra against a reference library. It is freely available from the NIST website: <http://chemdata.nist.gov/mass-spc/amdis/>.

RECOMMENDED OPERATING PROCEDURES FOR CWC-RELATED ANALYSIS**Section 1.** General**Part B.** Analytical Strategy**Chapter V.** False positive identification of Chemical Weapons Convention-related compounds**Table 3.** False positive identifications for Schedule 1.A.1

Scheduled Chemical	False Positive	RI Scheduled	RI False Positive	RI Difference
		1468	970	-498
		1324	830	-496
		1665	1060	-595
		1530	1205	-325

All scheduled chemicals in Table 3 show a strong spectral signal from the O-alkyl group of the Schedule 1.A.1 chemical. The problem illustrated in Figure 3 is more complex. For the two chemicals, the false positive match is the result of a random match between the unscheduled chemical's spectrum to one in the OCAD. There is no way to predict what the retention index of the match will be relative to the scheduled chemical. For the two example cases, the RI difference for the upper one (a) is over 100 RI units and thus not likely to produce a false positive in practice, but for the lower one, (b) is small enough to cause problems (ca. 30 RI units).

In the case of trichloronitromethane, shown in Figure 5, the RI values may not be used, as this chemical elutes well before any of the alkanes used as the RI standards. The best way to distinguish between trichloronitromethane and carbon tetrachloride is the absolute retention time. One problem associated with these chemicals is the elution in relation to the solvent of the sample and the following retention behaviour. Often it might be advisable to analyse trichloronitromethane using split injections. However, this is not possible if the concentration is too low.

All of the above cases have been based on a search of library spectra from the OCAD against the larger NIST library. This allows the OCAD to be tested against a very large set of possible compounds that might be found during an inspection, but it does not deal with the specific problems of experimental spectra. It is useful to look at a couple of additional examples where the data examined are not from a comparison of library spectra but from GC-EI/MS analysis experiments.

In the case shown in Figure 6, the spectrum of the unknown, extracted from a GC-EI/MS data file used in the analysis of a food sample for pesticides, matches the OCAD spectrum at a match factor of ca. 80 and the retention index is less than 20 from the OCAD value. For this case, the search of the NIST database does not produce any better match. The compound that is giving rise to the experimental spectrum is unknown – although it occurs

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in a number of food analysis samples. This particular data file is one of a set of over 20 files; most of them show this peak at 9.54 minutes. All spectra show a strong m/z 127 ion and in about half of the cases give match factors of over 80 for some of the schedule 1.A.1 compounds. In only one of the files, the match is made to one of the 1.A.1 chemicals that has an RI value close to the RI of the unknown. The experimental spectrum does not have a better match in the NIST database than the scheduled chemical, so the false positive cannot be resolved even by using the larger database.

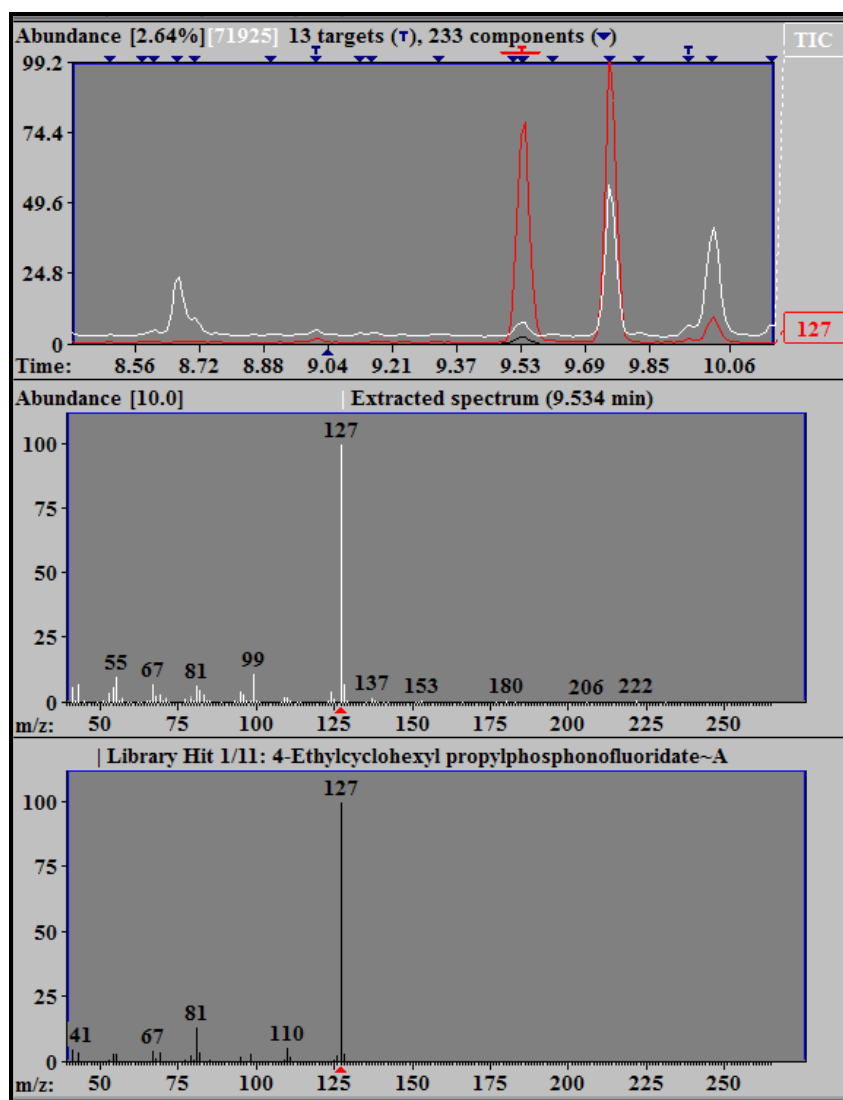


Figure 6. False positive from experimental data – unknown target (*Top/white*: Total ion chromatogram; *top/red*: extracted ion chromatogram for ion m/z 127; *middle*: spectrum of the analyte; *bottom*: first library hit from OCAD database).

The presence of only one or two strong ions in the spectrum means that as the concentration of the component becomes lower, the noise in the background, from which the signal is extracted, will increase (smaller signal-to-noise ratio). As a result, the small peaks that can often provide the ability to distinguish between spectra can no longer be reliably extracted from the background and false positives become more likely.

The second case shown in Figure 7 is an example of a problem that could be expected to occur in a routine industry inspection. Here, a very small impurity in a chemical appears to

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have a very good match factor (85) and a very similar retention index (difference of 5.1 RI units) with a schedule 1.A.3 compound.

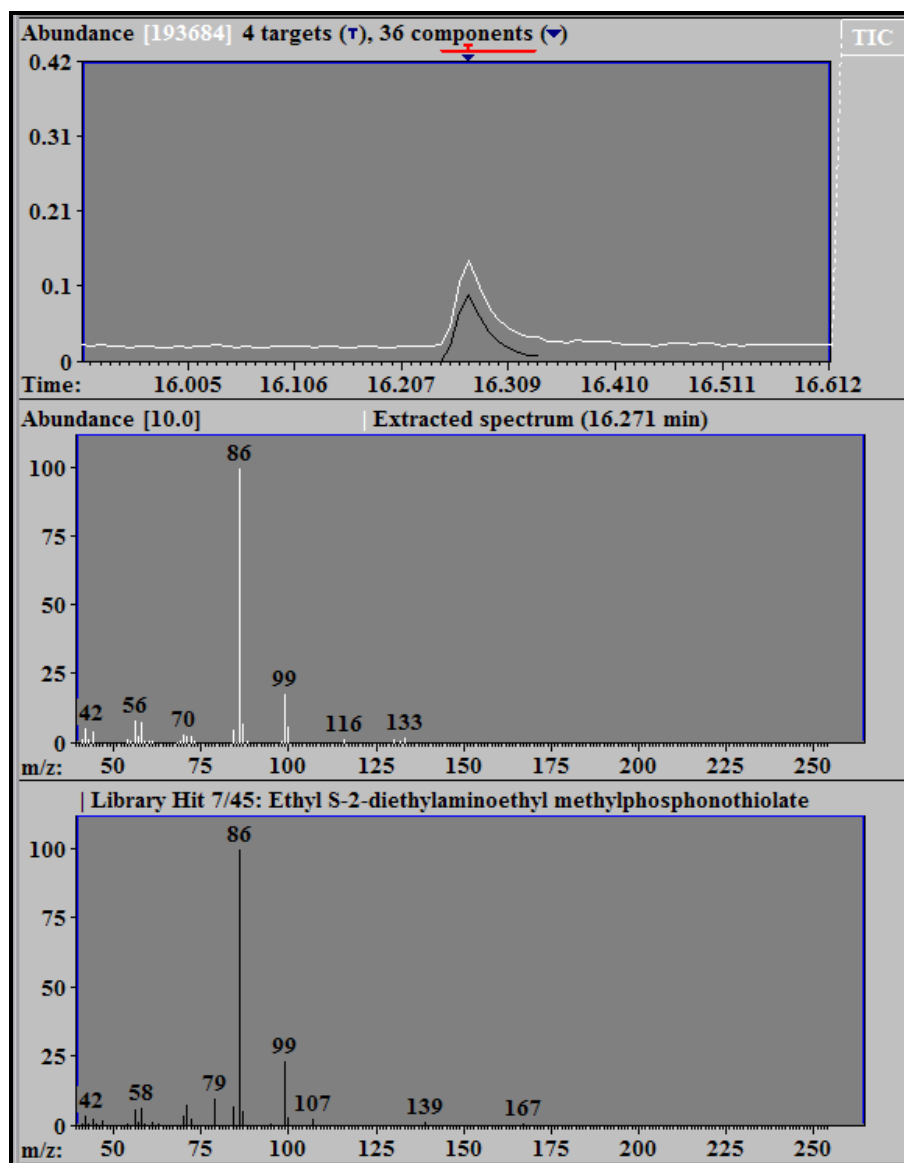


Figure 7. False positive with known target (*Top/white*: Total ion chromatogram; *middle*: spectrum of the analyte; *bottom*: first library hit from OCAD database).

However, when the spectrum is searched against the NIST database, the compound is identified as bis(2-diethylaminoethyl) sulphide, with a higher match factor than any scheduled chemical and the RI of the unscheduled chemical is less than 10 from the experimental value. The compound has the same characteristic diethylamino group that is present in the ethyl S-2-diethylaminoethyl methylphosphonothiolate, as shown in Figure 8. This particular compound will almost always give a false positive when compared to the OCAD, since both the spectrum and the retention index are essentially the same as the scheduled chemical, but since the spectrum is in the NIST database, the search against this database will confirm that this is not a scheduled, but a structurally related chemical. The spectra and structures of the unscheduled chemical and the scheduled chemical are shown in Figure 8. While many of the other chemicals discussed above are unlikely to be seen in

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the context of CW analysis, this compound, which is a minor oxidation product of common industrial chemicals, is a highly probable false positive.

Note! In an OPCW Proficiency Test, bis(2-diethylaminoethyl) sulfide (a non-scheduled chemical) would normally be a reportable chemical as it is an oxidation product of 2-diethylaminoethylthiol, which is a degradation product of ethyl S-2-diethylaminoethyl methylphosphonothiolate.

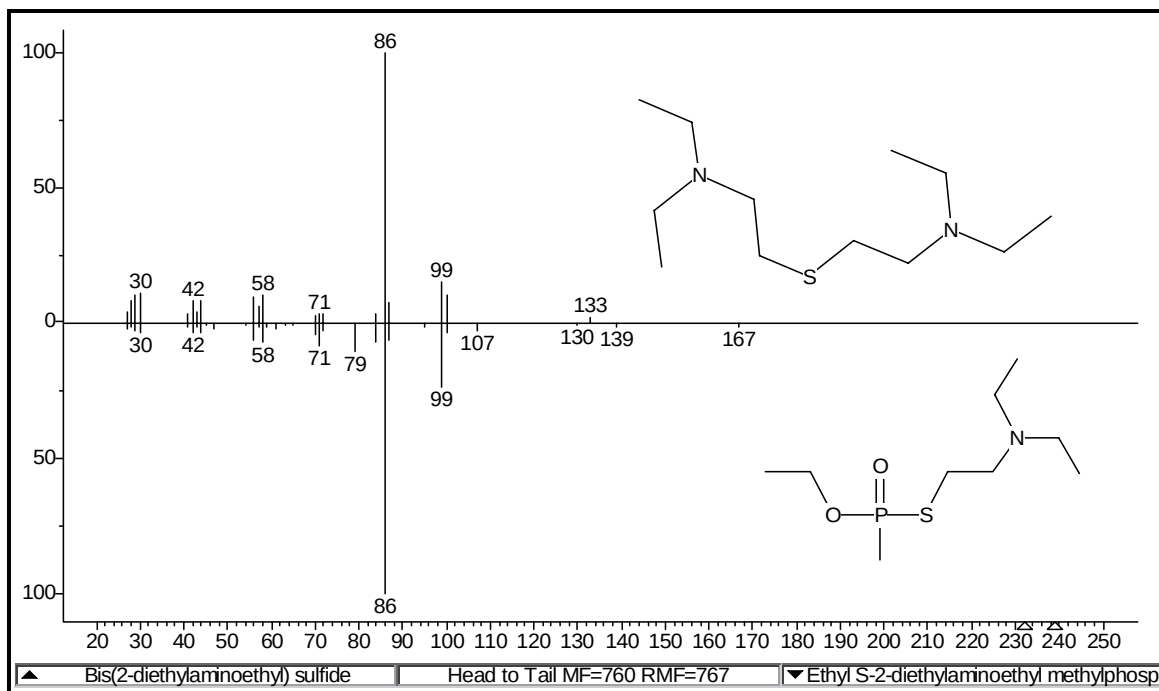


Figure 8. Comparison of a false positive and Scheduled chemical (MF=760, RMF=767 → NMF=762).

As noted above, the Schedule 1.A.3 false positives tend to be structurally related to the amine group in the scheduled chemical and not random matches, as noted above for some of the Schedule 1.A.1 chemicals. Because the anaminium ions ($R_2N^+=CH_2$) are so stable, these spectra are dominated by the same ion, so the spectral similarity is very high. In this case, many scheduled chemicals are found with matching spectra (29 with $MF \geq 80$) and several with RI values similar to the unknown.

While the lack of a definitive identification means that the risk of a false positive is high, in this case the general class of chemicals is identified even if the exact identity of the compound is not clear. In addition, a false positive is more likely to occur with a limited database. As noted for the cases above, if the experimental data had been searched with the full database, and the compound was in the database, the false positive would have been avoided. The notable exception to this is the spectrum shown in Figure 6, where no non-scheduled match exists for the data. Looking at some of the other data files in the series, it is likely that the compound may be a derivative of ethyl fumarate, which might be expected to occur in vegetables, but there is no definitive identification. It is cases such as these that will always be the limiting case for dealing with false positives using spectral data, even with retention information. Note also that in the other data files examined, no false positive result was present in the sense of an identification (considering both match factor and RI), but the match factor was always very close to the threshold for an

identification. A careful examination of these data files may have led an investigator to think that a CWC-related chemical was present in these samples.

7. Summary of GC–EI/MS identification

For the most part, the risks of false positive identification are very small, even with the limitations of using the OCAD, so long as both RI and MS data are used. The majority of the remaining risks can be eliminated – when possible – by using a larger set of spectral data to show that the best match is not the scheduled chemical but a non-scheduled chemical with a similar spectrum. This will still leave a very small number of cases where the spectral similarity and the retention similarity are sufficiently great that no clear cut way exists to eliminate the possibility for false positive identifications. The worst of these types of cases will occur when the compound yielding the false positive is not in any larger database and thus no way is available to show that the identification is incorrect.

8. Additional tools for reducing false positive risks

The discussion above has focused on a single technique – GC–EI/MS. This technique is typically the most powerful currently available since, for most chemicals of concern for chemical weapons analysis, the sensitivity of GC–EI/MS is very high (detection limits of <0.1 ppm are normal). The method can also be applied to the identification of most CWC-related chemicals when used with derivatisation techniques [see Chapter *Derivatisation for gas chromatography* (2-A-III)]. However, as noted above, some cases still exist where the risk of false positive is present. Several analytical techniques can be of great assistance in eliminating these risks.

To understand these methods, looking at the types of false positives that are encountered is useful. In both the Schedule 1.A.1 and 1.A.3 cases (where the bulk of the false positives are found), the false positives share one common feature – they do not contain phosphorus. In the very few cases where false positives might be seen in Schedule 1.A.2, these also do not contain phosphorus. For the other schedule 1 chemicals – Schedule 1.A.4 (the sulphur mustards), 1.A.5 (Lewisites) or 1.A.6 (the nitrogen mustards) – no false positives have been encountered. Therefore, the GC–EI/MS analysis is sufficient for these schedules.

A number of other analytical methods can be used for the analysis of chemical weapons related compounds. Two of these methods have special value in relation to the issue with false positives: nuclear magnetic resonance spectroscopy (NMR) and GC with flame photometric detector (GC–FPD).

NMR suffers from limitations on the sensitivity of the methods and thus may be of limited use if a very low level signal is producing the false positive. The ability of NMR to detect phosphorus-containing chemicals at low ppm levels makes it a very useful technique in resolving false positive matches in GC–EI/MS. Typically, in an OPCW Proficiency test, NMR can detect the number and types of the phosphorus-containing chemicals at normal spiking levels (1–10 µg/g). This helps in ruling out the impossible false positive matches. The main limitation of NMR is that its use is limited to off-site laboratories only and it may not be utilised on on-site inspections.

GC–FPD is a method that, while not being sufficiently specific for general identification of scheduled chemicals, it is perfect for resolving false positives. Flame photometric detectors (FPD) will give signals that are unique for chemicals containing specific

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elements. The normal FPD detectors that are used for CWC-related work are sensitive to either sulphur or phosphorus. These detectors are routinely used for screening, as discussed in [see Chapter *Screening* (1-B-I)], but can be invaluable in eliminating false positives.

While not possible within OPCW on-site operations, GC instruments can routinely be equipped for simultaneous determination of both phosphorus signals using FPD in the phosphorus mode, FPD(P), and mass spectra using the same chromatographic column. Some national mobile laboratories are equipped with such instruments. The sensitivity of the FPD(P) detectors is typically even greater than the sensitivity of the GC–EI/MS instruments; thus, the analysis of a sample with FPD(P) and GC–EI/MS in the same experiment can usually eliminate the false positives discussed above, since in all cases the chemicals giving the false positive results do not contain phosphorus. The only, and very rare, situation where a technique such as this might break down is the case where a low-level phosphorus chemical elutes from the GC column at about the same time as the false positive. In this rare case, confusion could arise.

9. Misidentification of scheduled chemicals

While the level of false positive identifications of non-scheduled chemicals as a relevant chemicals is low and, with care, even with just GC–EI/MS under field conditions, can be near zero, the same cannot be said for false identifications between scheduled chemicals. The degree of spectral similarity between some of the scheduled chemicals is so high that the spectral information alone has almost no assurance of providing a correct identification. For the identification of scheduled chemicals, the role of retention index is critical.

Table 4 gives the results of searching the OCAD with every spectrum in the indicated schedules. As expected, each spectrum will give a perfect match with itself, but when this hit is removed and the rest of the data is examined, it is possible to measure the number of other chemicals in the OCAD that could be mistaken for the target compound. A retention penalty can then be applied to the remaining elements on the hit list. The first column of table 4 shows the fraction of all spectra in the OCAD that, when searched without RI penalty, will give a MF ≥ 80 for a compound other than the correct compound. Subsequent columns show the same results but with various windows and penalties for the retention mismatch.

Table 4. Effect of RI Penalties on misidentification levels. The percentage listed in table shows the fraction of chemicals in the schedule in question showing a wrong match in OCAD with MF ≥ 80 .

RI window	-	± 100	± 30		± 20	± 10	
RI Penalty Level	0	1000	1000	10	10	10	1000
1.A.1	91%	73%	20%	48%	40%	40%	6%
1.A.2	90%	79%	12%	34%	17%	17%	0%
1.A.3	100%	81%	19%	54%	43%	43%	9%
2.B.4*	94%	73%	19%	47%	35%	35%	7%

* Only the simple dialkyl alkylphosphonates were considered for this analysis

This table shows that 91% of the schedule 1.A.1 spectra will match the spectrum of a different schedule 1.A.1 chemical with a match factor of 80 or above, with no retention

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index penalty. In addition, with a retention window of 30 units and a match factor penalty of 10 (out of 100) per window, 48% of the schedule 1.A.1 spectra will still have a MF > 80 for an another compound. Table 4 was constructed using library spectra to compare to the library. In this case, the first match is guaranteed to be the correct hit and was removed to find the number of spectra other than the correct spectrum that gave high match factors. However, when the same kind of comparison is done with experimental data, the first hit (just due to spectral matching) often will not be the correct hit.

Table 5 below presents an example of this type of experimental data. Here, the results are taken from an experimental analysis of a paint sample that had been spiked at a low level of chemical warfare agents. The value of the match factor (MF) is without RI penalty. The difference between the experimental and library value for the RI is shown separately. While the spectral similarity of all of these library spectra to the experimental spectra is very good, only by using the RI information can the correct identification be made. The correct match based on the RI and the spectral match is the third entry on the hit list. The identity of the compound is known in this case, but for a true on-site sample, the identity would be unknown. Small changes in the noise, response or tuning of the instrument could have easily made almost any of these spectra have match factors essentially equal to the match factor of the correct compound. The level of spectral similarity is by far the highest with the schedule 1.A.3 compounds, but similar – although less severe – problems occur with other schedules in the OCAD, as seen in Table 4.

Table 5. Results of spectral search for a CWC-related component in a sample spiked at a low level. The correct hit is shown in bold.

Name	MF	RI-RI(lib)
sec-Butyl S-2-diisopropylaminoethyl isopropylphosphonothiolate	89	-92.7
sec-Butyl S-2-diisopropylaminoethyl propylphosphonothiolate	89	-136.7
Isopropyl S-2-diisopropylaminoethyl isopropylphosphonothiolate	88	-0.7
Isobutyl S-2-diisopropylaminoethyl propylphosphonothiolate	87	-170.7
Isopropyl S-2-diisopropylaminoethyl propylphosphonothiolate	86	-47.7
Pentyl S-2-diisopropylaminoethyl methylphosphonothiolate	86	-183.7
Isobutyl S-2-diisopropylaminoethyl isopropylphosphonothiolate	86	-126.7
2-Methylcyclopentyl S-2-diisopropylaminoethyl methylphosphonothiolate	85	-254.7
1,2-Dimethylbutyl S-2-diisopropylaminoethyl ethylphosphonothiolate	84	-224.7
2-Methylbutyl S-2-diisopropylaminoethyl methylphosphonothiolate	84	-137.7
3-Methylbutyl S-2-diisopropylaminoethyl methylphosphonothiolate	84	-139.7
Cyclohexyl S-2-diisopropylaminoethyl ethylphosphonothiolate	84	-398.7

Note that even in the cases where the correct hit is first in the hit list, as it seen the example in Table 6, the spectral similarity can cause problems. In these experimental data, the first hit is correct, but both the first and second hit have very good match factors (96 and 95) as well as a very good retention index matches (-2.9 and +9.1). Thus while the first hit is the compound that was spiked into the sample, in the absence of prior knowledge, the confidence in the identification would not be high.

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Table 6. Results of spectral search for a CWC-related component in a spiked sample. The correct hit is shown in bold.

Name	MF	RI-RI(lib)
2-Ethylhexyl ethylphosphonofluoridate	96	-2.9
2-Propylpentyl ethylphosphonofluoridate	95	9.1
2-Methylbutyl ethylphosphonofluoridate	87	260.1
2-Methylpentyl ethylphosphonofluoridate	85	173.1
2-Ethylbutyl ethylphosphonofluoridate	84	170.1

The shaded column in Table 4 represents a reasonable level of retention penalty for normal analysis. This level of retention penalty still allows for 48% of the schedule 1.A.1, 34% of the schedule 1.A.2, 54% of the schedule 1.A.3 and 47% of the schedule 2.B.4 compounds to have MF ≥ 80 for compounds other than the correct compound. Note that the retention index window and penalty combination cannot be made too severe. The last column in the table shows the effect of a relatively narrow window (10) and a very high penalty (1000). This would mean that if the experimental retention index was only 11 units from the library data, the match factor would be penalised by 100 points (i.e., driven to zero). From the point of view of reducing misidentifications this may seem ideal, but as the example above shows, the experimental value for RI and the library value may differ even if the correct compound is identified. Experimental RI values may vary due to changes in the chromatographic column, slight differences in experimental conditions from those used to determine the library RI value, differences in the experimental matrix and concentration. As a result, uncertainty will always be associated with comparisons of experimental RI determinations with values from the OCAD. A level of at least ± 10 and possibly ± 20 retention index units must be used (latter is the OPCW-allowed window). Closing down the window in such a way that a difference of 15 units completely eliminates a match can cause correct identifications to be missed, resulting in false negatives.

10. Positive effects of false positive results

The range of compounds covered by the Chemical Weapons Convention is very large. The number of schedule 1.A.1, 1.A.2, 1.A.3 and 2.B.4 chemicals in the current OCAD is small compared to the total number of possible chemicals in these schedules. The number of scheduled chemicals is large because of the number of alkyl groups that satisfy the CWC requirement “*O-Alkyl ($\leq C_{10}$, incl. cycloalkyl) alkyl*” as attachments to the phosphorus atom in schedules 1.A.1, 1.A.2 and 1.A.3. The range of data covered by the current contents of the OCAD does include all of the common alkyl groups, as well as a substantial number of alkyl groups that are not likely to be seen in large scale production. However, examination of Table 4 shows that a reasonably high chance of finding a chemical in each of these schedules actually exists, even if it is not in the database. The table shows the probability of getting a match factor for a spectrum in the database for a compound known not to be in the database – which is a misidentification, but useful for the data that are not in the database. The table was generated by searching each spectrum in the OCAD against the OCAD and removing exact match (and any replicates of the same compound) and examining the resulting hit list. The cases where the spectrum was matched with NMF ≥ 80 give an estimate of the chance of finding any spectrum of that class of chemicals assuming the data currently in the OCAD represent an adequate statistical sampling of all of the data in that class. This assumption is valid for the schedule 1.A.1, 1.A.2 and 1.A.3. It

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is less valid value for the schedule 2.B.4 chemicals, where the number of theoretical chemicals is much larger.

The results in Table 4 show that if no RI penalty is applied, essentially a 100% chance of finding a schedule 1.A.3 chemical exists, even if it is not in the database, and a 90% chance for 1.A.1 and 1.A.2 chemicals. Even if a conservative RI penalty is applied (a window of ± 20 index units and a penalty level of 10), a 40% chance still exists of finding a 1.A.1 or 1.A.3 chemical and a 17% chance of finding a 1.A.2 chemical that is not in the database. Two points should be made here. First, a limited number of asymmetric data are available for the amine (i.e., different alkyl groups connected to the nitrogen) in the schedule 1.A.2 data and none in 1.A.3; thus, an analysis with the OCAD may miss such compounds in the sample. Second, many of the more unusual spectra in each set are those with O-methyl or small O-cycloalkyl groups (i.e., O-cyclopropyl, O-methylcyclopropyl and O-cyclobutyl). These are for the most part included in the OCAD, but the analysis discussed above removes them from the results listing. Since they are more likely to be spectrally different, their removal will cause the number of missed identifications to be higher than expected experimentally. Chemicals with more complex O-alkyl groups, which are far more likely to be missing from the OCAD, tend to have spectra far more similar to the spectra of compounds in the database (i.e., the O-4-methylhexyl compound may be missing, but we would expect it to be very spectrally similar to the O-4-methylpentyl or the O-hexyl, both of which are present). However, since we have all of these unusual compounds and the other compounds that are known to be missing from the OCAD do not look much like these, the calculated rate of missed identifications will be higher than is expected.

It should be pointed out that completely new classes of compounds may still be added, especially to the 2.B.4 schedule. Here, the Convention language states, “*Chemicals, except for those listed in Sched 1, containing a phosphorus atom to which is bonded one methyl, ethyl or propyl (normal or iso) group but not further carbon atoms*”, which allows for a wide range of compounds and new classes of these, such as those noted in Figure 9, and these will not be found without their spectra being in the OCAD.

Not only are the spectra of these compounds different from the corresponding acyclic compounds, they are also different from the spectra of other members of the set.

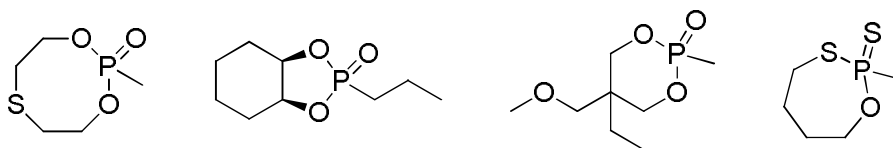


Figure 9. Compounds for which spectra and retention indices have been recently added to the OCAD.

11. Conclusion

The risk of false positive results – meaning the identification of a non-scheduled chemical as a scheduled chemical – is very low.

Spectral similarity between schedule 1.A.3 chemicals and non-scheduled chemicals containing the dialkylaminoethyl groups attached to a heteroatom (primarily to nitrogen, sulphur or oxygen) can give rise to spectra dominated by the same small number of ions as for the 1.A.3 chemicals. Because so many 1.A.3 chemicals are known, the range of

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retention indices covered by these chemicals is very large. Therefore, the use of just the retention parameters to resolve the false positives may not be sufficient. Since the issue here is a fundamental one related to the stability of the ions themselves, no real way is available to avoid the problem.

For the schedule 1.A.1 chemicals, two classes of problems are recognised. One is where the O-alkyl end of the molecule carries a significant portion of the charge, in which case the spectra may be very similar to the corresponding alcohol or alkene. In these cases, the use of retention information will completely eliminate the false positive since the scheduled chemicals will have retention indices 200 to 300 greater than the corresponding non-scheduled chemical. The second class is more difficult in that some chemicals are present that, on a random basis, have similar spectra to the schedule 1.A.1 chemicals. These are rare and predicting what the retention index difference will be is not possible.

In every case except one seen to date, the use of a larger database – such as the NIST database carried with the inspection equipment – would have resolved the problem of false positive. The analysis of more data files may yield a few more examples, but several thousands of different data files from a large variety of sources have not revealed any significant level of false positives. However, the OCAD should clearly be examined for its risk of producing false positives at an ongoing effort to ensure that new additions to the OCAD do not significantly raise the risk of false positives. The requirement that both the spectra and the retention index be measured before a compound can be added to the OCAD will keep the risk of false positive identifications very low, as it is currently.

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Chapter I.

Introduction

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1. Scope

This section describes generally the preparation of environmental samples for the analysis of chemical warfare agents (CWAs) and their degradation products.

2. General considerations

Chemical analysis of samples normally involves sample preparation followed by instrumental analysis. The sample preparation is the key and limiting step for the whole analytical process. In most cases, sample preparation is directed towards a certain instrumental analytical method. These two activities are linked together and sometimes completely integrated. Furthermore, in the case of the determination of analytes at low concentrations in a complicated matrix (target chemical analysis), additional clean-up might sometimes be necessary after the initial instrumental analyses have been carried out.

Well-planned, effective, and reasonably simple sample preparations suitable for the method of analysis are needed for analysis of CWAs and their degradation products. Proper preparation of samples requires a thorough understanding of the behavior of the various types of chemicals in sample matrixes, both before and during the sample preparation, and an awareness of the limitations of the chosen instrumental method of analysis. The sample type, the chemicals that are analysed, the amount of the chemicals as well as the background chemicals in samples, and the method used for analysis affect the selection of the sample preparation method.

Proper sample preparation is an important step in the chemical analysis and is a prerequisite for the whole identification process. It prevents erroneous results due to cross-contamination or other interferences being reported. To prevent these possibilities, it is necessary to always perform the same sample preparation method on one or more matrix or glassware blank samples.

Ideally, the recovery efficiency should be 100%, which, however, will seldom be obtained. For accurate quantitative analysis, a determination of the recovery will be necessary. This may be performed by spiking a blank background sample and carrying out the sample preparation method on the spiked sample. However, verification analysis is focused on

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unambiguous identification rather than on producing accurate quantitative results. Moreover, for accurate quantitative results, stable analytes are presumed, which is certainly not the case with some chemical warfare agents in environmental samples.

Due to a large variety of matrices (environmental samples: soil, decontamination solution, concrete, etc.; complex mixtures: diesel fuel, etc.) and target chemicals (CWC-scheduled chemicals, precursors, starting materials, analogues, analytical artefacts, degradation (break-down) products, additives), rather universal procedures should be used. The Recommended Operating Procedures (ROPs) are designed to be comprehensive enough to allow the analysis of all CWC-related chemicals. Accordingly, some of the procedures contain many sample preparation steps. It is also recommended that the ROPs should be used as a first choice in the analysis, with other approaches not excluded.

2.1 Safety

Handle CWC-related chemicals with great care. Be sure that decontamination solution, protective masks, and autoinjectors of nerve agent antidotes are available when toxic samples or samples of unknown toxicity are handled. These samples must always be prepared in a fume cupboard and no-one must ever work alone. During sample preparation, individual protective gear such as laboratory coat, chemically resistant protective gloves, and safety goggles must be used.

Clean the fume cupboard directly after sample preparation. Materials requiring decontamination must be treated with a proper decontamination solution and disposed of in designated waste containers.

2.2 Purity

The solvents should be as pure as possible. Their purity should preferably be checked in advance of the analysis. Also, the glassware used during the extraction and for the storage of the extracts (vials, bottles) must be thoroughly clean. Moreover, separate glassware should be used for samples, matrix, and glassware blank samples as well as reference standards. Appropriate labeling of the various extracts is another consideration.

2.3 Storing of samples

Store the samples in a freezer at -20 °C prior to sample preparation. Aqueous samples and wet samples are stored in a refrigerator at +4 °C to prevent the bottles and vials from getting broken, however.

The prepared sub-samples are also stored in a freezer at -20 °C (aqueous samples in refrigerator at +4 °C) after the sample preparation.

2.4 Warming of samples

Open the sample packages carefully in a fume cupboard, place sample bottles on the cupboard table, check that they are tightly closed, and let the samples warm up to room temperature slowly without external heating before opening the sample vessels.

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Let the prepared sub-samples also warm up to room temperature slowly without external heating before opening the sample vials to avoid moistening of the samples by absorption of the air humidity.

3. General procedures

Sample is divided into sub-samples, which are prepared to recover chemicals of different polarity and/or for different analysis methods. Usually, the matrix is extracted with organic solvent to recover the nonpolar CWC-related chemicals such as nerve and mustard agents, but, in view of the probable degradation (hydrolysis), it is also extracted with water to recover the water-soluble polar chemicals. It is always recommended to take a new sub-sample to each sample preparation procedure. This is not always possible, however. The sample size is too small or it is obvious that the sample is not homogenous. Then, the organic extraction must be performed before aqueous extraction to avoid hydrolysis of the agents because of sample preparation. General flowcharts of sample preparation of environmental sample are given in Figure 1.

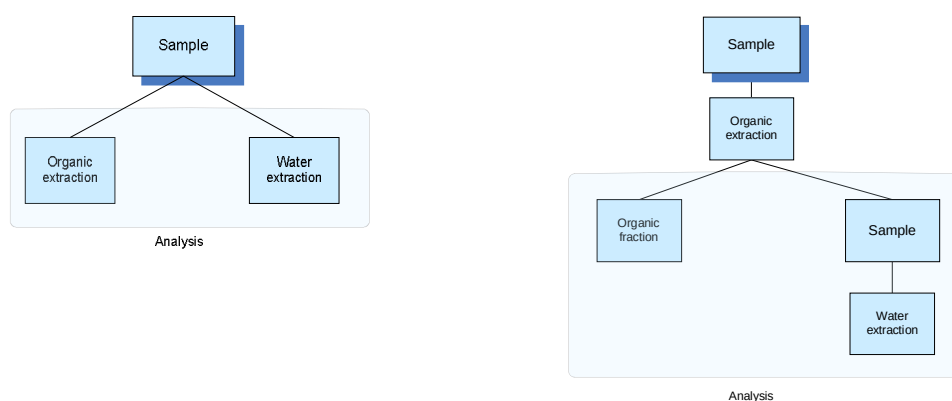


Figure 1. General flowcharts for sample preparation.

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Chapter II.

Concentration

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1. Scope

This section describes methods for the concentration of liquid samples containing chemical warfare agents (CWAs) and their degradation products. The analysis takes place by gas chromatography (GC) or liquid chromatography (LC) as an introduction technique for spectrometers. The concentration methods for analysis by nuclear magnetic resonance (NMR) spectroscopy are performed in the same ways in cases in which the concentration or evaporation of samples is needed.

2. Summary of the method

Extracts of different sample preparation procedures are often concentrated, especially organic extracts. The purpose is to increase the concentration of the analytes in extracts in order to increase the sensitivity of the analysis. When the sample size is small and the solvent is volatile, the samples are concentrated by mild nitrogen flow. For larger sample volumes and quantities, laboratory equipment such as a TurboVap[®] evaporator or a centrifuge evaporator should be used. These instruments are designed for the evaporation of liquid samples in glass vials and tubes.

Nonvolatile and low volatile polar chemicals are normally in an aqueous matrix, and removal of water is required before the derivatisation step for GC-based analysis. In addition, alcohol extracts have to be evaporated to dryness before silylation. The evaporation residue is dissolved in a suitable solvent. For evaporation of aqueous samples

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to dryness, equipment such as a TurboVap[®] evaporator, a centrifuge evaporator, or rotary evaporator should be used. In addition, for LC and NMR analysis, the aqueous extracts can be concentrated or the solvent can be changed in a similar manner.

3. Safety aspects and storing

Safety aspects of sample preparation and storing of samples are presented in Section 2, Part A, Chapter I: *Introduction*. Safety precautions of the used instruments are presented in the corresponding instruction manuals delivered by the manufacturers.

4. Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in this ROP are presented in Table 1.

Table 1. Equipment, supplies and chemicals needed to concentrate the samples.

Equipment	Supply	Chemical
Rotary evaporator	Pasteur pipettes	Nitrogen cylinder with micrometric pressure regulator and manometer
	Screw cap vials	
	Teflon tubing	
	Evaporation vessels for rotary evaporator 10–50 ml)	
TurboVap II [®] Concentration Workstation (Biotage)	Test tubes for TurboVap [®]	
TurboVap LV [®] Concentration Workstation (Biotage)		
Centrifuge evaporator [#]		

[#] e.g. Eppendorf[®] concentrator 5301 or Jouan[®] centrifuge evaporator RC1010

5. Procedure

Organic solvents are concentrated:

1. to increase the sensitivity of the analysis
2. to change the solvent

Concentration to dryness should be avoided.

Aqueous solvents and samples are concentrated:

1. to increase the sensitivity of the analysis
2. to change the solvent: the samples are evaporated to dryness for derivatisation
3. to change the solvent: the samples are evaporated to quite a small volume for LC and NMR analysis

5.1 Concentration of samples with mild nitrogen flow

In concentration with nitrogen flow, a stream of nitrogen removes the solvent vapor from the surface of the sample. Mild flow is needed to keep the analytes in the sample and evaporate the solvent away.

5.1.1 Concentration of organic solvents

The sample or the extract is concentrated usually 5- or 10-fold with mild nitrogen flow: 1 ml to 200 μ l or 100 μ l. Attach a clean Teflon tube to the nitrogen cylinder. Protect the end of the tube from contamination, e.g. with a Pasteur pipette. Place the tip of the Pasteur pipette just above the surface of the extract and adjust the nitrogen flow with pressure regulator to cause a gentle depression on the surface of the solution (Figure 1). Let the solution cool down during evaporation. The rate of the evaporation can be controlled by the degree of immersion of the Pasteur pipette in the vial. Concentration of 1 ml of dich, hexane or acetone extracts takes ca. 10 minutes.

Note! Do not concentrate the organic extract to dryness since certain volatile CWC-related chemicals such as sarin are lost from the sample and, on the other hand, certain less volatile CWC-related chemicals such as VX in residues of organic extracts are firmly adsorbed to glass surfaces.



Maria-Leena Rapinaja

Figure 1. Adjusting the mild nitrogen flow.

5.1.2 Concentration of aqueous samples

Concentration of water samples with mild nitrogen flow is time-consuming. For this reason, it is recommended to concentrate aqueous samples in the TurboVap[®] evaporator, in a centrifuge evaporator or in a rotary evaporator as described in the following sections. Small amounts of aqueous samples can be concentrated with mild nitrogen flow, but heating of the vial in a block heater or a sand bath at 40–50 °C is recommended to avoid prolonged evaporation.

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5.2 Concentration of samples in a TurboVap[®] evaporator

The TurboVap[®] evaporators direct the nitrogen flow into the sample tubes at a precise angle. The vortex created by the blowing gas travels down the tube to the solvent surface, where it increases the gas/solvent interface — providing faster evaporation than with conventional methods.

Two TurboVap[®] concentration workstations can be used for automatic evaporation of samples.

- The TurboVap II[®] evaporator (Figure 2A) allows evaporation of up to six samples and uses optical sensors to provide fast and efficient evaporation. Either 50 ml or 200 ml unit glassware with either 0.5 ml or 1.0 ml endpoint stems can be used.
- The TurboVap LV[®] evaporator (Figure 2B) provides simultaneous evaporation of up to 50 samples. Many interchangeable tube racks are available, ranging in volumes from 1.5 ml to 30 ml, giving the flexibility for automated low-volume sample preparation. The microprocessor control provides monitoring of the timed operation and water bath temperature.

When the TurboVap[®] evaporator is used, the shorter the concentration times, the better the recoveries. Nitrogen pressure should be kept as high as possible, but without splashing; thereby increasing the pressure as the volume decreases.

Bath temperatures for dichloromethane and acetone should be 40–45 °C and nitrogen pressures 10–20 psi.



Figure 2A. TurboVap II[®] concentration workstation (Biotage).



Figure 2B. TurboVap LV[®] concentration workstation (Biotage).

5.2.1 Concentration of organic solvents

The organic sample or extract is concentrated, but usually not to dryness. Concentration of 1 ml of dichloromethane, hexane or acetone extracts takes ca. 10 minutes.

Note! Do not concentrate the organic extract to dryness since certain volatile CWC-related chemicals such as sarin are lost from the sample and, on the other hand, certain less volatile CWC-related chemicals such as VX in residues of organic extracts are firmly adsorbed to glass surfaces.

The following procedure is recommended:

- Turn on the evaporator
- Preselect the temperature of the water bath to 30 °C or 40 °C
- Adjust the nitrogen pressure to ca. 10–20 psi
- Put the test tubes in the evaporator
- Select the position of test tubes to begin the evaporation

Bath temperatures of 40–45 °C for dichloromethane and acetone and nitrogen pressure of 10–20 psi are recommended.

5.2.2 Concentration of aqueous samples

The aqueous sample or extract is concentrated or evaporated to dryness if a derivatisation step is required. The sample is reconstituted in an appropriate solvent for derivatisation or for LC or NMR analysis.

The following procedure is recommended:

- Turn on the evaporator
- Preselect the temperature of the water bath to 50 °C or 60 °C
- Adjust the nitrogen pressure to ca. 10–20 psi
- Put the test tubes in the evaporator
- Select the position of test tubes to begin the evaporation

5.3 Concentration of samples in a centrifuge evaporator

A centrifugal evaporator is a device used for the efficient and gentle evaporation of various solvents from many samples at the same time. The instrument consists of a vacuum pump connected to a centrifuge chamber in which the samples are placed. Many systems also have a cold trap or solvent condenser placed in line between the vacuum pump and the centrifuge chamber to collect the evaporated solvents. Some systems also have a cold trap on the pump exhaust.

The pressure is lowered in the centrifuge using a vacuum pump, e.g. the samples are centrifuged before and during vacuum evaporation. The sample is forced down into the bottom of the tube and for this reason the chamber is heated to ensure evaporation. The heating level and the depth of the vacuum are selected based on the particular samples (analytes and solvent) evaporated. The advantage of centrifugation is that the finished sample is in a pellet at the bottom of the evaporation tube and not distributed over the sides of the evaporation tube.

Note! When using the centrifugal evaporator, always place a centrifuge tube of the same size with the same filling level as a counterbalance on the opposite side of the centrifuge rotor.

Centrifuge evaporators of different manufacturers can be used: Figure 3A shows an Eppendorf® concentrator 5301 and Figure 3B a Jouan® RC 1010 centrifuge evaporator.

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Figure 3A. Eppendorf® concentrator 5301.



Figure 3B. Jouan® centrifuge evaporator RC1010.

5.3.1 Concentration of organic solvents in Eppendorf® concentrator

The following procedure is recommended for organic solvents:

- Turn on the evaporator
- Preselect the temperature of the chamber to 30 °C
- Load the rotor
- Close the lid
- Select the depth of the vacuum and centrifugation parameters based on the particular samples (analytes and solvent)

5.3.2 Concentration of aqueous samples in Eppendorf® concentrator

The concentrator removes 0.5–1.0 ml of water within 60–100 minutes using the following procedure:

- Turn on the evaporator
- Preselect the temperature 50 °C or 60 °C
- Load the rotor
- Close the lid
- Select the depth of the vacuum and centrifugation parameters for aqueous solutions
- Once the sample is dry, remove tube from the unit, close the vial and let it cool to room temperature before derivatisation
- Derivatise the sample without delay

5.3.3 Concentration of aqueous samples in Jouan® centrifuge evaporator

The following procedure is recommended for aqueous samples and solvents:

- Turn on the evaporator
- Select heating level 1 and pulse mode
- Load the rotor
- Close the lid

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- Start the pump and evaporation
- Once the sample is dry, remove tube from the unit, close the vial and let it cool to room temperature before derivatisation
- Derivatise the sample without delay

The evaporation time depends on the size of the sample, the size of the tube (the width and height of the liquid pillar) and the number of the samples evaporated at the same time. To evaporate four samples of 10 ml in tubes of Ø i.d. 18 mm takes ca. 420 minutes (=7 hours). This means that the centrifuge evaporator is usually not the first choice to evaporate 5–10 ml of unknown aqueous sample to dryness because the evaporation time is unknown.

At the end of the evaporation, the dryness of the sample should be checked. The sample must be free of water since both the methylation and the silylation reagents react with water and consume the derivatisation reagent. On the other hand, to avoid the decomposition of analytes, evaporation should not be continued after the sample is dry.

Note! When stopping the evaporation in the middle of the evaporation or when checking the dryness of the sample, disturb the vacuum when the evaporator is pulsing to avoid cross-contamination of samples.

5.4 Concentration of samples in a rotary evaporator

The rotary evaporator removes the vapor phase of the evaporating solvent from the surface of the sample using the vacuum. To decrease the evaporation time, the sample is rotated. Rotating of the sample vessel increases the surface area because the liquid travels up the walls. The rotation also provides uniform heating and helps to prevent flash boiling.

A rotary evaporator is usually used for evaporation of larger samples than with nitrogen flow, a TurboVap[®] or a centrifuge evaporator. With a normal laboratory rotary evaporator, presented in Figure 4A, samples from 5–100 ml are usually evaporated.

It is recommended to use pear-shaped evaporation vessels (4B). Usually 2–10 ml of the aqueous or alcohol solution is evaporated to dryness in a 50 ml pear-shaped boiling flask having a ground joint of 29/32.



Figure 4A. Laboratory-scale rotary evaporator.



Figure 4B. A pear-shaped evaporation vessel.

5.4.1 Concentration of aqueous samples on a rotary evaporator

The following procedure is recommended for aqueous samples:

- Turn on the evaporator
- Set the temperature of the water bath to 50 °C or 60 °C
- Load the sample and adjust rotation speed to ca. 100 rpm
- Adjust the vacuum at 35 mbar and start pumping
- When the vacuum is reached, set the sample step by step as low as possible in the water bath
- Take care that the sample is not foaming out of the evaporation vessel when you adjust it in the water bath
- Once the sample is dry, remove the tube from the unit and let it cool to room temperature before derivatisation
- Derivatise the sample without delay

To evaporate 5–10 ml of water to dryness in the recommended conditions (35 mbar, 50 °C, 100 rpm) takes 20–30 minutes.

It is possible that alcohols and alkyl phosphonic acids are distilled with water as an azeotropic mixture. For this reason, it is recommended to clean the rotary evaporator with clean water between different samples or evaporate matrix blank samples (clean water) after each sample.

5.4.2 Concentration of alcohol solutions on a rotary evaporator

The following procedure is recommended for alcohol-based samples and extracts:

- Turn on the evaporator
- Set the temperature of the water bath to 50 °C or 60 °C
- Load the sample and adjust rotation speed to ca. 100 rpm
- Adjust the vacuum at 200 mbar and start pumping
- When the vacuum is reached, set the sample step by step as low as possible in the water bath
- Take care that the sample is not foaming out of the evaporation vessel when you adjust it in the water bath
- After ca. 15 minutes, adjust the pump to 35 mbar
- Once the sample is dry, remove tube from the unit and let it cool to room temperature before derivatisation
- Derivatise the sample without delay

6. Quality control

Quality control of sample preparation is presented in general Section 2, Part A, Chapter V: *Quality assurance and quality control in sample preparation.*

RECOMMENDED OPERATING PROCEDURES FOR CWC-RELATED ANALYSIS

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1. Scope

This chapter describes derivatisation procedures recommended for the gas chromatographic analysis of polar precursors/degradation products of toxic chemicals listed in Schedules 1 and 2 of the Chemical Weapons Convention (CWC). Also described are derivatisation procedures for reactive, corrosive, or highly volatile analytes such as Lewisite 1, phosgene, and some precursors to Schedule 1 chemicals. Most of the procedures described have been applied predominantly for identifying chemicals at concentrations above 1 part per million in a variety of environmental and man-made matrices, typical of routine or challenge inspections and OPCW Proficiency Tests. Some alternative derivatisation procedures are included that are better suited to identifying relevant chemicals at concentrations down to low parts per billion, e.g. in investigations of alleged use.

2. The requirement for derivatisation

The major requirement for derivatisation of CWC-related analytes is for the analysis of polar precursors/degradation products of Schedule 1 CW agents. These analytes chromatograph very poorly or degrade at the high temperatures required for gas chromatography. Examples are alkyl alkylphosphonic acids and alkylphosphonic acids, derived from the hydrolysis of organophosphorus nerve agents, and thiodiglycol and ethanolamines, which are precursors and hydrolysis products of sulfur and nitrogen mustards respectively.

Derivatisation is also used to reduce the reactivity and/or volatility of certain scheduled compounds prior to gas chromatography. Examples are the reactive Lewisites 1 and 2 (Schedule 1), which chromatograph poorly, the highly volatile chemicals

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perfluoroisobutylene (Schedule 2) and phosgene (Schedule 3), and reactive and corrosive precursors such as arsenic trichloride (Schedule 2).

For some applications, particularly the analysis of biomedical or environmental samples associated with investigations of alleged use, analyte concentrations may be in the low parts per billion range. In these cases, derivatisation may also be used to enhance the selectivity or sensitivity of detection. Examples are the conversion of alkyl alkylphosphonic acids and thiodiglycol to perfluorinated derivatives, to enable analysis by the very sensitive technique of negative ion chemical ionisation (NICI) mass spectrometry.

3. Overview of the methods

A detailed review of derivatisation methods of chemicals related to the CWC was published in 2003.¹

3.1 Silyl derivatives

The most broadly applicable procedure for the derivatisation of polar CW agent precursors/degradation products is conversion to trimethylsilyl (TMS) or *tert*-butyldimethylsilyl (TBDMS) esters or ethers. Conversion to TMS derivatives is the approved procedure for OPCW on-site analysis during inspections. For off-site analysis, some OPCW designated laboratories prefer to use TBDMS derivatives. These have the advantage of greater robustness with regard to traces of water and stability on storage.

Silylation can be used for alkyl alkylphosphonic acids (and their thio analogues), alkylphosphonic acids, degradation products derived from the side chain of V agents (*N,N*-dialkylaminoethane-2-thiols, *N,N*-dialkylaminoethane-2-ols), thiodiglycol and related mustard degradation products, ethanolamines derived from nitrogen mustards, and BZ and its degradation products benzilic acid and 3-quinuclidinol. For 2-(*N,N*-dialkylamino)ethanesulfonic acids, derived from oxidative decontamination of V-agents, only TBDMS derivatives appear to be stable. For thiodiglycol sulfoxide, TMS derivatisation is used to avoid Pummerer rearrangement or reduction.

The silylating agents and derivatives produced are sensitive to moisture. Consequently current recommended sample preparation procedures require concentration of aqueous extracts or solutions of polar precursors/degradation products to dryness prior to derivatisation. Residues are derivatised in an organic solvent usually with heating, particularly for the bulkier TBDMS derivatives, and analysed by gas chromatography using generic or selective detectors, or preferably by gas chromatography–mass spectrometry (GC–MS).

3.2 Methyl esters

Alkyl alkylphosphonic acids, alkylphosphonic acids, 2-(*N,N*-dialkylamino)ethanesulfonic acids, and benzilic acid are readily converted to their methyl or dimethyl esters with diazomethane. This provides a second GC–MS procedure for identification, particularly for laboratories that do not use LC–MS. An advantage of methyl esters is that they are much less sensitive to moisture than silyl esters. Also, their mass spectra are generally more

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easily interpreted and are more informative than those of silyl esters. Diazomethane does not react with non-acidic analytes such as thiodiglycol.

3.3 Perfluorinated derivatives

Perfluorinated derivatives are not generally used for the analysis of polar CW agent precursors/degradation products where concentrations are more than ca.1 part per million. They are not recommended for use in OPCW Proficiency Tests. Their main application has been for samples where analyte concentrations are in the low parts per billion range, e.g. in investigations of alleged use of CW, particularly for the analysis of biomedical samples such as urine or blood. Pentafluorobenzoyl and heptafluorobutyryl derivatives are used for the trace analysis of thiodiglycol.²⁻⁴ Pentafluorobenzyl esters are used for the trace analysis of phosphonic acids.⁵⁻⁸ Recently, 3,5-bis-trifluoromethylbenzyl derivatives have been advocated for phosphonic acids, on the basis of much simpler sample preparation compared to that currently used for pentafluorobenzyl esters.⁹ Trifluoroacetyl and heptafluorobutyryl derivatives have been explored for the environmental analysis of ethanolamines and thiodiglycol but have not found widespread use.¹⁰

3.4 Thioesters

Lewisite 1 (and to a lesser extent Lewisite 2) is an exception among Schedule 1 CW agents in that it is too reactive for routine analysis by GC. Trivalent arsenic forms strong bonds with sulphur, and this is exploited for derivatisation by converting analytes to thioesters. Alkyl or aryl thiols are used as derivatising agents for Lewisites 1 and 2, and their primary degradation products 2-chlorovinylarsonous acid and bis(2-chlorovinyl)arsinous acid, respectively. In contrast to the silylating and esterifying reagents described above, thiols are nucleophilic rather than electrophilic, and can be used in aqueous solution if required. A disadvantage of thioesters is that they do not distinguish agents from their degradation products, both converting to the same As(III) thioesters. Butanethiol (BT) is approved for on-site analysis of Lewisites and is the most commonly used reagent for derivatising intact and hydrolysed Lewisites. Dimercaptol (4-methylbenzene-1,2-dithiol, DMT) is also approved for on-site use but caution must be exercised when this reagent is used, particularly in proficiency tests, as it converts Lewisite 2 to the Lewisite 1 derivative.¹¹

Butanethiol is also used for the derivatisation of arsenic trichloride, and other arsenicals relevant to old and abandoned chemical weapons such as Adamsite, Clark 1 and Clark 2. Thiols can also be used for derivatising perfluoroisobutylene and phosgene.

4. Safety aspects

Most derivatisation reactions do not require specific precautions concerning safety. Samples to be analysed normally contain only small concentrations of toxic chemicals, and the reagents are treated as for any other volatile reactive chemical. Normal laboratory safety measures should be used during sample handling, such as protective gloves, goggles and fume hoods. Trimethylsilyl cyanide, if used, produces HCN on reaction with phosphonic acids and thiodiglycol etc. Thiol derivatising agents produce a stench and should always be confined to well-ventilated fume hoods.

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An exception is diazomethane if used for the conversion of phosphonic or benzoic acids to methyl esters. Diazomethane has relatively high toxicity (possibly carcinogenic) and has potential for detonation. Trimethylsilyldiazomethane, though not widely used, has been proposed as a safer derivatising agent for conversion of acids to methyl esters.¹²

5. Reagents and derivatives

Derivatising agents and their main applications are summarised in Table 1.

Table 1. Derivatives and reagents for the GC and GC–MS analysis of scheduled chemicals and their degradation products.

Derivative	Reagent	Analytes	Comments
trimethylsilyl	BSTFA or BSA + TMSCl or TMSCN	alkyl alkylphosphonic acids, alkylphosphonic acids, pinacolyl alcohol, thiodiglycol & homologues, thiodiglycol sulfoxide, ethanolamines,* BZ, benzoic acid, quinuclidinol, arsenic trichloride	Used in on-site analysis. Most broadly applicable derivative, mild derivatising conditions. TMSCN reagent produces fewer by-products (mainly HCN – care) and may give cleaner chromatograms.
<i>tert</i> -butyl-dimethylsilyl	MTBSTFA + 1% TBDMSCl	as above except for thiodiglycol sulfoxide	More robust towards moisture than TMS and more stable on storage. Higher temperature generally required for complete derivatisation. Produces a mixture of 4 derivatives (from Pummerer rearrangement) with thiodiglycol sulfoxide. TMS gives superior S/N with triethanolamine.
methyl ester	CH ₂ N ₂ or TMSCHN ₂ / MeOH or TMPAH	alkyl alkylphosphonic acids, alkylphosphonic acids, benzoic acid	Applicable only to acidic analytes. Trimethylsilyldiazomethane is safer to handle than diazomethane. ¹² TMPAH is used for thermally assisted methylation in the hot GC injection port; it may be used in on-site analysis for sequential methylation & silylation. ¹³
propyl ester	C ₃ H ₇ Br/K ₂ CO ₃	as above	Reported in the literature for direct derivatisation of phosphonic acids in aqueous solution prior to hollow fibre solid phase microextraction. Propyl

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			esters are not included in the OCAD.
pentafluoro-benzoyl	PFBzCl/ pyridine	thiodiglycol	Used for targeted trace analysis in biomedical samples, usually in combination with NICI-MS.
heptafluorobutyl	HFAA or HFBI	thiodiglycol	Used for targeted trace analysis in biomedical samples, with EI or positive CI-MS.
pentafluorobenzyl	PFBBBr/base	alkyl alkylphosphonic acids	Used for targeted trace analysis in biomedical samples, usually in combination with NICI-MS.
bis-3,5-trifluoro-methylpentafluoro-benzyl	3,5-(CF ₃) ₂ -C ₆ F ₃ CH ₂ N ₂	alkyl alkylphosphonic acids, alkyl phosphonic acids	Diazo reagent can be used in aqueous solution. Much simpler procedure than above, but reagent not available commercially.
trifluoroacetyl	TFAA or TFAI	thiodiglycol ethanolamines	Reported in the literature but have not been widely used.
butylthioester	BuSH (BT)	Lewisites 1 & 2, 2-chlorovinylarsonous acid, bis-(2-chlorovinyl)-arsinous acid	Used in on-site analysis. Can be used in the presence of water. Does not distinguish agent from degradation product. Ethyl and propyl thiols have also been used.
other thiol esters	HS(CH ₂) _n SH n=2,3 HSCH ₂ CH(SH)-CH ₂ OH (BAL) HSCH ₂ CO ₂ Me	Lewisites 1 & 2, and hydrolysis products	Dithiols used for analysis of Lewisite 1 hydrolysis product in biomedical samples.
dimercaptol ester	1,3,4-MeC ₆ H ₃ (SH) ₂	Lewisite 1 and 2-chlorovinylarsonous acid	Used in on-site analysis. Forms Lewisite 1 derivative with Lewisite 2.

* hydrolysis products/precursors of nitrogen mustards, *N*-methylbis(ethanolamine), *N*-ethylbis(ethanolamine), triethanolamine, and precursors or degradation products of V-agents, e.g.

HOCH₂CH₂N(iPr)₂

BSTFA = *N*,*O*-bis(trimethylsilyl)trifluoroacetamide

BSA = *N*,*O*-bis(trimethylsilyl)acetamide

TMSCl = trimethylsilyl chloride

TMSCN = trimethylsilyl cyanide

MTBSTFA = *N*-methyl-*N*-(tert-butyl)dimethylsilyl)trifluoro- acetamide

TBDMSCl = tert-butyl)dimethylsilyl chloride

BT = butanethiol

DMT = dimercaptotoluene

BAL = British anti-lewisite

PFBzCl = pentafluorobenzoyl chloride

PFBBBr = pentafluorobenzyl bromide

HFAA = heptafluorobutyric anhydride

HFBI = heptafluorobutylimidazole

TFAA = trifluoroacetic anhydride

TFAI = trifluoroacetylimidazole

TMPAH = trimethylphenylammonium hydroxide

OCAD = OPCW Central Analytical Database

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The equipment and supplies needed in derivatisations are given in Table 2 and the chemicals and solvents in Table 3. The solvents should be of high purity and intended for derivatisation.

Table 2. Equipment and supplies needed for derivatisation.

Equipment/supply	Derivatisation procedure
Adjustable pipettes or syringes	All
Reaction vials 1 ml	All
Ultrasonic bath	BSTFA, Diazomethylation
A block heater, sand bath, or laboratory oven	BSTFA, MTBSTFA/TBDMSCI, TMSCHN ₂ , PFBBBr, HFBI, PFBzCl, BT
Vortex apparatus	BSA/TMSCI, TMSCN,
Disposable syringes	Diazomethylation, BT
Disposable syringe filters 0.45 µm or 0.2 µm	Diazomethylation, BT
Spatula	PFBBBr, BT, DMT
Florisil cartridge 100 mg	PFBBBr
pH paper	BT, DMT
Centrifuge	DMT

Table 3. Chemicals needed for derivatisation.

Solvents/chemicals	Derivatisation procedure
Acetonitrile	BSTFA, TMSCN, TMSCH ₂ N ₂ , PFBBBr, HFBI, PFBzCl
Pyridine	BSA/TMCS, MTBSTFA/TBDMSCI
Acidic methanol	Diazomethylation
Methanol	TMSCHN ₂ , BT (arsenic trichloride)
<i>n</i> -Hexane	TMSCHN ₂ , BT, DMT
Dichloromethane	PFBBBr
10% Methanol in dichloromethane	PFBBBr
K ₂ CO ₃	PFBBBr
<i>n</i> -Heptane	HFBI
Water	HFBI
Toluene	PFBzCl
Triethylamine	BT (organic solvents)
Na ₂ SO ₄ (anhydrous)	BT, DMT
37% HCl	BT (soil and aqueous samples)
Acetone	DMT
0.01 M HCl	DMT
2 M HCl	DMT

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7. Recommended derivatising conditions

The following are procedures that are widely used in OPCW Proficiency Tests, on-site analysis or for biomedical samples. Conditions such as temperature, time, and relative amounts of derivatising agent may vary between laboratories.

7.1 Trimethylsilyl derivatives

7.1.1 BSTFA reagent

7.1.1.1 Organic solvents

Depending on the solvent, the organic sample (0.1–0.5 ml) may be derivatised directly in a 1 ml reaction vial or evaporated to dryness and redissolved in acetonitrile. The vial is capped using a screw cap fitted with a Teflon-coated silicone septum and BSTFA (10–50 µl) is added by injection from a syringe through the septum. The vial is typically heated at 60 °C for 30 min. The resulting solution is analysed directly by GC or GC–MS. Samples can be stored in a closed vial in a freezer at -20 °C, but care must be taken to exclude moisture, particularly on re-warming the vial to ambient temperature.

7.1.1.2 Aqueous samples or alcoholic solutions

Silylation-grade acetonitrile (0.5 ml) is added to the dry residue (concentrated from an extract or aqueous sample and purged with nitrogen to ensure dryness) in a 1 ml reaction vial. The vial is capped, using a screw cap fitted with a Teflon-coated silicone septum, and BSTFA (50 µl) is added by injection from a syringe through the septum. The vial is sonicated for 5 min to ensure mixing, and typically heated at 60 °C for 30 min. The resulting solution is analysed directly by GC or GC–MS. Samples can be stored in a closed vial in a freezer at -20 °C, but care must be taken to exclude moisture, particularly on re-warming the vial to ambient temperature.

7.1.2 BSA/TMSCl reagent

7.1.2.1 Organic solvents

Depending on the solvent, the organic sample (0.1–0.5 ml) may be derivatised directly in a 1 ml reaction vial or evaporated to dryness and redissolved in acetonitrile. The vial is capped using a screw cap fitted with a Teflon-coated silicone septum and 5:1 BSA/TMSCl reagent (20 µl) is injected from a syringe through the septum. The vial is vortexed for 30 seconds to mix the contents and allowed to stand for 45 min. The resulting solution is analysed directly by GC or GC–MS. Samples can be stored in a closed vial in a freezer at -20 °C, but care must be taken to exclude moisture, particularly on re-warming the vial to ambient temperature.

7.1.2.2 Aqueous samples or alcoholic solutions

Silylation-grade pyridine (80 µl) is added to the dry residue in a 1 ml reaction vial and the vial closed as above. 5:1 BSA/TMCS reagent (20 µl) is injected from a syringe through the septum, the vial is vortexed for 30 s to mix the contents, and allowed to stand at room temperature for 45 min. The resulting solution is analysed directly by GC or GC–MS. Samples can be stored in a closed vial in a freezer at -20°C, but care must be taken to exclude moisture, particularly on re-warming the vial to ambient temperature.

7.1.3 TMSCN reagent

7.1.3.1 Organic solvents

Depending on the solvent, the organic sample (0.1–0.5 ml) may be derivatised directly in a 1 ml reaction vial or evaporated to dryness and redissolved in acetonitrile. The vial is capped using a screw cap fitted with a Teflon-coated silicone septum and TMSCN (50 µl) is added by injection from a syringe through the septum. The vial is closed, vortexed for 1 min, and allowed to stand at ambient temperature for 5 min. The resulting solution is analysed directly by GC or GC–MS. Samples can be stored in a closed vial in a freezer at -20 °C, but care must be taken to exclude moisture, particularly on re-warming the vial to ambient temperature.

7.1.3.2 Aqueous samples or alcoholic solutions

TMSCN (50 µl) and silylation-grade acetonitrile (50 µl) are added to the dried residue in a 1 ml reaction vial. The vial is closed as above, vortexed for 1 min, and allowed to stand at ambient temperature for 5 min. The resulting solution is analysed directly by GC or GC–MS. Samples can be stored in a closed vial in a freezer at -20°C, but care must be taken to exclude moisture, particularly on re-warming the vial to ambient temperature.

7.2 Tert-butyldimethylsilyl derivatives

7.2.1 Organic solvents

Depending on the solvent, the organic sample (0.1–0.5 ml) may be derivatised directly in a 1 ml reaction vial or evaporated to dryness and redissolved in acetonitrile. The vial is capped using a screw cap fitted with a Teflon-coated silicone septum and MTBSTFA / 1% TBDMSCl (10–50 µl) is added by injection from a syringe through the septum. The vial is typically heated between 60 °C and 100 °C for 30 min. For sterically hindered analytes it is recommended that the higher temperature is used. Note that hard Teflon-faced rubber septa are recommended as losses may occur when heating vials sealed with soft Teflon-faced silicone septa. The resulting solution is analysed directly by GC or GC–MS. Samples can

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be stored in a closed vial in a freezer at -20 °C, but care must be taken to exclude moisture, particularly on re-warming the vial to ambient temperature.

7.2.2 Aqueous samples or alcoholic solutions

Silylation-grade acetonitrile (50 µl), silylation-grade pyridine (5 µl) and MTBSTFA / 1% TBDMSCl (50 µl) are added to the dried residue in a 1 ml reaction vial. The vial is capped using a Teflon-faced rubber septum, vortexed to mix the contents and heated, typically between 60 °C and 100 °C for 30 min. For sterically hindered analytes it is recommended that the higher temperature is used. Note that hard Teflon-faced rubber septa are recommended as losses may occur when heating vials sealed with soft Teflon-faced silicone septa. The resulting solution is analysed directly by GC or GC–MS. Samples can be stored in a closed vial in a freezer at -20 °C, but care must be taken to exclude moisture, particularly on re-warming the vial to ambient temperature.

7.3 Methyl esters

7.3.1 Diazomethane reagent

Diazomethane in ether solution is generated from *N*-methyl-*N*-nitroso-*N'*-nitroguanidine or *N*-methyl-*N*-nitroso-*p*-toluenesulfonamide (Diazald) using a commercial generation kit, observing the recommended safety precautions.¹⁴⁻¹⁸ Diazomethane in ether can be stored in a closed vial in a freezer for several days provided it retains its yellow colour. An alternative procedure is to generate gaseous diazomethane and methylate a solution of the analyte in acidic methanol solution using nitrogen for gas transfer.¹⁹⁻²⁰ Samples can be stored in a closed vial in a freezer at -20 °C.

Note! *Incomplete derivatisation can lead to rapid column deterioration and contamination of the GC injector.*

7.3.1.1 Organic solvents

The organic sample (0.5–1 ml) in a 4 ml vial is methylated with a large excess of diazomethane, typically for 10–15 min at ambient temperature.

7.3.1.2 Aqueous samples

The dry residue is dissolved in dry acidic (HCl) methanol (2 ml) by sonicating for 5 min. If necessary, the solution is filtered through a 0.45 or 0.2 µm disposable filter, and methylated with a large excess of diazomethane in ether, typically for 10–15 min at ambient temperature. (Note that the acids are isolated in a dry solution prior to derivatisation to avoid excessive consumption of the reagent).

7.3.2 Trimethylsilyldiazomethane reagent

TMSCHN₂ (TMSDAM) is a safer and more stable reagent than diazomethane, and is commercially available as a 2 M solution in hexane. To the dry residue in acetonitrile (40 µl) in a reaction vial is added a mixture of methanol (30 µl), hexane (80 µl), and 2.0 M TMSDAM in hexane (30 µl). The reaction vial is capped and heated at 80 °C for 20 min; the solution can be injected directly into the GC.

7.4 Pentafluorobenzyl esters

Pentafluorobenzyl (PFB) esters are used mainly for trace-level detection of alkyl alkylphosphonic acids in biomedical fluids (usually blood, plasma, or urine).

The residue obtained from SPE of urine (1 ml) (e.g. from an Oasis HLB polymeric cartridge) is dissolved in acetonitrile (500 µl) and derivatised with pentafluorobenzyl bromide (PFBBBr) (10 µl) in the presence of potassium carbonate (20 mg) in a sealed 1 ml reaction vial at 90 °C for 45 min. The reaction mixture is carefully concentrated to dryness under a stream of nitrogen at 50 °C and re-dissolved in dichloromethane (300 µl). The derivatised sample is added to a 1 ml / 100 mg Florisil SPE cartridge, pre-conditioned with dichloromethane (1 ml), the cartridge is washed with dichloromethane (1 ml) to remove excess PFBBBr, and the derivatised analyte eluted with 10% methanol in dichloromethane (2 x 0.75 ml). The eluate is evaporated to dryness under nitrogen and redissolved in dichloromethane (100 µl) for analysis by GC–MS (NICI).

7.5 Bis-Heptafluorobutyryl derivative of thiodiglycol

The bis-heptafluorobutyryl (HFB) derivative is used for the trace analysis of thiodiglycol in biomedical samples, usually urine. The dried residue from SPE of urine (1 ml) (e.g. from an Oasis HLB polymeric cartridge) in a 1 ml reaction vial is dissolved in acetonitrile (20 µl), heptafluorobutyrylimidazole, HFBI (30 µl) is added, and the sample heated at 50 °C for 30 min. After cooling, water (200 µl) is added to destroy excess derivatising agent. The acetonitrile / water mixture is extracted with heptane (2 x 300 µl), the two extracts combined, and concentrated to approximately 50 µl under a stream of nitrogen.

7.6 Bis-Pentafluorobenzoyl derivative of thiodiglycol

Pyridine (50 µl) and pentafluorobenzoyl chloride (PFBzCl) (10 µl) are added to the dried residue from SPE (e.g. from an Oasis HLB polymeric cartridge) of urine in a 1 ml reaction vial. The mixture is vortexed and allowed to stand at room temperature for 5 minutes. Toluene (440 µl) is added, the sample is vortexed, centrifuged, and the supernatant analysed directly.

7.7 Thioesters

7.7.1 Butanethiol (BT) reagent

7.7.1.1 Organic solvents

Dichloromethane or hexane extracts may be derivatised directly by addition of BuSH (BT) in hexane or neat. The solution is allowed to stand at ambient temperature for ca.15 min.

Alternatively, to derivatise intact/hydrolysed Lewisite* organic samples/extracts (50 µl) can be taken in a micro vial, to which 50 µl is added of a 0.1% solution of butanethiol in hexane and 10 µl of triethylamine. The vial is capped, shaken vigorously, and heated at 60 °C for 10 minutes. The resulting solution is analysed directly by GC or GC–MS.

Note! *In organic solvents and extracts, the same procedure with butanethiol is also used for the derivatisation of cyanogen chloride, arsenic oxide, Adamsite, Clark I, and Clark II.*

In addition, to derivatise **arsenic trichloride** in organic samples and extracts, the following optimised procedure can be recommended: * Organic sample/extract (50 µl) is taken in a micro vial, to which 50 µl of methanol is added. The vial is capped, shaken vigorously. 50 µl of a 0.1% solution of butanethiol in hexane and 10 µl of triethylamine is added. The vial is capped again, shaken vigorously, and heated at 60 °C for 10 minutes. The resulting solution is analysed directly by GC or GC–MS.

7.7.1.2 Soil samples

Lewisites

Soil, typically 5 g, is weighed into a suitable extraction vessel and a solution of 1 mg/ml butanethiol (BT) in hexane (10 ml) is added. If the sample contains appreciable amounts of moisture, an equal weight of anhydrous sodium sulphate is added to the sample and mixed well prior to addition of solvent. Extraction is usually performed by mechanical shaking, typically for 10 minutes. After allowing suspended material to settle, the solvent layer is removed using a Pasteur pipette and transferred to a suitable clean vessel. Extraction with a second portion of solvent is then performed if required. Extracts are combined and an additional drying step with anhydrous sodium sulphate may be performed. Aliquots of extract are filtered if necessary (0.2 or 0.45 µm PTFE filters are recommended) and may be concentrated to small volume under a gentle stream of nitrogen prior to analysis by GC–MS.

Lewisite hydrolysis products

For extractive derivatisation of hydrolysis products of Lewisites I and II, soil samples are acidified with 1 M hydrochloric acid and extracted with a solution of 1 mg/ml butanethiol in hexane as above. The breakdown products are detected as the same derivatives as those from intact Lewisites. Any intact Lewisites must be extracted, as above, from samples prior to acidification.

* The method is used in the Spiez laboratory.

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Alternatively, to derivatise intact/hydrolysed Lewisites in acidic conditions[†] 0.5 g of soil can be weighed into a 6 ml sample vial with Teflon-lined screw cap, to which 0.5 ml of 37% hydrochloric acid (HCl) is added. 1 ml of a 0.1% solution of butanethiol in hexane is added. The vial is capped, shaken vigorously, and heated at 60 °C for 30 minutes under repeatedly vigorous shaking. An aliquot of 150 µl of the organic layer is transferred to a micro vial. The solution is concentrated to 50 µl with mild nitrogen flow in order to remove HCl residues and analysed directly by GC or GC–MS.

7.7.1.3 Aqueous samples

An aliquot of an aqueous sample (5 ml) is transferred to a 15 ml centrifuge tube and acidified with 1 M HCl (1 ml). A solution of 1 mg/ml butanethiol in hexane (1 ml) is added, the tube is capped, and the mixture shaken, typically for 10 min. If the volume of water is <5 ml, the amount of reagent is adjusted proportionally. The phases are allowed to separate (1 ml of hexane is added to facilitate if necessary) and the upper organic layer transferred to a 4 ml vial. Combined extracts are dried, either by addition of anhydrous sodium sulphate and shaking, or by passing through a small column of anhydrous sodium sulphate contained in a Pasteur pipette. Dried extracts are filtered, 0.2 or 0.45 µm PTFE filters are recommended, and may be concentrated to small volume under a gentle stream of nitrogen prior to analysis by GC–MS.

Alternatively, to derivatise intact/hydrolysed Lewisite[†] in acidic conditions an aliquot of an aqueous sample (50 µl) can be taken in a micro vial, to which 50 µl of 37% hydrochloric acid (HCl) is added. 100 µl of a 0.1% solution of butanethiol in hexane and the vial is capped, shaken vigorously, and heated at 60 °C for 30 minutes under repeatedly vigorous shaking. An aliquot of 50 µl of the organic layer is transferred to a micro vial. The solution is concentrated to 25 µl with mild nitrogen flow in order to remove HCl residues and analysed directly by GC or GC–MS.

7.7.2 Dimercaptotoluene (DMT) reagent

7.7.2.1 Organic solvents

Organic sample can be derivatised with DMT for analysis of PFIB (1,1,3,3,3-pentafluoro-2-(trifluoromethyl)-1-propene).[†] Organic sample/extract (50 µl) is taken in a micro vial, to which 50 µl of a 5 mg/ml of 3,4-dimercaptotoluene (DMT) in acetone is added. The vial is capped, shaken, and allowed to stand for 10 minutes. 50 µl of hexane is added and the sample is shaken vigorously for 30 seconds every 5 minutes for 30 minutes. The resulting solution is analysed directly by GC or GC–MS.

7.7.2.2 Soil samples

Soil (ca. 5 g) is weighed into a 25 ml vial, 0.1 M hydrochloric acid (15 ml) is added, the vial is capped and shaken gently to ensure that the soil is fully wetted. The pH of the

[†] The method is used in the Spiez laboratory.

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aqueous layer is rechecked and adjusted if necessary to ≤ 2 with 2 M hydrochloric acid. The pH is checked and adjusted again as necessary until it remains stable at ≤ 2 for a period of at least 30 min. A fresh solution (not older than 2 months) of dimercaptol in acetone (5 mg/ml, 0.1 ml) is added using a glass syringe. The vial is capped and shaken vigorously for a few seconds, and allowed to stand for 10 min to allow the reaction to complete. Hexane (1 ml) is pipetted into the vial, the vial is capped tightly and shaken vigorously for 30 seconds every 5 min for 30 min. The sample is decanted into a 15 ml centrifuge tube and centrifuged for 5 min. The hexane layer is transferred by pipette to a 4 ml vial, a small amount of anhydrous sodium sulphate is added, the vial is capped and shaken, and allowed to stand for 30 min. If necessary, the dried extract is filtered into another 4 ml vial prior to GC–MS analysis.

7.7.2.3 Aqueous samples

An aliquot of water sample or aqueous extract (10 ml) is transferred to a 25 ml centrifuge tube and the pH adjusted if necessary to ≤ 2 with 2 M hydrochloric acid. The pH-adjusted sample is filtered if necessary and transferred to a 25 ml vial. A fresh solution (not older than 2 months) of dimercaptol in acetone (5 mg/ml, 0.1 ml) is added using a glass syringe. The vial is capped and shaken vigorously for a few seconds, and allowed to stand for 10 min to allow the reaction to complete. Hexane (1 ml) is pipetted into the vial, the vial is capped and shaken vigorously for 30 seconds every 5 minutes, for a total of 30 min. The hexane layer is allowed to separate; if necessary, it is withdrawn, transferred to a centrifuge tube and centrifuged for 5 minutes. The hexane layer is transferred by pipette to a 4 ml vial, a small amount of anhydrous sodium sulphate is added, the vial is capped and shaken, and allowed to stand for 30 min. If necessary, the dried extract is filtered into another 4 ml vial prior to GC–MS analysis

8. Applicability

Silyl derivatives, methyl esters and thioesters are recommended for use in OPCW Proficiency Tests, and generally for the analysis of samples where concentrations are likely to be above ca. 100 parts per billion. Mass spectra for some derivatives, TMS, TBDMS, and methyl esters, butyl thioesters of Lewisites 1 and 2, and the dimercaptol ester of Lewisite 1, are included in the OCAD.

The First OPCW Confidence Building Exercise on Biomedical Sample Analysis, held in 2009, demonstrated that silyl derivatives and methyl esters are well-suited to the identification of CW degradation products or metabolites at spiking levels of 100 ng/ml (in clean synthetic urine). However, these derivatives gave poor signal-to-noise ratios at analyte spiking levels of 10 ng/ml, and selected ion or selected reaction chromatograms usually showed several additional peaks across displayed chromatographic windows. At these lower concentrations, perfluorinated derivatives were clearly superior in terms of selectivity and sensitivity of detection, although they may be more restricted in terms of the number of ions or transitions available for monitoring.²¹

Derivatives for the trace analysis of 2-chlorovinylarsonous acid in urine have not been directly compared. The most sensitive procedure reported uses *in situ* derivatisation with 1,3-propanedithiol, extraction of the derivatised analyte by SPME, and analysis by GC–MS/MS.

9. Advantages / Disadvantages

The clear advantage of derivatisation is that it extends the range of analytes amenable to gas chromatographic analysis. Some derivatives may enhance the mass spectrometric properties in terms of structurally characteristic high-mass ions. For trace analysis, derivatisation allows a degree of tailoring of the properties to suit different methods of detection, for example perfluorinated derivatives for use with NICI–MS.

Derivatisation can be a source of error. Residual traces of water or large amounts of extraneous materials extracted from the matrix can suppress derivatisation, or produce a complex chemical background. Most derivatising agents are reactive electrophiles; many environmental contaminants are nucleophiles that will also react with the derivatising agent.

Polar analytes are most often present in aqueous samples, or aqueous extracts of complex matrices such as soil. Derivatisation of polar analytes using current ROPs requires concentration of aqueous solutions to dryness, a procedure that is time consuming and which can be an additional source of error. Residual traces of water may react with the reagent and the derivative, and losses of analyte may occur on evaporation of aqueous solutions if the analyte has a degree of volatility; examples are thiodiglycol and pinacolyl alcohol.

Derivatisation may introduce artefacts. An example recently suspected in an OPCW Proficiency Test was the partial O-dealkylation of a V agent during derivatisation of the sample with a silylating agent. Lewisite 2 is converted to the Lewisite 1 derivative if dimercaptol is used as derivatising agent. In the case of methyl esters of alkyl alkylphosphonic acids or alkylphosphonic acids, the esters themselves are important analytes, some of them being precursors or by-products in nerve agent production. It is therefore important to distinguish methyl esters as derivatives from methyl esters as analytes. The OPCW laboratory requires laboratories participating in proficiency tests to demonstrate that any analyte identified as a methyl or other alkyl ester after derivatisation was not present as that derivative in the original sample.

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1. Introduction

Sample clean-up methods are important tools for successful and reliable analysis and especially important with chromatographic separations such as GC and LC and also for electrophoresis. Solid matrices, such as soil, and liquid matrices, such as wastewater, can contain hundreds of chemicals. Chromatograms of these may become complex, making identification difficult. This is true especially when the analyte concentration is much lower than the concentration of the interferences. This means that sample clean-up is needed before reliable analysis. Another case is when the sample contains very high boiling material, which can cause different problems, e.g. injection port or the front edge of the column can be polluted, which can cause adsorption of the analytes before the chromatographic separation.

Many methods to clean the samples also have to be used in analysis of chemical warfare agents (CWAs) and their degradation products. This chapter presents the most commonly used methods: filtration, centrifugation, solvent exchange and solid phase extraction (SPE) methods used for cation exchange and for isolation of water-soluble chemicals. Solid phase microextraction (SPME) and liquid phase microextraction (LPME) methods, which are used for extraction of CWC-related chemicals, are presented in Section 2, Part A, Chapter III: *Aqueous liquid samples*, however. Some of these recommendations are not clear recommendations, but discuss essential issues to use the method properly.

2. Safety aspects and storing

Safety aspects of sample preparation and storing of samples are presented in Section 2, Part A, Chapter I: *Introduction*.

3. Filtration

Filtration is an easy method to clear samples and extracts before analysis and before the next sample preparation step, e.g. before concentration or evaporation to dryness. Soil extracts and solid material extracts are usually filtered through traditional filter paper in a filtration funnel. Whatman 4 (pore size 20–25 µm) has proved to be a good choice for first filtration. It offers fast filtering with a good retention of coarse particles and gelatinous precipitates and is useful as a rapid filter for cleaning up extracts.

Sometimes finer filtering is needed. The sample or extract can be filtered with a filter paper of smaller pore size. Whatman 1 with pore size of 11 µm still offers medium retention and flow rate. However, membrane filters with a pore size of 0.45 or 0.5 µm can also be used, especially when only a portion (<1 ml) of the sample or extract has to be filtered.

When membrane filters are used, it should be noted that different membranes have different resistance to chemicals. In addition, hydrophilic membranes must be used with hydrophilic samples and hydrophobic membranes with hydrophobic samples, respectively. To easily resolve the last-mentioned problem of having many different types of filters to different applications, it is recommended to use regenerated cellulose membrane filters with which both aqueous and organic samples and extracts can be filtered (e.g. Millex LRC and Whatman RC).

4. Centrifugation

Centrifugation is a fast and efficient method to separate a heterogeneous mixture of solid and liquid, e.g. to separate the extraction solvent of the soil sample from the matrix before filtration. After successful centrifugation, the solid precipitate is settled at the bottom of the test tube and the solution above it, and the centrifugate is clear. If the centrifugate is not clear, you can always increase the time of centrifugation or the centrifugal force.

It should be noted that water-organic solvent emulsions can also efficiently be separated into two phases by centrifugation.

Note! *Centrifuge always has to be in balance. This means that in the opposite holder there has to be another sample of the same weight.*

Usually, a centrifugal force of 2000 G is recommended. The amount of revolutions per minute RPM (speed) for a specific rotor can be calculated using the following equation:

$$RPM = 1000 \times \sqrt{\frac{RCF}{11.17 \times R_{\max}}}$$

RCF = Relative centrifugal force, e.g. 2000 G

$R_{\max}$ = Maximum radius from the axis of rotation in centimeters

Special centrifuge tubes can be used. However, these are not usually needed, if the centrifugal force, ca. 2000 G, is used. The samples can be centrifuged in normal screw cap glass sample bottles and vials. In that case, special care must be exercised with these containers. In addition, the use of adapters by which small tubes are fitted in the rotor cavities are recommended.

Note! *Take care that a glass bottle/vial is not jiggling during centrifugation and not in contact with the metal surface.*



Figure 1. Hettich Rotofix 32A centrifuge.

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5. Solvent exchange

Solvent exchange is used to make the sample more suitable for chromatographic separation, spectrometric analysis and derivatisation, and also to clean the sample from matrix components:

- Polar solvent can be evaporated totally or partly to get peaks of a polar analyte sharper and to avoid peak splitting when nonpolar column is used for analysis in GC
- Organic solvent is changed to be more suitable for LC–MS analysis
- Buffers and components of sample disturbing ionisation are evaporated for LC–MS analysis
- Organic solutions are concentrated and extracted with acetonitrile to remove hydrocarbon background: see section 5.1
- Water is evaporated to dryness before methylation
- Water and alcohols are evaporated to dryness before silylation

Note! SPE is also a solvent exchange method when the analytes of interest are trapped in the solid phase and extracted with another solvent. The sample is usually concentrated in tandem.

Methods for solvent exchange by concentration and evaporation are presented in Section 2, Part A, Chapter II: *Concentration*.

5.1 Acetonitrile extraction of organic solutions to remove hydrocarbon background

Hydrocarbon background is removed from organic liquid by partitioning the analytes from nonpolar hydrocarbon solvent to relatively polar aprotic solvent. The method is applicable to organic solvents immiscible in acetonitrile.

Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in acetonitrile extraction are presented in Table 1.

Table 1. Equipment, supplies and chemicals needed in acetonitrile extraction of organic solutions.

Equipment	Supply	Chemical
Vortex shaker (option)	Pasteur pipettes	<i>n</i> -Hexane
	Sample vials 1.5 and 4 ml	Acetonitrile
	Adjustable pipette(s) or measuring cylinders [#]	Dichloromethane

[#] to divide the sample/extract and to measure solvents

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Procedure

Take 1 ml of organic liquid into a screw capped sample vial. Add 300 µl of high-purity acetonitrile and shake for 10 minutes in vortex shaker or manually. Allow the mixture to stand for 20 minutes, preferably at ca. 0 °C. Draw the acetonitrile portion from bottom and transfer into another sample vial. Repeat the process twice and combine the acetonitrile extracts. Concentrate the acetonitrile extract to ca. 100 µl and add 100 µl of dichloromethane. Analyse the extract with GC and GC–MS. This organic extract is analysed for nonpolar CWC-related chemicals.

Also analyse the extract after silylation (Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*). The silylated organic extract is analysed for polar CWC-related chemicals.

Note! *The method is applicable to organic liquids immiscible in acetonitrile like hydrocarbons and silicon oil. In case of solvents miscible with acetonitrile, the organic liquid can be concentrated with mild nitrogen flow (not to dryness) and reconstituted with n-hexane. Thereafter, the acetonitrile extraction can be performed.*

Evaluation of method

The acetonitrile extraction method of organic solutions is simple and robust. It has been tested with samples provided in OPCW Proficiency Tests. Recovery of very nonpolar phosphonates (e.g. dioctyl or didecyl alkylphosphonates) may be lower than their more polar, smaller homologs (e.g. dipropyl alkylphosphonates). Other polar aprotic solvents (e.g. dimethylformamide), which are immiscible in hexane, can be also used.¹

6. Solid phase extraction methods

6.1 Cleaning of hydrocarbon background with silica cartridges

Hydrocarbon and other relatively nonpolar background chemicals are removed from an organic liquid sample and organic extracts using a silica SPE (Si) cartridge by partitioning the analytes from a nonpolar hydrocarbon solvent to a polar Si sorbent. The sorbent is washed with a nonpolar solvent and the analytes are extracted out of the sorbent with a polar solvent.

Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in SPE cleaning of hydrocarbon background are presented in Table 2.

Procedure

Elute a portion of 200–500 µl of the organic liquid sample slowly (a drop in second) through a Si cartridge. Wash the cartridge with 2 ml of *n*-hexane and extract with methanol to an end volume of 500 µl.

This sample is analysed for volatile CWC-related chemicals (soluble in organic solvents).

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Alternatively, a Si cartridge can be preconditioned with 1 ml of *n*-hexane, the organic liquid sample can be diluted before elution with an equal volume of *n*-hexane, and 2 ml of methanol can be used for extraction.

Note! The quality of Si cartridges may vary between different manufacturers and batches of cartridges. Usually, manufactures do not recommend using cartridges over two years old.

Table 2. Equipment, supplies, and chemicals needed in cleaning of hydrocarbon background using Si cartridges.

Equipment	Supply	Chemical
Vacuum manifold (option)	Pasteur pipettes	<i>n</i> -Hexane
	Sample vials 1.5 and 4 ml	Methanol
	Adjustable pipette(s) or measuring cylinders [#]	
	Si (or NH ₂) SPE cartridges (100 mg)	
	Disposable syringes 2, 5, or 10 ml	
	Adapters for SPE (optional)	
	Stop cocks for SPE (optional)	

[#] to divide the sample/extract and to measure solvents

Evaluation of method

The most nonpolar CWC-related chemicals such as mustard gas do not efficiently retain on Si cartridge, and the amount that has been retained is extracted out of the cartridge with the washing solvent (*n*-hexane). This means that these chemicals are not present in the methanol extract of a Si cartridge.

The cartridge can be extracted with a more nonpolar solvent such as acetonitrile or ethyl acetate instead of methanol. However, the more polar CWC-related chemicals such as VX are not extracted with these solvents, but methanol is needed to release it from the cartridge. For this reason, methanol is recommended as the extraction solvent.

Do not throw away the sample eluted through the cartridge, the washing solvent(s), and the cartridge itself before you have analysed the extract and are satisfied with the results of this purification step.

Aminopropyl NH₂ cartridges can be used similarly to remove hydrocarbon background from organic samples and extracts.

6.2 Ion exchange methods

6.2.1 Cation exchange with strong cation exchange cartridges (SCX) to remove metallic cations from aqueous samples

Cations are removed from an aqueous sample or extract with a cation exchange cartridge to avoid salt formation of organic bases and acids during sample preparation.

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Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in removing of metallic cations are presented in Table 3.

Table 3. Equipment, supplies and chemicals needed in cleaning of metallic cations from aqueous samples with SCX cartridges.

Equipment	Supply	Chemical
Vacuum manifold (option)	Pasteur pipettes	Methanol
	Sample vials 1.5, 4, 8, and 20 ml	Clean water
	Adjustable pipette(s) or measuring cylinders [#]	0.1 M and 2 M NH ₄ OH solutions
	SCX SPE cartridges (100 or 500 mg)	
	Disposable syringes 2, 5, or 10 ml	
	Adapters for SPE (optional)	
	Stop cocks for SPE (optional)	
	pH paper (1–11)	

[#] to divide the sample/extract and to measure solvents

Procedure

Measure the pH of the aqueous sample or extract. The pH of the sample has to be ca. 7 or higher when the cation exchange is started. If the aqueous sample is acidic, neutralise it with dilute ammonium hydroxide solution before elution into the cartridge.

Condition a SCX cartridge by following the instructions given by the manufacturer (e.g. elute 300 µl of high-purity methanol and 2 ml of high-purity water through the cartridge). Do not let the SPE extraction material dry. Push 2–10 ml of the aqueous sample or extract slowly through the adsorbent bed using a plastic syringe adjustable to the cartridge. Measure the pH of the effluent. The pH of the effluent is determined to test the completeness of the cation exchange, which is repeated if necessary. The sample should be more acidic after cation exchange than before it.

Separate a part of the effluent (0.5–1 ml) and filter through a disposable syringe filter for LC–MS and CE analysis.

Evaporate the rest to dryness for GC and GC–MS analysis as presented in Section 2, Part A, Chapter II: *Concentration*, and derivatise the residue as presented in Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*.

This sample is analysed for polar alkylphosphonic and alkylthiophosphonic acids.

Note! Amino alcohols remain in the cation exchanger and cannot be recovered.

Evaluation of method

Cation exchange has demonstrated its usefulness in international comparison and proficiency tests to get polar water-soluble alkylphosphonic and alkylthiophosphonic acids well recovered after derivatisation.² Kataoka *et al.* have clearly shown that the

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improvement of yields of the acids is related to the decrease of concentrations of divalent metal cations in aqueous soil extracts after cation exchange.³

6.2.2 Anion exchange with strong anion-exchange cartridges for isolation of alkylphosphonic and alkylthiophosphonic acids from aqueous samples

Alkylphosphonic and alkylthiophosphonic acids are extracted from an aqueous sample or extract with a strong anion-exchange cartridge.

Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in isolating of alkylphosphonic and alkylthiophosphonic acids are presented in Table 4.

Table 4. Equipment, supplies and chemicals needed in isolating of alkylphosphonic and alkylthiophosphonic acids from aqueous samples.

Equipment	Supply	Chemical
Vacuum manifold (option)	Pasteur pipettes	Methanol
	Sample vials 1.5, 4, 8, and 20 ml	Clean water
	Adjustable pipette(s) or measuring cylinders [#]	0.1 M and 2 M NH ₄ OH solutions
	Oasis [®] mixed mode anion-exchange MAX (30 mg, 1 ml, Waters)	Acetone
	Disposable syringes 2, 5, or 10 ml	0.1 M HCl solution in methanol
	Adapters for SPE (optional)	
	Stop cocks for SPE (optional)	
	pH paper (1–11)	

[#] to divide the sample/extract and to measure solvents

Procedure

Place the cartridge over the vacuum manifold and condition an Oasis[®] MAX cartridge with 1–2 ml of methanol followed by 1 ml of clean water. Elute 1 ml of aqueous sample or extract through the cartridge under gravity. Wash the cartridge with 2 ml of clean water. If necessary, wash with 0.5 ml of high-purity methanol to remove significant quantities of polyethylene glycols (PEGs). Completely dry the sorbent by applying a vacuum for 2 min. Finally, extract with 1 ml 0.1 M of acidic methanol (methanol acidified with aqueous HCl) under gravity. Evaporate the eluate to dryness on a rotary evaporator or with mild nitrogen flow at room temperature as described in Section 2, Part A, Chapter II: *Concentration*.

Before analysis with GC–MS, the residue is dissolved in 100–200 µl of high-purity acetonitrile and silylated as described in Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*.

This sample is analysed for polar alkylphosphonic and alkylthiophosphonic acids.

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If silica-based Accubond II (Agilent; 200 mg) or strong anion-exchange (SAX) cartridges are used, the procedure remains the same as mentioned above except the conditioning and washing steps. The cartridges are conditioned with 200 µl of acetone followed by 3 ml of methanol and then 3 ml of water. Methanol washing must not be performed. In addition, take special care that the cartridge does not get dry after conditioning.

Note! The extraction is based on ionic interactions and is pH-dependent; hence sample pH must be higher than 6 and eluent (acidic methanol) pH must be ca. 2.

Evaluation of method

The method has been duly validated with OPCW proficiency test samples and can extract acidic CWC-related chemicals with high efficiency. The MAX cartridge is slightly better than the silica-based SAX cartridge, as the recoveries of alkylphosphonic acids are higher. Furthermore, higher PEGs (molecular weight ≥ 300) can be effectively removed by methanol washing. As an advantage, drying the MAX cartridge after conditioning and washing does not decrease the recoveries of alkylphosphonic acids.⁴

6.2.3 Cation exchange with strong cation-exchange cartridges (SCX) for isolation of amino alcohols from aqueous samples

Amino alcohols are extracted from an aqueous sample or extract with a strong cation exchange SCX cartridge.

Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in isolating of amino alcohols are presented in Table 5.

Table 5. Equipment, supplies, and chemicals needed in isolating of amino alcohols from aqueous samples.

Equipment	Supply	Chemical
Vacuum manifold (option)	Pasteur pipettes	Methanol
	Sample vials 1.5, 4, 8, and 20 ml	Clean water
	Adjustable pipette(s) or measuring cylinders [#]	10% NH ₄ OH in methanol
	SCX SPE cartridges (500 mg, 3 ml)	
	Disposable syringes 2, 5, or 10 ml	0.1 M HCl solution in methanol
	Adapters for SPE (optional)	
	Stop cocks for SPE (optional)	
	pH paper (1–11)	

[#] to divide the sample/extract and to measure solvents

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Procedure

Condition the strong cation-exchange SCX cartridge with 2–3 ml of methanol and 3–5 ml of clean water. Do not let the cartridge dry. Elute 1–2 ml of aqueous sample or extract (preferably under gravity) through the cartridge. After the sample has completely eluted through the cartridge, wash the cartridge with 3 ml of clean water. Purge the cartridge with nitrogen stream for 1 minute to remove water from the sorbent. Finally, extract with 2 ml of 10% ammonium hydroxide in methanol. Separate a portion of eluate, filter through a 0.45 µm disposable syringe filter, and analyse with LC–MS.

Evaporate the rest of the eluate to dryness with mild nitrogen flow at room temperature or on a rotary evaporator as described in Section 2, Part A, Chapter II: *Concentration*. The evaporation residue is dissolved in 100–200 µl of high-purity acetonitrile. A small portion of this acetonitrile can be filtered as described above for LC–MS analysis, and the rest of it is silylated as described in Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*.

These samples are analysed for polar amino alcohols.

Note! The extraction efficiency depends on pH and, hence, the sample pH must be lower than 8 and the extraction solvent (10% NH₄OH) pH must be above 10.

Evaluation of method

The method is highly sensitive and can extract traces of amino alcohols. Aqueous ammonium hydroxide solution may also be used for extraction; however, methanolic solution is recommended to get the maximum recoveries. Methanol helps in disrupting secondary hydrophobic interactions between carbon chain of amino alcohols and SCX material.⁵

7. Quality control

Quality control of sample preparation is presented in Section 2, Part A, Chapter V: *Quality assurance and quality control in sample preparation*.

8. References

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Chapter V.

Quality assurance and quality control in sample preparation

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Section 1, Part A, Chapter IV: *Quality control* and Section 4, Chapter III: *Chain of custody*, discuss the importance of coding of samples and chain of custody (documentation showing the full process of acquisition, transfer, handling and disposal of samples) in quality assurance of chemical analysis. Safety aspects of sample preparation and storing of samples are presented in Section 2, Part A, Chapter I: *Introduction*. This chapter describes different means for assuring and controlling the quality of sample preparation.

1. Plan the sample preparation carefully and make it efficient and simple.
2. Follow the recommended operating procedures in all sample preparations.
3. Check the purity of solvents (also the concentrated solvent) and reagents before sample preparation with the same methods used for analysis.
4. Go through the whole sample preparation procedure without samples, using the same solvents, same reagents, and similar glassware to investigate the background generated during the procedure (method blank).
5. If possible, prepare at least two parallel samples.
6. Determine the approximate recovery of the sample preparation method by spiking the blank matrix with a known amount of the compounds of interest and repeating the sample preparation (this assumes that a clearly marked blank sample of the same matrix has been delivered with the samples).
7. Use separate glassware for all samples, blind samples and blank matrix samples to avoid cross-contamination.
8. Pay attention to clear and unambiguous record-keeping to ensure unbroken chain of custody.

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Chapter I. Organic samples

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1. Scope

Organic samples are samples in organic solvents. They can be collected directly from the inspected sites or can be organic solvent extracts of aqueous, soil, wipe, active charcoal, concrete, paint, rubber, other polymeric samples and air sampling tubes. A wide range of organic solvents or their mixtures can be used. This chapter describes methods for the preparation of organic samples for the analysis of chemical warfare agents (CWAs) and their degradation products, where chromatographic methods such as gas chromatography (GC), liquid chromatography (LC) as an analysis technique or an introduction technique to spectrometers are used as the separation technique. The sample preparation methods required for analysis by nuclear magnetic resonance (NMR) spectroscopy are also presented.

2. Summary of the method

A general flowchart of sample preparation of organic sample is given in Figure 1. An organic sample is divided into several sub-samples, which are treated in manners described in this chapter. Three separate sub-samples of untreated organic sample are analysed directly by GC and GC–MS as well as with minimum sample preparation with LC–MS and NMR. Three other separate sub-samples are derivatised: methylated, silylated and derivatised with thiols for identification of the Lewisites. Still, a separate sub-sample is cleaned up, if needed, with normal-phase silica Si solid phase extraction (SPE). A sub-sample can also be analysed after solvent exchange using LC–MS for analysis and one sub-sample is treated for NMR analysis. Finally, one sub-sample can be extracted after solvent exchange with acetonitrile using liquid-liquid extraction. After extraction, a part of this sample is analysed with GC–MS, while another part is evaporated almost to dryness and derivatised for GC–MS analysis.

3. Safety aspects and storing

Safety aspects of sample preparation and storing of samples are presented in Section 2, Part A, Chapter I: *Introduction*.

4. Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in this ROP are presented in Table 1.

5. Procedure

5.1 Division of the sample

If the amount of the original organic sample is 5 ml or bigger, it is always better to take a fresh portion of sample for each step to assure good recovery of chemicals of interest.

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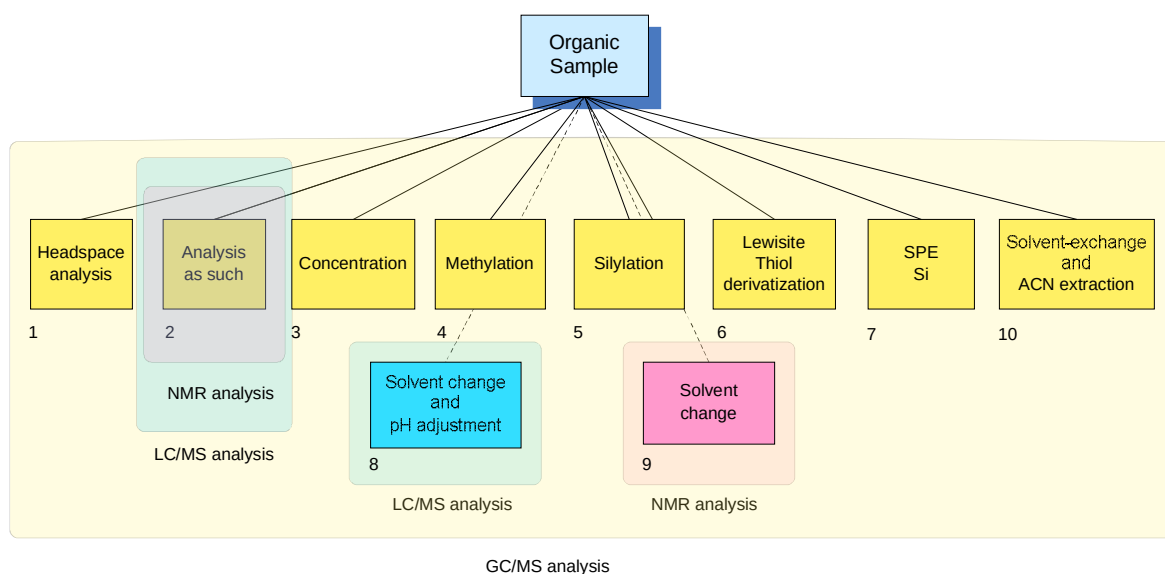


Figure 1. Flowchart of organic samples.

Table 1. Equipment, supplies and chemicals needed to prepare organic samples.

Equipment	Supply	Chemical
Shaker (option)	Pasteur pipettes	<i>n</i> -Hexane
Oven	pH paper	Acetonitrile
	Adjustable pipette(s) or measuring cylinders [#]	Deuterated solvents (dichloromethane- <i>d</i> ₂)
	Disposable syringes 2 or 5 ml	
	Disposable syringe filters 0.45 µm	
	Sample vials 1.5, 4, and 8 ml	MeOH
	Headspace vials 10 ml with silicone PTFE septa	Derivatisation reagents
	Gas tight syringe 500 µl	
	Silica (Si) cartridge (optional)	

[#] to divide the sample and to measure solvents

5.2 Headspace analysis

The most volatile CWC-related chemicals can be analysed from the headspace of a sub-sample of the organic liquid sample (Fraction 1, Figure 1) using GC or GC–MS. The composition of the sample solvent* can be analysed if it is unknown.

5.2.1 Headspace analysis without heating to measure the solvent

Adjust the GC–MS to manual split injection mode with split ratio of ca. 1:20, set the oven isothermal run at 40 °C and adjust the scan range to 20–300 amu. Open the organic liquid

* The solvent can also be analysed diluting the sample and injecting it into GC–MS without solvent delay time.

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sample, take a clean headspace syringe and take 100–500 µl of the headspace sample. Check that the GC–MS is ready for the run. Inject the sample and start the run.

5.2.2 Headspace analysis with heating[†]

The sample is placed in a 10 ml headspace vial. The vial is sealed using a silicone-PTFA septum and an aluminium cap and is kept at 80 °C for 10–15 minutes. The gas phase is drawn and analysed with GC and GC–MS.

5.3 Direct analysis

One sub-sample of the organic sample (e.g. 200 µl) is analysed directly with GC, GC–MS, and LC–MS (Fraction 2, Figure 1).

The original sample is analysed for volatile CWC-related chemicals, but since the procedure does not contain sample clean-up steps, large amounts of background chemicals in the sample may make the reliable identification of the target chemicals difficult.

Similarly, a small portion can be analysed with NMR. 0.8 ml of the organic liquid is diluted with 0.2 ml of corresponding deuterated solvent and transferred into an NMR tube.

5.4 Concentration

If the direct analysis of the organic liquid sample (Fraction 2) shows very small amounts of possible analytes of interest, take a portion of 1–2 ml of the sample for concentration. Concentrate the sample with mild nitrogen flow (Section 2, Part A, Chapter II: *Concentration*) (Fraction 3, Figure 1) to 0.1 or 0.2 ml, respectively.

Note! Do not concentrate the organic extract to dryness since certain volatile CWC-related chemicals such as sarin are lost from the sample and, on the other hand, certain less volatile CWC-related chemicals such as VX in residues of organic extracts are firmly adsorbed to glass surfaces.

The concentrated organic liquid sample is analysed for volatile CWC-related chemicals.

5.5 Methylation

A portion of the organic liquid sample is methylated as described in Section 2, Part A, Chapter III: *Derivatisation for gas chromatography* (Fraction 4, Figure 1).

If the sample is cloudy after methylation, filter it with a disposable syringe filter before analyses with GC based methods. This sample is analysed for polar, non- or slightly volatile CWC-related chemicals such as alkylphosphonic and alkylthiophosphonic acids.

[†] This type of headspace analysis is used in AMMS laboratory to measure the solvent and to analyse highly volatile CWC-related chemicals in a low-volatile solvent.

5.6 Silylation

A portion of the organic liquid sample is silylated as described in Section 2, Part A, Chapter III: *Derivatisation for gas chromatography* (Fraction 5, Figure 1).

This sample is analysed for polar, non- or slightly volatile CWC-related chemicals such as thiodiglycol, amino alcohols, and alkylphosphonic and alkylthiophosphonic acids.

5.7 Derivatisation to analyse Lewisites

Another sub-sample of the original sample is derivatised with butanethiol for GC–MS analysis of Lewisite 1 and 2, and Lewisite oxide (Fraction 6, Figure 1). The derivatisation procedure is presented in Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*.

5.8 Cleaning of hydrocarbon background with silica cartridges

One sub-sample of organic sample can be cleaned up of hydrocarbon and other relatively nonpolar background chemicals by Silica (Si) solid phase extraction (SPE), partitioning the analytes from a nonpolar hydrocarbon solvent to a polar Si sorbent. The sorbent is washed with a nonpolar solvent and the analytes are extracted out of the sorbent with a polar solvent (Fraction 7, Figure 1). The procedure for Si cleaning is described in Section 2, Part A, Chapter IV: *Sample clean-up methods*.

Si cleaned extracts can be concentrated, if necessary (Section 2, Part A, Chapter II: *Concentration*). Si cleaned extracts can also be analysed after silylation or methylation (Sections 5.6 and 5.5 above).

5.9 Solvent exchange for LC analysis

For LC and LC–MS analysis, the solvent of the organic liquid sample may have to be changed to be more suitable for analysis. For this reason, one sub-sample is concentrated with mild nitrogen flow as described in Section 2, Part A, Chapter II: *Concentration*. The residue is dissolved with a solvent suitable to the solvent chemistry of the LC analysis (pH and amount of aqueous solvent) (Fraction 8, Figure 1).

5.10 Solvent-exchange and acetonitrile extraction to remove hydrocarbon background

The solvent of the organic sample can be changed to *n*-hexane. One sub-sample of the original sample is concentrated 10-fold with mild nitrogen flow as described in Section 2, Part A, Chapter II: *Concentration*. The residue is reconstituted with *n*-hexane (Fraction 10, Figure 1).

Note! Take care that the organic sample is **not evaporated to dryness** since that will result in the loss of volatile CWC-related chemicals.

This sub-sample (in hexane) can then be liquid-liquid extracted with acetonitrile to remove hydrocarbon background. The analytes are partitioned from nonpolar hydrocarbon solvent to acetonitrile, which is a relatively polar aprotic solvent. The method is applicable to

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organic solvents immiscible in acetonitrile and presented in Section 2, Part A, Chapter IV: *Sample clean-up methods*.

The acetonitrile extract is usually analysed for nonpolar CWC-related chemicals. The extract or the concentrated extract can also be analysed after silylation (Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*).

The acetonitrile extract can be analysed with LC and LC–MS directly or by adding same volume of water. In addition, acetonitrile extract can also be analysed with NMR. 0.5 ml of the extract is diluted with 0.1 ml of deuterated dichloromethane and transferred into a NMR tube.

6. Quality control

Quality control of sample preparation is presented in Section 2, Part A, Chapter V: *Quality assurance and quality control in sample preparation*.

7. Evaluation of method

Generally, organic samples can be analysed using GC and GC–MS without sample preparation, but sometimes it is difficult to detect the CWC-related chemicals with direct GC–MS analysis in presence of masking/background components, especially in complex hydrocarbon background.¹ Clean-up methods such as SPE or liquid-liquid extraction are often used to remove the background chemicals from the sample matrix. Changing the solvent into *n*-hexane and extracting it with acetonitrile can eliminate the hydrocarbons from the organic samples effectively. If time permits, the mixed solution of hexane and acetonitrile can be stored at -20 °C in freezer during the extraction process in order to achieve a better separation of the two layers. This cooling process will result in a cleaner background and better recoveries of the analytes.¹ Except Silica SPE clean-up, C₁₈ SPE has also been used to remove hydrocarbon background.² However, the clean-up procedures might cause the loss of the target chemicals, so analysis of untreated original sample is always necessary. Organic samples are also analysed after derivatisation (e.g. silylation and/or methylation) for the presence of polar nonvolatile CWC-related chemicals.³

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1. Scope

This section describes methods for the preparation of soil samples for the analysis of chemical warfare agents (CWA) and their degradation products, where chromatographic methods such as gas chromatography (GC), liquid chromatography (LC), ion chromatography (IC), or capillary electrophoresis (CE) as analytical techniques or introduction techniques to spectrometers, are used as the separation technique. The sample preparation methods required for analysis by nuclear magnetic resonance (NMR) spectroscopy are also presented.

2. Summary of the method

A general flowchart of sample preparation of soil samples is given in Figure 1. The soil sample is divided into sub-samples, which are treated in manners described in this section. Sub-samples of soil samples are extracted with organic solvents, under neutral and alkaline conditions, using traditional liquid-liquid extraction. A separate sub-sample of the original soil sample or the remaining soil fraction after neutral extraction is extracted with water. This water fraction or part of it is evaporated and derivatised. Another fraction of this water extract can also be analysed after filtration and pH change using LC. Similarly, a new sub-sample of the original soil sample is extracted again with water or the remaining water fraction of the first water extract is cation-exchanged. After ion exchange, this sub-sample can be analysed after pH adjustment with LC, IC and CE, but for the GC analysis, a part of it is evaporated to dryness and derivatised. Finally, one sub-sample is extracted-derivatised for identification of Lewisites.

3. Safety aspects and storing

Safety aspects of sample preparation and storing of samples are presented in Section 2, Part A, Chapter I: *Introduction*.

4. Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in this ROP are presented in Table 1.

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Table 1. Equipment, supplies and chemicals needed to prepare soil samples.

Equipment	Supply	Chemical
Shaker (option)	Pasteur pipettes	Dichloromethane
Analytical balance*	pH paper	Deuterated dichloromethane
Ultrasonic bath	Spoon*	Clean water
	Adjustable pipette(s) or measuring cylinders [#]	Deuterated water
	Filter paper	Methanol
	Filtration funnel	Triethylamine
	Disposable syringes 5 or 10 ml	HCl
	Disposable syringe filters 0.45 or 0.5 µm	Anhydrous sodium sulfate
	Sample vials 1.5, 4, 8, and 20 ml	

* to weigh the drying agent

[#] to divide the aqueous sample and to measure solvents

5. Procedure

5.1 Division of the sample and pH measurement

If possible, the original sample is divided into four or more portions of 1–10 g for different extractions.

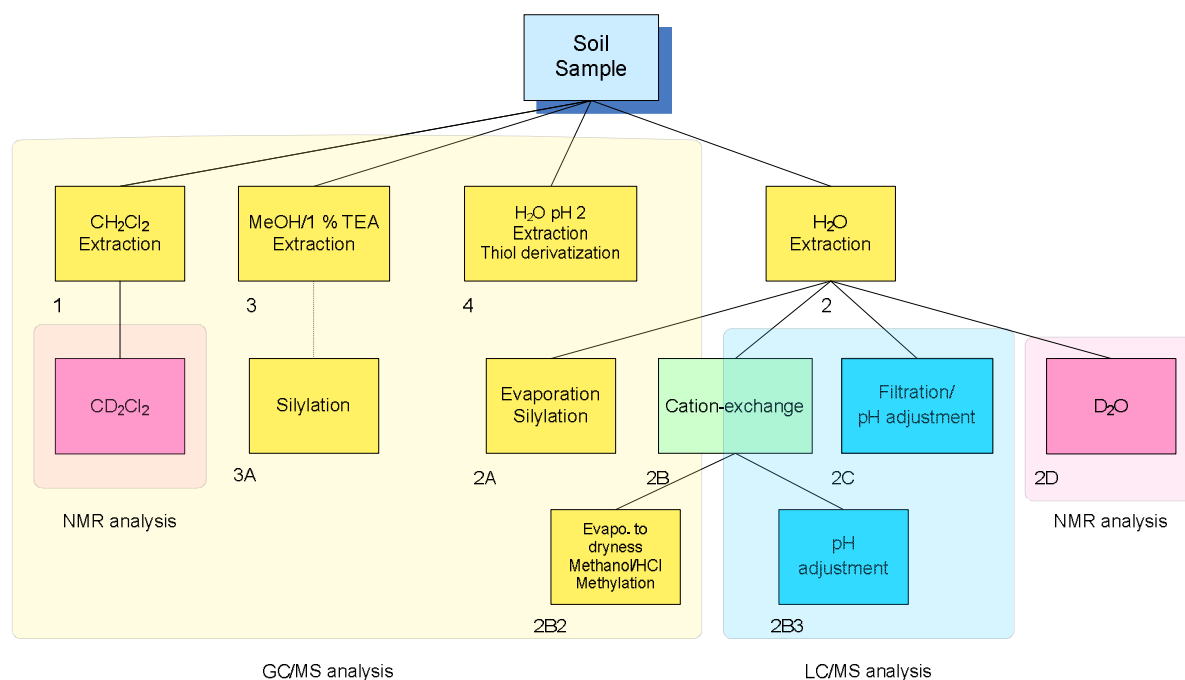


Figure 1. Flowchart of soil samples.

5.2 Organic extraction

Homogenise the soil sample with the help of a spoon or spatula, if necessary (the agents are assumed to be homogeneously distributed in the soil). Take ca. 1–10 g of soil and place

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it in a screw-capped glass bottle. Add the same volume in millilitres (ml) of the high-purity organic solvent as the amount of the sample in grams (g) (e.g. 5 ml to 5 g of soil) and close the bottle tightly. Usually, dichloromethane is used as extraction solvent.* Extract the soil by sonication, shaking, tumbling, or agitation for 10 min. Sonication may reduce recoveries by creating new active sites for analytes in the soil. Quickly centrifuge the extract, decant the supernatant and filter, if necessary, through a filter paper or a disposable 0.45 µm syringe filter. If you want to filter the whole extract with a disposable syringe filter, it is recommended to centrifuge for 5 min at 5000 G. Otherwise, centrifugation for 3 minutes at 2000 G is usually enough.

Note! *General instructions for filtration and centrifugation are described in Part A, Chapter IV: Sample clean-up methods.*

Repeat the extraction process once or twice. Combine the extracts and dry the combined extract with anhydrous sodium sulfate by adding the drying agent to the extract, if needed. Usually ca. 1 g of the drying agent is a proper amount for 5 ml of the final extract. The dried extract is analysed with GC and GC–MS as such (Fraction 1, Figure 1) or, where necessary, after approximately 2–10-fold concentration (Section 2, Part A, Chapter II: *Concentration*) (Fraction 1A).

Note! *Do not concentrate the organic extract to dryness since certain volatile CWC-related chemicals such as sarin are lost from the sample and, on the other hand, certain less volatile CWC-related chemicals such as VX in residues of organic extracts are firmly adsorbed to glass surfaces.*

Note! *If the amount of the original soil sample is 15 g or bigger, it is always better to take a fresh sub-sample to each extraction step to assure good recovery of chemicals of interest.*

This organic extract is analysed for nonpolar CWC-related chemicals.

5.2.1 LC sample

For LC and LC–MS analysis, the solvent has to be changed to be more suitable for LC analysis. For this purpose, the dichloromethane extract has to be concentrated with mild nitrogen flow as described in Section 2, Part A, Chapter II: *Concentration*. The residue is dissolved in a solvent suitable to the solvent chemistry of the LC analysis (pH and amount of aqueous solvent) (Fraction 1B).

5.2.2 NMR sample

Dichloromethane extracts can be analysed with NMR. One portion of the extract is diluted with deuterated dichloromethane and transferred into an NMR tube (Fraction 1C).

If the sample and/or solvent background are considered too high for successful NMR analysis, the solvent can be changed. Part of the dichloromethane extract is concentrated

* Other organic solvents (ethyl acetate, chloroform, *n*-hexane....) can also be used.

near to dryness with mild nitrogen flow (Section 2, Part A, Chapter II: *Concentration*), and the residue is dissolved with deuterated solvent.

5.3 Aqueous extraction

Take the same soil sample as above, or a new 1–10 g portion of soil. Place it in a screw-capped glass bottle. Add the same volume in millilitres (ml) of deionised, redistilled water as the amount of the sample in grams (g) (e.g. 5 ml to 5 g of soil) and close the bottle tightly. Extract similarly to the first extraction. Rapidly centrifuge the extract for 5 min at 2000 G, decant the supernatant and filter, if necessary, through a filter paper or a disposable 0.45 µm syringe filter. If you want to filter the whole extract with a disposable syringe filter, it is recommended to centrifuge for 5 min at 5000 G. Repeat the extraction process once or twice. Measure the pH of the combined extracts. Divide the extract into four sub-samples.

This extract is analysed for polar water-soluble CWC-related chemicals (Fraction 2, Figure 1).

Note! *General instructions for filtration and centrifugation are described in Part A, Chapter IV: Sample clean-up methods.*

Note! *Sonication may reduce recoveries of the polar CWC-related chemicals by creating new active sites for analytes in the soil.*

5.3.1 Evaporation and derivatisation for GC and GC–MS analysis

A portion of a water extract of the soil sample is evaporated to dryness (without cation-exchange) and silylated. Evaporation and derivatisation procedures are presented in Section 2, Part A, Chapter II: *Concentration* and Chapter III: *Derivatisation for gas chromatography* (Fraction 2A, Figure 1).

The derivatised sample is analysed for polar CWC-related chemicals such as thiodiglycol and amino alcohols. Polar alkylphosphonic and alkylthiophosphonic acids can also be recovered from this fraction, but since the procedure does not include a cation-exchange step, a large amount of inorganic cations in the sample may mean that the recoveries of smaller acids such as methyl or ethylphosphonic acid may be low, or the chemicals may not be recovered at all.

5.3.2 Cation-exchange, evaporation and derivatisation for GC and GC–MS analysis

Another portion of a water extract of the soil sample is cation exchanged to remove inorganic cations, alkali and earth alkali metal ions such as Na⁺, Ca²⁺, Mg²⁺, etc. to help the derivatisation procedure for the GC analysis and to remove ion background for LC, CE, IC and sometimes also for NMR analysis (Fraction 2B, Figure 1). The pH of the eluate is determined in order to check the efficiency of the cation-exchange.

The cation-exchanged sample or the most part of it (see Section 5.3.3) is evaporated to dryness and methylated for GC and GC–MS analysis. Cation-exchange, evaporation and derivatisation procedures are presented in Section 2, Part A, Chapter IV: *Sample clean-up methods*, Chapter II: *Concentration*, and Chapter III: *Derivatisation for gas chromatography* (Fraction 2B2, Figure 1).

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Cation-exchanged samples are analysed for polar alkylphosphonic and alkylthiophosphonic acids.

Note! Amino alcohols remain in the cation-exchanger and cannot be recovered.

5.3.3 LC sample

Filter part of the water extract through a disposable 0.45 µm syringe filter and adjust to a suitable pH for LC, CE and IC analysis. (Fraction 2C, Figure 1)

To avoid salt formation of organic bases and acids, a portion of a cation-exchanged water extract should be analysed similarly (Fraction 2B3, Figure 1) directly or after concentration.

5.3.4 NMR sample

One portion of the water extract is diluted with deuterated water and transferred into an NMR tube (Fraction 2D, Figure 1); e.g. ca. 0.8 ml of the cation-exchanged water extract of soil sample is diluted with ca. 0.2 ml of deuterated water and transferred into an NMR tube.

Cation-exchange can also be useful to remove background chemicals, e.g. paramagnetic metal ions from aqueous samples.

If the sample ion background is considered too high for successful NMR analysis, the water can be changed to deuterated water. Part of the water extract or cation-exchanged extract (e.g. 0.8 ml) is concentrated near to dryness (Section 2, Part A, Chapter II: *Concentration*), and the residue is dissolved in D₂O.

5.3.5 Solid phase microextraction and hollow fiber microextraction

Water extracts of soil samples can be analysed for water-soluble degradation products of CWC-related chemicals as an aqueous sample using solid phase microextraction (SPME) and hollow fiber liquid phase microextraction HF-LPME (Section 2, Part B, Chapter III: *Aqueous liquid samples*).

5.4 Alkaline extraction

Take the same soil sample as above or a new 1–10 g portion of soil. Place it in a screw-capped glass bottle. Add the same volume in millilitres (ml) of methanol containing 1% of triethylamine (TEA) as the amount of the sample in grams (g) (e.g. 5 ml to 5 g of soil) and close the bottle tightly. Extract similarly as the organic extraction. If necessary, dry the extract with anhydrous sodium sulphate. The extract is analysed by GC and GC–MS as such (Fraction 3, Figure 1) or after concentration.

A portion of the extract is evaporated and silylated as presented in Section 2, Part A, Chapter II: *Concentration* and Chapter III: *Derivatisation for gas chromatography*. (Fraction 3A, Figure 1).

This fraction is analysed for alkaline CWC-related chemicals such as VX, BZ, and amino alcohols.

5.4.1 LC analysis

The TEA/methanol extract may cause problems for LC–MS analysis. For this reason, the TEA/methanol solvent can be changed to be more suitable for LC–MS analysis by concentration and solvent change (Section 2, Part A, Chapter II: *Concentration*, and Chapter IV: *Sample clean-up methods*).

Another possibility is to take the same sample as above (after organic and/or water extraction) or, if available, a new 1–10 g portion, and extract the sample with 0.5 M NH₄OH/MeOH as in the first extraction. After centrifugation, if necessary, filter the extract through a 0.45 µm disposable filter. Analyse the extract as such or after concentration with LC–MS.

5.5 Extraction/derivatisation to analyse Lewisites with GC–MS

Another sub-sample of 1–10 g of the original sample is placed in a screw-capped glass bottle and prepared for analysis of the sample for Lewisite 1 and 2, and Lewisite oxide (Fraction 4, Figure 1). The thiol derivatisation procedure is presented in Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*.

6. Instrumental extraction

Organic, water, and alkaline extraction can be performed in a pressurised solvent extraction (PSE) instrument designed for extraction of solid and semisolid sample matrices. PSE reduces solvent consumption and sample preparation time. Solvent is pumped into an extraction vessel containing the sample and is heated and pressurised. The pressurised solvent at high temperature accelerates the extraction process by increasing the solubility of the analyte in the solvent and also increasing the kinetic rate of desorption of the analyte from the sample matrix. High temperature makes possible the high solubility and high diffusion rate while the high pressure keeps the solvent below its boiling point. At high pressures and high temperatures, the solvents penetrate the solid samples at a much higher rate. The extraction process is fast and efficient. In addition to PSE, the method is also called accelerated solvent extraction (ASE) and pressurised fluid extraction. The extraction parameters (extraction pressure, temperature, time and number of cycles) have to be optimised to obtain the best efficiency of the extraction process.^{1,2}

7. Quality control

Quality control of sample preparation is presented in Section 2, Part A, Chapter V: *Quality assurance and quality control in sample preparation*.

8. Evaluation of method

Analysis of non-polar chemicals from soil samples using GC and GC–MS is usually trouble-free. In OPCW proficiency tests, most of the problems have been faced in sample preparation of water extracts for water-soluble chemicals before GC analysis, because the water must be displaced and the analytes derivatised.³

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Only a few results of analysis of real soil samples have been published, but identification of chemicals has been possible after several years from the collection of samples. The methods used have been quite similar with this ROP.

D'Agostino analysed soil samples in the late 1980s from Iran/Iraq, suspected to have been contaminated with mustard. Mustard and 17 other mustard related impurities were identified after chloroform extraction.⁴ Also Black *et al.* analysed soil samples collected from a Kurdish village after a chemical attack. GC–MS using headspace analysis, solvent extraction with organic solvent (dichloromethane or ethyl acetate) and thermal desorption methods successfully confirmed the presence of sulphur mustard and 21 related compounds.⁵ A year later they published results of identification of sulphur mustard and/or thiodiglycol in six soil samples collected also from Kurdistan. In addition to these chemicals, isopropyl methylphosphonic acid and methylphosphonic acid, the hydrolysis products of the nerve agent sarin, were detected in six different soil samples. The neat agent was identified in dichloromethane extract and the hydrolysis products in water extracts, which were evaporated and silylated for GC–MS and GC-MS/MS analysis.⁶

Several years later aqueous extracts of the same soil samples were successfully analysed with LC–APCI–MS/MS for thiodiglycol and methylphosphonic acid.⁷ D'Agostino *et al.* have also used a LC–MS (LC–ESI–MS/MS) method to identify thiodiglycol and nine longer chain diols in aqueous extracts of soil samples. These samples were taken from a former mustard storage site.⁸

Recoveries of aqueous extraction may be low especially for the most polar degradation products and depend on the soil type. Kataoka *et al.* determined recoveries of four alkyl phosphonates on 21 different soil types. Compared to the recoveries measured directly by CE, the yields obtained after silylation and GC–MS analysis were low. The efficiencies of derivatisation depended on the amount of alkali earth metals extractable from soils as well as high total carbon content of the soil.⁹

As with aqueous water samples, SPME methods have been tested to analyse both the neat agents and water-soluble CWC-related chemicals extracted from soil samples.^{10–17}

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1. Scope

This section describes methods for the preparation of environmental aqueous liquid samples for the analysis of chemical warfare agents (CWAs) and their degradation products, where chromatographic methods such as gas chromatography (GC), liquid chromatography (LC), ion chromatography (IC) or capillary electrophoresis (CE) as an analysis technique or an introduction technique to spectrometers, are used as the separation technique. The sample preparation methods required for analysis by nuclear magnetic resonance (NMR) spectroscopy are also presented.

2. Summary of the method

A general flowchart of sample preparation of aqueous liquid sample is given in Figure 1. An aqueous liquid sample is divided into sub-samples, which are treated in manners described in this section. Sub-samples of aqueous samples are neutralised and extracted with organic solvents, under neutral and alkaline conditions, using traditional liquid-liquid extraction or Extrelut (Merck) extraction cartridges. The neutral unpreserved aqueous sub-sample is alternatively extracted with a C₁₈ solid-phase extraction (SPE) cartridge. A separate sub-sample of the original aqueous sample or the remaining aqueous fraction after neutral extraction is evaporated and derivatised. Similarly, a new sub-sample of the original aqueous sample or the remaining aqueous fraction after neutral extraction is cation-exchanged. After ion-exchange, this sub-sample can be analysed after pH adjustment with LC, IC and CE, but for the GC analysis, a part of it is evaporated to dryness and derivatised. A sub-sample can also be analysed after filtration and pH change using LC. Finally, one sub-sample is derivatised for identification of Lewisites and one sub-sample is treated for NMR analysis.

In addition, procedures for the extraction of aqueous samples using Solid-Phase Microextraction (SPME), Hollow Fiber Liquid Phase Microextraction (HF-LPME) and Single Drop Microextraction (SDME) for GC–MS analysis are presented. These procedures serve as complementary analysis to further screen for CWAs and related chemicals after the described procedures have been performed. Moreover, the use of HFs for extraction of O-alkyl alkylphosphonic acids for LC–MS analysis is presented.

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3. Safety aspects and storing

Safety aspects of sample preparation and storing of samples are presented in Section 2, Part A, Chapter I: *Introduction*.

4. Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in this ROP are presented in Table 1.

Table 1. Equipment, supplies and chemicals needed to prepare aqueous liquid samples.

Equipment	Supply	Chemical
Shaker (option)	Pasteur pipettes	Dichloromethane
Analytical balance*	pH paper	Deuterated dichloromethane
Ultrasonic bath	Spoon*	Deuterated water
	Adjustable pipette(s) or measuring cylinders [#]	0.1 M and 2 M NH ₄ OH solutions ^{&}
	Disposable syringes 5 or 10 ml	0.1 M and 2 M HCl solutions
	Disposable syringe filters	0.1 M and 2 M formic acid ^{&}
	Sample vials 1.5, 4, 8, and 20 ml	Na ₂ SO ₄ drying cartridges (option)
	Extrelut or Chemelut columns (option)	Ethyl acetate (option for Extrelut and C ₁₈ SPE extraction)
	C ₁₈ SPE cartridges 500 mg (option)	

* to weight the drying agent

[#] to divide the aqueous sample and to measure solvents

[&] NH₄OH solutions and formic acid should be used for pH adjustment when LC–MS is used for analysis

5. Procedure

5.1 Division of the sample and pH measurement

If the amount of the original aqueous sample is 15 ml or bigger, it is always better to take a fresh portion of sample for each extraction step to assure good recovery of chemicals of interest. If possible, divide the original sample into five or more sub-samples (1–10 ml) for different analytical purposes, and transfer the sub-samples to screw-capped glass vials.

Measure the pH of one sub-sample.

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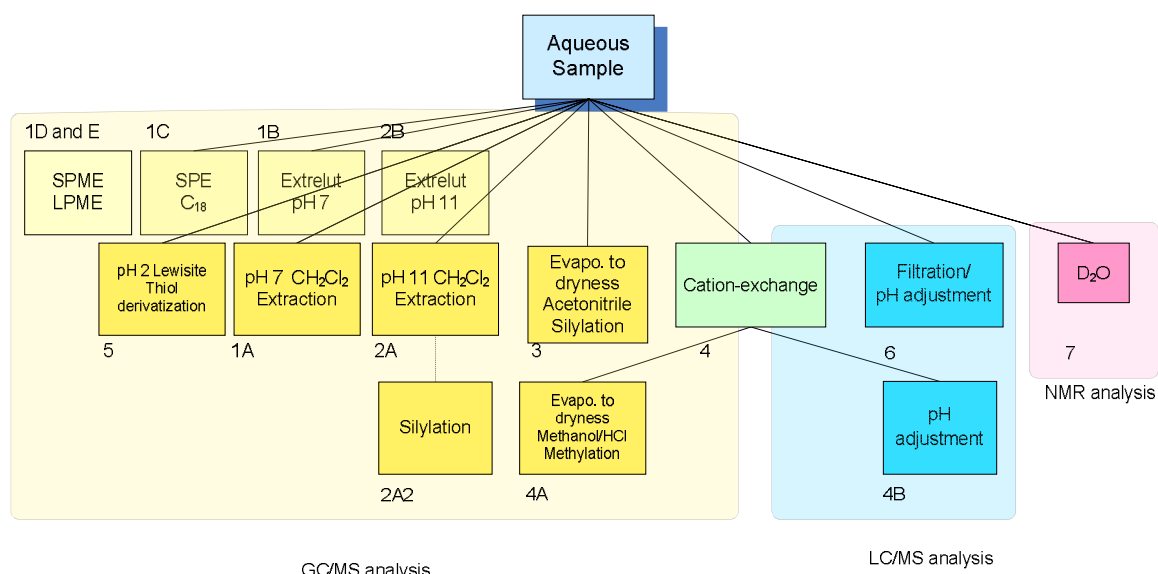


Figure 1. Flowchart of aqueous liquid samples.

5.2 Liquid-liquid extractions

If the aqueous sample is acidic, neutralise it with dilute ammonium hydroxide solution and if it is alkaline, neutralise with dilute hydrochloric acid. Extract the neutralised sample with two portions of organic solvent. Liquid-liquid extract the neutralised sample with two portions of water-immiscible high purity organic solvent using a one-to-two (1:2) volume ratio of solvent to sample. Usually, dichloromethane is used as extraction solvent and liquid-liquid extraction times are two to five minutes. Combine the extracts and dry the combined extract with anhydrous sodium sulfate by adding the sulfate to the extract. Usually ca. 1 g of the drying agent is a proper amount for 5 ml of the final extract. The dried extract is analysed with GC and GC-MS as such (Fraction 1A, Figure 1) or, where necessary, after approximately 2–10-fold concentration (Section 2, Part A, Chapter II: Concentration).

Note! Do not concentrate the organic extract to dryness since certain volatile CWC-related chemicals such as sarin are lost from the sample and, on the other hand, certain less volatile CWC-related chemicals such as VX in residues of organic extracts are firmly adsorbed to glass surfaces.

This organic extract is analysed for non-polar CWC-related chemicals.

Make the aqueous fraction of the first step or another sub-sample of the original sample alkaline with dilute ammonium hydroxide solution. Repeat the same liquid extraction procedure as in the first step. Analyse the dried extract as such with GC and GC-MS (Fraction 2A in Figure 1) or, where necessary, after concentration (Section 2, Part A, Chapter II: Concentration).

Analyse the extract also after silylation (Fraction 2A2 in Figure 1)(Section 2, Part A, Chapter III: Derivatisation for gas chromatography).

The alkaline organic extract and the silylated alkaline organic extract are analysed for alkaline CWC-related chemicals.

5.2.1 LC sample^{*}

For LC and LC–MS analysis the solvent has to be changed to be more suitable for LC analysis. For this reason dichloromethane extract has to be concentrated with mild nitrogen flow as described in Section 2, Part A, Chapter II: *Concentration*. The residue is dissolved with a solvent suitable to the solvent chemistry of the LC analysis (pH and amount of aqueous solvent).

5.2.2 NMR sample[†]

Both the neutral and alkaline dichloromethane extracts can be analysed with NMR. For example, ca. 0.8 ml of the extract is diluted with ca. 0.2 ml of deuterated dichloromethane and transferred into a NMR tube (Fraction 7, Figure 1).

If the sample and/or solvent background are considered too high for successful NMR analysis the solvent can be changed. Part of the dichloromethane extract (e.g. 0.8 ml) is concentrated near to dryness with mild nitrogen flow (Section 2, Part A, Chapter II: *Concentration*), and the residue is dissolved with deuterated solvent.

5.3 Extrelut extractions

Alternatively, both the neutralised and alkaline liquid-liquid extraction steps (Fractions 1B and 2B, Figure 1) can be performed using Extrelut[®] NT 1 (Merck) or other corresponding extraction columns (Varian Chemelut[®]). Columns contain specially processed wide-pore diatomaceous earth, a chemically inert matrix for use within a pH range of 1–13, and acts as a holder for the sample. The columns are used according to the instructions for use delivered by the manufacturer.

1 ml of a sub-sample is pH-adjusted and applied to the column. The column is left to stand for 10–15 minutes. Next, 6 ml of immiscible organic solvent (e.g. ethyl acetate) is passed through the column, to extract the analytes of interest, to get an end volume of 3–5 ml. Concentrate the extract, if necessary (Section 2, Part A, Chapter II: *Concentration*).

The neutral organic extract is analysed for nonpolar and the alkaline organic extract for alkaline CWC-related chemicals.

5.4 Non-polar solid phase extraction

Aqueous samples can be extracted by C₁₈ solid phase extraction (SPE, Fraction 1C, Figure 1). The pH is measured and adjusted to pH 7 with 1 M HCl or 1% v/v NH₄OH solution (adjust to alkaline pH for the extraction of alkaline chemicals). The C₁₈-SPE cartridge (500 mg) is conditioned with 2 ml of methanol followed with 5 ml of clean water, according to the instructions delivered by the manufacturer. Without letting the extraction material dry, a sub-sample (5 ml) is eluted through the cartridge by slowly pushing the aqueous sample through the adsorbent bed using a plastic syringe adjustable to the

^{*} This type of solvent exchange for LC–MS analysis is used at VERIFIN to measure e.g. VX and dialkyl alkylphosphonates in dichloromethane extracts

[†] These types of samples are prepared at VERIFIN to analyse dichloromethane extracts with NMR

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cartridge (sample can also be sucked through the cartridge under vacuum). Push or suck air through the cartridge to remove most of the water deposited in the cartridge.

Note! *The quality of C_{18} cartridges may vary between different manufacturers and batches of cartridges. Usually manufacturers do not recommended to use cartridges over two years*

Elute the SPE extract through Na_2SO_4 drying cartridge with ethyl acetate (or an appropriate solvent) to an end volume of 1 ml. If necessary, concentrate the dried extract (Section 2, Part A, Chapter II: *Concentration*).

The pH 7 SPE extract is analysed for non-polar and pH 11 SPE extract for alkaline CWC-related chemicals.

5.5 Derivatisation without cation-exchange

The aqueous fraction from the neutral or alkaline liquid–liquid extraction or another sub-sample of the original aqueous sample is evaporated to dryness (without cation-exchange) and derivatised. Evaporation and derivatisation procedures are presented in Section 2, Part A, Chapter II: *Concentration* and Chapter III: *Derivatisation for gas chromatography* (Fraction 3, Figure 1).

The derivatised sample is analysed for polar CWC-related chemicals such as thiodiglycol and aminoalcohols. Polar alkylphosphonic and alkylthiophosphonic acids can also be recovered from this fraction, but since the procedure does not contain a cation-exchange step, a large amount of inorganic cations in the sample may mean that the recoveries of smaller acids such as methyl or ethylphosphonic acid are low, or the chemicals may not be recovered at all.

5.6 SCX cation-exchange and derivatisation

The aqueous fraction from the neutral or alkaline liquid–liquid extraction or another sub-sample of the original aqueous sample is cation exchanged to remove inorganic cations, alkali and earth alkali metal ions such as Na^+ , Ca^{2+} , Mg^{2+} , etc..., which are usually present in environmental aqueous samples, to help the derivatisation procedure of the GC analysis and to remove background for LC, CE, IC and sometimes also for NMR analysis.

The cation-exchanged sample or most of it is evaporated to dryness and derivatised for GC and GC–MS analysis. Cation-exchange, evaporation and derivatisation procedures are presented in Section 2, Part A, Chapter II: *Concentration*, Chapter III: *Derivatisation for gas chromatography* and Chapter IV: *Sample clean-up methods* (Fraction 4A, Figure 1).

Cation-exchanged samples are analysed for polar alkylphosphonic, alkylthiophosphonic acids and benzoic acid.

Note! *Aminoalcohols remain in the cation-exchanger and cannot be recovered.*

Part of the cation-exchanged sample portion (0.5–1 ml) is separated, filtered through a disposable syringe filter and adjusted for suitable pH for LC, CE, and IC analysis (Fraction 4B, Figure 1).

Cation-exchange can also be useful to remove paramagnetic metal ions from aqueous samples. 0.8 ml of the cation-exchanged sample is diluted with 0.2 ml of deuterated water and transferred into an NMR tube.

5.7 Extraction/derivatization to analyse Lewisites with GC–MS

The aqueous fraction from the neutral or alkaline liquid–liquid extraction or another sub-sample of the original sample is prepared for analysis of the sample for Lewisite 1 and 2, and Lewisite oxide (Fraction 5, Figure 1). The derivatisation procedure is presented in Section 2, Part A, Chapter II: *Concentration* and Chapter III: *Derivatisation for gas chromatography*.

5.8 LC–MS sample preparation

Aqueous sample can be analysed with LC and LC–MS as such with minor sample preparation. If a pre-column is not used, the sample is usually filtered before injection. A sample portion (200 µl) is filtered through a disposable syringe filter (Fraction 6, Figure 1).

As mentioned in Section 5.6 above, cation-exchanged aqueous samples are also analysed with LC–MS.

***Note!** Depending on the chromatographic methods for analysis, it can be useful to adjust the pH of the sample before injection. For pH adjustment, volatile buffers such as ammonium hydroxide and formic acid should be used.*

5.9 NMR sample preparation

Aqueous sample can be analysed with NMR as such with minor sample preparation. 0.8 ml of the cation-exchanged sample is diluted with 0.2 ml of deuterated water and transferred into a NMR tube.

If the sample background is considered too high for successful NMR analysis, the water can be changed to deuterated water. Part of the cation-exchanged sample (e.g. 0.8 ml) is concentrated near to dryness (Section 2, Part A, Chapter II: *Concentration*) and the residue is dissolved with D₂O.[‡]

As mentioned in Section 5.6 also cation-exchanged aqueous samples are analysed with NMR.

6. Solid phase microextraction of aqueous samples

This section describes the procedures for the extraction of aqueous samples using Solid-Phase Microextraction (SPME) followed by analysis with GC–MS. These procedures serve as complementary analysis to further screen for CWAs and related chemicals after the procedures presented above have been performed.

For other matrix types such as soil and other solid materials, a water extraction can first be done on the materials to obtain an aqueous extract. The extract can then be treated as an aqueous sample to proceed with the SPME technique.

Solid phase microextraction is a simple and effective adsorption/desorption technique, to analyse both volatile and non-volatile chemicals in liquid samples by gas chromatography. SPME eliminates the need of solvents and concentration and provides linear results for

[‡] This type of sample preparation is used at VERIFIN

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wide concentrations of analytes. By controlling the polarity and thickness of the coating on the fiber, maintaining consistent sampling time and adjusting several other extraction parameters, an analyst can ensure highly consistent, quantifiable results from low concentrations of analytes.¹

A variety of SPME fibers are commercially available.² The 65 μm Polydimethylsiloxane/Divinylbenzene (PDMS/DVB) fiber has been evaluated to be the most efficient fiber in the extraction of the chemical warfare agents investigated while the 75 μm Carboxen/Polydimethylsiloxane (CAR/PDMS) fiber has been evaluated as the best fiber for most of the degradation products investigated.^{3,4}

Note! The working pH range of the PDMS/DVB and CAR/PDMS fibers is 2–11. Adjust the pH of the aqueous sample to that region, if needed.

The equipment, supplies and chemicals needed in SPME are presented in Table 2.

Table 2. Equipment, supplies and chemicals needed for SPME.

Equipment	Supply	Chemical
Magnetic stirrer with adjustable speed	Stir bars (teflon)	NaCl
SPME holder/fiber assembly	PDMS/DVB fibers 65 μm	Na ₂ SO ₄
Timer	CAR/PDMS fibers 75 μm	0.1 M HCl solution
	pH paper	0.1 M NaOH solution
	Spatula	MTBSTFA
	Adjustable pipette(s) or measuring cylinders [#]	BSTFA
	Sample vials 4 ml	

[#] to divide the aqueous sample and to measure solvents

6.1 Conditioning of the fiber

Fibers have to be conditioned before use. Turn on the split vent of the GC injection port and set a carrier gas flow of at least 50 ml/min. Insert the fiber into the injection port and condition it according to the recommendations of the manufacturer.² The PDMS/DVB is conditioned at 250 °C for 30 min while the CAR/PDMS at 300 °C for 1–2 h, respectively.

6.2 GC–MS analysis

Immediately before every sample, an injection of the blank fiber is performed to ascertain the chemical background of the fiber. The SPME fiber is desorbed at the GC injection port either manually using the GC in the manual injection mode or automatically using a sampler capable to operate in SPME mode. The fiber is desorbed by exposing the fiber at the injection port for 5 min at the start of the run. After 5 min, the fiber is retracted and removed from the injection port. The GC injector temperature is maintained at 250 °C (for PDMS/DVB) or at 300 °C (for CAR/PDMS) throughout the run.

The chemical background of the fiber should not be significant. This means that in the fiber blank run, there should be less than 20 peaks that have signal-to-noise greater than 10:1. If the blank run does not fulfill this demand, the conditioning and a new blank run should be repeated. The quality control test (QC test) of the instrument is tested similarly

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as with other samples. The use of narrow bore liners (0.75 mm) can be recommended but it is not necessary.

6.2.1 Extraction of volatile chemicals

SPME without salting is performed by immersing the conditioned PDMS/DVB fiber into 3 ml of aqueous sample in a 4 ml vial. Extraction is carried out with stirring for 30 min. The fiber is then desorbed at the GC injection port for 5 min (injector 250 °C) split vent closed.

SPME with salting is performed by immersing the conditioned PDMS/DVB fiber into 3 ml of aqueous sample in a 4 ml vial, to which NaCl has been added to saturation. Extraction is carried out with stirring for 30 min. The fiber is then desorbed at the GC injection port for 5 min (injector 250 °C) split vent closed.³

This sample fraction is analysed for non-polar CWC-related chemicals.

6.2.2 Extraction of acidic non-volatile chemicals

For SPME with in-situ derivatisation under acidic conditions, the conditioned CAR/PDMS fiber is first exposed to the headspace of MTBSTFA (100 µl in a 2 ml vial) for 5 min. Next, the fiber is immersed into 3 ml of aqueous sample in a 4 ml vial, to which the pH has been adjusted to 2 using 0.1 M HCl and NaCl has been added to saturation. Extraction is carried out with stirring for 30 min. The fiber is again exposed to the headspace of MTBSTFA for 15 min. The fiber is then desorbed at the GC injection port for 5 min (injector 300 °C) split vent closed.⁴

This derivatised sample fraction is analysed for acidic polar CWC-related chemicals such as alkylphosphonic and alkylthiophosphonic acids.

6.2.3 Extraction of alkaline non-volatile chemicals

For SPME with in-situ derivatisation under alkaline conditions, the conditioned CAR/PDMS fiber is first exposed to the headspace of BSTFA (100 µl in a 2 ml vial) for 5 min. Next, the fiber is immersed into 3 ml of aqueous sample in a 4 ml vial, to which the pH has been adjusted to 10 using 0.1 M NaOH and Na₂SO₄ has been added to saturation. Extraction is carried out with stirring for 30 min. The fiber is again exposed to the headspace of BSTFA for 15 min. The fiber is then desorbed at the GC injection port for 5 min (injector 300 °C) split vent closed.⁵

This derivatised sample fraction is analysed for polar CWC-related chemicals such as thiodiglycol and aminoalcohols.

7. Hollow fiber liquid phase microextraction of aqueous samples

This section describes the procedures for the extraction of aqueous samples using Hollow Fiber Liquid Phase Microextraction (HF-LPME) followed by analysis with GC-MS. Similar to SPME, these procedures serve as complementary analysis to further screen for CWAs and related chemicals after the presented procedures have been performed. In addition, HFs can be used for extraction of O-alkyl alkylphosphonic acids for LC-MS analysis using liquid-liquid-liquid microextraction (Section 7.6).

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For other matrix types such as soil and other solid materials, a water extraction can first be done on the materials to obtain an aqueous extract. The extract can then be treated as an aqueous sample to proceed with the HF-LPME technique.

The HF-LPME device consists of a 10 µl microsyringe and a 1.2 cm hollow fiber membrane attached to the needle tip. The equipment, supplies and chemicals needed in HF-LPME are presented in Table 3.

Table 3. Equipment, supplies and chemicals needed for HF-LPME.

Equipment	Supply	Chemical
Magnetic stirrer with adjustable speed	Stir bars (teflon)	NaCl
Timer	Accurel Q 3/2 PP hollow fibers i.d 600 µm; wall 200 µm; pore size 0.2 µm	Na ₂ SO ₄
Ultrasonic bath	HF syringe (bevel tip needle) ^{&}	Chloroform
	Adjustable pipette(s) or measuring cylinders [#]	Toluene
	Sample vials 1, 2 and 4 ml	Trichloroethylene
	Retort stand to hold the syringe	Acetone
	pH paper	1 M HCl solution
	Spatula	1 M NaOH solution
	Tweezers to secure HF to needle tip	MTBSTFA
		BSTFA

[&]e.g. 10 µl Hamilton gas tight syringe (needle length 5.1 cm; o.d. 0.71 mm; i.d. 0.15 mm) or 10 µl SGE gas tight syringe (needle o.d. 0.63 mm)

[#] to divide the aqueous sample and to measure solvents

7.1 Conditioning of the fiber

Fibers have to be conditioned before use. Cut the hollow fiber membrane into 1.2 cm (3.2 cm in the procedure in Section 7.6) lengths. Clean these pieces ultrasonically in acetone to remove the contaminants and dry in air to allow the solvent to evaporate prior to use.

Note! Each conditioned hollow fiber should be used only once for extraction.

7.2 Extraction of volatile chemicals using chloroform as extraction solvent

30% (w/v) of NaCl is added to 3 ml of the neutral aqueous sample in a 4 ml vial. A 5 µl volume of chloroform is drawn into the syringe before a hollow fiber is affixed onto the tip of the syringe needle. The fiber is immersed in solvent for three seconds to immobilise the solvent in the pores of the hollow fiber prior to extraction. Upon immersion of the hollow fiber into the aqueous sample, the syringe plunger is depressed to fill the hollow fiber with solvent. Extraction is carried out with stirring for 30 min. After extraction, the analyte-enriched solvent is drawn back into the syringe and the hollow fiber is discarded. A volume of 1 µl of solvent is injected into the GC-MS for analysis.⁶

This sample fraction is analysed for non-polar CWC-related chemicals.

7.3 Extraction of volatile chemicals using trichloroethylene as extraction solvent

Draw an aliquot (8.0–10 µl) of organic solvent (typically trichloroethylene) into microsyringe. Insert the needle of microsyringe into a hollow fiber segment and inject the organic solvent into needle so that the channel of the fiber is filled. Immerse the hollow fiber into trichloroethylene for 20–40 s for further impregnation of its wall. Thus, the lumen of hollow fiber is filled with trichloroethylene, the extracting solvent.

Remove the hollow fiber from the solvent and immerse into 5.0 ml of aqueous sample containing analytes. The second end of hollow fiber is kept open so that extraction could also take place at open end of fiber. Fix the syringe barrel with the retort stand. After immersing the fiber in sample, place the stir bar into sample and mount on a magnetic stirrer. Stir the sample (500–1000 rpm or as desired) to allow the extraction of analytes from aqueous to organic phase. After extraction (typically 15 min), retract the solvent in fiber into the syringe and discard the hollow fiber membrane. Inject 1 µl aliquot of extractant in to GC–MS for analysis.⁷

This sample fraction is analysed for non-polar CWC-related chemicals.

7.4 Extraction of acidic non-volatile chemicals

The sample pH is adjusted to pH = 0 using 1 M HCl followed by the addition of 30% (w/v) of NaCl to 3 ml of the sample in a 4 ml vial. A solvent-derivatising agent mixture is prepared by mixing chloroform and MTBSTFA in equal amounts. The extraction procedure is similar to that described previously. Extraction is carried out with stirring for 45 min.⁷

This derivatised sample fraction is analysed for acidic polar CWC-related chemicals such as alkylphosphonic and alkylthiophosphonic acids.

7.5 Extraction of alkaline non-volatile chemicals

In a typical extraction of basic compounds, the sample pH is adjusted to 12 using 1 M NaOH followed by the addition of 30% (w/v) of Na₂SO₄ to 3 ml of the sample in a 4 ml vial. The extraction of basic compounds follows that for the acidic compounds except that 1 µl of extract is injected into the GC together with another microliter of derivatising agent. Extraction is carried out with stirring for 20 min.⁵

This derivatised sample fraction is analysed for polar CWC-related chemicals such as thiodiglycol and aminoalcohols.

Note! *Either MTBSFA or BSTFA can be used as the silylation reagents. However, MTBSTFA gives better results for the acidic degradation products while BSTFA gives better results for the alkaline degradation products.*^{5,8}

7.6 Extraction of O-alkyl alkylphosphonic acids for LC–MS analysis

Using liquid-liquid-liquid microextraction and a hollow fiber (HFLLEME), alkyl alkylphosphonic acids are extracted from acidified aqueous phase to organic phase present in pores of the hollow fiber, and then back extracted into the alkaline acceptor phase present in the lumen of the hollow fiber for LC–MS analysis. The equipment, supplies and chemicals needed in HF procedure are presented in Table 4.

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Table 4. Equipment, supplies and chemicals needed for HFLLME.

Equipment	Supply	Chemical
Magnetic stirrer with adjustable speed	Stir bars (teflon) 5 x 2 mm	1-Octanol
Timer	Accurel Q 3/2 PP hollow fibers i.d. 600 µm; wall 200 µm; pore size 0.2 µm	1 M HCl solution
Ultrasonic bath	50 µl syringe (flat tip needle)	1 M NaOH solution
	Adjustable pipette(s) or measuring cylinders [#]	Acetone
	Sample vials 6 ml	
	Retort stand to hold the syringe	
	pH paper	

[#] to divide the aqueous sample and to measure solvents

In this procedure, 3.2 cm hollow fiber segments are used (internal volume of the segment should be ca. 10 µl). Take 6.0 ml of sample in a sample vial. Place the sample vial on a magnetic stirrer and place a stir bar in sample vial. Add HCl solution to water sample so that its pH remains less than 1 (it become the donor phase). Make the acceptor phase by dissolving and diluting the NaOH in water so that its pH ranges from 12–14. Withdraw 8 µl of this acceptor phase into a 50-µl microsyringe fitted with a flat tip needle. Insert the needle of syringe into the hollow fiber and introduce acceptor solution into it. The pH of the acceptor phase should always be kept high enough so that it does not allow the acids to diffuse out from acceptor phase to organic phase. Immerse the fiber in 1-octanol for 20 s to impregnate the pores of hollow fiber. Subsequently, insert the fiber with syringe needle into aqueous sample (donor phase) and fix the syringe on the retort stand. Switch on the magnetic stirrer and stir at 900 rpm for 60 min. After extraction, switch off the stirring and take the syringe fiber assembly out of the sample solution. Withdraw exactly 8 µl of acceptor phase into syringe and inject into 2 µl sample loop for LC–MS analysis. Discard the used fiber and use a fresh one for the next extraction.⁹

This sample fraction is analysed with LC–MS for acidic O-alkyl alkylphosphonic acids.

8. Single Drop Microextraction of aqueous samples

In the absence of hollow-fiber, Single Drop Microextraction (SDME) can be performed. This section describes the procedures for the extraction of aqueous samples using SDME followed by analysis using GC–MS. The equipment, supplies and chemicals needed are presented in Table 5.

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Table 5. Equipment, supplies and chemicals needed for SDME.

Equipment	Supply	Chemical
Magnetic stirrer with adjustable speed	Stir bars (teflon)	Chloroform
Timer	5 µl microsyringe*	Toluene
	Adjustable pipette(s) or measuring cylinders [#]	
	Sample vials 2, 4 and 8 ml	
	Retort stand to hold the syringe	

* e.g. 10 µl Hamilton gas tight syringe (needle length 5.1 cm; o.d. 0.71 mm; i.d. 0.15 mm)

[#] to divide the aqueous sample and to measure solvents

Take the water sample (2–5 ml) in a screw capped septum sealed vial. Place a stir bar into the sample and mount on a magnetic stirrer. Fill the 5-µl microsyringe with 1–2 µl of extracting solvent (e.g. toluene or chloroform) and insert in vial by piercing the septum. Immerse the needle tip into the solution, ca. 1 cm below the surface of the liquid. Gently push 1 µl extractant out of needle so that it forms a drop at the tip of the needle, keeping it suspended at the needle tip. Stir the solution at 300 rpm for 30 minutes. When the extraction is complete, suck back the drop in the microsyringe and remove the needle from the sample vial. Depress the plunger of the syringe to 0.5 µl position and wipe the tip of needle to remove any possible water contamination. Inject the extracted liquid into GC–MS for analysis.¹⁰

This sample fraction is analysed for non-polar CWC-related chemicals.

9. Quality control

Quality control of sample preparation is presented in Section 2, Part A, Chapter V: *Quality assurance and quality control in sample preparation*.

10. Evaluation of method

Analysis of non-polar chemicals from aqueous samples using GC and GC–MS is usually trouble-free. Most non-polar CWAs hydrolyse in aqueous solution and efforts should be paid on analysis of degradation products in water samples, however. In this respect, surprisingly few results of analysis of real aqueous samples have been published. D'Agostino *et al.* used LC–MS for the analysis of aqueous samples collected at a former mustard destruction site. Both thiodiglycol and the corresponding hydrolysis product of O-mustard were identified in these samples.¹¹ A few years later they analysed (also with LC–MS) a snow sample that was accidentally contaminated with an organophosphorus chemical warfare agent during the destruction of a chemical munition. Sarin, its hydrolysis products and a number of related compounds were identified.¹²

Aqueous liquid sample has been the other of the most popular sample types in OPCW proficiency tests. Most of the problems in these tests have been faced in sample preparation of water-soluble chemicals before GC analysis, because the water must be displaced and the analytes derivatised.¹³ Isolation of highly polar analytes from an aqueous matrix prior to derivatisation is problematic. Liquid–liquid extraction is inefficient, SPE may be

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unworkable for some matrices, and concentration of aqueous solutions to dryness is time-consuming and a source of error. For these reasons, approaches such as ion-pair SPE^{14,15} and derivatisation directly in the aqueous medium or on a solid support have been tested. New miniaturised extraction techniques such as surfactantless microextraction and HF-LPME have been used to isolate the derivatives formed in aqueous media.^{16,17} Similarly, HF-LPME has been explored in on-support derivatisation where a mixture of solvent and derivatising agent was held in the hollow fiber for in-situ derivatisation-extraction of water-soluble CWC-related chemicals from aqueous samples.^{5,8}

HF-LPME method is simple and requires small amount of organic solvents. Main advantages of the method over SPME are its cost effectiveness and no carryover of analytes. SDME in turn requires careful handling as drop may dislodge during stirring. HF-LPME is robust and can extract analytes from aqueous samples containing suspended particles. Method has been tested with proficiency test samples.^{7,8} On the other hand, SPME has been the most tested method of these modern extraction techniques and a review of its use in analysis of CWC related chemicals was published in 2007.¹⁸

Alkyl alkylphosphonic acids can be extracted with HF from aqueous sample for LC-MS analysis down to the concentration of 1–10 ng/ml. Addition of 10% NaCl in water sample (the donor phase) further enhances the sensitivity of the method. This method is of special importance when nitrogenous and non-polar impurities are present in the sample. Applicability of the method has been tested with proficiency test samples. If the spiking concentration of analytes is too high (more than 0.1 mg/l), the sample can be diluted before subjecting to liquid-liquid-liquid microextraction.⁹

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1. Scope

This section describes a method for the collection of air samples in Tenax[®] tubes and their analysis by thermal desorption–gas chromatography–mass spectrometry (TD–GC–MS). Specific sample preparation is not required and the samples are analysed on-line. Thus, the analysis of sample tubes requires specific analytical equipment; sampling and analysis procedures are described here together. Other methods using SPME fibers, canisters or Tedlar bags can also be used for sampling CWA in air, but are not described in this chapter.

Sampling on Tenax[®] tubes is very efficient for the collection of most volatile CWA (sarin, soman, tabun, mustard gas,¹ and chloropicrin). However, because of the low specific surface area (25 m²/g), Tenax[®] is not suitable for the trapping of very highly volatile chemicals. It also cannot be used for thermolabile and chemicals of very low volatility such as V agents and polar degradation products.

Because sample preparation is not required before analysis, solvent or derivatising agent is not added into the sample. Thus, no risk of interference or reactivity with relevant chemicals can occur during sample analysis. In addition, results are obtained more quickly (within 30 minutes).

The higher the sampled volume is, the bigger the mass of collected chemicals in the tube. This means that the concentration of chemicals in the tube is significant and sensitivity is good. Therefore, TD–GC–MS can be used for the detection and identification of CWA in air both at high and low concentration levels (down to 10 µg/m³).

It should be also noted that collected Tenax[®] samples could be extracted with organic solvent instead of thermal desorption. Chlorinated solvents such as dichloromethane must not be used, however.

2. Summary of the method

Air samples are collected into sampling tubes packed with at least 200 mg of Tenax[®] GR, with a handheld pump working at a constant air flow (50–200 ml/min). It is strongly recommended to collect onto two sample tubes at the same time in order to overcome possible problems during sampling or analysis. Recommended sampling volume is 0.2–2.0 liters depending on the concentration. If the concentration level is not known, the sampling volume must be adjusted to 2 liters.

After sampling, tubes are capped with ¼-inch brass Swagelok[®]-type caps and stored at low temperature (<4 °C) before analysis.

To verify the correct performance of the analytical system, a quality control run (QC test, see below) is carried out before each sample batch, followed by a blank run to verify the absence of CWC-related chemicals.

Collected chemicals are thermally desorbed from the sample tube and eluted to the analytical column using a two-stage thermal desorption system interfaced to a GC–MS.

A 25–30-m-long non- or semi-polar fused silica capillary column (0.25 mm i.d. and 0.5–1 µm film thickness) is used. Adjust the GC–MS EI scan range to 29–400 amu. The chemicals are identified by comparing the measured EI spectra to those in relevant libraries or to spectra of authentic reference chemicals or interpreting the spectra.

3. Safety aspects

The thermal desorber requires specific safety precautions. During the thermal desorption, only a part of sample is transferred into the GC–MS. The rest of the sample – possibly containing a large amount of toxic chemicals – is flushed through the split valve to the laboratory. Therefore, the thermal desorber must be installed under a filtering hood, or alternatively, the split valve exit flow must be connected to a charcoal filter.

In addition, the hot parts of the instrument, e.g. valve and transfer line (>200 °C) must be taken into account during maintenance operations.

GC–EI/MS safety recommendations are described in Section 3, Part B, Chapter II: *Gas chromatography–electron ionisation/mass spectrometry (GC–EI/MS)*.

4. Instrumentation, reagents and chemicals

4.1 Thermal desorber

A two-step thermal desorption system must be necessarily used in order to maintain capillary column efficiency.

The following requirements are recommended for optimal results:

- coupling with the gas chromatograph must be performed with a heated deactivated fused-silica transfer line
- tube desorption temperature must be adjustable from 50 °C to 350 °C
- cold trap temperature must be down to 0 °C during primary desorption and must reach at least 200 °C in a few seconds during secondary desorption
- valve must be temperature controlled and manufactured from an inert material.

4.2 Tenax® tubes

Tenax® GR is a composite material of 70% of Tenax® TA* polymer and 30% of graphitised carbon. The resulting material gives higher breakthrough volume for most compounds and superior trapping of volatile compounds. It also presents a low affinity for water, avoiding hydrolysis and minimising disturbances of the gas chromatographic analysis, for example, damage of the stationary phase. The properties of Tenax® GR are presented in Table 1.

Despite its relatively low specific surface area, Tenax® GR is sufficient for collection of most monitored CWA in ambient air. However, breakthrough volumes of volatile CWA at 20 °C (Table 2) show that it is not suitable for sampling of very highly volatile chemicals (hydrogen cyanide,² cyanogen chloride, phosgene, arsine).

* Tenax® TA is a porous polymer resin based on 2,6-diphenylene-oxide

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Table 1. Tenax[®] GR properties.

Property	Specification
Temperature limit	350 °C
Density	0.55 g/ml
Specific surface area	24.1 m ² /g
Pore volume	2.4 ml/g
Average pore size	200 nm
Mesh size	60/80 mesh

Table 2. Breakthrough volume of some volatile CWA at 20 °C on Tenax[®] GR.

Chemical	Volatility (g/m ³)	Concentration (ppm)	Breakthrough volume (l/g)
Sarin	16 at 20 °C	7.5	72
Chloropicrin	165 at 20 °C	10	21
Hydrogen cyanide	1080 at 25 °C	200	0.08
Cyanogen chloride	6132 at 25 °C	200	0.1

Stainless steel tubes are preferred rather than glass tubes, which are more fragile. The tube size depends on the thermal desorption system.

Commercial Tenax[®] tubes are recommended. They usually contain at least 200 mg of Tenax[®] GR. Some tubes are individually manufactured with an engraved identification number for QA/QC.

4.3 GC–EI/MS

Any modern GC–EI/MS instrument, capable of a scan speed of at least 1 scan/s and covering a mass range of at least m/z 20–500 can be used.

A 25–30 m non-polar[†] or a semi-polar[‡] fused silica capillary column with 0.25 mm inner diameter and 0.5–1 µm film thickness allows correct retention and separation of most CWC-related chemicals.

4.4 Sampling pumps

It is recommended that the pumps meet the following requirements:

- small and light in weight (less than 1 kg)
- equipped with a nickel-cadmium battery allowing long operating life
- allow constant flow rates of 0.0–0.5 l/min through the sampling tube

[†] Stationary phase consisting of 95%-methyl-5%-phenylpolysiloxane (e.g.: HP-5, Rtx-5, BP-5, SE-54). Also polyarylene-phase, which is virtually equivalent can be used

[‡] Stationary phase consisting of 14%-cyanopropylphenyl-86%-methylpolysiloxane (e.g.: Rtx-1701, BP-10, OV-1701)

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- the flow rate must not deviate more than 5% from the initial set value throughout the sampling period
- as an option, the pump can have a timer allowing run time and delayed start time programs
- if possible, sampling pump must also have a regular calibration schedule

4.5 Reagents and chemicals

The reagents and chemicals needed in this ROP are presented in Table 3.

Table 3. The reagents and chemicals needed for TD–GC–EI/MS analysis.

Test	Chemical	Concentration	Purity
QC test sample	Trimethylphosphate (CAS 512-56-1)	10 ng/μl	≥99%
	2,6-Dimethylphenol (CAS 576-26-1)	10 ng/μl	≥99%
	5-Chloro-2-methylaniline (CAS 95-79-4)	10 ng/μl	≥99%
	Tri- <i>n</i> -butylphosphate (CAS 126-73-8)	10 ng/μl	≥99%
	Dibenzothiophene (CAS 132-65-0)	10 ng/μl	≥99%
	Malathion (CAS 121-75-5)	10 ng/μl	≥99%
	Methyl stearate (CAS 112-61-8)	10 ng/μl	≥99%
	<i>n</i> -Alkanes (C ₈ –C ₂₄ , even members)	10 ng/μl	≥99%
Sensitivity test sample	<i>n</i> -Pentadecane (CAS 629-62-9)	10 ng/μl	≥99%
Carrier gas			
	Helium		≥99.995%

5. Sampling

5.1 Tube conditioning

After analysis, Tenax[®] GR tubes must be carefully conditioned, regenerated, to eliminate all traces of chemicals still adsorbed on the sorbent that could cause false positive results during the analysis.

The conditioning procedure consists of heating the tubes to a higher value than the normal desorption temperature under purified helium or nitrogen flow.

Optimal conditioning is achieved by heating Tenax[®] GR tubes at 310 °C for at least 30 minutes with a 100 ml/min gas flow. Of course, the regeneration time can be longer, but does not need to exceed 2 hours.

Tube conditioning can be performed with a thermal desorber or with a specific tube regeneration device (Figure 1), which allows regeneration of up to 20 tubes simultaneously.

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After conditioning, clean tubes are sealed with ¼-inch brass Swagelok®-type screw caps fitted with combined PTFE ferrules or in air-tight containers (depending on the tube manufacturer). Tubes can be stored in this way at ambient temperature for 6 months. Beyond 6 months, it is advisable to proceed with reconditioning of the tubes.



Figure 1. MARKES tube conditioner TC20.

5.2 Tube testing

For good results and repeatability, it is recommended to control the quality of Tenax® packing in the sampling tubes. As the sorbent is not visible in a stainless steel tube, a control can be performed by measuring the air flow rate through the tube with a flowmeter. This control must be performed in the laboratory.

Repeat the operation to all Tenax® GR tubes without readjusting the pump flow rate. Write down the obtained value of each tube.

Connect the air pump to one side of the tube and the flow meter to the other side. Adjust the pump flow rate to the required value.

Calculate the average flow rate F_{av} of all the measured tubes. Discard all tubes having flow rate values $\pm 5\%$ of the average value.

$$\text{Flow rate of accepted tube} = F_{av} \pm 5\% \text{ ml/min}$$

5.3 Air sampling on Tenax® tubes

5.3.1 Flow adjustment

Flow adjustment must be performed before sampling in order to determine with accuracy the collected volume for quantitative analysis. This step can be achieved in the laboratory.

At least two Tenax® GR tube samples must be collected at the same sampling location, preferably in parallel or alternatively, one after the other. For parallel sampling, use a dual channel tube holder connected to the pump with a short piece of tygon tubing. Connect the tubes chosen randomly among the batch of tubes to both channels of the holder. Connect a flowmeter to one channel and precisely adjust the airflow rate through the tube by tightening of the screw at the dual channel tube holder with a screwdriver.

Repeat the operation for the other channel.

5.3.2 Sampling procedure

Remove the caps from the sampling tube and connect it to the sampling pump with tygon tubing. Note the sampling tube direction on the tube, because the desorption must always be performed in the reverse direction to that from which the sample entered. Most stainless steel tubes have a groove that must be directed to the opposite of the pump during sampling.

Because of the high sensitivity of modern mass spectrometers, collected volumes are not critical. Recommended volumes are 0.2 liter for high concentration (more than 1 mg/m³), 1 liter for medium concentration (between 0.1 mg/m³ and 1 mg/m³) and 2 liters for low concentration (less than 0.1 mg/m³). In the absence of information on the concentration level, the sampling volume must be adjusted to 2 liters.

Optimal trapping is obtained with 50–200 ml/min air flow depending on the amount of Tenax® in the tube. High amount of humidity in the sampling tube may cause problems for the desorption and analysis. If samples must be collected during rainfall, protect the tube so that the rain will not be in direct contact.

After sampling tubes are sealed with ¼-inch brass Swagelok®-type caps and transferred to the laboratory preferably in a cooler. They must be stored in the laboratory at low temperatures (<4 °C) before analysis.

5.4 Field (transport) blanks

Field blanks are conditioned Tenax® GR tubes, which are transported with the tubes to be used for sample collection. Once at the monitoring location, all the processes carried out on the sampling tubes (uncapping, recapping, storage, etc.), except sample collection itself, should also be carried out on the field blanks. They provide important controls for all tube handling, storage, and transport procedures.

5.5 Storage

Sampled tubes must preferably be stored at low temperature (<4 °C) before analysis. However, care should be taken to ensure that the atmosphere inside the refrigerator or

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freezer is as clean as possible – for example, it should not contain solutions in volatile solvents such as dichloromethane or ethyl acetate – in order to avoid contamination of the tubes.

In these storage conditions, the tubes can be stored for two weeks without loss of their integrity.

Once the tubes have been allowed to reach the storage temperature, they should be briefly removed from cold storage and the brass caps should be retightened.

To prevent condensation, the tubes removed from the refrigerator/freezer should be allowed to equilibrate at room temperature before being removed from the storage container and/or uncapped and analysed.

6. Sample analysis

6.1 Thermal desorption conditions

Recommended conditions for thermal desorption are given in Table 4.

Table 4. Recommended conditions for thermal desorption.

Thermal desorber parameter	Specification	Note
Trap composition	Tenax [®] TA, Tenax [®] GR, or Carbotrap [®]	Never use a molecular sieve
Transfer line temperature	220 °C	
Valve temperature	220 °C	
Primary desorption temperature	280 °C	
Primary desorption time	5 min	
Trap temperature	-30 °C	
Secondary desorption temperature	30 °C	
Secondary desorption time	15 min	Can be equivalent to the run time
Helium pressure (If working at constant pressure)	15–20 psi, depending on the column	
Helium flow (If working at constant flow)	1.2–1.5 ml/min, depending on the installed column	Optimum column efficiency
Purge time	1 min	1 min
Outlet split	10 ml/min	Minimum allowed
Desorb flow	50 ml/min	

The outlet split flow must be necessarily adjusted to at least 10 ml/min. Lower values lead to broadening of peaks without sensitivity improvement.

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The percentage of sample introduced into the column is defined as follows:

$$\%sample = \frac{ColumnFlow}{ColumnFlow + OutletSplitFlow}.$$

6.2 Gas chromatography

Recommended GC conditions are given in Table 5.

Table 5. Recommended GC conditions.

GC parameter	Specification	Note
Injection mode	Split mode (Thermal desorber)	
Sample mass introduced in the column	Around 10% of total sample	Defined by the split and the column flow
Temperature program for GC oven	40 °C (1 min) – 10 °C/min – 300 °C for 5–10 min	
GC–MS interface temperature	200–250 °C	

6.3 Mass spectrometry

Recommended MS conditions are given in Table 6. The given conditions correspond to the requirements of MS conditions used for recording MS spectra for the OCAD MS-library.

Table 6. Recommended MS conditions in TD–GC–EI/MS analysis.

MS parameter	Specification	Note
Solvent delay time	0 min	No solvent in the sample
Mass range	From m/z 29 to 450	
Scan rate	≥ 1 scan/s	
Electron energy	70 eV	
Ion source temperature	Refer to instrument manual	
Electron multiplier voltage	Refer to instrument manual	Adjusted often during instrument tuning

6.4 Interferences caused by Tenax® GR

As with all porous polymers, degradation of Tenax® GR is constantly observed during analysis where Tenax® GR oxidises generating benzaldehyde, benzophenone, phenol, and benzoic acid³ in the presence of ozone. Other compounds such as pentanal, 2- and 3-hexanone, hexanal, 3-heptanone, and heptanal⁴ can be detected at lower concentration levels.

Nitrogen oxides, which are more predominant in an urban location, can also produce aromatic chemicals.

7. Quality control

The reliable identification of CWC-related chemicals requires that the data produced by the instrument is correct. Therefore, it is essential to continuously run quality control tests to confirm correct and stable instrument performance during the sample analysis.

The QC tests must be run at the beginning and at the end of each tube batch and, in the case of a long sample batch, at least after every fifteenth sample.

7.1 Instrument performance test

The performance of a TD–GC–EI/MS instrument is checked by carrying out a column performance test sample run, in which the mass spectrometric behaviour of test compounds and the condition of the column, transfer line, and cold trap are confirmed to be acceptable.

The test is performed using a test solution containing the following compounds at 10 ng/μl each in a solvent, which is not well retained by the sorbent in use.

- trimethylphosphate
- 2,6-dimethylphenol
- 5-chloro-2-methylaniline
- tri-*n*-butylphosphate
- dibenzothiophene
- methyl stearate
- *n*-alkanes (C₈–C₂₄, even members)

The purge step that occurs just before the primary desorption allowed eliminating a large part of the solvent in the tube particularly if the solvent is poorly retained.

Methanol or ethanol are the most common solvents when Tenax[®] sorbent is used (see Table 7), but other solvents such as *n*-hexane or ethyl acetate can also be used.

Table 7. Breakthrough volumes of some typical solvents.

Solvent	Breakthrough volume (l) on 200 mg Tenax/l
Methanol	0.07
Ethanol	0.36
Acetone	1.2
<i>n</i> -Hexane	6.4
Ethyl acetate	7.2

5 μl of test solution is introduced with a syringe into a conditioned Tenax[®] tube. The tube is analysed by TD–GC–MS/EI using the conditions in Tables 4, 5, and 6 (the oven rate can be increased to 15 °C/min and the solvent delay must be set to 3 minutes).

The test is accepted if the following criteria are met:

- Absolute retention times of the seven compounds are within 15 s in comparison with the reference test (performed after column installation)
- Peak height ratio of 2,6-dimethylphenol to 5-chloro-2-methylaniline must be between 1 and 4

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- Trimethylphosphate asymmetry measured at ½ height must be below 4
- Intensity ratios of the isotopic ions are within the following limits: (the values are expressed as percentages of the less abundant isotopic ions relative to the more abundant ones ± the acceptable error limits)
 - m/z 143/141 ($^{37}\text{Cl}/^{35}\text{Cl}$) for 5-chloro-2-methylaniline is $33\pm 3.0\%$
 - m/z 186/184 ($^{34}\text{S}/^{32}\text{S}$) for dibenzothiophene is $5.5\pm 1.0\%$

n-Alkanes series can be used for calculation of retention indexes (see Section 3, Part B, Chapter I of ROP for GC analysis).

7.2 Sensitivity test

The sensitivity of the mass spectrometer is checked using a test solution containing *n*-pentadecane at 10 ng/μl in a non-retained solvent.

2 μl of the test solution is introduced with a syringe into a conditioned tube. The tube is analysed by TD–GC–MS/EI using the conditions in Tables 4, 5, and 6 (the oven rate can be increased to 15 °C/min and the solvent delay must be set to 3 minutes).

The test is accepted if the signal-to-noise (S/N) ratio for *n*-pentadecane is >10:1 in the total ion chromatogram (TIC). The evaluation is done visually or automatically if available.

7.3 Blank system

Blank tests are empty tubes used to check for possible contamination of equipment (trap, transfer line, GC column, MS source). Analysis must be performed before each sample batch with the same TD–GC–MS/EI parameters as the sample tubes (see Tables 4, 5, and 6). If significant pollution is detected, maintenance operations must be realised to eliminate all traces in the system.

7.4 Blank test

Blank tests are conditioned Tenax[®] tubes prepared in the same way and at the same time as tubes to be used for sampling, but are never removed from the laboratory. Blank tests are used to check for possible contamination of inherent sorbent stability. Analysis must be performed before each sample batch with the same TD–GC–MS/EI parameters as the sample tubes (see Tables 4, 5, and 6).

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1. Scope

This section describes methods for the preparation of concrete, paint, rubber, and other polymeric, as well as wipe samples, when chemical warfare agents (CWAs) and their degradation products are analysed with chromatographic methods such as gas chromatography (GC), liquid chromatography (LC), ion chromatography (IC), or capillary electrophoresis (CE) as the separation technique or an introduction technique to spectrometers. The sample preparation methods required for analysis by nuclear magnetic resonance (NMR) spectroscopy are also presented. Chapter V.A. describes the preparation of concrete samples, Chapter V.B. preparation of paint, rubber, and other polymeric samples, and Chapter V.C. preparation of wipe samples.

Other solid material samples such as wood, metal, cloth, garment, wool, vegetation (leaves, grass), etc. can be prepared in a similar way as the described sample types. However, no detailed ROP has been written, because analysis of such types of samples has not been tested in any international comparison (round-robin) and proficiency tests and are, thus, not validated.

Common facts for preparation of all types of solid material are listed in the following paragraphs.

2. Preparation of solid material samples

Division of solid samples can be problematic and may cause loss of analytes of interest because of heterogeneity of the analytes in the sample material. Physical conditions (e.g. hardness) and too small an amount of available sample can make division of the sample impossible.

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The solid material sample is commonly extracted successively with an organic and an aqueous solvent.

2.1 Organic extraction

Place the sample into a small glass vial or use the original sample container as an extraction vessel. Check that the selected solvent does not dissolve the sampling matrix to avoid matrix interferences. Pour the selected organic solvent (acetone, ethyl acetate, dichloromethane, or deuterated analogues if NMR is used for analysis) over the sample matrix. Increasing the surface by cutting or crushing the sample into smaller pieces will enhance the efficiency. The amount of solvent used will depend on the sample. In general, care must be taken to soak, cover, or rinse the material with the solvent. As a rule, 1 g of a material needs to be extracted with a few ml of a solvent.

Extraction by agitating, shaking, rinsing, or sonicating for 3–10 minutes is usually sufficient to obtain good extraction efficiencies. Filtration of the extract through a membrane filter is recommended. If concentration of the extract is required, apply mild nitrogen flow, but not to concentrate to dryness (Section 2, Part A, Chapter II: *Concentration*).

This organic extract is analysed for nonpolar CWC-related chemicals.

The extract should also be analysed with GC after silylation (Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*), because it might be possible that the extract contains polar CWC-related chemicals soluble in the used solvent.

If the organic extract is analysed with LC and/or LC–MS, the solvent (e.g. acetone, dichloromethane or ethyl acetate) can be exchanged or diluted with a solvent more suitable for LC analysis. After concentration with mild nitrogen flow as described in Section 2, Part A, Chapter II: *Concentration*, the residue is dissolved in a solvent suitable to the mobile phase chemistry of the LC analysis (pH and the amount of aqueous solvent).

The organic extract can be analysed with NMR directly if deuterated solvent was used for the extraction. In other cases, 0.8 ml of the extract is diluted with 0.2 ml of the corresponding deuterated solvent and transferred into an NMR tube.

2.2 Aqueous extraction

Repeat the procedure on the sample using an aqueous or aqueous soluble solvent (e.g. water, acetonitrile or methanol).

The aqueous extract is analysed for polar CWC-related chemicals such as alkylphosphonic and alkyl thiophosphonic acids, thiodiglycol, and amino alcohols.

For GC analysis of polar CWC-related chemicals, derivatisation is needed. Evaporation and derivatisation procedures are presented in Section 2, Part A Chapter II: *Concentration*, and Chapter III: *Derivatisation for gas chromatography*.

Water, acetonitrile and methanol extracts can be analysed with LC and LC–MS as such without derivatisation. If a pre-column is not used, the extract is usually filtered through a disposable syringe filter before injection.

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Polar extracts can be analysed with NMR directly if deuterated solvent was used for the extraction. In other cases, 0.8 ml of the extract is diluted with 0.2 ml of deuterated solvent and transferred into an NMR tube.

2.3 Extraction/derivatisation to analyse Lewisites with GC–MS

Both organic and aqueous extracts can be prepared for analysis of Lewisites 1 and 2, as well as Lewisite oxide. The derivatisation procedure is presented in Section 2, Part A Chapter III: *Derivatisation for gas chromatography*.

3. Quality control

If available, perform the entire extraction procedure with matrix blank material. Quality control is presented in Section 2, Part A, Chapter V: *Quality assurance and quality control in sample preparation*.

4. Evaluation of method

Evaluation and references of the specific methods for concrete, polymeric samples and wipe samples are summarised in their individual chapters V.A., V.B., and V.C. Considering other types of solid material samples, Black *et al.* published their experiences of analysis of real wool samples, which were collected from a Kurdish village three months later from a chemical attack in 1988 and clothing and metal fragment samples that were collected four years later.^{1,2} The wool samples were extracted with dichloromethane and the clothing and metal samples in addition with water. The water extracts were silylated after evaporation to dryness. Nerve agent sarin as well as its polar hydrolysis products were identified from the extracts of the metal fragments, although the samples were collected four years after the attack.

5. References

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Chapter V.A. Concrete samples

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1. Summary of the method

If necessary, the sample is homogenised by crushing, and thereafter first extracted with organic solvent (e.g. acetone or dichloromethane), subsequently water, and 1 M HCl. Extractions can be done successively on the same sample or on different samples. The extracts are prepared for analysis according to the type of extract. A general flowchart describing the sample preparation of concrete samples is given in Figure 1.

Note! Concrete is an alkaline matrix and rapid degradation of most chemical warfare agents deposited on the untreated surface would be expected. Accordingly, the sample preparation procedure includes an efficient extraction step with acidic water for polar degradation products.

2. Safety aspects and storing

Safety aspects of sample preparation as well as storage of samples are presented in Section 2, Part A, Chapter I: *Introduction*.

3. Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in this ROP are listed in Table 1.

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Table 1. Equipment, supplies and chemicals needed to prepare concrete samples.

Equipment	Supply	Chemical
Ultrasonic bath	Pasteur pipettes	Acetone ^{&}
Centrifuge	Adjustable pipette(s) or measuring cylinders [#]	Dichloromethane ^{&}
	Disposable syringes 5 or 10 ml	Water
	Disposable syringe filters 0.45 µm or 0.5 µm	Acidic water
	Sample vials 1.5, 4, 8, 20, or 50 ml	1 M HCl
	Tweezers for transferring the sample matrix in extraction vial	
	Mortar or such a like supplies to crush the original sample to smaller pieces	
	Option: Screw capped centrifuge tubes 20 or 50 ml	

[#] to measure solvents

[&] Possible organic extraction solvents. Deuterated analogs are used if NMR is used for analysis.

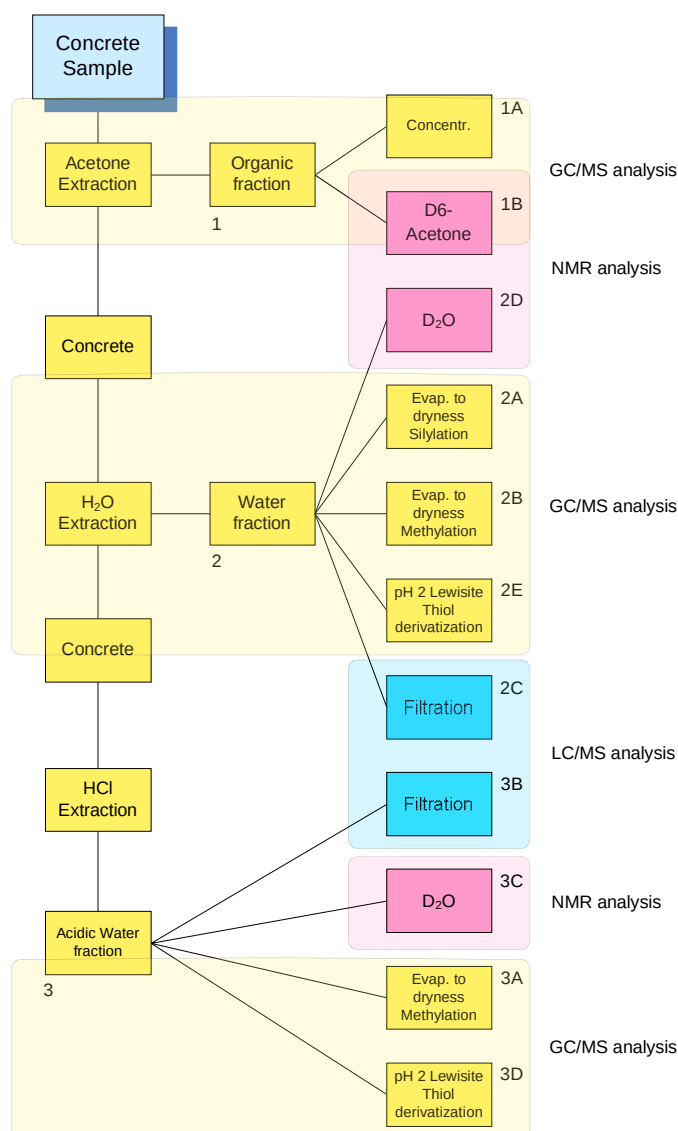


Figure 1. Flowchart of concrete samples.

4. Procedures

4.1 Organic extraction

Transfer ca. 2–5 g of concrete into a glass vial/bottle or use the original sample container as an extraction vessel. If sample pieces are large, homogenising by crushing will be necessary before extraction; small pieces may be extracted without crushing. Add ca. 2–5 ml of acetone or dichloromethane.

Note! Use more solvent if the whole sample is not covered by it.

Close the bottle tightly and sonicate for 30 min. Monitor the temperature of the water bath and add ice, if necessary, to maintain ambient temperature.

Quickly transfer the solvent into another screw-capped vial or centrifuge tube, close the tube, centrifuge for 3 min at 2000 G, and filter the supernatant through a 0.45 µm filter. Repeat the procedure with another portion of solvent and combine the extracts (Fraction 1, Figure 1).

If concentration of the extract is required, apply a gentle stream of nitrogen, but precautions should be taken to prevent the sample from going to dryness (Section 2, Part A, Chapter II: *Concentration*) (Fraction 1A, Figure 1).

This organic extract is analysed for non-polar CWC-related chemicals.

The extract should also be analysed by GC after silylation, because it might be possible that the extract also contains polar CWC-related chemicals that have a tendency to be extracted by the selected solvent (Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*).

4.2 Extraction with water

After the first extraction, extract the same sample or another 2–5 g portion of concrete in a similar way using pure (deionised) water as the solvent (Fraction 2, Figure 1).

The water extract is analysed for polar CWC-related chemicals such as alkylphosphonic and alkylthiophosphonic acids, thiodiglycol, and amino alcohols.

For GC analysis of these chemicals, derivatisation is usually needed. Water extracts must be evaporated to dryness before silylation and methylation. Evaporation and derivatisation procedures are presented in Section 2, Part A Chapter II: *Concentration*, Chapter III: *Derivatisation for gas chromatography* and Chapter IV: *Sample clean-up methods* (Fractions 2A and 2B, Figure 1).

4.3 Extraction with acidic water

After the second extraction, extract the same sample or another 2–5 g portion of concrete in a similar way with 1 M HCl solution. Handle the third extract the same way as the second extract (Fraction 3, Figure 1).

The acidic water extract is analysed for polar acidic CWC-related chemicals such as alkylphosphonic and alkylthiophosphonic acids.

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For GC analysis of these chemicals derivatisation is usually needed. Water extracts must be evaporated to dryness before methylation and silylation. Evaporation and derivatisation procedures are presented in Section 2, Part A, Chapter II: *Concentration* and Chapter III: *Derivatisation for gas chromatography* (Fraction 3A, Figure 1).

4.4 Liquid Chromatography–Mass Spectrometry (LC–MS) sample preparation

Aqueous extracts can be analysed with LC and LC–MS as such without derivatisation. If a pre-column is not used, the extract is usually filtered through a disposable syringe before injection (Fraction 2C and 3B, Figure 1).

***Note!** Depending on the chromatographic method used for analysis, it can be useful to adjust the pH of the sample for optimal chromatographic performance before injection. For pH adjustment, volatile buffers as ammonium hydroxide and formic acid should be used to avoid signal suppression when LC–MS is used for analysis.*

If organic extracts are subjected to analysis by LC or/and LC-/MS analysis, the solvent (e.g. acetone or dichloromethane) can be exchanged to or diluted with a solvent more suitable for optimised LC analysis. After concentration with mild nitrogen flow as described in Section 2, Part A, Chapter II: *Concentration*, the residue is dissolved in a solvent suitable for the mobile phase chemistry of the LC analysis (pH and amount of aqueous solvent).

4.5 Nuclear Magnetic Resonance (NMR) sample preparation

For NMR analysis, 0.8 ml of the extract is diluted with 0.2 ml of deuterated solvent and transferred into a NMR tube (Fraction 1B, 2D and 3C, Figure 1).

If the sample and/or solvent background are considered too high for successful NMR analysis, all of the solvent can be changed to NMR solvent. Part of the extract (e.g. 0.8 ml) is concentrated/evaporated to near-dryness (Section 2, Part A, Chapter II: *Concentration*), and the residue is dissolved with the corresponding deuterated solvent.

4.6 Extraction/derivatisation to analyse Lewisites with GC–MS

The water/acidic water extract is, in addition to the analytes previously mentioned, prepared for analysis of the sample for Lewisites 1 and 2, as well as Lewisite oxide (Fraction 2E and 3D, Figure 1). The derivatisation procedure required for Lewisites is presented in Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*.

5. Quality control

If available, perform the entire extraction procedure with matrix blank concrete material. Quality control of sample preparation is presented in Section 2, Part A, Chapter V: *Quality assurance and quality control in sample preparation*.

6. Evaluation of method

This ROP of concrete samples was established after the third interlaboratory comparison (round-robin) test held in 1992,^{1,2,3} but thereafter, the method has not been further evaluated or tested in any proficiency tests arranged by OPCW. D'Agostino *et al.*

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published their results of analysis of acetonitrile extracts of concrete samples of the this round-robin test.^{4,5} Later concrete samples have been extracted with acetonitrile or deuterated chloroform when degradation of mustard gas and sarin on concrete or cement have been investigated.^{6,7,8,9}

Based on the results of the round-robin test, in which this ROP is based on, as well as based on the mentioned later publications, it is impossible to say which solvent produces the best extraction efficiencies. Concrete samples should simply be extracted with an organic solvent in which the material is not soluble (e.g. acetonitrile and acetone). To increase the recovery of polar CWC-related chemicals water should be chosen for the second extraction. Because concrete is an alkaline matrix and rapid degradation of most chemical warfare agents deposited on the untreated surface would be expected, the sample preparation procedure also includes an additional efficient extraction step with acidic water for the most polar acidic degradation products.

It is not totally necessary to repeat the extraction procedures. However, to maximise the extraction of minor amounts of chemicals the extraction can be repeated and the combined extracts gingerly concentrated, when necessary. Centrifugation to separate the solid material from the extraction solvent is neither usually needed, but simple decantation of the solvent is enough to separate the phases.

Note! *Division of a concrete sample can be problematic and may cause loss of analytes of interest because of heterogeneity of the analytes in the sample material.*

7. References

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Chapter V.B. Paint, rubber and other polymeric samples

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1. Summary of the method

The paint, rubber, or other polymeric sample is extracted successively with organic and aqueous solvents. A general flowchart of sample preparation of paint, rubber, and other polymeric samples is given in Figure 1.

2. Safety aspects and storing

Safety aspects of sample preparation as well as storage of samples are presented in Section 2, Part A, Chapter I: *Introduction*.

3. Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in this ROP are listed in Table 1.

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Chapter V.B. Paint, rubber and other polymeric samples

Table 1. Equipment, supplies and chemicals needed to prepare paint, rubber, or other polymeric samples.

Equipment	Supply	Chemical
Ultrasonic bath	Pasteur pipettes	Acetone ^{&}
Centrifuge	Adjustable pipette(s) or measuring cylinders [#]	Ethyl acetate ^{&}
	Disposable syringes 5 or 10 ml	Dichloromethane ^{&}
	Disposable syringe filters 0.45 µm or 0.5 µm	Water
	Sample vials 1.5, 4, 8, ,20, 50, or 100 ml	
	Tweezers for transferring the sample matrix in extraction vial	

[#] to measure solvents

[&] Possible organic extraction solvents. Deuterated analogs are used if NMR is used for analysis.

Note! The selected solvent must not dissolve the sampling medium.

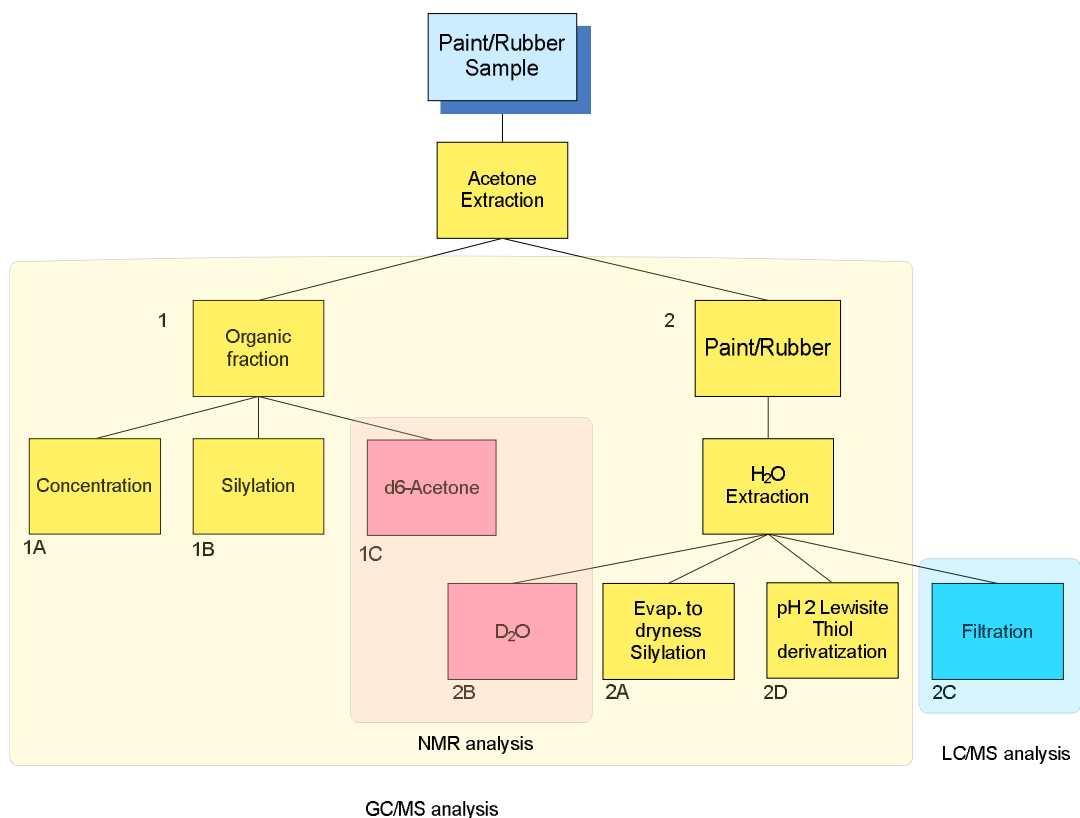


Figure 1. Flowchart of paint, rubber, and other polymeric samples.

4. Procedures

4.1 Organic extraction

If required, crush or cut the sample into smaller pieces and transfer the sample into a small glass vial or use the original sample container as an extraction vessel. Before extraction, use a small piece of sample to check that the intended organic solvent does not dissolve the sample. Pour the selected solvent (acetone, ethyl acetate, dichloromethane, or deuterated analogues if NMR is used for analysis) over the sample matrix. As a general rule, use solvent volume twice the volume of the sampling matrix (e.g. 10 ml) or enough to cover the sample. Sonicate the sample for 10 minutes and quickly transfer the solvent to a new container. If necessary, discard particles and debris by centrifuging briefly. (Fraction 1, Figure 1).

If concentration of the extract is required, apply a gentle stream of nitrogen, but caution should be taken to prevent the sample from going to dryness (Section 2, Part A, Chapter II: *Concentration*), (Fraction 1A, Figure 1).

This organic extract is analysed for nonpolar CWC-related chemicals.

The extract should also be analysed by GC after silylation, because it might be possible that the extract contains also polar CWC-related chemicals that have a tendency to be extracted by the selected solvent (Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*).

4.2 Water extraction

After the first extraction, extract the same sample, or another if available, in a similar way but using pure (deionised) water. (Fraction 2, Figure 1).

Note! *Extraction of a new piece of sample with water could lead to erroneous conclusions regarding the presence of degradation products (e.g. degradation products can be unintentionally formed by adding water to the sample).*

The water extract is analysed for polar CWC-related chemicals such as alkylphosphonic and alkylthiophosphonic acids, thiodiglycol, and aminoalcohols.

For GC analysis of polar CWC-related chemicals, derivatisation is usually needed. Water extracts must be evaporated to dryness before silylation. Evaporation and derivatisation procedures are presented in Section 2, Part A, Chapter II: *Concentration*, Chapter III: *Derivatisation for gas chromatography* and Chapter IV: *Sample clean-up methods* (Fraction 2A, Figure 1).

4.3 Liquid Chromatography/Mass Spectrometry (LC–MS) sample preparation

Water extract can be analysed with LC and LC–MS as such without derivatisation. If a precolumn is not used, the extract is usually filtered through a disposable syringe before injection (Fraction 2C, Figure 1).

Note! *Depending on the chromatographic method used for analysis, it can be useful to adjust the pH of the sample for optimal chromatographic performance before injection.*

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For pH adjustment, volatile buffers as ammonium hydroxide and formic acid should be used to avoid signal suppression when LC–MS is used for analysis.

If organic extracts are subjected to analysis by LC or/and LC–MS analysis, the solvent (e.g. acetone, dichloromethane or ethyl acetate) can be exchanged to or diluted with a solvent more suitable for optimised LC analysis. After concentration with mild nitrogen flow as described in Section 2, Part A, Chapter II: *Concentration*, the residue is dissolved in a solvent suitable for the mobile phase chemistry of the LC analysis (pH and amount of aqueous solvent).

4.4 Nuclear Magnetic Resonance (NMR) sample preparation

Both extracts can be analysed directly with NMR if the extraction was done with deuterated solvents. In other cases, 0.8 ml of the extract is diluted with 0.2 ml of the corresponding deuterated solvent and transferred into an NMR tube (Fractions 1C and 2B, Figure 1).

If the sample and/or solvent background are considered too high for successful NMR analysis, all of the solvent can be changed to NMR solvent. Part of the extract (e.g. 0.8 ml) is concentrated to near-dryness with a gentle nitrogen flow (Section 2, Part A, Chapter II: *Concentration*), where after the residue is dissolved with deuterated solvent.

4.5 Extraction/derivatisation to analyse Lewisites with GC–MS

Both organic and aqueous extracts can be prepared for analysis of the sample for Lewisites 1 and 2 as well as Lewisite oxide (Fraction 2D, Figure 1). The derivatisation procedure required for Lewisites is presented in Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*.

5. Quality control

If available, perform the entire extraction procedure with matrix blank paint, rubber, or other polymeric material. Quality control of sample preparation is presented in Section 2, Part A, Chapter V: *Quality assurance and quality control in sample preparation*.

6. Evaluation of method

This ROP of paint, rubber, and other polymeric samples was established after the third interlaboratory comparison (round-robin) test.^{1,2,3} Wils *et al.* (1992) have shortly described their results of analysis of paint and rubber samples using dichloromethane or thermal desorption for analysis of mustard gas.⁴ The ROP method has later been successfully tested in the trial proficiency test for rubber and paint samples, in the first proficiency test for polymer samples, and in the third proficiency test for paint samples. OPCW arranged these tests. Two groups have published their results of analysis of paint and rubber samples of these tests.^{5,6,7} *n*-Hexane or acetone were used for organic extraction and water, if used, for aqueous extraction, respectively. These organic solvents were used because paint or polymeric materials are less soluble in acetone than in chlorinated solvents or ethyl acetate. In addition, to increase the recovery of the polar CWC-related chemicals organic extraction should be followed by water extraction as Greasy *et al.* have also shown.⁸

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It should be noted that many CWC-related chemicals diffuse deep into polymeric materials relatively quickly, meaning that a low recovery would be estimated by noninvasive solvents recommended in this method. In the recent 29th OPCW proficiency test, it was shown that the recovery of mustard gas and analogues from a flooring PVC material could be enhanced from below 20% to more than 90% using an extraction method based on total solubilisation of the polymer. However, to avoid too much background from the matrix, it is for qualitative analysis recommended to extract the analytes by rinsing or wiping of the polymeric surface with a suitable solvent. Collecting paint and rubber samples in the battlefield to prove use and/or presence of prohibited chemicals is considered as a realistic sampling scenario also for military purposes.⁹

There is usually no need to repeat the extraction procedures. However, to maximise the extraction of minor amounts of chemicals, the extraction can be repeated and the combined extracts gingerly concentrated, when necessary. Centrifugation to separate the solid material from the extraction solvent is not usually needed, but simple decantation of the solvent is enough to separate the phases.

Note! *Division of a polymeric sample can be problematic and may cause loss of analytes of interest because of heterogeneity of the analytes in the sample material.*

7. References

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Chapter V.C. Wipe samples

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Note! During an on-site inspection or in similar situations, there may be a need to sample liquids or wipe solid surfaces at the sampling site using sampling media such as cotton buds, filter paper, or glass fiber beds. The matrix selected for the sampling should be such as to cause no interference with the later analysis. The background coming from the sampling matrix must be carefully determined.

1. Summary of the method

The material wiped onto or absorbed in the wipe sampling matrix, is extracted successively with organic and polar solvents. A general flowchart of sample preparation of wipe samples is given in Figure 1.

2. Safety aspects and storing

Safety aspects of sample preparation as well as storage of samples are presented in Section 2, Part A, Chapter I: *Introduction*.

3. Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in this ROP are listed in Table 1.

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Table 1. Equipment, supplies and chemicals needed to prepare wipe samples.

Equipment	Supply	Chemical
Ultrasonic bath	Pasteur pipettes	Acetone ^{&}
Centrifuge	Adjustable pipette(s) or measuring cylinders [#]	Ethyl acetate ^{&}
	Disposable syringes 5 or 10 ml	Dichloromethane ^{&}
	Disposable syringe filters 0.45 µm or 0.5 µm	Water
	Sample vials 1.5, 4, 8, 20, 50, or 100 ml	
	Tweezers for transferring the sample matrix in extraction vial	

[#] to measure solvents

[&] Possible non-polar extraction solvents. Deuterated analogs are used if NMR is used for analysis.

Note! The selected solvent must not dissolve the sampling medium.

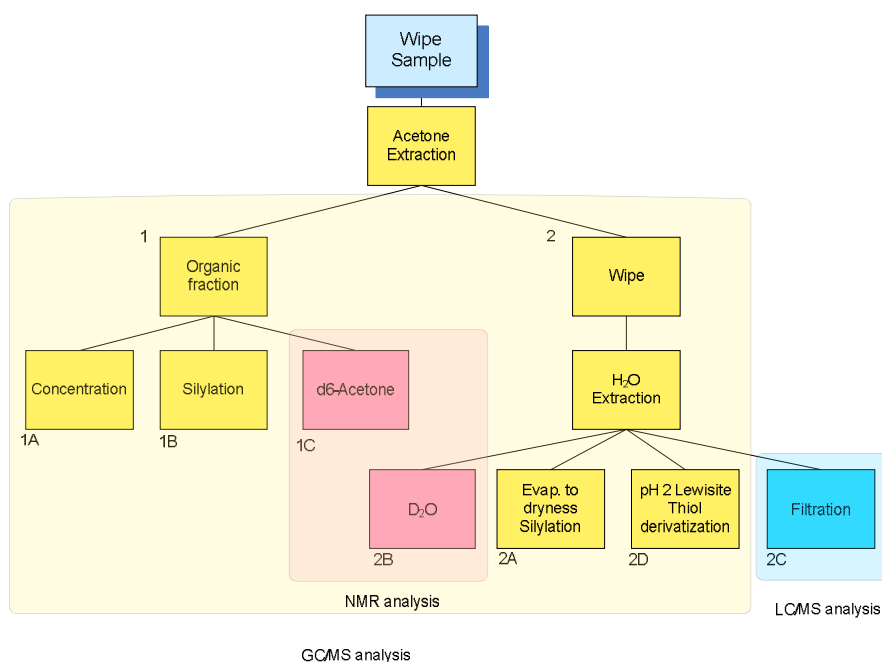


Figure 1. Flowchart of wipe samples.

4. Procedures

4.1 Organic extraction

Place the sample into a small glass vial or use the original sample container as an extraction vessel. Before extraction, use a small piece of the sample to check that the intended organic solvent does not dissolve the sampling matrix. Pour the selected solvent (acetone, ethyl acetate, dichloromethane, or deuterated analogues if NMR is used for analysis) over the sample matrix. As a general rule, use solvent volume twice the volume of the sampling matrix or enough to cover the sample. Sonicate the sample for 3 minutes and quickly transfer the solvent to a new container. If necessary, discard particles and debris by centrifuging briefly (Fraction 1, Figure 1).

If concentration of the extract is required, apply a mild nitrogen flow, but caution should be taken to prevent the sample from going to dryness (Section 2, Part A, Chapter II: *Concentration*), (Fraction 1A, Figure 1).

This organic extract is analysed for non-polar CWC-related chemicals.

The extract should also be analysed by GC after silylation, because it might be possible that the extract contains also polar CWC-related chemicals that have a tendency to be extracted by the selected solvent (Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*).

4.2 Second extraction

Repeat the procedure on the matrix using water or a solvent soluble to water (acetonitrile or methanol), (Fraction 2, Figure 1).

The polar extract is analysed for polar CWC-related chemicals such as alkylphosphonic and alkylthiophosphonic acids, thiodiglycol, and aminoalcohols.

For GC analysis of these chemicals, derivatisation is usually needed. Acetonitrile extracts can be silylated as such, methanol extracts can be methylated as such, but water and methanol extracts must be evaporated to dryness before silylation. Evaporation and derivatisation procedures are presented in Section 2, Part A, Chapter II: *Concentration*, Chapter III: *Derivatisation for gas chromatography* and Chapter IV: *Sample clean-up methods* (Fraction 3, Figure 1).

Note! *The use of alcoholic solvents may lead to esterification and trans-esterification products.*

4.3 Liquid Chromatography/Mass Spectrometry (LC–MS) sample preparation

Aqueous, acetonitrile, and methanol extracts can be analysed with LC and LC–MS as such without derivatisations. If a pre-column is not used, the extract is usually filtered through a disposable syringe filter before injection (Fraction 2C, Figure1).

Note! *Depending on the chromatographic method used for analysis, it can be useful to adjust the pH of the sample for optimal chromatographic performance before injection. For pH adjustment, volatile buffers as ammonium hydroxide and formic acid should be used to avoid signal suppression when LC–MS is used for analysis.*

If organic extracts are subjected to analysis by LC or/and LC–MS analysis, the solvent (e.g. acetone, dichloromethane, or ethyl acetate) can be exchanged to or diluted with a solvent more suitable for optimised LC analysis. After concentration with mild nitrogen flow as described in Section 2, Part A, Chapter II: *Concentration*, the residue is dissolved with a solvent suitable to the mobile phase chemistry of the LC analysis (pH and amount of aqueous solvent).

4.4 Nuclear Magnetic Resonance (NMR) sample preparation

Both organic and aqueous extracts can be analysed with NMR directly if deuterated solvent was used in the extraction. In other cases, 0.8ml of the extract is diluted with 0.2 ml of deuterated solvent and transferred into a NMR tube (Fraction 1C and 2B, Figure1).

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If the sample and/or solvent background are considered too high for successful NMR analysis, all of the solvent can be changed to NMR solvent. Part of the extract (e.g. 0.8 ml) is concentrated to near-dryness with a gentle nitrogen flow (Section 2, Part A, Chapter II: *Concentration*), after which the residue is dissolved in deuterated solvent.

4.5 Extraction/derivatisation to analyse Lewisites with GC–MS

Both organic and aqueous extracts can be prepared for analysis of the sample for Lewisites 1 and 2 as well as Lewisite oxide (Fraction 2D, Figure 1). The derivatisation procedure required for Lewisites is presented in Section 2, Part A, Chapter III: *Derivatisation for gas chromatography*.

5. Quality control

If available, perform the entire extraction procedure with matrix blank wipe material. Quality control of sample preparation is presented in Section 2, Part A, Chapter V: *Quality assurance and quality control in sample preparation*.

6. Evaluation of method

The existing ROP of wipe samples was established after the second interlaboratory comparison (round-robin) test.^{1,2,3} Wils *et al.* published their results of both wipe (cotton bud) and air samples delivered in this test.⁴ Wipe samples were extracted with ethyl acetate and deuterated chloroform.

The ROP of the wipe sample has later been successfully tested once in the fourth official proficiency test arranged by OPCW. Nassar *et al.* published their successful experiences of analysis of water extracts of wipe samples using CE for analysis. In the same article, similar analysis of real wipe samples taken in the field site after nerve agent exposure was reported.⁵ In fact, taking wipe samples in battlefield conditions to prove use and/or presence of prohibited chemicals is considered as a very realistic sampling scenario also for military purposes.⁶

Based on the results of the round-robin test, in which this ROP is based on, as well as based on the mentioned later publications, it is impossible to say which organic solvent produces the best extraction efficiencies. The wipe should be simply extracted with an organic solvent in which the wipe material is not soluble. Acetone is usually a good candidate for organic extraction, because polymeric wipe materials are less soluble in acetone than in chlorinated solvents or ethyl acetate. To increase the recovery of the polar CWC-related chemicals, organic extraction should be followed by water extraction of the sample.⁷

Usually, there is no need to repeat the extraction procedure. However, to maximise the extraction of minor amounts of chemicals, the extraction can be repeated and the combined extracts gingerly concentrated, when necessary. In addition, centrifugation to separate the solid material from the extraction solvent is not usually needed, but simple decantation of the solvent is enough to separate the phases.

Note! *Division of the wipe sample can be problematic and may cause loss of analytes of interest because of heterogeneity of the analytes in the sample material.*

7. References

1. M. Rautio (ed.), “International Interlaboratory Comparison (Round-Robin) Test for the Verification of Chemical Disarmament, F.2. Testing of Procedures on Simulated Industry Samples”, The Ministry for Foreign Affairs of Finland, Helsinki, 1991.
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Chapter VI. Active charcoal samples

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1. Scope

This section describes methods how trapped compounds are desorbed from an active charcoal matrix and analysed with chromatographic methods such as gas chromatography (GC), liquid chromatography (LC), ion chromatography (IC), or capillary electrophoresis (CE) as an analysis technique or as an introduction technique to spectrometers. Nuclear magnetic resonance (NMR) spectroscopy can also be used for analysis.

2. Summary of the method

The compounds trapped on the active charcoal are extracted with organic solvent. If the origin and nature of the active charcoal sample material are known, the sample or a few grains of it may also be analysed by thermal desorption. The flowchart of sample preparation of active charcoal sample is given in Figure 1.

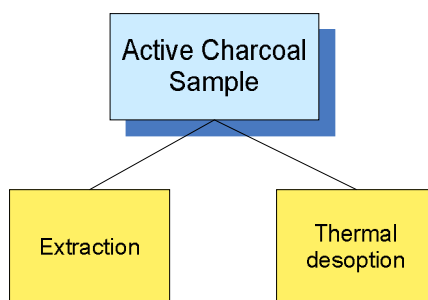


Figure 1. Flowchart of active charcoal samples.

Normally, active charcoal of unknown origin is not recommended for air or liquid sampling since it may contain substantial amounts of water and trace amounts of doping chemicals.

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3. Safety aspects and storing

Safety aspects of sample preparation and storing of samples are presented in Section 2, Part A, Chapter I: *Introduction*.

4. Equipment, supplies and chemicals

The equipment, supplies and chemicals needed in this ROP are presented in Table 1.

Note! *The origin and background of active charcoal samples should always be ascertained, because the charcoal may have been doped with various chemicals to modify its properties for a specific purpose. This may cause degradation of the sample compounds and extraction problems.*

Table 1. Equipment, supplies and chemicals needed to prepare active charcoal samples.

Equipment	Supply	Chemical
Ultrasonic bath	Pasteur pipettes	Acetone ^{&}
Centrifuge	Adjustable pipette(s) or measuring cylinders [#]	Dichloromethane ^{&}
	Disposable syringes 5 or 10 ml ^{\$}	Carbon disulfide ^{&}
	Disposable syringe filters 0.45 µm or 0.5 µm	
	Sample vials 1.5, 4, 8, and 20 ml	

[#] to measure solvents

[&] desorption solvent. Deuterated analogs are used if NMR is used for analysis

^{\$} filter paper and a filtering funnel or cotton wool and Pasteur pipette can be used also

5. Procedures

5.1 Extraction

Place the active charcoal material in a glass vial. Add the extracting solvent (twice the volume of the charcoal), cap the vial, and sonicate the mixture for three minutes. Suitable solvents are acetone, carbon disulfide, dichloromethane. If NMR is used for analysis, deuterated solvents (e.g. acetone-*d*₆ or dichloromethane-*d*₂) can be used.

Where necessary, remove the supernatant by brief centrifugation and quick filtration. If concentration of the extract is required, do it under a gentle stream of nitrogen, but not to dryness (Section 2, Part A, Chapter II: *Concentration*).

This organic extract is analysed for non-polar CWC-related chemicals.

5.2 Thermal desorption

If the origin and nature of the active charcoal sample material, as well as the sampling procedure, are known, the sample or a few grains of it may also be analysed by thermal desorption.

5.3 LC–MS sample preparation

For LC and LC–MS analysis, the solvent of the extract can be changed to be more suitable for analysis. For this reason, dichloromethane extract especially has to be concentrated with mild nitrogen flow as described in Section 2, Part A, Chapter II: *Concentration*. The residue is dissolved with a solvent suitable to the solvent chemistry of the LC analysis (pH and amount of aqueous solvent).

5.4 NMR sample preparation

The extract can be analysed with NMR as if deuterated solvent was used. In other cases, ca. 0.8 ml of the extract is diluted with ca. 0.2 ml of deuterated solvent and transferred into a NMR tube.

If the sample and/or solvent background are considered too high for successful NMR analysis, all of the solvent can be changed to NMR solvent. Part of the extract (e.g. 0.8 ml) is concentrated almost to dryness with mild nitrogen flow (Section 2, Part A, Chapter II: *Concentration*), and the residue is dissolved with deuterated solvent.

6. Quality control

Quality control of sample preparation is presented in Section 2, Part A, Chapter V: *Quality assurance and quality control in sample preparation*.

7. Evaluation of method

This recommendation is based on the results obtained in the second international interlaboratory comparison (round-robin) test held in 1990.^{1,2,3} This means that the ROP method for active charcoal samples has not been validated in interlaboratory or proficiency tests. In addition, only a few, quite old, successful uses of charcoal in battlefield conditions have been reported.^{5,6,7} Furthermore, active charcoal is not recommended for the collection of organophosphorus chemicals (nerve agents) because it may induce their decomposition.⁴ In general, the recoveries of CWC-related chemicals tend to be lower from active charcoal than from other air sampling materials. Therefore, active charcoal as a strong adsorbent requires more effective extraction methods than, e.g., Tenax[®] tubes.

The use of active charcoal as a sampling material of CWC-related chemicals is overall problematic. The materials used in filters, protective clothing, respiratory canisters, etc. are usually impregnated with chemicals according to the needs of the use. These chemicals have often catalytic and chelating properties, which mean that the chemicals desorbed from such charcoal materials might be different from those originally adsorbed. This problem arises especially if the charcoal is analysed by thermal desorption. In this respect, it is essential for successful sample pre-treatment that the origin and contents of active charcoal material should be known. Unfortunately, this will seldom be possible, for both military and commercial reasons.

8. References

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Section 2. Sample preparation

Part B. Established methods

Chapter VI. Active charcoal samples

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Chapter I. Introduction

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1. Scope

This chapter introduces the techniques that are used for the identification of scheduled chemicals in verification activities associated with the Chemical Weapons Convention (CWC). Analysis in support of verification may be performed on-site at an inspected facility, or off-site in Designated Laboratories in cases where on-site analysis is equivocal or in cases of alleged use of chemical weapons (CW) in which more extensive analysis is required.

Analytical methods can be broadly divided into those that combine a separation technique with identification, and those that analyse an aliquot or extract of the sample without separating the components. Section 3 on analytical methods is divided into five parts:

- GC-based techniques
- LC-based techniques
- Other separation techniques
- Spectroscopy-based techniques
- Other analysis (saxitoxin and ricin)

2. GC-based techniques

Most CW agents have a degree of volatility, and historically the separation technique of choice has been gas chromatography (Chapter “*Gas chromatography*”, 3-B-I). Agents are usually chromatographed directly without derivatisation. Exceptions are Lewisites I and II and phosgene because of their reactivity and, in the case of phosgene, very high volatility. Gas chromatography combined with element selective detection (GC-FPD, GC-NPD, GC-AED) is useful for screening, but the major technique for identification of volatile scheduled chemicals is gas chromatography combined with mass spectrometry using electron ionisation (GC-EI/MS) (Chapter “*Gas chromatography-electron ionisation/mass spectrometry*”, 3-B-II). In support of verification, the Organisation for the Prohibition of Chemical Weapons (OPCW), in cooperation with Member States, has compiled an extensive database of electron ionisation (EI) spectra (OCAD). GC-EI/MS may be used as the sole technique for on-site analysis. For analysis at the parts per million level, simple bench top quadrupole and ion trap instruments are commonly employed, although mass

spectra acquired on ion traps may show some differences, e.g. from self-protonation. Time of flight (TOF) instruments, employing high acquisition rates, are seeing increasing application in GC–MS analysis. Additional GC–MS analysis using chemical ionisation (GC–CI/MS) (Chapter “*Gas chromatography-chemical ionisation/mass spectrometry*”, 3-B-III) is used to confirm the molecular mass of an analyte for off-site analysis and in OPCW Proficiency Tests.

Some scheduled precursors, and most environmental degradation products of CW agents, are more polar than the agents, and much less volatile. These require conversion, e.g., to trimethylsilyl (TMS), *tert*-butyldimethylsilyl (TBDMS) or methyl ester derivatives, in order to facilitate analysis by GC–MS. In the case of aqueous solutions, this usually requires isolation of the analyte or complete removal of the water, processes that are both time consuming and prone to error.

GC–tandem mass spectrometry (GC–MS/MS) is not used extensively for identification at the parts per million levels typical of OPCW Proficiency Tests. It is a key technique in trace analysis, particularly for biomedical samples, where analyte levels may be in the low parts per billion range. Applications will be described in the chapter entitled “*Gas chromatography-tandem mass spectrometry*” (3-B-IV; not in 2011 edition).

High resolution MS allows the determination of the molecular formula of unknowns, e.g. in OPCW Proficiency Tests, and may be used to increase selectivity in trace analysis. As MS instrumentation is evolving, the use of high resolution magnetic sector instruments is declining, whilst the use of other types of instruments capable of high resolution is increasing. The chapters “*Gas chromatography-high resolution mass spectrometry: sector instruments, selective reaction monitoring*” (3-B-V) and “*Gas chromatography-high resolution mass spectrometry*” (3-B-VI) describe the use of sector instruments and GC–TOF/MS for high resolution applications.

3. LC-based techniques

An alternative approach to the analysis of more polar chemicals is liquid chromatography-mass spectrometry (LC–MS) (Chapter “*Liquid chromatography separation for mass spectrometry*”, 3-C-1; not in 2011 edition). This has the advantage that it can be applied directly to aqueous solutions. Modern LC–MS is performed almost universally using one of two atmospheric pressure ionisation (API) techniques, electrospray ionisation (ESI) and atmospheric pressure chemical ionisation (APCI) (Chapter “*Liquid chromatography-API mass spectrometry*”, 3-C-II). Single stage LC–MS provides molecular mass information, similar to GC–CI/MS, and provides a very rapid screening procedure in OPCW Proficiency Tests. Additional structural information can be obtained by LC–tandem mass spectrometry (LC–MS/MS). The increasing use of LC–TOF/MS is described in the chapter entitled “*Liquid chromatography-tandem mass spectrometry*” (3-C-III). LC–MS and LC–MS/MS are now widely used by Designated Laboratories. These techniques are essential for the analysis of biomedical samples, and for the identification of saxitoxin and ricin.

4. Other separation techniques

Capillary electrophoresis (CE) and ion chromatography are used by only a small number of Designated Laboratories. Both are used primarily for the screening and identification of alkyl alkylphosphonates and alkylphosphonic acids. The chapter entitled “*Capillary electrophoresis*” (3-D-I) describes the use of CE in combination with UV detection. The

chapter called “*Ion chromatography*” (3-D-II) describes the use of ion chromatography (IC) with conductivity detection or ESI/MS.

5. Spectroscopy-based techniques

The principle technique used without separating the components of the sample is nuclear magnetic resonance spectrometry (NMR). Advances in instrumentation and techniques, including 2D and correlation techniques, have increased the sensitivity of NMR and the ability to discriminate between individual chemicals in mixtures. NMR is widely used in OPCW Proficiency Tests, and is particularly useful in identifying isomers. ^1H , ^{31}P , ^{19}F , 1D ^1H – ^{31}P heteronuclear single-quantum correlation (HSQC) and 2D ^1H – ^{31}P correlation techniques are described in the chapters entitled “*Basic ^1H NMR techniques*”, “*Basic ^{31}P NMR techniques*”, “*Basic ^{19}F NMR techniques*”, “*Nonselective 1D ^1H – ^{31}P HSQC technique*” and “*2D ^1H – ^{31}P correlation techniques*” (3-E-I to 3-E-V).

Fourier transform infrared spectroscopy (FTIR) has applications in on-site analysis and, in combination with GC, occasional application in off-site analysis. These applications will be described in the chapter entitled “*FTIR*” (3-E-VI; not in 2011 edition).

6. Other analysis (saxitoxin and ricin)

The Scientific Advisory Board of the OPCW (through its Temporary Working Group on Sampling and Analysis) and the Technical Secretariat of the OPCW are now beginning to address the identification of the two toxins included in Schedule 1 of the CWC, saxitoxin and ricin. These cannot be analysed by GC–MS; rather, LC–MS/MS is the major method of identification. In order to meet the requirement for at least two identification techniques, additional methods are required, such as immunoassays for ricin in the chapter entitled “*Analysis of Ricin: ELISA*” (3-F-IV). The identification of saxitoxin and ricin are described in the chapters “*Analysis of saxitoxin: LC–MS/MS*” (3-F-I) and “*Analysis of Ricin: MALDI–MS*” (3-F-VI; not in 2011 edition) and “*Analysis of Ricin: LC–MS/MS*” (3-F-VII; not in 2011 edition).

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Section 3. Analytical methods

Part A. Introduction

Chapter I Introduction

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Chapter I. Gas chromatography (GC)

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1. Scope

This chapter covers the gas chromatographic (GC) analysis of chemical warfare agents, their precursors, and degradation products or derivatives, i.e. chemicals related to the Chemical Weapons Convention, CWC. Generally, the procedure is for identifying and screening chemicals above the 0.1 mg/kg concentration level. However, the procedure can also be utilised for identifying and screening relevant chemicals down to low µg/kg concentration levels with some selective detectors. This procedure alone gives in most cases only tentative identification and final identification should be done by spectrometric methods.

2. Summary of the method

Chemicals extracted from samples into organic solvent are separated on properly selected columns and detected with appropriate detectors. Before and at the end of each sample batch, a quality control run to verify the correct performance of the instrument, and a blank run to verify the absence of CWC-related chemicals is carried out. Identifications are made with absolute retention times against the absolute retention times of authentic reference chemical or with retention indices (RI) measured against *n*-alkane series C₈–C₂₄ and comparing the measured RI values to those in the library.

3. Safety aspects

GC analysis does not require any specific precautions concerning safety. Samples to be analysed normally contain only small concentrations of toxic chemicals and the used detectors destroy the sample molecules. Normal laboratory safety measures like protective gloves and goggles during sample handling in manual injections should be used.

Hot parts of instruments, e.g. hot septum nut (>250 °C), must be taken into account during operation and maintenance of instruments.

4. Instrumentation

4.1 Gas Chromatograph

Any modern, preferably a two-channel GC instrument, can be used.

4.2 Injectors

Injection technique plays an important role in chromatographic separation. On-column and splitless injections are the most useful techniques for trace analysis. On-column injection has some advantages over other techniques, especially in the analysis of thermally labile compounds such as chloropicrin and compounds with adsorptive properties such as VX homologues and dialkyl methylphosphonates. However, for environmental samples such as soil and plant extracts and biological samples, on-column injection may lead to rapid contamination of the analytical column. For the on-column technique, a separate injector is needed while splitless and split injections can be carried out with the same injector. The split injection technique provides good separation efficiency for volatile compounds when the concentrations of these are high enough, but discrimination against high boiling components is evident.

4.3 GC columns

The non-polar phases tend to have better characteristics in terms of resistance to oxygen, separation efficiency of the column and higher maximum temperatures. Laboratories should have at least two or three columns of different polarity available for situations when the target chemical elutes together with a background chemical on standard SE-54-type (5%-phenyl-95%-dimethylpolysiloxane) phase material. Non-polar OV-1 (100% dimethylpolysiloxane) and medium-polar OV-1701 (14%-cyanopropylphenyl-86%-dimethylpolysiloxane) phase materials are suitable for solving such co-elution problems. A polar CW-20M (polyethyleneglycol) column can only be used in limited applications for the analysis of the most polar compounds. Some of the CWC-related compounds do not elute through CW-20M column due to their strong retention and to the quite low maximum temperature allowed for the phase material.

The inner diameter of a column has a direct influence on retention, efficiency, and capacity of the column. The on-column injection technique requires an internal diameter (i.d.) of at least 0.25 mm. This is also a good choice for MS analysis, as it facilitates a proper adjustment of the carrier gas flow. Narrower bore columns are typically used in various kinds of fast GC applications. Columns of i.d. 0.25–0.32 mm can be considered most suitable for the GC analysis of CWC-related compounds.

Thin stationary phase film columns (below 0.15 μm) should be avoided because they tend to give poor peak shapes for strongly adsorptive compounds such as thioglycol, dialkyl methylphosphonates, and aminoalcohols, especially at low concentration levels. Thick film columns (above 2 μm) are ideal for the separation of very volatile compounds such as hydrogen cyanide, phosgene, and cyanogen chloride. However, they are unsuitable for higher molecular weight compounds such as BZ or VX. Standard (0.25–0.33 μm) film thickness offers the best compromise between resolution and sample capacity.

The selection of the length of the column depends on the required resolution and analysis time. Short columns (10–15 m) are useful for fast GC and for screening purposes.

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Intermediate column lengths of 25–30 m, providing sufficient separation power and reasonable analysis time, are recommended.

4.4 Detectors

Critical properties of detectors are sensitivity, selectivity, linearity of response, reproducibility and reliability of operation. Spectrometric detectors, MS and FTIR, are described in more detail in other sections.

Precise recommendations for the choice of a detector cannot be given, because it depends much on the requirements of the analytical task like the use of RI values for identification and the need for sample screening. Despite the great separation power of capillary columns, the analysis of trace compounds in some complex matrices may be difficult if only universal detectors are used. Many selective detectors are available for monitoring CWC-related compounds containing heteroatoms. Chromatograms are simplified with these detectors and the sensitivity of analysis is increased. The following detectors are useful for CWC-related compounds:

Flame ionisation detector (FID) is the most popular universal detector, having both a wide linear range and great reliability. The detection limits of CWC-related compounds are from 0.1 ng to 1 ng depending on its structure. Sensitivity can be poor for some chemicals such as hydrogen cyanide. In addition, phosgene and chloropicrin because of their thermal decomposition cannot be detected as such with the FID.

Phosphorus-selective detectors are of greatest importance for verification analysis. **The nitrogen-phosphorus detector (NPD)** is very sensitive and selective towards compounds containing phosphorus and/or nitrogen, reaching detection limits of 10–50 pg for phosphorus and 50–100 pg for nitrogen. **Flame photometric detector (FPD)** is the most widely used detector for sulphur-containing compounds. Besides sulphur, it is selective for phosphorus, which makes it a suitable for both nerve agents and sulphur-containing mustards. Detection limits are ca. 100–500 pg in phosphorus mode and ca. 500–1000 pg in sulphur mode. Two-channel FPD detectors operating simultaneously in S- and P-mode are commercially available.

Some CWC-related compounds containing chlorine atoms, e.g. vesicants, are detectable with **the electron capture detector (ECD)**. However, the sensitivity of ECD for some compounds is rather low depending on the electron affinity of the compounds. The ECD can be considered as moderately sensitive but not selective enough, especially for environmental and biological samples, many of which contain irrelevant ECD-active compounds, interfering the detection of the CWC-related compounds.

FID, NPD and FPD are the most reliable, easy to set up and most commonly used non-spectrometric detectors for the detection of CWC-related chemicals. By using two detectors in parallel, it is possible to increase the information obtained from a single GC run. For screening it is recommended that the same stationary phase is used with selective detectors and spectrometers.

One of most attractive detectors for screening purposes is **the atomic emission detector (AED)**, which is particularly well-suited in CWC analysis, where extensive screening for a large variety of chemicals containing various heteroatoms (P, S, N, As, F, Cl) is often required, and where other GC detectors may be insufficient.

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Large variations in detection limits are observed between elements when using AED. Better sensitivity is reached for carbon, phosphorus, sulphur, arsenic and silicon. Sensitivity for nitrogen and chlorine is moderate and poor for oxygen and fluorine. Sensitivity for Sarin is ca. 1 ng detected with carbon line at 193 nm, 2 ng with phosphorus line at 186 nm and only 50 ng with fluorine line at 690 nm. Sensitivity for mustard gas is ca. 2 ng detected with sulphur line at 181 nm and chlorine line at 479 nm.

With AED, it is also possible to calculate an approximate total or partial empirical formula for unknown compounds.

Large variety of gases (nitrogen, hydrogen, oxygen, mixed nitrogen/methane 90/10 v/v) can be used as reagent gases to improve sensitivity and very high purity helium is required as carrier and plasma gas. Maintenance operations such as discharge tube replacement have to be made periodically in order to maintain a good AED performance.

In addition to GC-MS, GC-AED offers the most powerful combination of techniques for obtaining data for unambiguous qualitative characterisation of CWC-related chemicals.

4.5 Reagents and chemicals

The reagents and chemicals needed for GC analysis are given in Table 1.

Table 1. Chemicals and reagents needed for GC.

Chemical	Concentration	Purity
QC test sample in dichloromethane		
Trimethylphosphate (CAS 512-56-1)	10 ng/μl	p.a. grade or better
2,6-Dimethylphenol (CAS 576-26-1)	10 ng/μl	p.a. grade or better
5-Chloro-2-methylaniline (CAS 95-79-4)	10 ng/μl	p.a. grade or better
Tri- <i>n</i> -butylphosphate (CAS 126-73-8)	10 ng/μl	p.a. grade or better
Dibenzothiophene (CAS 135-65-0)	10 ng/μl	p.a. grade or better
Malathion (CAS 121-75-5)	10 ng/μl	p.a. grade or better
Methyl stearate (CAS 112-61-8)	10 ng/μl	p.a. grade or better
<i>n</i> -Alkanes (C ₈ –C ₂₄ , even members)	10 ng/μl	p.a. grade or better
Carrier gas		
Helium		≥99.995%
Detector gases		
According to the demands of the manufacturer of detectors		
Solvents		
Dichloromethane (for cleaning of syringes)		p.a. grade or better
Ethyl acetate (for cleaning of syringes)		p.a. grade or better

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5. Experimental

5.1 Recommended GC conditions

Recommended GC conditions are given in Table 2. The given conditions correspond to the conditions used for measuring RI-values for the OPCW Central Analytical Database (OCAD).

Table 2. Recommended GC conditions.

GC parameter	Recommended conditions	Notes
Septum purge flow	2–4 ml/min	
Injection mode	Splitless	
Injection volume	1 µl	Large volume injector liner
Splitless time	1 min	Same as the isothermal time of the starting temperature in the oven temperature program
Split ratio after splitless time	≥ 1:50	Recommended to be reduced to 1:20 after 2 min to save carrier gas if gas saver function is available in the GC
Injector temperature	250–280 °C	
Detector temperature	280–300 °C	At least as high (if possible) as the end temperature of the oven temperature program
Detector gas flows		According to the demands of the manufacturer of detector
Helium linear flow rate	30 cm/s at 40 °C	Optimum column efficiency
Carrier gas flow mode	Constant pressure	Recommended if RIs are used for identification
Temperature program for GC oven	40 °C (1 min) – 10 °C/min – 280–300 °C for 5–10 min	

Run conditions may be modified depending on the type of samples or used injector, column and detector. For example concentrated samples (>100 ng/µl) should be injected in split injection mode with split ratio 1:20 or higher to avoid column overloading and to achieve better separation of low boiling chemicals (**Note! OCAD RI-values are not valid with split-injection**). For splitless injections, concentrated samples must be diluted.

For standard GC analysis, it is recommended to use a 25–30 m long SE-54-type fused silica capillary column (0.25 mm i.d. and film thickness 0.25 µm), preferably so called low bleed MS column. If the column is split between two detectors the use of three-way press-fit connector, a Y-splitter, is recommended.

6. Quality control

The reliable identification of CWC-related compounds requires that the data produced by the instrument is correct. Therefore, it is essential to run quality control tests on regular basis to confirm correct and stable instrument performance during the sample analysis and to check any possible contamination.

6.1 Instrument performance test

6.1.1 Column test

The goal of this test is to control the performance of the column (See Table 2). This test can also be used for estimation the sensitivity of the instrument. The QC test must be run at the beginning and at the end of each sample batch and, in the case of a long sample batch, at least after every fifteenth sample.

The peak shape of trimethylphosphate can be used to estimate the column activity. The acidic and basic properties of the column can be estimated by comparing the peak heights and/or peak areas of phenol and aniline peaks. Poor peak shape or reducing peak height of 5-Chloro-2-methylaniline indicates an acidic nature of the column that is not good for basic compounds. Poor peak shape or reducing peak height of 2,4-dimethylphenol indicates a basic nature of the column and the column is not good for acidic compounds.

The column is satisfactory if the following criteria of the QC-test chemicals are met:

- Absolute retention times of detected compounds are within ± 15 s retention index window compared to the QC-test performed after the column installation.
- The trimethylphosphate peak asymmetry measured at half height of peak is below 4.
- The peak height ratio of 2,6-dimethylphenol / 5-chloro-2-methylaniline is between 0.5 and 2
- Signal-to-noise (S/N) ratio for each test chemical is $>10:1$ in the chromatogram.

Figure 1 shows typical chromatograms of a QC test sample indicating good instrument performance obtained from the FID, NPD and FPD(S).

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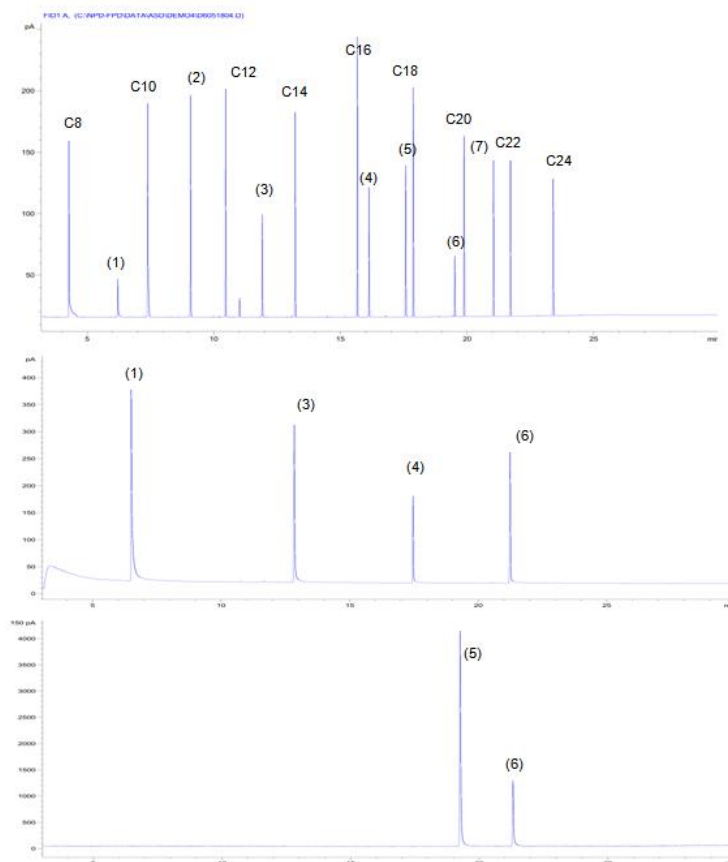


Figure 1. Analysis of the OPCW QC test sample (10 ppm) obtained from top) FID, middle) NPD, and bottom) FPD(S). Column: 30 m x 0.25 mm I.D. x 0.25 μ m SE-54 (95% dimethyl-5% diphenyl polysiloxane); Carrier gas: Helium, constant pressure, gas flow 30 cm/s; splitless injection with splitless time of 1 min; oven temperature program: 40 °C (1 min) – 10 °C/min – 280 °C (5 min). Peak labels: (1) trimethylphosphate, (2) 2,6-dimethylphenol, (3) 5-chloro-2-methylaniline, (4) tri-*n*-butylphosphate, (5) dibenzothiophene, (6) malathion and (7) methyl stearate.

6.1.2 Sensitivity test

If QC test solution is not used for estimation of the sensitivity of the instrument or better sensitivity evaluation is needed, the use of specific tests (Table 3) for controlling sensitivity of FID, FPD, NPD, ECD and AED is recommended.

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Table 3. Tests for sensitivity control.

Detector	Mode	Compound	Concentration
FID	Universal	<i>n</i> -pentadecane	1 ng/μl
FPD	Sulphur	dibenzothiophene	1 ng/μl
	Phosphorus	malathion	0.5 ng/μl
NPD	Nitrogen / phosphorus	malathion	0.5 ng/μl
ECD	Halogen	lindane	1 ng/μl
AED	Phosphorus (178 or 186 nm)	triethylphosphate	2 ng/μl
	Sulphur (181 nm)	dibenzothiophene	1 ng/μl
	Chlorine (479 nm)	lindane	5 ng /μl
	Nitrogen (174 nm)	azobenzene	5 ng /μl
	Arsenic (189 nm)	triphenylarsine	2 ng/μl
	Carbon (193 nm)	<i>n</i> -pentadecane	0.5 ng/μl
	Silicon (252 nm)	tetraethylorthosilicate	0.5 ng/μl

Solvent for test chemicals can be hexane or ethyl acetate. Chlorinated or nitrogen containing solvent (dichloromethane, chloroform, acetonitrile) must be avoided because of their incompatibility with some detectors (ECD, NPD)

The criteria for acceptable detector sensitivity are met if each test compound gives a peak having a signal-to-noise ratio of 10.

6.2 Blank tests

A system blank (blind test) is used to check any possible contamination coming from the sample preparation or from the instrument. This test is run at least in the beginning and at the end of each sample series and sometimes before every sample.

If blank test shows contamination, a solvent blank run with the same solvent used in sample preparation must be carried out to eliminate the possibility of contamination of the equipment (e.g. contamination from the syringe or the memory effect of the equipment). **Note!** *This is extremely important for silylated samples. The memory effect is possible, for example, in the analysis of alkyl phosphonic acids. If the derivatisation of these chemicals has not been complete, the non-derivatised acids are adsorbed on the injector liner. A silylation reagent may in such case react in situ with the adsorbed acids in the liner yielding false positive results.*

If CWC-related chemicals are found in any blank chromatogram, the blank tests are repeated after elimination of the possible contamination source. All contamination coming from glassware, solvents, standard solutions (or authentic reference solutions), syringe, injector system, or columns must be eliminated before new sample analysis is started.

7. Analysis of samples

7.1. Sequence of analysis

In the beginning of an analysis batch, a QC test and a system blank test are run. When all tests have been accepted, samples can be analysed. Generally, 1.0 μl of the sample is injected on each run.

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7.2 Screening

The use of selective detectors for screening of samples is presented in Section 1, Part B, Chapter I and the use of extracted ion chromatograms of selective ions for screening with GC–EI/MS are presented in Section 3, Part B, Chapter II.

7.3 Identification

The peak of an identified chemical and the peak of a corresponding reference chemical must have a signal-to-noise ratio of at least 5:1. For non-spectrometric GC techniques, if there are peaks in the chromatogram within the retention time window of ± 0.1 min of the peak of the identified chemical, standard addition of the reference chemical should be used to confirm the identity of the identified peak in the chromatogram. Add ca. 50% of the estimated original amount of the chemical of interest into the sample to see a clear increase of the peak of interest.

7.3.1. Absolute retention times

Absolute retention times are the most useful identification parameters if reference chemicals are available. A pure standard of the identified chemical is run as a reference as soon as possible after the sample run. The identification is accepted if the absolute retention time of the GC peak of interest in the sample falls within a window ± 0.1 min of the retention time of reference chemical.

7.3.2. Retention indices

The use of RI-values for identification is particularly advantageous when reference chemicals are not available. The ease of use of RIs is increased significantly if a computer program for calculation of RI values is available. It is recommended to calculate RI values with external standard method using the *n*-alkanes in the QC test run as retention index standards with AMDIS software*. However, AMDIS operates only with GC–MS instruments. For FID, the RI values must be calculated manually according to the following equation.

$$RI = 100C_n + 100(C_{n+i} - C_n) \cdot \frac{t_{R(x)} - t_{R(n)}}{t_{R(n+i)} - t_{R(n)}}$$

where :

C_n and C_{n+i} = carbon numbers of the C-standards eluting before and after the compound to be identified

$t_{R(x)}$ = retention time of the compound to be identified

$t_{R(n)}$ and $t_{R(n+i)}$ = retention times of C_n and C_{n+i} alkanes

* Available at National Institute of Standards and Technology (NIST) website (<http://chemdata.nist.gov>)

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C-series, the sample and solvent blank must be run by the same physical instrument and under identical chromatographic conditions. The identification is accepted if the calculated RI value falls within a window of ± 20 RI units of the RI value contained in the OCAD library or in the Validation Group Working Database (VGWD) library.

8. Evaluation of method

8.1 Area of applicability

GC method can be applied to all chemical warfare agents, their degradation products, precursors or derivatives, except saxitoxin and ricin. Extremely volatile (e.g. DF), non-volatile (e.g. phosphonic acids), and reactive (e.g. lewisite 1) CWC-related chemicals can be analysed after derivatisation. The method is suitable for analysing and identifying any chemical at the 0.1 mg/kg concentration level with FID and at the 0.01 mg/kg concentration level with selective detectors.

8.2 Advantages/Disadvantages

Advantages of the method are good separation of chemicals, sensitivity, and possibilities for selective detection. Modern instruments are easy to use and AMDIS software provides automated RI analysis. GC screening with selective detectors before GC-MS identification is important in producing useful information from complex mixtures.

The disadvantage of using GC alone is that identification is based on retention data giving only tentative identification and for final identification a spectrometric method is required as a second technique.

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Chapter II.

Gas chromatography–electron ionisation/mass spectrometry (GC–EI/MS)

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1. Scope

This chapter describes the use of gas chromatography–electron ionisation mass spectrometry (GC–EI/MS) in screening and identifying chemical warfare agents, their precursors, degradation products or derivatives, i.e., chemicals related to the Chemical Weapons Convention (CWC). Generally, this procedure is used to identify chemicals above the 1 mg/kg concentration level. However, the procedure can also be used to identify relevant chemicals down to low µg/kg concentration levels.

2. Summary of the method

GC–EI/MS measurements are carried out with modern GC–EI/MS instruments operating in full scanning mode. Samples are analysed with a GC–MS instrument equipped with a 25–30 m capillary column with SE-54-type stationary phase material. Before each sample batch, a quality control run (QC test, see below) is carried out to verify the correct performance of the instrument, and a blank run to verify the absence of CWC-related chemicals. Screening of relevant chemicals is carried out with the Automated Mass Spectral Deconvolution and Identification System (AMDIS) with an appropriate target library or by selecting characteristic ions of chemical families or individual chemicals (extracted ion chromatograms, EICs) from the total ion chromatograms (TICs). Relevant chemicals are identified by comparing the measured EI spectra to those in relevant libraries or to spectra of authentic reference chemicals or interpretation of spectra.

If needed, linear retention indexes are measured against *n*-alkane series C₈–C₂₄ [included in the QC test; see the Section 3, Part B, Chapter I, “Gas chromatography (GC)”].

3. Safety aspects

GC–EI/MS analysis does not require any specific precautions concerning safety. Samples to be analysed normally contain only small concentrations of toxic chemicals, and electron ionisation will destroy the sample molecules. Normal laboratory safety measures such as

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wearing protective gloves and goggles while handling samples in manual injections should be used.

Hot parts of the instrument, e.g., the hot septum nut (>250 °C), must be taken into account while operating and maintaining the instrument.

4. Instrumentation

4.1 GC–EI/MS instrument

Any modern GC–EI/MS instrument capable of a scanning speed of at least 1 scan/s and covering a mass range of at least m/z 40–500 can be used.

4.2 GC columns

A 25- to 30-m-long SE-54-type* fused silica capillary column (0.20–0.25 mm i.d. and film thickness 0.25 µm), preferably the low-bleed MS column, is used.

4.3 Reagents and chemicals

The reagents and chemicals needed for GC–EI/MS analysis are given in Table 1.

Table 1. Chemicals and reagents needed for GC–EI/MS.

Chemical	Concentration	Purity
QC test sample in dichloromethane		
Trimethylphosphate (CAS 512-56-1)	10 ng/µl	p.a. grade or better
2,4-Dimethylphenol (CAS 576-26-1)	10 ng/µl	p.a. grade or better
5-Chloro-2-methylaniline (CAS 95-79-4)	10 ng/µl	p.a. grade or better
Tri- <i>n</i> -butylphosphate (CAS 126-73-8)	10 ng/µl	p.a. grade or better
Dibenzothiophene (CAS 135-65-0)	10 ng/µl	p.a. grade or better
Malathion (CAS 121-75-5)	10 ng/µl	p.a. grade or better
Methyl stearate (CAS 112-61-8)	10 ng/µl	p.a. grade or better
<i>n</i> -Alkanes (C ₈ –C ₂₄ , even members)	10 ng/µl	p.a. grade or better
Carrier gas		
Helium		≥99.995%
Solvents		
Dichloromethane (for cleaning syringes)		p.a. grade or better
Ethyl acetate (for cleaning syringes)		p.a. grade or better

* SE-54-type column means a stationary phase consisting of 95%-methyl-5%-phenylpolysiloxane. Polyarylene-type columns that are virtually equivalent to the SE-54-type columns can also be used.

5. Experimental

5.1 Recommended GC conditions

The recommended GC conditions are given in Table 2. The given conditions correspond to the conditions used for measuring retention indexes (RIs) in the OCAD RI library:

Table 2. Recommended GC conditions.

GC parameter	Recommended conditions	Notes
Septum purge flow	2–4 ml/min	
Injection mode	Splitless	
Injection volume	1 µl	Large volume injector liner
Splitless time	1 min	Same as the isothermal time of the starting temperature in the oven temperature program
Split ratio after splitless time	≥ 1:50	Recommended to be reduced to 1:20 after 2 min to save carrier gas if the gas saver function is available in the GC
Injector temperature	250–280 °C	For PTV injector temperatures, see the corresponding instrument manual
GC–MS interface temperature	280–290 °C	At least as high as the end temperature of the oven temperature program
Helium linear flow rate	30 cm/s	Optimum column efficiency
Carrier gas flow mode	Constant pressure	Needed if RIs are measured
Temperature program for GC oven	40 °C (1 min) – 10 °C /min – 280 °C for 5–10 min	

Run conditions may be modified depending on the sample type. For example, concentrated samples (>100 ng/µl) should be injected in the split injection mode with a split ratio of 1:20 or higher to avoid column overloading and to achieve better separation of low-boiling chemicals. For splitless injections, concentrated samples must be diluted.

Note! OCAD RI values are not valid with split-injection.

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5.2 Recommended MS conditions

The recommended MS conditions are given in Table 3. The given conditions correspond to the requirements for the MS conditions used for recording MS spectra for the OCAD MS library:

Table 3. Recommended MS conditions in GC–EI/MS analysis.

MS parameter	Conditions	Notes
Solvent delay time	3 min (>3 min for silylated samples)	Depends on the sample solvent
Mass range (m/z)	From ≤ 40 to ≥ 500	
Scan rate	≥ 1 scan/s	
Electron energy	70 eV	
Ion source temperature	Refer to instrument manual	
Electron multiplier voltage	Refer to instrument manual	Adjusted often during instrument tuning

5.3 Recommended tuning and mass calibration of the MS

Use the tuning method recommended for the instrument. An automatic tuning and calibration procedure should be used, if available. Tuning and calibration should be performed once a week or more often if indicated by the QC test. Printouts of tuning and calibration reports should be archived for further reference.

6. Quality control

6.1 Instrument performance test

Reliable identification of CWC-related compounds requires that the data produced by the instrument is correct. Therefore, quality control tests must be run continuously to confirm the instrument is performing correctly and stably while analysing the samples.

The performance of a GC–EI/MS instrument is controlled by carrying out a QC test run, in which the mass spectrometric behavior of the test compounds and the adsorption behavior of the column are confirmed to be acceptable. The QC test must be run at the beginning and the end of each sample batch and in the case of a long sample batch at least after every fifteenth sample.

The instrument is acceptable for use if the MS and GC data produced from each test compound meet the following requirements:

1. The signal-to-noise (S/N) ratio for each test chemical $>10:1$ in the total ion chromatogram (TIC). The evaluation is done visually or automatically if available (the sensitivity of the GC–EI/MS equipment can be estimated with the QC test sample containing each test chemical at the 10 ng/ μ l concentration level).
2. The m/z values of the major ions of each test chemical are correct and no extra ions above 5% are present (the mass range to be printed must extend 10 m/z beyond the molecular mass, so that the whole relevant mass spectrum is registered).

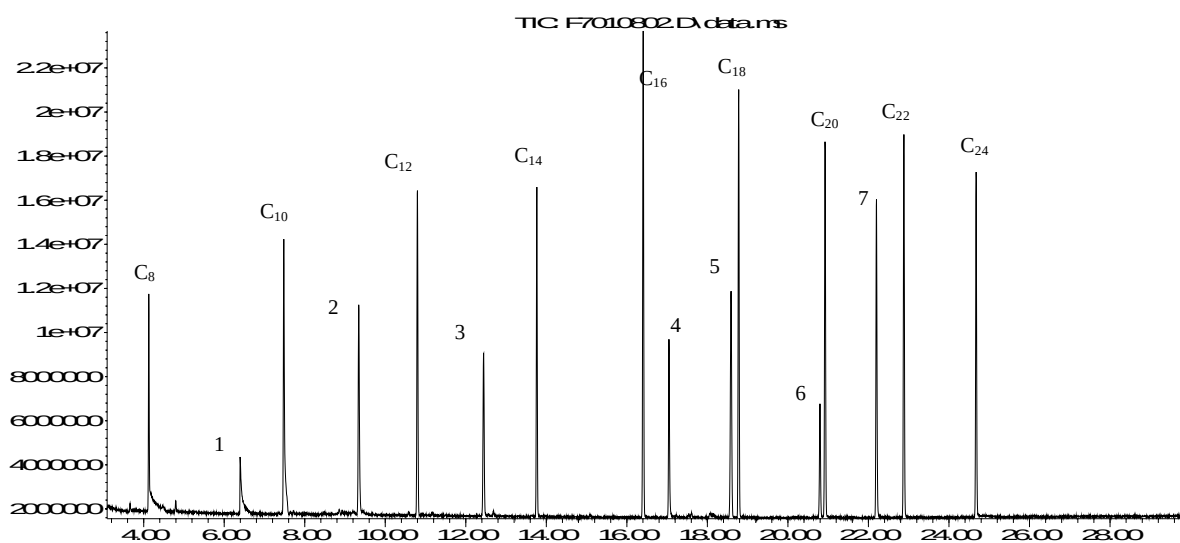
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- The intensity ratios of the isotopic ions are within the following limits (the values are expressed as percentages of the less abundant isotopic ions relative to the more abundant ones $\pm$ the acceptable error limits):
 - m/z 143/141 ($^{37}\text{Cl}/^{35}\text{Cl}$) for 5-chloro-2-methylaniline is $33\pm 3.0\%$
 - m/z 186/184 ($^{34}\text{S}/^{32}\text{S}$) for dibenzothiophene is $5.9\pm 1.0\%$.
- The peak height of trimethylphosphate is at least 20% of that of dibenzothiophene. The evaluation is done visually using the TIC.

Abundance



Time-->

Figure 1. Analysis of the OPCW QC test solution (10 ppm) showing good instrument performance on a 30 m x 0.25 mm i.d. x 0.25 μm SE-54 (95% dimethyl-5% diphenyl polysiloxane) column; Carrier gas: Helium, constant pressure, gas flow 30 cm/s; splitless injection with splitless time of 1 min; oven temperature program: 40 $^{\circ}\text{C}$ (1 min) – 10 $^{\circ}\text{C}/\text{min}$ – 280 $^{\circ}\text{C}$ (5 min). Peak labels (1) trimethylphosphate, (2) 2,6-dimethylphenol, (3) 5-chloro-2-methylaniline, (4) tri-n-butylphosphate, (5) dibenzothiophene, (6) malathion and (7) methyl stearate.

6.2 Blank tests

A system blank run (blind test) using the same solvents/reagents used in sample preparation is highly recommended to check any possible contamination coming from the sample preparation or the instrument.

Note! This is extremely important for silylated samples as injecting silylation reagent-containing samples can produce in-situ silylation in the injector. Additionally, the system blank should contain the same silylation reagent as the sample to prove the absence of instrument contamination.

If CWC-related chemicals are found in any system blank chromatogram, the blind tests are repeated after the possible contamination source has been identified and eliminated. All contamination from glassware, solvents, standard solutions (or authentic reference solutions), syringes, the injector system, or columns must be eliminated before sample analysis is started.

7. Analysis of samples

7.1. Sequence of analysis

In the beginning of an analysis batch, a QC test and a blank test are run. When all tests are acceptable, the samples can be analysed. Blank tests in between sample series are recommended. Generally, 1.0 µl of the sample is injected. If retention indices are used, they can be calculated with an external method using the *n*-alkanes in the QC test run as retention index standards. AMDIS software[†] is recommended for calculating the retention index (refer to the GC ROP for detailed instructions). According to the current OPCW rules, a blank run is required before each sample analysis!

7.2 Screening

If phosphorus, nitrogen and sulphur selective detectors are used to screen the samples, extracted ion chromatograms (EICs) can be used to locate those chemicals, which are not detectable with these selective detectors. A list of these chemicals and the specific ions for screening are given in Table 4. Appendix 1 contains the specific ions of the most important groups of CWC-related chemicals.

In addition, AMDIS screening should be used for all samples analysed with GC–EI/MS.

The normal screening procedures for CWC-related samples is discussed in the Section 1, Part B, Chapter I, “Screening”.

Table 4. Specific ions for chemicals not detected with phosphorus, nitrogen and sulphur selective detectors.

Chemical	Specific ions for screening (<i>m/z</i>)
PFIB (Note! head space or splitless injection)	93, 181, 200
Chloropicrin	82, 117, 119, 121
Pinacolyl alcohol	45, 57, 87, 102
Methyl derivative of benzilic acid	77, 105, 183
TMS derivative of benzilic acid	255, 329
TMS derivative of BZ	110, 255
DMT derivatives of Lewisite I and Lewisite II	229, 290
DMT derivative of Arsenic trichloride	107, 121, 229
1,3-butylthiol derivative of Lewisite I	164, 204, 314
1,3-butylthiol derivative of Lewisite II	203, 229, 286

Note! PFIB, HCN and phosgene analysed without derivatisation may be screened using the head space or split injection technique.

[†] Available at the National Institute of Standards and Technology (NIST) website (chemdata.nist.gov)

7.3 Identification

The final identification must always be approved by an experienced analyst.

Identification can be based on comparison with an authentic reference chemical or by comparing the recorded spectrum with the reference spectrum in the OPCW Central Analytical Database (OCAD) (when identifying isomers having similar EI/MS spectra, the retention index (RI) may be used to distinguish which isomer is in question). Commercial databases may also be used for identification (e.g., NIST, Wiley). If a reference chemical or reference spectra are not available, the identification can be done with spectral interpretation.

The recorded spectrum (reference and unknown) must be at least ten mass units above the molecular weight of the proposed molecule. No ions having mass-to-charge the ratio above the molecular weight (excluding the isotope ions) must be present. Ions having intensities above 10% in relative scale and not belonging to the reference spectrum must not be present. The signal-to-noise ratio of a recorded TIC peak must be greater than five (visual inspection). If identification is based on spectral interpretation, a spectrum of a closely related chemical together with a spectral interpretation and a fragmentation pathway must be provided.

***Note!** If extra peaks are present in the spectrum, use extracted ion chromatograms by selecting the impurity ion together with two ions, which belong to the spectrum of an authentic reference chemical. If the ion profiles are different, try to extract as much pure spectrum as possible of the chemical of interest. If the target chemical elutes exactly together with a background chemical, a more selective column, such as OV-1701 or Carbowax, should be used to separate the chemicals.*

8. Method evaluation

8.1 Area of applicability

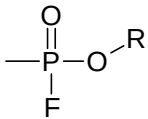
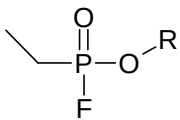
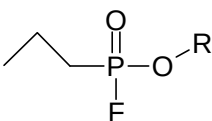
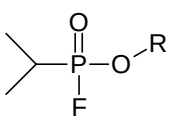
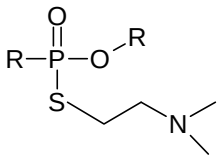
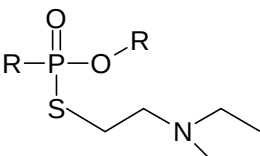
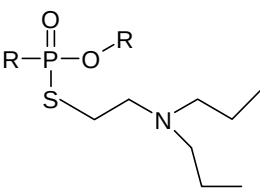
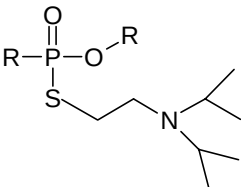
The GC–EI/MS method can be applied to all chemical warfare agents, their degradation products, precursors or derivatives, except saxitoxin and ricin. Extremely volatile (e.g., alkyl phosphonyldifluorides), nonvolatile (e.g., phosphonic acids) and reactive (e.g., Lewisite I) CWC-related chemicals can be analysed after being derived to more volatile derivatives. The method is suitable for analysing and identifying any chemical at the 1 mg/kg level in full scan mode and at the 0.01 mg/kg level in selective ion monitoring (SIM mode, target analysis only).

8.2 Advantages/Disadvantages

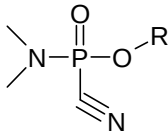
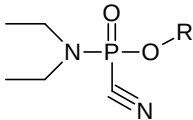
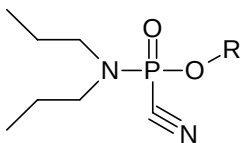
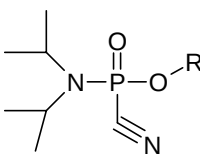
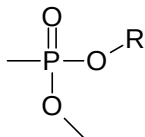
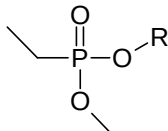
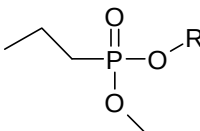
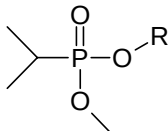
The advantages of the method are reliability and sensitivity. Modern instruments are easy to use, and AMDIS software provides automated data analysis and identification. Thousands of chemicals using commercial libraries or OCAD can be identified within a very short period.

The disadvantage in identification using GC–EI/MS is the similarity of the isomers' mass spectra. This problem may be solved by simultaneously using the retention index data of possible isomers found from the OCAD.

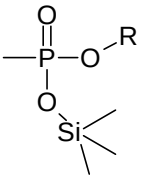
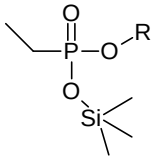
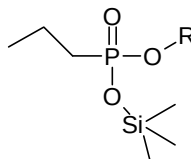
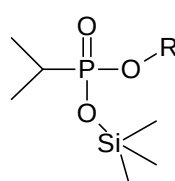
RECOMMENDED OPERATING PROCEDURES FOR CWC-RELATED ANALYSIS**Section 3.** Analytical methods**Chapter II.** Gas chromatography–electron ionisation/mass spectrometry (GC–EI/MS)**Appendix 1.** Ion list of chemical warfare agents, their precursors or analytical derivatives for GC–EI/MS.

Structure	Name*	Ion (abundance)
<i>Sarin-type chemicals</i>		
	Alkyl methylphosphonofluoridates	99 (100%) 81 (5–30%)
	Alkyl ethylphosphonofluoridates	113 (100%)
	Alkyl propylphosphonofluoridates	127 (100%)
	Alkyl isopropylphosphonofluoridates	127 (100%)
<i>VX-type chemicals</i>		
	O-Alkyl S-2-dimethylaminoethyl alkylmethylphosphonothiolates	58 (100%) 71 (30–90%)
	O-Alkyl S-2-diethylaminoethyl alkylmethylphosphonothiolates	86 (100%) 99 (20–50%)
	O-Alkyl S-2-dipropylaminoethyl alkylmethylphosphonothiolates	114 (100%) 127 (10–30%)
	O-Alkyl S-2-diisopropylaminoethyl alkylmethylphosphonothiolates	114 (100%) 127 (10–30%)

RECOMMENDED OPERATING PROCEDURES FOR CWC-RELATED ANALYSIS**Section 3.** Analytical methods**Chapter II.** Gas chromatography–electron ionisation/mass spectrometry (GC–EI/MS)

Structure	Name*	Ion (abundance)
<i>Tabun-type chemicals</i>		
	Alkyl N,N-dimethylphosphoramido-cyanidates	135 (100%) 70 (20–100%) 108 (10–40%)
	Alkyl N,N-diethylphosphoramido-cyanidates	147 (100%) 163 (10–80%)
	Alkyl N,N-dipropylphosphoramido-cyanidates	161 (100%) 191 (10–80%)
	Alkyl N,N-diisopropylphosphoramido-cyanidates	133 (100%) 175 (10–90%)
<i>Alkylphosphonates</i>		
	Alkyl methyl methylphosphonates	111 (100%) 93 (30–80%) 137 (2–50%)
	Alkyl methyl ethylphosphonates	125 (100%) 107 (20–70%) 79 (10–50%)
	Alkyl methyl propylphosphonates	139 (50–100%) 121
	Alkyl methyl isopropylphosphonates	139 (100%)

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Structure	Name*	Ion (abundance)
<i>TMS derivatives of alkylphosphonates</i>		
	Alkyl trimethylsilyl methylphosphonates	153 (100%) 169 (10–80%)
	Alkyl trimethylsilyl ethylphosphonates	167 (100%) 183 (10–80%)
	Alkyl trimethylsilyl propylphosphonates	181 (100%) 197 (10–80%)
	Alkyl trimethylsilyl isopropylphosphonates	181 (100%) 197 (10–80%)

* Alkyl ≠ methyl

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Chapter II. Gas chromatography–electron ionisation/mass spectrometry (GC–EI/MS)

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Chapter III.

Gas chromatography–chemical ionisation/mass spectrometry (GC–CI/MS)

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1. Scope

This chapter deals with the application of gas chromatography–chemical ionisation/mass spectrometry (GC–CI/MS) to determine molecular weight and identify relevant chemicals, hereafter chemical warfare agents, their precursors and degradation products regarding the schedules of the Convention for the Prohibition of Chemical Weapons. Chemical ionisation is essential for confirming the structures of relevant chemicals after performing experiments with gas chromatography–electron ionisation/mass spectrometry (GC–EI/MS). Generally, this technique is applied to identify chemicals above the 1 mg/kg concentration level. However, the technique can also be used to identify relevant chemicals below 1 mg/kg concentration levels.

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Chapter III. Gas chromatography–chemical ionization/mass spectrometry (GC–CI/MS)

2. Summary of the method

GC–CI/MS experiments are carried out with modern GC–MS instruments used in chemical ionisation. The principle of the technology is based on a quadrupole or ion trap mass analyser operating in full scanning mode. Samples are introduced in the GC typically equipped with a 25- to 30-m capillary column, generally coated with the same chemical phase used in previous experiments run in GC–EI/MS.

A quality control run (QC test sample) is performed to check the GC–MS performance and to ensure the reliability of the analysis in each sample batch.

Every relevant chemical, previously detected in GC–EI/MS, is examined in GC–CI/MS. Relevant chemicals are identified by comparing them with authentic reference chemicals when available or by interpreting the spectra.

Determining the molecular weight is of key importance to elucidate the structure. Methane, ammonia and isobutane are widely used as reagent gases to confirm the molecular weight of CWC-related chemicals in GC–CI/MS experiments.

Methane (proton affinity (PA) ca. 135 kcal/mol) is the most common reagent gas for CWC scheduled compounds.^{1–3} Even if the $[M+H]^+$ quasi-molecular ion is not systematically observed, positive chemical ionisation (PCI) with methane allows information about the structure of relevant chemicals to be obtained that complements the information obtained in electron ionisation.

For instance, because of the relatively huge difference in the proton affinity (PA) between methane and dialkyl alkylphosphonates (ca. 65 kcal/mol), many fragmentations can be observed in their CI mass spectra. These fragmentations can differ from those observed in the EI mass spectra. Indeed, odd-electron species (i.e., M^+) can dissociate following different pathways than those occurring from the even-electron ions (i.e., $[M+H]^+$).

Extensive fragmentation, the non-systematic presence of the $[M+H]^+$ ion and the two adduct ions $[M+C_2H_5]^+$ and $[M+C_3H_5]^+$ are characteristic of positive chemical ionisation using methane as a reagent gas.

Because of methane's low proton affinity, this reagent gas can protonate other organic compounds (backgrounds) together with relevant chemicals present at low concentrations in complex mixtures. It can be difficult to extract a good-quality mass spectrum for the target chemicals.

Isobutane (PA ca. 192 kcal/mol) is a reagent gas softer than methane. Isobutane is more selective and causes less fragmentation.

The simple fragmentation pattern, large abundance of the $[M+H]^+$ ion and the presence often observed of $[M+C_3H_3]^+$, $[M+C_3H_5]^+$ and $[M+C_4H_9]^+$ are characteristic of positive chemical ionisation using isobutane⁴ as a reagent gas.

Ammonia (PA ca. 204 kcal/mol) is a more specific reagent gas due to its high proton affinity. Ammonia is recommended for confirming the molecular weight of organophosphorus compounds (OPC)^{5–8} and for chemicals containing sulfur and nitrogen atoms with a proton affinity higher than 204 kcal/mol.

If the difference of $(PA_{OPC} - PA_{NH_3})$ is $\leq \pm 10$ kcal/mol, the $[M+NH_4]^+$ ion is stable enough to be detected.

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Negative chemical ionisation^{9,10} is particularly recommended for analysing halogenated compounds. However, α deprotonation of organophosphorus compounds is possible and can yield $\text{RH}\ddot{\text{C}}\text{--P(=O)}(\text{OR})_2$ as well as high acid reagents (e.g., $\text{HCl} \rightarrow \text{Cl}^-$).

Elimination can lead to the formation of $\text{RCH}_2\text{--P(=O)}(\text{OR}')\text{O}^-$ instead of deprotonation that could yield $[\text{M-H}]^-$. The difference in behavior allows complementary dissociative processes to be obtained, and thus, specific ions can be obtained.¹¹

Chemical ionisation (CI) with liquid reagents

Liquid CI reagents^{12,13} offer several important benefits over pressurised gases. These benefits include safety, convenience, cost and ease of use.

Liquid CI uses the room temperature vapor pressure of the liquid as the reagent gas. This is possible only with an internal source ion trap and not an external source instrument be it an ion trap or a quadrupole. Liquid CI is highly selective because of the wide variety of common reagents available in a typical laboratory. The different types of liquids used give specific adduct ions particular to the reagent and the compound class of the analyte.

There is a list of the typical CI reagents (methanol, acetonitrile, acetone, including deuterated forms). Acetonitrile has become the most common reagent because it reacts extremely well with hydrocarbon¹⁴ and any compound containing nitrogen, phosphorus and/or sulphur.

Acetonitrile's (ACN) molecular weight is 41, and the CI reagent is the $\text{M}+1$, which is mass 42. The adduct ion is $\text{M}+13$, which is an attachment of CH .

3. Safety aspects

Analysing chemical warfare agents requires specific safety precautions be adopted. For the bulk of the molecules arriving in the ion source of the mass spectrometer, only a fraction is ionised and thus destroyed. The majority of the samples go through the instrumentation directly to the primary pump. Although quantities injected into the GC–MS are at low concentration levels, after several experiments, precautions have to be taken when the oil of the primary pump has to be changed. That is why the operations mentioned above should be carried out with masks and butyl gloves under a suction and filtering hood. In addition, some parts of the GC–MS are hot (e.g., the GC's injector and the GC–MS's transfer line). This is why precautions should be taken to avoid damage or injury during maintenance.

4. Instrumentation

4.1 GC columns

Generally, the 25- to 30-m capillary column used in GC–CI/MS is coated with the same chemical phase used in previous experiments by GC–EI/MS.

Note! According to the Proficiency Test criteria, the same phase column must be used in GC–EI/MS and GC–CI/MS, if the target chemical is identified against a library instead of an authentic reference chemical.

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Chemical	Concentration	Purity	PCI CH ₄	PCI C ₄ H ₁₀	PCI NH ₃	PCI ACN	NCI
QC-test mixture in n-hexane or ethyl acetate							
Trimethylphosphate (CAS 512-56-1)	10 ng/μl	p.a. grade or better	X	X	X	X	X
2,6-Dimethylphenol (CAS 576-26-1)	10 ng/μl	p.a. grade or better	X	X	X	X	X
5-Chloro-2- methylaniline (CAS 95-79-4)	10 ng/μl	p.a. grade or better	X	X	X	X	X
Tri- <i>n</i> -butylphosphate (CAS 126-73-8)	10 ng/μl	p.a. grade or better	X	X	X	X	X
Dibenzothiophene (CAS 135-65-0)	10 ng/μl	p.a. grade or better	X	X	X	X	X
Malathion (CAS 121-75-5)	10 ng/μl	p.a. grade or better	X	X	X	X	X
Methyl stearate (CAS 112-75-5)	10 ng/μl	p.a. grade or better	X	X	X	X	X
<i>n</i> -alkanes (C8–C4, even members)	10 ng/μl	p.a. grade or better	X			X	
QC-test mixture in iso-octane							
Benzophenone (CAS 119-61-9)	10–50 fold the value recommended by the GC–MS manufacturer	p.a. grade or better	X				
Octafluoronaphthalene (CAS 313-72-4)	see benzophenone	p.a. grade or better					X
Reagents gas				Purity			
Methane				≥99.995%			
Ammonia				≥99.995%			
Isobutane				99.995%			
Acetonitrile				≥99.995%			
Carrier gas							
Helium				≥99.995%			
Solvents							
Dichloromethane (for cleaning syringes)				p.a. grade or better			
Ethyl acetate (for cleaning syringes)				p.a. grade or better			
Methanol (for cleaning syringes)				p.a. grade or better			
Hexane (for cleaning syringes)				p.a. grade or better			

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5. Experimental

5.1 Recommended GC conditions

The recommended GC conditions are given in Table 2. Normally, these conditions are the same as the ones in GC–EI/MS.

According to the Proficiency Test criteria, the GC conditions in GC–EI/MS and in GC–CI/MS must be the same if the target chemical is identified against a library instead of an authentic reference chemical.

If the target chemical is compared against an authentic reference chemical, the GC conditions can be different.

Table 2. Recommended GC conditions.

GC parameter	Recommended conditions	Notes
Septum purge flow	2–4 ml/min	
Injection mode	Splitless	
Injection volume	1 µl	Large volume injector liner
Splitless time	1 min	Same as the isothermal time of the starting temperature in the oven temperature program
Split ratio after splitless time	≥1:50	Recommended to be reduced to 1:20 after 2 min to save carrier gas if gas saver function is available in the GC
Injector temperature	250–280 °C	
GC–MS interface temperature	280–290 °C	At least as high as the end temperature of the oven temperature program
Helium linear flow rate	30 cm/s	Optimum column efficiency
Carrier gas flow mode	Constant pressure or constant flow	
Temperature program for GC oven	40 °C (1 min) – 10 °C/min – 280 °C for 5–10 min	

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5.2 Recommended MS conditions

The recommended MS conditions are given in Table 3. The given conditions correspond to the Proficiency Test criteria.

Table 3. Recommended MS conditions in GC–CI/MS analysis.

MS parameter	Conditions	Notes
Solvent delay time	3 min (longer for silylated samples)	Depends on the sample solvent
Mass range	From $m/z \leq 80$ to ≥ 600	
Scan rate	≥ 1 scan/s	
Electron energy	Refer to instrument manual	
Ion source temperature	Refer to instrument manual	Generally temperature is $< 200\text{ }^{\circ}\text{C}$
Electron multiplier voltage	Refer to instrument manual	Adjusted often during instrument tuning

5.3 Recommended tuning and mass calibration of the MS

Use the recommended tuning method for the instrument. The automatic tuning and calibration procedure should be used, if available.

According to the quality assurance system implemented in the laboratory, tuning and calibration routines will be conducted periodically. Tuning and calibration reports will be kept to assure the traceability of the results.

6. Quality control

6.1 Instrument performance test

Identifying CWC-related compounds reliably requires that the data is correct. Therefore, running quality control tests periodically to confirm that the instrument works correctly and the performance of the instrument remains stable during the sample analysis is essential.

The performance of a GC–CI/MS instrument is assessed by carrying out a QC test run, in which the mass spectrometric behavior of the test compounds and the adsorption behavior of the column are confirmed to be acceptable.

The instrument is accepted for use if the MS and GC data produced from each test compound is in accordance with the following requirements:

1. Signal-to-noise (S/N) ratio for each test chemical $>10:1$ in the total ion chromatogram (TIC). The evaluation is done visually or automatically if available.
2. The m/z values of the major ions of each test chemical are correct and no extra ions above 5% are present (the mass range to be printed must be extended until 60 m/z beyond the molecular weight so that the overall relevant mass spectrum is registered).

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3. The relative ion abundances deviation regarding the abundances of the main ions could be around $<\pm 50\%$, compared with the values of the reference spectra.
4. The intensity ratios of the isotopic ions are within the following limits (the values are expressed as percentages of the less abundant isotopic ions relative to the more abundant ones $\pm$ the acceptable error limits):
 - a) In positive mode (PCI):

With methane or isobutane as the reagent gas:

- m/z 144/142 ($^{37}\text{Cl}/^{35}\text{Cl}$) for 5-chloro-2-methylaniline is $33\pm 3.0\%$
- m/z 187/185 ($^{34}\text{S}/^{32}\text{S}$) for dibenzothiophene is $5.9\pm 1.0\%$

With ammonia as the reagent gas:

- m/z 144/142 ($^{37}\text{Cl}/^{35}\text{Cl}$) for 5-chloro-2-methylaniline is $33\pm 3.0\%$
- m/z 186/184 ($^{34}\text{S}/^{32}\text{S}$) for dibenzothiophene is $5.9\pm 1.0\%$

With acetonitrile as the reagent gas:

- m/z 144/142 ($^{37}\text{Cl}/^{35}\text{Cl}$) for 5-chloro-2-methylaniline is $33\pm 3.0\%$
- m/z 187/185 ($^{34}\text{S}/^{32}\text{S}$) for dibenzothiophene is $5.5\pm 1.0\%$

- b) In negative mode (NCI) with methane, ammonia or isobutane as the reagent gas:

- m/z 142/140 ($^{37}\text{Cl}/^{35}\text{Cl}$) for 5-chloro-2-methylaniline is $33\pm 3.0\%$
- m/z 185/183 ($^{34}\text{S}/^{32}\text{S}$) for dibenzothiophene is $7.5\pm 1.5\%$

Methyl stearate is recommended to check the stability of the retention parameters. The absolute retention times of methyl stearate must be within ± 15 s.

Trimethylphosphate (preferably dimethylmethylphosphonate- d_9) is recommended to ensure the performances of the capillary column. The trimethylphosphate asymmetry could be ≤ 10 measured at $\frac{1}{2}$ height. It is particularly recommended for checking the adsorption behavior of dialkyl alkylphosphonates.

Note! *Manufacturers recommend using benzophenone to check the GC–MS performance in positive chemical ionisation with methane as the reagent gas. Octafluoronaphthalene is recommended for checking the GC–MS in negative chemical ionisation.*

These compounds can be used at a concentration 10 to 50 times the value recommended by the manufacturer. The signal-to-noise ratio will be generally expressed in RMS (Root Mean Square).

Estimating of the GC–CI/MS sensitivity is performed with the QC test sample in a concentration level of 10 ng/ μl and/or 10- to 50-fold the concentration level recommended by the manufacturer for benzophenone and octafluoronaphthalene.

Figure 1 shows a typical TIC of a QC test sample indicating good instrument performance under positive chemical ionisation with methane (quadrupole mass spectrometer).

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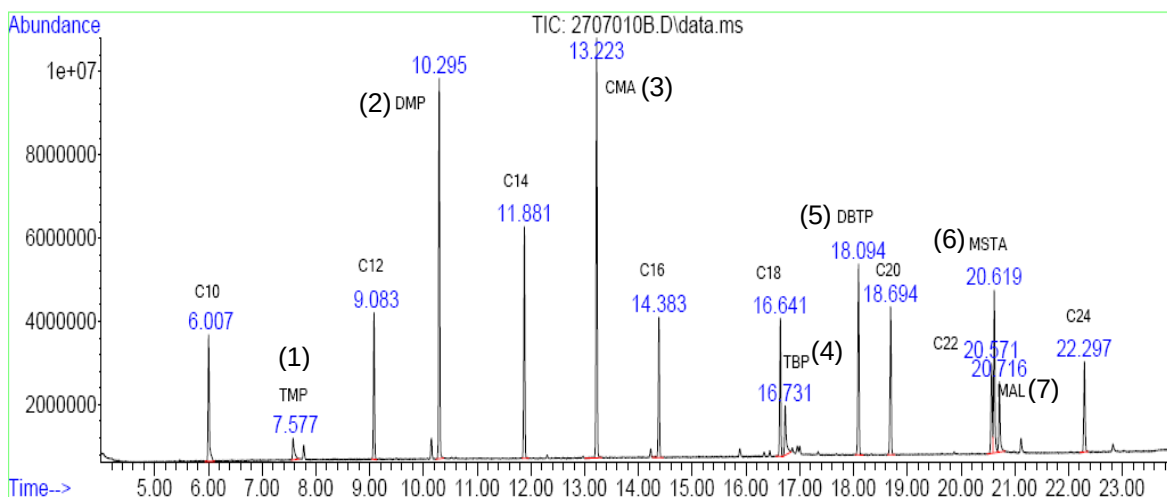


Figure 1. Analysis of the QC test sample (10 ppm) on a 30 m x 0.25 mm i.d. x 0.25 μ m OV1701-MS (86%-dimethyl-14%-cyanopropylphenyl polysiloxane) column; carrier gas: Helium, constant flow, 1.35 ml/min; splitless injection with splitless time of 1 min; oven temperature program: 40 °C (1 min) – 10 °C/min – 280 °C (5 min). Peak labels: (1) trimethylphosphate, (2) 2,6-dimethylphenol, (3) 5-chloro-2-methylaniline, (4) tri-*n*-butylphosphate, (5) dibenzothiophene, (6) methyl stearate and (7) malathion.

Figure 2 shows a typical TIC of a QC test sample indicating good instrument performance under positive chemical ionisation with acetonitrile (ion trap mass spectrometer).

Note! The inversion in the retention times of malathion and methyl stearate depends on the column's stationary phase.

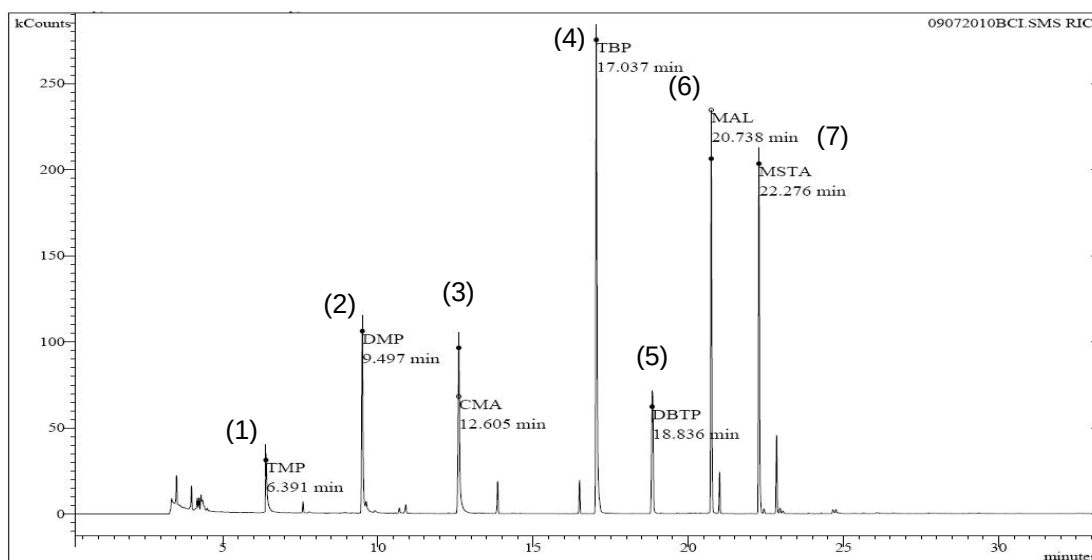


Figure 2. Analysis of the QC test sample (10 ppm) on a 30 m x 0.25 mm i.d. x 0.25 μ m Factor Four DB5-MS (95%-dimethyl-5%-diphenyl polysiloxane) column; carrier gas: Helium, constant pressure, 10 psi; splitless injection with splitless time of 1 min; oven temperature program: 40 °C (1 min) – 10 °C/min – 280 °C (5 min). Peak labels: (1) trimethylphosphate, (2) 2,6-dimethylphenol, (3) 5-chloro-2-methylaniline, (4) tri-*n*-butylphosphate, (5) dibenzothiophene, (6) malathion and (7) methyl stearate.

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6.2 Blank tests

A solvent blank run must be conducted between two analysed samples to avoid cross contamination of the GC–MS, and thus to ensure the reliability of the measurements. If CWC-related compounds are found in the blank chromatogram, the solvent blank test is repeated until the contamination source is completely eliminated.

After a silylated sample is analysed, some memory effects could remain due to the presence of non-volatile chemicals in the injector. In this case, a solvent used in sample preparation containing a silylation agent is injected to remove contamination.

7. Analysis of samples

7.1. Sequence of analysis

In the beginning of an analysis batch, a QC test sample and a blank test are run. When both tests have been accepted, the samples can be analysed. Generally, 1.0 µl of the sample is injected in the splitless mode.

During sample analysis, the QC test sample is run every 15 to 20 injections to validate the GC–MS performance.

At the end of the batch samples, the QC test sample is run to validate the GC–MS performance.

7.2 Screening

Screening is typically performed in GC and GC–EI/MS systems. GC–CI/MS analysis is generally performed to confirm the molecular weight of the relevant chemicals previously identified with GC–EI/MS. However, screening in GC–CI/MS is recommended to help in identifying CWC-related chemicals. Some scheduled chemicals are not detected with phosphorus, sulfur and nitrogen selective detectors. These chemicals and their specific ions for GC–CI/MS screening are listed in Table 4. Appendix 1 contains the specific ions of the most important groups of CWC-related chemicals. The ion intensity depends on the reagent gas, the repeller voltage and the ion source temperature, and therefore, the intensity scale is not included in Table 4.

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Chemical	MW	Specific ions for screening (m/z)				
		PCI				NCI
		CH ₄	NH ₃	C ₄ H ₁₀	ACN	
PFIB	200	<u>181</u>	Predicted: <u>181</u> , 218 [M+NH ₄] ⁺	Predicted: <u>181</u>		<u>200</u> [M] ⁺
Chloropicrin	163	83, 85, <u>117</u> , <u>119</u> , <u>121</u>	<u>117</u> , <u>119</u> , <u>121</u>	<u>117</u> , <u>119</u> , <u>121</u>	<u>117</u> , <u>119</u> , 158,160	<u>46</u> , <u>117</u> , <u>119</u> , <u>121</u> ,128,130, 164 [M+H] ⁺
Pinacolyl alcohol	102	<u>85</u> , 101 [M-H] ⁺	<u>120</u> [M+18] ⁺ , 137 [M+35] ⁺	<u>85</u> , 101 [M-H] ⁺	<u>85</u>	85, <u>101</u> [M-H] ⁺
Methyl derivative of benzilic acid	242	105, 183, <u>225</u> , 253	183, <u>225</u> , 242, <u>260</u> [M+18] ⁺	183, <u>225</u>	183, <u>225</u>	183, <u>213</u> , 227, <u>241</u> [M-H] ⁺
TMS derivative of benzilic acid	372	255, <u>283</u> , 311, <u>357</u>	<u>283</u> , <u>300</u> , 255	<u>283</u> , 255	114, <u>283</u> , 357	210, <u>299</u> , 371

Note! PFIB, HCN and phosgene analysed without derivatisation may be screened using the head space or split injection technique.

7.3 Identification

The final identification must always be approved by an experienced analyst. The recorded spectra (reference chemicals if available and relevant chemicals) must be presented from ≤80 to at least 60 mass units above the mono-isotopic molecular weight to include all expected adducts, including those when isobutane is the reagent gas. The isotopic ions ratio must be consistent with the proposed structure.

All background ions present in the mass spectrum of the relevant chemical with an intensity >10% must be labeled as background ions. In this case, the extracted ion chromatogram (EIC) should be shown by selecting an impurity ion with one or two characteristic ions of the relevant chemical. In most cases, the profile ions are not perfectly overlapped, and a good-quality spectrum can be extracted.

There are two possibilities for confirming the structure of CWC-related chemicals in GC–CI/MS in a Proficiency Test.

- The best way to confirm the structure and the identification is to compare the GC–CI/MS mass spectrum of a relevant chemical against the GC–CI/MS mass spectrum of a reference chemical. The retention time of the relevant chemical and the reference chemical must remain within ±0.1 min.
- GC–CI/MS data without comparison to similarly obtained data from a reference chemical, is accepted only as a supporting technique to GC–EI/MS data. Where no GC–EI/MS data has been submitted or the GC–EI/MS data is insufficient, GC–CI/MS data without comparison to similarly obtained data from a reference chemical will be rejected.

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- o There must be a retention parameter match between peaks in GC–EI/MS and GC–CI/MS, for which spectrometric data is presented. Chromatograms must be recorded on the same phase type and column brand, using identical chromatographic conditions; specifically, the length, internal diameter, film thickness, flow control and injector temperature should be the same. The acceptable retention time window for the peak in the CI chromatogram compared to the peak in the EI chromatogram is ± 0.2 min.
- o The quasi-molecular ion (e.g., $[M+H]^+$, $[M-H]^-$, $[M-H]^+$) must be indicated, and its intensity must be above 10% of the base peak.
- o Intense ions ($>20\%$ of the base peak) at m/z above the assigned quasi-molecular ion must be explained.

8. Method evaluation

8.1 Area of applicability

The GC–CI/MS method can be applied to all volatile chemical warfare agents,¹⁵ their degradation products¹⁶ and their precursors, except saxitoxin and ricin. Non-volatile polar CWC-related chemicals with alcohol¹⁷ and/or acid groups (e.g., thiodiglycol,¹⁸ triethanolamine, alkylphosphonic acids¹⁸) can be derivatised with silylating reagents (e.g., BSTFA, MTBSTFA); chemicals with acid groups (e.g., alkylphosphonic acids, benzoic acid) can be derivatised with methylating reagents (e.g., diazomethane).

CWC-related chemicals with arsenic atoms are typically derivatised with thiol reagents (e.g., propanethiol, butanethiol, DMT¹⁹).

This technique can be used in full scan mode for chemicals with a concentration around the mg/kg level. The technique can be used in Selected Ion Monitoring (SIM) mode to search target chemicals with a concentration around the 0.1 mg/kg²⁰ level, although the SIM mode is not accepted in Proficiency Test reports.

8.2 Advantages/Disadvantages

Advantages

The GC–CI/MS is a reliable and sensitive identification technique if the mass spectrometer is regularly calibrated and if the sensitivity of the GC–MS is checked with the use of test compounds (e.g., QC test sample). Many reagent gases can be used to confirm the molecular weight of CWC-related chemicals. Positive and negative chemical ionisation often reveals several characteristic ions in addition to those obtained under electron ionisation (EI). Positive and negative chemical ionisation contributes to the structure elucidation of CWC-related chemicals.

Disadvantages

The ion intensities in the CI spectra are different depending on the ion source temperature, the repeller voltage and the pressure of the reagent gas. That is why CI databases do not exist.

To switch from operating in the EI to the CI mode in a quadrupole system, the ionisation source has to be changed. This requires venting of the MS system. Approximately two hours are required to get the vacuum and an additional 6 hours or more for the reagent gas

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to stabilise. The task is time-consuming. An ion trap system does not require such a long downtime.

Isobutane rapidly contaminates the ion source, requiring frequent cleaning and maintenance of the ion source.

Ammonia is toxic and corrosive. Use of ammonia requires special maintenance and safety precautions (e.g., the oil of the primary pump must be changed more regularly).

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APPENDIX 1. Ion list of phosphorus-containing chemical warfare agents, their precursors or analytical derivatives for positive GC–CI/MS.

Structure	Name [♦]	Specific ions for screening (<i>m/z</i>)			
		PCI CH ₄	PCI NH ₃	PCI iC ₄ H ₁₀	PCI ACN
Sarin-type chemicals ²¹					
	Alkyl methylphosphono-fluoridates	99 *[M+H] ⁺	^{*99} *[M+H] ⁺ [M+18] ⁺	^{*99} [M+H] ⁺	
	Alkyl ethylphosphono-fluoridates	113 *[M+H] ⁺	^{*113} *[M+H] ⁺ [M+18] ⁺	^{*113} [M+H] ⁺	
	Alkyl propylphosphono-fluoridates	127 *[M+H] ⁺	^{*127} [M+H] ⁺ [M+18] ⁺	^{*127} [M+H] ⁺	
	Alkyl isopropylphosphono-fluoridates	127 *[M+H] ⁺	^{*127} [M+H] ⁺ [M+18] ⁺	^{*127} [M+H] ⁺	
VX-type chemicals					
	Alkyl S-2-dimethylaminoethyl alkylphosphonothiolate	72 [M+H] ⁺	^{*72} ^{*106} [M+H] ⁺	[M+H] ⁺	
	Alkyl S-2-diethylaminoethyl alkylphosphonothiolate	100 [M+H] ⁺	^{*86} ^{*100} ^{*134} [M+H] ⁺	[M+H] ⁺	100 134 [M+H] ⁺
	Alkyl S-2-dipropylaminoethyl alkylphosphonothiolate	128 [M+H] ⁺	^{*114} ^{*128} ^{*162} [M+H] ⁺	[M+H] ⁺	114 128 [M+H] ⁺

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Structure	Name [†]	Specific ions for screening (<i>m/z</i>)			
		PCI CH ₄	PCI NH ₃	PCI iC ₄ H ₁₀	PCI ACN
	Alkyl S-2-diisopropylaminoethyl alkylphosphonothiolate	128 [M+H] ⁺	*114 *128 *162 [M+H] ⁺	[M+H] ⁺	114 128 [M+H] ⁺
Tabun-type chemicals					
	Alkyl N,N- dimethylphosphoramido cyanide	136 [M+H] ⁺	[M+H] ⁺ [M+18] ⁺	[M+H] ⁺	
	Alkyl N,N-diethylphosphoramido cyanide	164 [M+H] ⁺	[M+H] ⁺ [M+18] ⁺	[M+H] ⁺	
	Alkyl N,N- dipropylphosphoramido cyanide	192 [M+H] ⁺	[M+H] ⁺ [M+18] ⁺	[M+H] ⁺	
	Alkyl N,N- diisopropylphosphoramido cyanide	192 [M+H] ⁺	[M+H] ⁺ [M+18] ⁺	[M+H] ⁺	
Alkylphosphonates					
	Alkyl methyl methylphosphonate	*111 [M+H] ⁺	*111 [M+H] ⁺ *[M+18] ⁺	*111 [M+H] ⁺	111 [M+H] ⁺
	Alkyl methyl ethylphosphonate	*125 [M+H] ⁺	*125 [M+H] ⁺ *[M+18] ⁺	*125 [M+H] ⁺	125 [M+H] ⁺
	Alkyl methyl propylphosphonate	*139 [M+H] ⁺	*139 [M+H] ⁺ *[M+18] ⁺	*139 [M+H] ⁺	139 [M+H] ⁺
	Alkyl methyl isopropylphosphonate	*139 [M+H] ⁺	*139 [M+H] ⁺ *[M+18] ⁺	*139 [M+H] ⁺	139 [M+H] ⁺
	Dialkyl methylphosphonate	*97 [MH-(R-H)] ⁺ [M+H] ⁺	*97 [MH-(R-H)] ⁺ [M+H] ⁺ *[M+18] ⁺	*97 [MH-(R-H)] ⁺ [M+H] ⁺	*97 [M+H] ⁺

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Structure	Name [♦]	Specific ions for screening (m/z)			
		PCI CH ₄	PCI NH ₃	PCI iC ₄ H ₁₀	PCI ACN
	Dialkyl ethylphosphonate	^{*111} [MH-(R-H)] ⁺ [M+H] ⁺	^{*111} * [MH-(R-H)] ⁺ [M+H] ⁺ *[M+18] ⁺	^{*111} [MH-(R-H)] ⁺ [M+H] ⁺	111, 153 [M+H] ⁺
	Dialkyl propylphosphonate	^{*125} [MH-(R-H)] ⁺ [M+H] ⁺	^{*125} * [MH-(R-H)] ⁺ [M+H] ⁺ *[M+18] ⁺	^{*125} [MH-(R-H)] ⁺ [M+H] ⁺	125 [M+H] ⁺
	Dialkyl isopropylphosphonate	^{*125} [MH-(R-H)] ⁺ [M+H] ⁺	^{*125} * [MH-(R-H)] ⁺ [M+H] ⁺ *[M+18] ⁺	^{*125} * [MH-(R-H)] ⁺ [M+H] ⁺	125 [M+H] ⁺
TMS derivatives of alkylphosphonates					
	Alkyl trimethylsilyl methylphosphonate ²²	153 169 [M+H] ⁺	[M+H] ⁺ *[M+18] ⁺	^{*169} [M+H] ⁺	169 [M+H] ⁺
	Alkyl trimethylsilyl ethylphosphonate	167 183 [M+H] ⁺	[M+H] ⁺ *(M+18) ⁺	^{*183} [M+H] ⁺	183 [M+H] ⁺
	Alkyl trimethylsilyl propylphosphonate	181 197 [M+H] ⁺	[M+H] ⁺ *(M+18) ⁺	^{*197} [M+H] ⁺	197 [M+H] ⁺
	Alkyl trimethylsilyl isopropylphosphonate	181 197 [M+H] ⁺	[M+H] ⁺ *(M+18) ⁺	^{*197} [M+H] ⁺	197 [M+H] ⁺

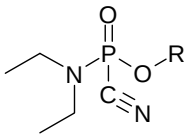
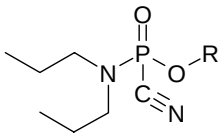
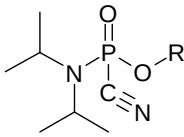
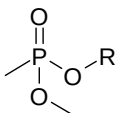
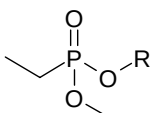
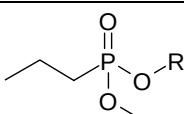
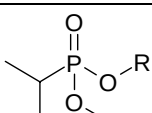
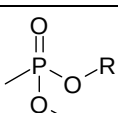
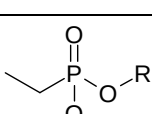
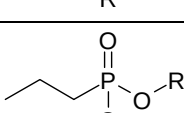
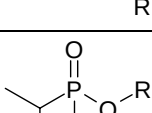
[♦]R ≠ methyl. *Not systematically present. R' = methyl, ethyl, propyl or isopropyl.

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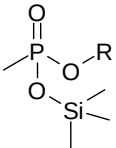
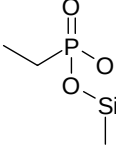
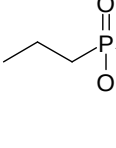
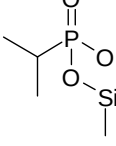
APPENDIX 2. Ion list of phosphorus-containing chemical warfare agents, their precursors or analytical derivatives for negative GC–CI/MS.

Structure	Name ♦	Ion $[m/z]$
Sarin-type chemicals		
	Alkyl methylphosphonofluoridates	*97 [M-H] ⁻
	Alkyl ethylphosphonofluoridates	*111 [M-H] ⁻
	Alkyl propylphosphonofluoridates	*125 [M-H] ⁻
	Alkyl isopropylphosphonofluoridates	*125 [M-H] ⁻
VX-type chemicals		
	Alkyl S-2-dimethylaminoethyl alkylphosphonothiolate	[M-H] ⁻ * [(M-H)-(R-H)] ⁻
	Alkyl S-2-diethylaminoethyl alkylphosphonothiolate	[M-H] ⁻ * [(M-H)-(R-H)] ⁻
	Alkyl S-2-dipropylaminoethyl alkylphosphonothiolate	[M-H] ⁻ * [(M-H)-(R-H)] ⁻
	Alkyl S-2-diisopropylaminoethyl alkylphosphonothiolate	[M-H] ⁻ * [(M-H)-(R-H)] ⁻
Tabun-type chemicals		
	Alkyl N,N-dimethylphosphoramidocyanidate	133 * [M-H] ⁻

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Structure	Name ♦	Ion [<i>m/z</i>]
	Alkyl N,N-diethylphosphoramidocyanidate (<i>predicted ions</i>)	161 * [<i>M-H</i>] [−]
	Alkyl N,N-dipropylphosphoramidocyanidate (<i>predicted ions</i>)	189 * [<i>M-H</i>] [−]
	Alkyl N,N-diisopropylphosphoramidocyanidate (<i>predicted ions</i>)	189 * [<i>M-H</i>] [−]
Alkylphosphonates		
	Alkyl methyl methylphosphonate	[<i>M-H</i>] [−]
	Alkyl methyl ethylphosphonate	[<i>M-H</i>] [−]
	Alkyl methyl propylphosphonate	[<i>M-H</i>] [−]
	Alkyl methyl isopropylphosphonate	[<i>M-H</i>] [−]
	Dialkyl methylphosphonate	* [(<i>M-H</i>)-(R-H)] [−] [<i>M-H</i>] [−]
	Dialkyl ethylphosphonate	* [(<i>M-H</i>)-(R-H)] [−] [<i>M-H</i>] [−]
	Dialkyl propylphosphonate	* [(<i>M-H</i>)-(R-H)] [−] [<i>M-H</i>] [−]
	Dialkyl isopropylphosphonate	* [(<i>M-H</i>)-(R-H)] [−] [<i>M-H</i>] [−]
TMS derivatives of alkylphosphonates		

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Structure	Name ♦	Ion [<i>m/z</i>]
	Alkyl trimethylsilyl methylphosphonate	[M-H]⁻
	Alkyl trimethylsilyl ethylphosphonate	[M-H]⁻
	Alkyl trimethylsilyl propylphosphonate	[M-H]⁻
	Alkyl trimethylsilyl isopropylphosphonate	[M-H]⁻

♦ R ≠ methyl. * Not systematically present. R' = methyl, ethyl, propyl or isopropyl.

Note! Major ions in appendices 1 and 2 are in bold. The ion intensity depends on the reagent gas, the repeller voltage and the ion source temperature, and therefore, the intensity scale is not included in these appendixes.

An asterisk by an ion name means that the ion is not necessarily present on its corresponding CI mass spectra.

[M+29]⁺ and [M+41]⁺ adduct ions are not mentioned in appendix 1, but they can be present on their corresponding PCI mass spectra with methane.

[M+39]⁺, [M+41]⁺ and [M+57]⁺ adduct ions are not mentioned, but they can be observed on their corresponding PCI mass spectra with isobutane.

The [M+13]⁺ adduct ion is not mentioned, but it can be observed on its corresponding PCI mass spectra with acetonitrile.

When none of the ions appears in bold, it means that the major ion is not systematically the same for chemicals that belong to the same family. For instance, on the PCI mass spectra of dialkyl alkylphosphonates with methane, the protonated molecule [M+H]⁺ is not systematically the base peak.

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Chapter V.

Gas chromatography–high resolution mass spectrometry (GC–HRMS): sector instruments, selective ion monitoring mode

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1. Scope

This chapter describes the use of gas chromatography–high resolution mass spectrometry (GC–HRMS) using magnetic sector instruments in selective ion monitoring mode (SIM) in the identification of chemical warfare agents, their precursors, degradation products, or derivatives, i.e., chemicals related to the Chemical Weapons Convention (CWC). This method is used for volatile chemicals in an organic solvent when interfering background chemicals prevent the analysis and identification of relevant chemicals using low resolution GC–EI/MS.

2. Summary of the method

GC–HRMS measurements using electron ionisation can be carried out with modern GC–HRMS magnetic sector instruments operating in selected ion monitoring (SIM) mode for target analysis. Samples are analysed with a GC–MS instrument equipped with a 25–30 m capillary column with SE-54-type stationary phase material. For GC–HRMS SIM, reference material (e.g., perfluorokerosine, PFK) is introduced into the ion source, and the instrument is tuned to high resolution ($R > 8000$). Before each sample batch, a quality control run (QC test, see below) is carried out to verify the correct performance of the instrument, using a SIM method created for the QC test sample. After a passed QC test, a specific SIM method for the sample of interest is created, and the sample is analysed. QC tests should be run daily before any sample(s) of interest are analysed.

The target analysis of chemicals of interest using GC–HRMS SIM is carried out by selecting the exact masses of characteristic ions for the individual chemicals (at least three ions must be selected for each target chemical). A chemical is identified when each selected ion produces a peak with the correct ion abundance ($\pm 20\%$) in comparison with the reference EI spectrum.

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3. Safety aspects

GC–HRMS analysis does not require any specific precautions concerning safety. Samples to be analysed normally contain only small concentrations of toxic chemicals, and the electron ionisation will destroy the sample molecules. Normal laboratory safety measures, such as the use of protective gloves and goggles during sample handling in manual injections, should be followed.

Hot parts of instruments, e.g., hot septum nuts (>250 °C), must be taken into account during the operation and maintenance of instrument.

4. Instrumentation

4.1 GC–HRMS instrument

Any modern GC–HRMS magnetic sector instrument that is capable of at least a resolution of 8000 and covers a mass range of at least 40–500 m/z can be used.

4.2 GC columns

A 25–30 m-long SE-54-type* fused silica capillary column (0.20–0.25 mm i.d. and film thickness 0.25 µm), preferably a so-called low bleed MS column, is used.

***Note!** As the GC–HRMS SIM data is normally almost clear of background chemicals, the actual amount of background introduced to the system or any possible column bleed will not be seen. It is essential to the QC test to verify the condition of the GC column during and after the GC–HRMS work.*

* A SE-54-type column means a stationary phase consisting of 95%-methyl-5%-phenylpolysiloxane. Also, polyarylene-type columns, which are virtually equivalent to the SE-54-type columns, can be used.

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4.3 Reagents and chemicals

The reagents and chemicals needed for the quality control (QC) of GC–HRMS analysis are given in Table 1.

Table 1. Chemicals and reagents needed for testing GC–HRMS SIM.

Chemical	Concentration	Purity	Ions for HRMS SIM (<i>m/z</i>)
QC-test mixture in dichloromethane			
Trimethylphosphate (CAS 512-56-1)	10 ng/μl	p.a. grade or better	110.013, 140.025
2,4-Dimethylphenol (CAS 576-26-1)	10 ng/μl	p.a. grade or better	107.050, 122.073
5-Chloro-2-methylaniline (CAS 95-79-4)	10 ng/μl	p.a. grade or better	106.066, 141.035, 143.032
Tri- <i>n</i> -butylphosphate (CAS 126-73-8)	10 ng/μl	p.a. grade or better	155.047, 211.110
Dibenzothiophene (CAS 135-65-0)	10 ng/μl	p.a. grade or better	152.066, 184.035, 186.031
Malathion (CAS 121-75-5)	10 ng/μl	p.a. grade or better	158.058, 173.081
Methyl stearate (CAS 112-61-8)	10 ng/μl	p.a. grade or better	255.232, 298.287
Carrier gas			
Helium		≥99.995%	
Solvents			
Dichloromethane (for cleaning of syringes)		p.a. grade or better	
Ethyl acetate (for cleaning of syringes)		p.a. grade or better	

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5. Experimental

5.1 Recommended GC conditions

Recommended GC conditions are given in Table 2. The given conditions correspond to the conditions used in other GC–MS methods.

Table 2. Recommended GC conditions.

GC parameter	Recommended conditions	Notes
Septum purge flow	2–4 ml/min	
Injection mode	Splitless	
Injection volume	1 µl	Large volume injector liner
Splitless time	1 min	Same as the isothermal time of the starting temperature in the oven temperature program
Split ratio after splitless time	≥1:50	Recommended to be reduced to 1:20 after 2 min to save carrier gas if gas saver function is available in the GC
Injector temperature	250–280 °C	
GC-MS interface temperature	280–290 °C	At least as high as the end temperature of the oven temperature program
Helium linear flow rate	30 cm/s	Optimum column efficiency
Carrier gas flow mode	Constant pressure	Needed if retention indices are measured
Temperature program for GC oven	40 °C (1 min) – 10 °C /min – 280 °C for 5–10 min	

Run conditions may be modified depending on the type of samples. For example, concentrated samples (>100 ng/µl) should be injected in split injection mode, with a split ratio of 1:20 or higher, to avoid column overloading and to achieve a better separation of low-boiling chemicals. For splitless injections, concentrated samples must be diluted.

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5.2 Recommended MS conditions

Recommended MS conditions are given in Table 3. The given conditions correspond to the requirements for MS conditions used in recording MS spectra for the OCAD MS library.

Table 3. Recommended MS conditions in GC–HRMS analysis in SIM mode.

MS parameter	Conditions	Notes
Solvent delay time	3 min (>3 min for silylated samples)	Depends on the sample solvent
Mass range	selected accurate masses	At least three ions/target chemical, with mass accuracy given to three decimals
Scan rate	≥1 cycle/s (1 cycle = total dwell time of the selected ions)	If several target chemicals are monitored from a single run, different SIM sets for different retention time windows corresponding to the target chemicals are required
Mass resolution	≥8000	
Electron energy	70 eV	
Ion source temperature	Refer to instrument manual	
Electron multiplier voltage	Refer to instrument manual	Adjusted often during instrument tuning

5.3 Tuning and mass calibration of the HRMS SIM

Follow the instructions given by the instrument manufacturer. Modern instruments have automatic tuning and calibration procedures that must be carried out for each individual SIM measurement. **Perfluorokerosine (PFK) or some other reference chemical recommended by the instrument instructions is used for tuning and calibration, and the reference material must be present in the ion source during HRMS SIM measurements.** Check that the correct reference file, which contains the reference ions of the reference chemical used, has been selected before you begin tuning.

6. Quality control

6.1 Instrument performance test

The reliable identification of CWC-related compounds requires that the data produced by the instrument is correct. Therefore, it is essential to continually run quality control tests to confirm correct and stable instrument performance during the sample analysis.

The performance of a GC–HRMS instrument is controlled by carrying out a QC test run, in which the mass spectrometric behaviour of test compounds and the adsorption behaviour of the column are confirmed to be acceptable. The QC test must be run at the beginning and at the end of each sample batch, and in the case of a long sample batch, it must be run

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at least after every fifteenth sample. The selected ions for each test chemical are given in Table 1.

For the QC test, three different SIM sets should be created:

1. m/z 110.013, 140.025 for trimethylphosphate
 m/z 107.050, 122.073 for 2,6-dimethylphenol and
 m/z 106.066, 141.035, 143.032 for 5-chloro-2-methylaniline
2. m/z 155.047, 211.110 for tri-*n*-butylphosphate,
 m/z 152.066, 184.035, 186.031 for dibenzothiophene and
 m/z 158.058, 173.081 for malathion
3. m/z 255.232, 298.287 for methyl stearate

The retention time window for each set can be checked by running the same QC test with the full scan GC–EI/MS method under same run conditions and observing the corresponding retention times.

The instrument is accepted for use if the MS and GC data produced from each test compound meet the following requirements:

1. The signal-to-noise (S/N) ratio for each test chemical >10:1 for all selected ion chromatograms. The evaluation is done visually.
2. The intensity ratios of the isotopic ions are within the following limits: (the values are expressed as percentages of the less abundant isotopic ions relative to the more abundant ones $\pm$ the acceptable error limits)
 - m/z 143.032/141.035 ($^{37}\text{Cl}/^{35}\text{Cl}$) for 5-chloro-2-methylaniline is $32.4\pm 3.0\%$
 - m/z 186.031/184.035 ($^{34}\text{S}/^{32}\text{S}$) for dibenzothiophene is $4.4\pm 1.0\%$

6.2 Blank tests

A system blank (blind test) is highly recommended for each target chemical to check for any possible contamination coming from the sample preparation or the instrument. If the blind test shows contamination, a solvent blank run with the same solvent used in the sample preparation must be carried out to eliminate the possibility of contamination of the equipment (e.g., contamination from the syringe or the memory effect of the equipment).

Note! *This is extremely important when silylated target chemicals are analysed.*

If target chemicals are found in any blank chromatogram, the blank tests are repeated after the elimination of the possible contamination source. All contamination coming from glassware, solvents, standard solutions (or authentic reference solutions), syringes, injector systems, or columns must be eliminated before sample analysis is begun.

7 Analysis of samples

7.1. Sequence of analysis

In the beginning of an analysis batch, a QC test (always) and blank tests are run. When all tests have been performed successfully, samples can be analysed. Generally, 1.0 μl of the sample is injected.

7.2 Identification

The final identification must always be approved by an experienced analyst.

A chemical is identified when each selected ion produces a peak with the same retention time. Additionally, each ion must have a correct ion abundance ($\pm 20\%$) in comparison with the reference EI spectrum.

8. Evaluation of method

8.1 Area of applicability

The GC–HRMS SIM method can be applied to all chemical warfare agents, their degradation products, precursors, or derivatives, except saxitoxin and ricin. Extremely volatile (e.g., DF), non-volatile (e.g., alkylphosphonic acids), and reactive (e.g., Lewisite I) CWC-related chemicals can be analysed after their derivatisation to volatile derivatives. The method is suitable for analysing and identifying any chemical at the 1 mg/kg level.

8.2 Advantages/Disadvantages

The advantages of the method are reliability and selectivity in analysing extremely complex sample mixtures. Modern HRMS sector instruments are easy to use with automatic tuning and calibration.

The disadvantage of identification using GC–HRMS SIM is that in practice, only a limited number of target chemicals can be analysed from a single run. Also, the selection of at least three characteristic ions for certain relevant chemicals may be problematic.

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1. Scope

This chapter describes the use of gas chromatography–high resolution mass spectrometry (GC–TOF) in screening and identifying chemical warfare agents, their precursors, degradation products or derivatives, i.e., chemicals related to the Chemical Weapons Convention, CWC. The descriptions in chapters II and III of GC–EI/MS and GC–CI/MS, respectively, are for the most part valid for GC–TOF. This chapter will focus on issues that differ from *low-resolution methods*.

2. Summary of the method

GC–TOF instruments differ from other GC–MS instruments by being non-scanning instruments. Instead, ions are measured in the time frame at a frequency of 15–30 kHz with either a TDC (Time-to-Digital Converter) or a Continuous Averager (ADC, also known as DSA or ITR). The detector is normally not an electron multiplier, but a micro-channel plate (MCP). The measurements are summed over time giving the apparent scan rate. This gives the possibility of having either very fast “scan rates” or high resolution. Here we will focus on the high-resolution GC–TOF.

The high-resolution GC–TOF has most of the characteristics of the modern GC–EI/MS and GC–CI/MS described in chapters II and III, respectively. One major difference is that the dynamic range of the GC–TOF is lower than the dynamic range of the typical quadrupole-based GC–MS. The dynamic range could, depending on the instrument, be enhanced by alternating scans with two different sensitivity settings (Dynamic Range Enhancement) resulting in a halving of the apparent scan rate.

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3. Safety aspects

GC–TOF analysis does not require any specific precautions concerning safety. Samples to be analysed normally contain only small concentrations of toxic chemicals, and the ionisation will destroy the sample molecules. Normal laboratory safety measures such as wearing protective gloves and goggles while handling samples in manual injections should be used.

Hot parts of the instrument, e.g., the hot septum nut (>250 °C), must be taken into account while operating and maintaining the instrument.

4. Instrumentation

4.1 GC–TOF instrument

Any modern GC–TOF instrument with a resolution of 5000–10,000 and covering a mass range of at least m/z 40–500 can be used.

4.2 GC columns

A 25- to 30-m-long SE-54-type* fused silica capillary column (0.20–0.25 mm i.d. and film thickness 0.25 µm), preferably the low-bleed MS column, is used.

4.3 Reagents and chemicals

The reagents and chemicals needed for GC–TOF analysis are the same as those given in Table 1 of chapters II and III for GC–EI/MS and GC–CI/MS, respectively.

5. Experimental

5.1 Recommended GC conditions

The recommended GC conditions are given in Table 1.

The run conditions may be modified depending on the sample type. Due to the lower dynamic range of the GC–TOF, the split injection mode is used more frequently. For example, samples with concentrations <10 ng/µl are typically done with a 1:5 split ratio. More concentrated samples (>100 ng/µl) should be injected in the split injection mode with a split ratio of 1:50 or higher to avoid column overloading and to achieve better separation of low-boiling chemicals. For splitless injections, most samples must be diluted.

* SE-54-type column means a stationary phase consisting of 95%-methyl-5%-phenylpolysiloxane. Polyarylene-type columns that are virtually equivalent to the SE-54-type columns can also be used.

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Table 1. Recommended GC conditions.

GC parameter	Recommended conditions	Notes
Septum purge flow	2–4 ml/min	
Injection mode	Split (1:5) Splitless	Only for diluted samples
Injection volume	1 µl	Large volume injector liner
Splitless time	1 min	Same as the isothermal time of the starting temperature in the oven temperature program
Split ratio after splitless time	≥1:50	Recommended to be reduced to 1:20 after 2 min to save carrier gas if gas saver function is available in the GC
Injector temperature	250–280 °C	
GC–MS interface temperature	280–290 °C	At least as high as the end temperature of the oven temperature program
Helium linear flow rate	30 cm/s	Optimum column efficiency
Carrier gas flow mode	Constant pressure or constant flow	
Temperature program for GC oven	40 °C (1 min) – 10 °C/min – 280 °C for 5–10 min	

5.2 Recommended MS conditions

The recommended MS conditions are given in Table 2. The given conditions correspond to the requirements for the MS conditions used for recording MS spectra for the OCAD MS library:

Table 2. Recommended MS conditions in GC–TOF analysis.

MS parameter	Conditions	Notes
Solvent delay time	3 min (longer for silylated samples)	Depends on the sample solvent
Mass range (<i>m/z</i>)	From ≤40 to ≥500	
Apparent scan rate	≥5 scan/s	
Electron energy	70 eV	
Ion source temperature	Refer to instrument manual	
MCP voltage	Refer to instrument manual	Adjusted often during instrument tuning
Calibration gas	Refer to instrument manual	To achieve accurate mass, calibration gas should be used during runs

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5.3 Recommended tuning and mass calibration of the MS

Use the tuning method recommended for the instrument. An automatic tuning and calibration procedure should be used, if available. Tuning should be carried out once a week or more often if indicated by the QC test. Mass calibration should be carried out before each batch of samples to achieve accurate mass. Printouts of tuning and calibration reports should be archived for future reference.

6. Quality control

6.1 Instrument performance test

Reliably identifying CWC-related compounds requires that the data produced by the instrument is correct. Therefore, quality control tests must be run continuously to confirm the instrument will perform correctly and stably while analysing samples.

The instrument performance test for GC–TOF analysis is the same as given in 6.1 of chapters II and III for GC–EI/MS and GC–CI/MS, respectively.

6.2 Blank tests

A system blank (blind test) should be performed to check for any possible contamination in the sample preparation or the instrument. If the blind test shows contamination, a solvent blank run with the same solvent used in sample preparation must be performed to eliminate the possibility of contamination of the equipment (e.g., contamination from the syringe or the memory effect of the equipment). **Note!** *Extremely important for silylated samples.*

If CWC-related chemicals are found in any blank chromatogram, the blank tests are repeated after the possible contamination source has been eliminated. All contamination from glassware, solvents, standard solutions (or authentic reference solutions), syringes, an injector system or columns must be eliminated before sample analysis is started.

7. Analysis of samples

7.1. Sequence of analysis

In the beginning of an analysis batch, a QC test (always) and a blank test are run. When all tests have been accepted, the samples can be analysed. Generally, 1.0 µl of the sample is injected. If retention indices are used, they can be calculated with an external method using the *n*-alkanes in the QC test run as retention index standards. Automated Mass Spectral Deconvolution and Identification System (AMDIS) software[†] should be used to calculate the retention index (refer to the GC ROP for detailed instructions). According to the current OPCW rules, a blank run is required before each sample analysis!

[†] Available at the National Institute of Standards and Technology (NIST) website (chemdata.nist.gov)

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7.2 Screening

If phosphorus, nitrogen and sulphur selective detectors are used to screen the samples, extracted ion chromatograms (EICs) can be used to locate those chemicals, which are not detectable with these selective detectors. A list of these chemicals and the specific ions for screening with GC–EI/MS are given in Table 3. Chapter III has a corresponding table for GC–CI/MS.

Appendix 1 of chapters II and III contains the specific ions of the most important groups of phosphorus-containing CWC-related chemicals for GC–EI/MS and GC–CI/MS, respectively.

AMDIS screening should also be used for all samples analysed with GC–EI/MS. Vendor-specific deconvolution software might be useful instead of AMDIS.

Table 3. Specific ions for chemicals not detected with phosphorus, nitrogen and sulphur selective detectors.

Chemical	Specific ions for screening (<i>m/z</i>)
PFIB	181, 200, 93
Chloropicrin	82, 117, 119, 121
Pinacolyl alcohol	57, 45, 87, 102
Methyl derivative of benzilic acid	183, 105, 77
TMS derivative of benzilic acid	255, 329
TMS derivative of BZ	255, 110
DMT derivatives of Lewisite 1 and Lewisite 2	229, 290
DMT derivative of Arsenic trichloride	229, 107, 121
1,3-butylthiol derivative of Lewisite 1	164, 204, 314
1,3-butylthiol derivative of Lewisite 2	203, 229, 286

Note! PFIB, HCN and phosgene analysed without derivatisation may be screened using the head space or split injection technique.

7.3 Identification

The final identification must always be approved by an experienced analyst.

The identification considerations for GC–TOF analysis are the same as given in 7.3 of chapters II and III for GC–EI/MS and GC–CI/MS, respectively.

8. Method evaluation

8.1 Area of applicability

The area of applicability for GC–TOF analysis is the same as given in 8.1 of chapters II and III for GC–EI/MS and GC–CI/MS, respectively. One major difference is that selective ion monitoring (SIM) cannot be conducted at the instrumental level. Instead of SIM, extracted ion chromatograms (EICs) are very useful due to the high resolution, which

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allows the mass chromatogram window to be narrower, e.g., 0.1 Da, reducing background and giving a better signal-to-noise ratio.

8.2 Advantages/Disadvantages

The advantages and disadvantages of GC–TOF analysis are generally the same as given in 8.2 of chapters II and III for GC–EI/MS and GC–CI/MS, respectively.

One extra advantage is the possibility of using accurate masses for calculating the elemental compositions of molecules and fragments.

One specific disadvantage is the low dynamic range.

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Chapter II.

Liquid chromatography–atmospheric pressure ionisation/ mass spectrometry

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1. Scope

This chapter describes the use of liquid chromatography – atmospheric pressure ionisation mass spectrometry (LC–API/MS) in the screening and identification of degradation products of chemical warfare agents in aqueous samples or in aqueous extracts of solid samples. LC–API/MS is also applicable to some intact chemical warfare agents. The procedures described are generally suitable for identifying chemicals at concentrations above 1 mg/kg or 1 mg/l and may be applicable at lower concentrations. The most widely used atmospheric pressure ionisation methods are electrospray ionisation (LC–ESI/MS) and atmospheric pressure chemical ionisation (LC–APCI/MS). A detailed review of the application of these techniques to CWC-related analytes has been published.¹

2. Summary of the method

LC–ESI/MS and LC–APCI/MS for analysis at parts-per-million levels are performed using quadrupole or ion trap instruments operating in full scanning mode, or using time-of-flight instruments. Samples are most commonly analysed using a reversed phase (C18 or C8) LC column.^{2–4} Column dimensions are left to personal preference, but 150 x 2.1 mm columns packed with 3 or 5 µm particles are broadly applicable. Some laboratories use capillary LC combined with ESI,^{5–6} but this is experimentally more demanding. Both ESI and APCI are broadly applicable to CWC-related chemicals, e.g. degradation products of nerve agents and vesicants, although some analytes will be more easily detected and identified using one of these ionisation methods. ESI is used for thermally labile and highly polar ionic analytes such as saxitoxin and high molecular mass materials such as ricin.

A recent modification has been hydrophilic interaction liquid chromatography (HILIC) for improved separation of highly polar chemicals. HILIC is performed using mobile phases with high organic content (usually acetonitrile) and is highly compatible with ESI.

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Analytes elute from the column in the order opposite to that in reverse phase LC; analytes poorly retained in reverse phase LC are strongly retained in HILIC.^{7–9}

Before analysis of samples, a quality control run (QC test, see below) is carried out to verify correct instrument performance. Screening of relevant chemicals is performed by selecting ions characteristic of chemical classes or individual chemicals (extracted ion chromatograms, EICs) from the total ion chromatograms (TICs). Identification of relevant chemicals is made by comparison of the measured spectra to those in relevant libraries, if available, or to spectra of authentic reference chemicals, or by interpretation of spectra.

3. Safety aspects and storing

LC–API/MS analysis does not require any specific safety precautions. Samples to be analysed usually contain low concentrations of intact chemical warfare agents or their less toxic breakdown products. However, some components of the mobile phases used are hazardous. Normal laboratory safety measures such as protective gloves and eyewear should be used.

4. Instrumentation

4.1 LC–API/MS instrument

Any modern LC–API/MS instrument can be used. It should have a scan-speed capacity of at least 1 scan/s and cover a mass range of at least m/z 50–400.

4.2 LC columns

A 100–250 mm long C18 or C8 column (1.0–4.6 mm i.d.) packed with 3 or 5 μm particles is commonly used.

4.3 Reagents and chemicals

The reagents and chemicals needed for LC–API/MS analysis are given in Table 1.

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Table 1. Chemicals and reagents needed for LC–API/MS.

Chemical/reagent	Concentration	Purity
Mobile phase		
Water		18M Ω or LC–MS grade
Methanol		$\geq 99.9\%$ LC–MS grade
Acetonitrile		$\geq 99.9\%$ LC–MS grade
Ammonium formate (or acetate)	0.01–0.05 M	$\geq 99\%$ MS grade
Formic acid	0.05–0.5%	puriss p.a. $\sim 98\%$
Trifluoroacetic acid	0.02–0.1%	$\geq 99\%$ HPLC grade
QC test mix in water (Dstl)		
Methylphosphonic acid (CAS 993-13-5)	10 $\mu\text{g/ml}$	$\geq 98\%$
N-Methyldiethanolamine (CAS 105-59-9)	10 $\mu\text{g/ml}$	$\geq 99\%$
Ethyl methylphosphonate (CAS 1832-53-7)	10 $\mu\text{g/ml}$	98%
Thiodiglycol (CAS 111-48-8)	10 $\mu\text{g/ml}$	$\geq 99\%$
Pinacolyl methylphosphonate (CAS 616-52-4)	10 $\mu\text{g/ml}$	97%
QC test mix in water (VERIFIN)		
Butyl dihydrogen phosphate (CAS 85391-12-4)	13 $\mu\text{g/ml}$	p.a. grade or better
Butylphosphonic acid (CAS 3321-64-0)	17 $\mu\text{g/ml}$	p.a. grade or better
Methionine (CAS 7005-18-7)	15 $\mu\text{g/ml}$	p.a. grade or better
2-(Methylpropylamino)ethanol (CAS 2893-45-0)	6 $\mu\text{g/ml}$	p.a. grade or better
Dibutyl hydrogen phosphate (CAS 107-66-4)	20 $\mu\text{g/ml}$	p.a. grade or better
Tributyl phosphate (CAS 126-73-8)	10 $\mu\text{g/ml}$	p.a. grade or better

Note: all purities are as quoted for Sigma-Aldrich products.

5. Experimental

5.1 Recommended LC conditions

LC conditions depend on the dimensions and type of LC column. Recommended LC conditions as used by Dstl and VERIFIN are given in Table 2.

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Table 2. Recommended LC conditions for LC–API/MS.

LC parameter	Recommended conditions	Comments
Dstl method		
Column	150 x 2.0 or 2.1 mm C18; 3 or 5 µm particle size	
Injection volume	5–25 µl	10 µl normally used Manual syringe or auto-sampler
Mobile phase flow	0.2 ml/min	
Mobile phase	A: 0.02 M ammonium formate/water B: 0.02 M ammonium formate/methanol	This mobile phase is generally applicable but others may be used to obtain different selectivity
Mobile phase gradient	5% B (0–5 min) to 90% B (15–20 min)	May be extended to 95% B or longer hold time at end of run used
VERIFIN method		
Column	150 x 4.6 mm C18; 5 µm particle size	
Injection volume	20 µl	Manual syringe or auto-sampler
Mobile phase flow	1 ml/min	
Mobile phase	A: 0.02 M ammonium acetate/water; pH 8.6 (adjusted with NH ₄ OH) B: methanol + NH ₄ OH (same amount as in A)	This mobile phase is generally applicable but others may be used to obtain different selectivity
Mobile phase gradient	5% B (0–5 min) to 100% B (5 min)	

Conditions may be modified depending on the sample type and other constituents of the sample. For example, samples containing low polarity chemicals may require a longer gradient or a less retentive column, e.g. C8. Some separations of relevant chemicals from interfering background chemicals may require use of alternative mobile phase compositions. The mobile phases in Table 2 are applicable to both positive and negative ion API/MS.

If hydrophilic interaction liquid chromatography (HILIC) is used, the mobile phase has a high organic content (usually acetonitrile). Chromatographic conditions are usually isocratic or employ a slow gradient with increasing aqueous content in the mobile phase.^{7–9}

5.2 Recommended MS conditions

Recommended MS conditions, which are instrument dependent, are given in Table 3. The conditions given are suitable for both ESI and APCI.

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Table 3. Recommended MS conditions for LC–API/MS.

MS parameter	Conditions	Comments
Thermo instrument (DSTI)		
ESI Spray voltage	3–5 kV	
APCI Vaporiser temperature	350–450 °C	
APCI Corona current	4–6 µA	
Sheath gas	35–50 arb. units*	Nitrogen
Auxiliary or sweep gas*	5–20 arb. units*	Nitrogen
Source CID	5–25 V	Higher voltages will give more fragmentation
Scan range	<i>m/z</i> from ≤50 to ≥400	
Scan time	1 s	Scan time in ion traps is usually set by automatic gain control
Polarity	+ve or -ve ion	- ve ion scans are selective for acidic chemicals
Micromass instrument (VERIFIN)		
Corona voltage	3.5 kV (+ve ion) 2.5 kV (-ve ion)	
Cone voltage	25 V (+), 30V (-)	
APCI vaporiser (probe) temperature	450 °C	
Drying gas	300 l/h	Nitrogen
Sheath gas	50 l/h	Nitrogen
Scan range	<i>m/z</i> 60–600	
Scan time	2 s	

* as used on Thermo instruments; nomenclature and values will differ for other types.

All MS source parameters (including voltages not included in Table 3) are optimised during instrument tuning (see below). Parameters that may be optimised will depend on the instrument type; refer to the manufacturer's instructions.

5.3 Recommended tuning and mass calibration of the MS

The instrument is tuned and calibrated according to the manufacturer's instructions. Automatic tuning and calibration procedures are used if available. Full tuning and calibration is carried out following instrument servicing. Additional mass calibration is performed if indicated by the QC test. MS source conditions are optimised by infusing a solution of one of the QC test components (e.g. pinacolyl methylphosphonate) into an appropriate flow and composition of mobile phase. Parameters are adjusted, usually automatically, to maximise response to a selected ion, e.g. *m/z* 181 for pinacolyl methylphosphonate in positive ion mode or *m/z* 179 in negative ion mode. Separate optimisations are performed for positive and negative ion modes. Optimisation should be performed in this way before commencing analysis of a batch of samples.

6. Quality control

6.1 Instrument performance test

The reliable identification of CWC-related chemicals requires that the data is correct. Therefore, it is essential to continuously run quality control tests to confirm that the instrument works correctly and the performance of the instrument remains stable during the sample analysis.

The performance of an LC–API/MS instrument is controlled by performing a QC test run, in which the chromatographic and mass spectrometric behaviour of test compounds is confirmed to be acceptable. The QC test must be run at the beginning and end of each sample batch and at least once daily.

6.1.1 QC test used by Dstl

The instrument is accepted for use if the MS and LC data produced from each test meet the following requirements.

Signal-to-noise (S/N) ratio for pinacolyl methylphosphonate in the total ion chromatogram (TIC) is $\geq 25:1$.

Peak asymmetry for pinacolyl methylphosphonate should be ≤ 1.8 . The evaluation is made by taking measurements from a print out of the TIC.

All other test chemicals in the QC test mix (Table 1) should give peaks with $S/N > 10:1$.

No peaks with $S/N > 5:1$, including any that are not due to QC test chemicals, should be present in the total ion chromatogram between 0 min and the retention time of pinacolyl methylphosphonate.

The m/z values of the major ions of each test chemical are correct and no additional ions above 5% relative abundance are present.

6.1.2 QC test used by VERIFIN

The instrument is accepted for use if the MS and LC data produced from each test meet the following requirements.

Requirement for LC (150 x 4.6 mm; 5 μ m column)

Retention times for the test mixture are in the following windows:

1. Butyl dihydrogen phosphate	1.9 ± 0.3 min
2. Butylphosphonic acid	2.2 ± 0.3 min
3. Methionine	2.4 ± 0.3 min
4. 2-(Methylpropylamino)ethanol	3.9 ± 0.3 min
5. Dibutyl hydrogen phosphate	6.1 ± 0.3 min
6. Tributyl phosphate	7.3 ± 0.3 min

Abundances of peaks (3, 4, 5 and 6) in TIC for positive mode and of peaks (1, 2, 3 and 5) in TIC for negative mode are $> 1.0 \cdot 10^5$.

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Requirement for MS calibration

For positive mode, calibration is accepted when the ions m/z 99, 131, 155, 187, 211, 243, 267 and 533 in the spectrum of tributyl phosphate deviate <0.4 mass units from integer values.

For negative mode, calibration is accepted when the ions m/z 79, 153 and 209 in the spectrum of dibutyl hydrogen phosphate deviate < 0.4 mass units from integer values.

Requirement for MS resolution

Resolution is accepted for positive ion mode when the following requirements are met:

The abundance of the ^{34}S isotopic ion (m/z 152) in the spectrum of methionine is in the window $5.1\% \pm 0.6\%$ units, and ^{34}S isotopic ions m/z 106 and 135 are visible.

The abundance of the ^{13}C isotopic ion (m/z 268) in the spectrum of tributyl phosphate is in the window $13.4\% \pm 2.5\%$ units, and ^{13}C isotopic ions m/z 156, 212, and 534 are visible.

Resolution is accepted for negative ion mode when the following requirements are met:

The abundance of the ^{34}S isotopic ion (m/z 150) in the spectrum of methionine is in the window $5.1\% \pm 0.6\%$ units.

The abundance of the ^{13}C isotopic ion (m/z 210) in the spectrum of dibutyl hydrogen phosphate is in the window $9.0\% \pm 1.5\%$ units.

6.2 Sensitivity test

The sensitivity of the LC–API/MS instrument is estimated with the QC test sample containing each test chemical.

6.3 Blank test

A solvent (water) blank run must be performed to eliminate the possibility of contamination of equipment (e.g. contamination from the syringe, auto-sampler or memory effect of the equipment). As the QC test mix may contain relevant chemicals, it is particularly important to run water blank following a QC test run. This blank should show no evidence of any peaks due to QC test chemicals. If such peaks are evident the blank run, or runs, should be repeated until no peaks are detected.

If CWC-related chemicals are found in the blank chromatogram, the blank test is repeated after elimination of the contamination source. All contamination from glassware, solvents, standard solution (or authentic reference solution), syringe, auto-sampler or columns must be eliminated before sample analysis is started.

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7. Analysis of samples

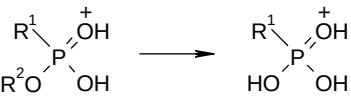
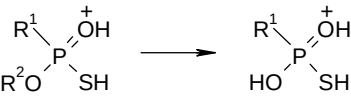
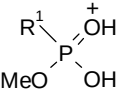
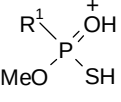
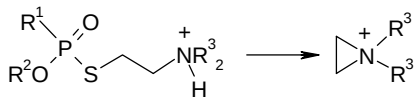
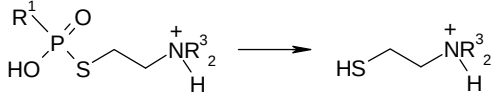
7.1 Sequence of analysis

A QC test and water blank are always run at the beginning of an analysis batch. When both tests have been accepted samples can be analysed. Generally, 10 or 20 µl of the sample is injected. Water blank should be run before injection of a new sample in the sequence. Blanks need not be run between multiple injections of the same sample.

7.2 Screening

Extracted ion chromatograms (EICs) may be used to locate chemicals or classes of chemicals which may be masked by interfering chemicals in the total ion chromatogram (TIC). Specific ions are given in Tables 4 and 5.

Table 4. Specific ions for screening using API(+)/MS.

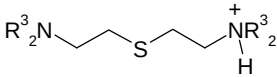
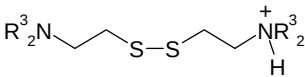
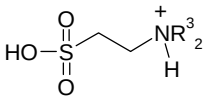
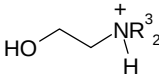
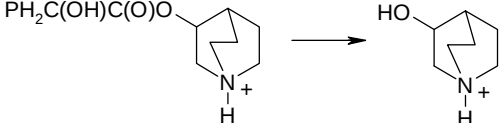
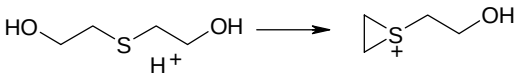
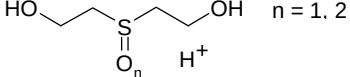
Group	Abundant ions		
	R ¹ = Me	R ¹ = Et	R ¹ = Pr
Alkyl alkylphosphonates R ² ≠ Me	97	111	125
			
Alkyl alkylthiophosphonates R ² ≠ Me*	113	127	141
			
Methyl alkylphosphonates	111	125	139
			
O-Methyl alkylthiophosphonates*	127	141	155
			
O-Alkyl S-2-(dialkylamino)ethyl alkylphosphonothiolates (V agents)*	72	100	128
			
S-2-(Dialkylamino)ethyl alkylphosphonothiolates	106	134	162
			

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Group	Abundant ions		
	R ¹ = Me	R ¹ = Et	R ¹ = Pr
Bis(dialkylaminoethyl) sulfides*	177	233	289
			
Bis(dialkylaminoethyl)disulfides*	209	265	321
			
2-(Dialkylamino)ethanesulfonic acids*	154	182	210
			
2-Dialkylaminoethanols	90	118	146
			
BZ, 3-quinuclidinol	338, 128		
			
Thiodiglycol*	123, 105**		
			
Thiodiglycol sulfoxide, thiodiglycol sulfone*	139 (n = 1) 155 (n = 2)		
			

* For sulfur-containing analytes note also ³⁴S ions.

** Also useful for screening for related compounds containing an HOCH₂CH₂SCH₂CH₂ moiety

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Table 5. Specific ions for screening using APCI(-)/MS.

Group	Abundant ions		
	R ¹ = Me	R ¹ = Et	R ¹ = Pr
Alkyl alkylphosphonates R ² ≠ Me (& alkylphosphonates)	95	109	123
$\begin{array}{c} \text{R}^1 \\ \diagup \\ \text{P}=\text{O} \\ \diagdown \\ \text{R}^2\text{O} \quad \text{O}^- \end{array} \longrightarrow \begin{array}{c} \text{R}^1 \\ \diagup \\ \text{P}=\text{O} \\ \diagdown \\ \text{HO} \quad \text{O}^- \end{array}$			
O-Alkyl alkylthiophosphonates R ² ≠ Me	111	125	139
$\begin{array}{c} \text{R}^1 \\ \diagup \\ \text{P}=\text{S} \\ \diagdown \\ \text{R}^2\text{O} \quad \text{O}^- \end{array} \longrightarrow \begin{array}{c} \text{R}^1 \\ \diagup \\ \text{P}=\text{S} \\ \diagdown \\ \text{HO} \quad \text{O}^- \end{array}$			
Methyl alkylphosphonates	109	123	137
$\begin{array}{c} \text{R}^1 \\ \diagup \\ \text{P}=\text{O} \\ \diagdown \\ \text{MeO} \quad \text{O}^- \end{array}$			
O-Methyl alkylthiophosphonates*	125	139	153
$\begin{array}{c} \text{R}^1 \\ \diagup \\ \text{P}=\text{S} \\ \diagdown \\ \text{MeO} \quad \text{O}^- \end{array}$			
S-2-(Dialkylamino)ethyl alkylphosphonothiolates*†			
$\begin{array}{c} \text{R}^1 \\ \diagup \\ \text{P}=\text{O} \\ \diagdown \\ \text{O}^- \end{array} \longrightarrow \begin{array}{c} \text{R}^1 \\ \diagup \\ \text{P}=\text{O} \\ \diagdown \\ \text{O}^- \text{S-CH}_3 \end{array}$	125	139	153
$\begin{array}{c} \text{R}^1 \\ \diagup \\ \text{P}=\text{O} \\ \diagdown \\ \text{O}^- \end{array} \longrightarrow \begin{array}{c} \text{R}^1 \\ \diagup \\ \text{P}=\text{S} \\ \diagdown \\ \text{O}^- \text{OH} \end{array}$	111	125	139
$\begin{array}{c} \text{R}^1 \\ \diagup \\ \text{P}=\text{O} \\ \diagdown \\ \text{O}^- \end{array} \longrightarrow \begin{array}{c} \text{R}^1 \\ \diagup \\ \text{P}=\text{O} \\ \diagdown \\ \text{O}^- \end{array}$	77	91	105
2-(Dialkylamino)ethanesulfonic acids*	152	180	208
$\begin{array}{c} \text{O} \\ \parallel \\ \text{O}^- \text{S} \\ \parallel \\ \text{O} \end{array} \longrightarrow \begin{array}{c} \text{O} \\ \parallel \\ \text{O}^- \text{S} \\ \parallel \\ \text{O} \end{array}$			
BZ		227, 183	
$\text{BZ} \longrightarrow \begin{array}{c} \text{Ph} \quad \text{OH} \\ \diagdown \quad \diagup \\ \text{C} \\ \diagup \quad \diagdown \\ \text{Ph} \quad \text{CO}_2^- \end{array} \quad \begin{array}{c} \text{Ph} \quad \text{O}^- \\ \diagdown \quad \diagup \\ \text{C} \\ \diagup \quad \diagdown \\ \text{Ph} \quad \text{H} \end{array}$			

* For sulfur-containing analytes note also ³⁴S ions.

† These fragment ions are not necessarily formed in all instruments

7.3 Identification

The final identification must always be approved by an experienced analyst.

Identification can be based on interpretation of a spectrum or comparison with a library spectrum if available. Confirmation of identity is preferably by comparison with a spectrum from an authentic reference chemical. Commercial (or OPCW) databases of LC–API/MS spectra of CWC-related chemicals are not currently available, although many such spectra have been published in scientific journals.

Recorded spectra (unknown and reference) must extend to at least 60 mass units above the molecular mass of the proposed molecule. No ions having a mass-to-charge ratio above the molecular mass (excluding isotope and adduct ions) must be present. Ions having intensities above 10% relative abundance and not belonging to the reference spectrum must not be present. The S/N ratio of a peak in the TIC or EIC must be >5. If identification is based on spectral interpretation, a spectrum of a closely related chemical together with a fragmentation pathway must be provided.

Note. If extra peaks are present in the spectrum, use extracted ion chromatograms by selecting the impurity ion together with two ions present in the spectrum of the authentic reference chemical. If the ion profiles are different, try to extract as pure a spectrum as possible of the chemical of interest. If the target chemical co-elutes exactly with a background chemical, a column possessing different selectivity or a different mobile phase should be used to separate the chemicals.

8. Evaluation of method

8.1 Area of applicability

LC–API/MS methods can be applied to many degradation products of chemical warfare agents and to some intact agents that are more or less stable in aqueous solution. Sensitivity varies depending on the type of chemical and may be better with either ESI or APCI methods. Negative ion methods are selective for acidic chemicals.

8.2 Advantages / disadvantages

Advantages of the method are speed of analysis of aqueous samples or extracts, removing the need for evaporation and derivatisation, which are required for GC-MS methods. API usually provides confirmation of the molecular mass and, where fragment ions are present, these generally result from simple neutral losses.

ESI spectra may be more complex than APCI spectra owing to the greater tendency to form adduct ions with solvent and metal ions such as Na⁺. Where there is a substantial chemical background, ion suppression can occur, which is also more prevalent with ESI.

Disadvantages for identification are the low content of structural information in API spectra, the limited 'fingerprint' compared to GC–EI/MS, plus the lack of a spectral database. For confirmation of identification MS/MS is recommended, and is essential for differentiating isomers other than by known retention time^{10–11}. For trace analysis LC–

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MS/MS is recommended rather than single-stage LC/MS, because the reduction in noise usually gives substantially enhanced S/N ratios.

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1. Scope

This chapter describes the use of liquid chromatography–tandem mass spectrometry (LC–MS/MS) for the identification of non-volatile chemical warfare agents and their polar or semi-polar degradation products; i.e., chemicals related to the Chemical Weapons Convention, CWC. The scope of analysable compounds by LC–MS/MS is similar to the scope of analysable compounds by LC–MS [see Chapter “Liquid chromatography–atmospheric pressure ionisation/mass spectrometry” (3-C-II)]. In general, the procedure is applied to identify chemicals present at above 1 mg/kg concentration level. However, the procedure can also be used to identify relevant chemicals down to low µg/kg concentration levels.

Note 1! Not all LC–MS/MS instruments are described in this version (e.g., Q–TOF and Orbitrap instruments are not covered). Others will probably be added in future editions of this Chapter.

Note 2! Some features typical of most instruments from different manufacturers are referred to throughout the text. These are the features that the authors are aware of at the time of the writing of this Chapter. It is possible that similar features are also available on other manufacturers’ systems.

2. Summary of the method

LC–MS/MS measurements are carried out with modern LC–MS/MS instruments like triple quadrupole (QqQ) or quadrupole–time-of-flight (Q–TOF; in this Chapter also QqTOF for clarity) or ion trap (linear or 3D) mass analysers operating in MS/MS mode. Samples are analysed with a LC–MS/MS instrument equipped with a similar LC column as would be used for a LC–MS experiment. Before each sample batch, a quality control run (QC test, see in Chapter 3-C-II) is carried out to verify the correct performance of the instrument (LC column and ionisation performance). A quality control run, especially for a MS/MS experiment, is necessary to verify proper calibration of the equipment with respect to frequency for isolation and excitation (see paragraph 6).

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LC–MS/MS techniques have two applications. The first case is where LC–MS has already been carried out and the chemical of interest is identified. In this case, LC–MS/MS is used for confirmation by the generation of characteristic product ions to strengthen the identification. Indeed, LC–MS with a soft ionisation source (APCI or ESI) often gives relatively few diagnostic ions and in most cases only the protonated or deprotonated ion and/or the sodium adduct are present.

Note 3! *In-source CID could be used to fragment the quasi-molecular ions. This method, however, would fragment all co-eluting chemicals and would result in a mixture of fragment ions. In-source CID can be used for screening purposes [see Chapter “Screening” (1-B-I), paragraph 5.2].*

In the second case, LC–MS/MS could be used for screening and identification of relevant chemicals by selecting characteristic product ions of chemical families or individual chemicals (extracted ion chromatograms, EICs) from the total ion chromatograms (TICs) obtained by automated MS/MS experiments done on the more intense ions (this option is currently available on modern instruments, especially ion traps). In the case of a QqQ or a QqTOF, multiple reaction monitoring (MRM) and/or precursor ion scan experiments could be performed. These types of experiments (see paragraph 4.1.1.3) allow to selection of only characteristic product ions beyond the collision cell and identification of the precursor ion.

Identification of relevant chemicals is made by comparison of the measured LC–MS/MS spectra to authentic reference chemical spectra or by spectrum interpretation.

3. Safety aspects

LC–MS/MS analyses do not require any specific safety precautions. Samples for analysis normally contain degradation products and are automatically injected. In some cases, a sample could contain small concentrations of toxic chemicals, if the hydrolysis does not go to completion. Normal laboratory safety measures, such as wearing protective gloves and goggles during sample handling in manual injections, should be used.

Hot parts (>250 °C) of the instrument (e.g., some metallic interfaces in the external ionisation source) must be taken into account during operation and maintenance of instrument.

4. Instrumentation

4.1 LC–MS/MS instrument

Any modern LC–MS/MS instrument, capable of a scan speed of at least 1 scan/sec and covering a mass range of at least m/z 50–600, can be used. This type of instrument allows a good quality of spectra and enables new applications. In practice, the mass range covered by the first quadrupole for QqQ or by the ion trap begins from m/z 50 due to the presence of water, nitrogen and carbon dioxide in the sample.

Triple quadrupole and 3D or 2D ion trap systems are unit resolution instruments over the entire mass range, and correspond to a resolving power of 1000 (for QqQ) or 5000 (for ion trap) for the majority of systems and a resolving power of 2000 (for QqQ) and of 10,000 (ITMS) for the most efficient instruments.

Two types of MS/MS instruments are available. First are the sequential MS/MS instruments, where the MS/MS experiment is performed in the same place but at different times (e.g., ion traps). Second are the spatial MS/MS instruments, where the MS/MS experiment is performed at different site within the instrument (e.g., triple quadrupole and Q–TOF instruments).

4.1.1 Characteristics of the instruments

4.1.1.1 Triple quadrupole mass spectrometer (QqQ)

The most common tandem-in-space mass spectrometers are triple quadrupole instruments that consist of two quadrupole mass analyzers separated by an RF-only (radio frequency) quadrupole collision cell.

The main characteristics of a triple quadrupole instrument are the following:

Transmission (impact on sensitivity and stability of signals)

The ratio of the number of ions having identical properties entering and exiting the quadrupole is generally 1:20. Triple quadrupole instruments are generally more sensitive for detecting target analytes, even in complex mixtures, and are therefore better as quantitation than as screening tools.

Scan speed (impact on sampling rate and quantitation results)

Today, quadrupole mass spectrometer systems routinely scan at m/z 10,000 per second. Therefore, a comprehensive scan, even one of short duration (e.g., an LC peak with the width of 1 s) would nevertheless capture each ion ten times or more in each second. Nevertheless, triple quadrupoles are relatively simple, easy-to-use, highly utilitarian instruments that offer a variety of interfaces at a relatively low cost, which explains their popularity.

4.1.1.2 Ion trap mass spectrometer (ITMS)

The main characteristics of ion traps are the following:

Analytical cycle time (impact on resolution, sampling rate, quantitation results)

In an ion trap, the analytical cycle time consists of the following major events:

- ICC (Ion Charge Control, Bruker) or AGC (Automatic Gain Control, Thermo Scientific) pre-scan in order to ensure that the trap is always filled with the optimum number of charges for mass accuracy and sensitivity;
- Ion injection (usually the rate-determining step);
- Isolation and activation of the precursor ion within the trap;
- Scan of the ions out of the trap (mass analysis).

The ion injection time (filling the trap) accounts for the largest proportion of the cycle time, particularly for low-level components. The use of an ICC/AGC pre-scan allows a maximum injection time to ensure optimum sensitivity and mass accuracy for a widest range of sample concentrations.

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Low mass cut-off (has an impact on the structural information obtained by MS/MS and on sensitivity)

The low mass cut-off (LMCO) corresponds to the smallest m/z value of ions with a stable trajectory in the ion trap. Usually, only the product ions with an $m/z > 1/4$ (3D ion trap) or $m/z > 1/5$ (linear trap used with resonant excitation) of the precursor ion have a stable trajectory and are trapped and detected.

Note! Linear ion trap instruments from Thermo Scientific can be used for an experiment called “Pulsed Q Collision Induced Dissociation (PQD)”. This type of experiment can allow measurement of ions with $m/z > 1/6.5$.¹

Different MS/MS experiments can be run depending of the type of MS/MS instrument.

4.1.2 Tandem mass spectrometry scan modes for spatial instruments²

Product Ion Scan (also called Fragment Ion Scan or Daughter Ion Scan)

Firstly, ions of a chosen m/z are selected on the first quadrupole (Q1). Secondly, these ions collide inside the collision cell and fragment. Thirdly, product ions resulting from the fragmentation of the selected precursor ions are analyzed by the third quadrupole (Q3).

For example, for analysis of diethyl methylphosphonate (DEMP):

Selection on Q1 of precursor ions m/z 153 ($[M+H]^+$), and then fragmentation of m/z 153 in the collision cell. The product ions m/z 125 (loss of $CH_2=CH_2$) and m/z 97 (losses of two $CH_2=CH_2$) are generated and detected in Q3.

Precursor Ion Scan (also called Parent Scan)

All ions generated from the sample are initially scanned using the first quadrupole (Q1 or Q). Then, these ions collide inside the collision cell and fragment. Finally, Q3 (in QqQ) or TOF (QqTOF) is focused on a selected m/z . All of the precursor ions that produce ions with the selected m/z through reaction or fragmentation are therefore detected.

For example, for analysis of alkyl methylphosphonates:

Selection on Q3 of product ions m/z 95 (in negative mode). Scanning on Q1 of all precursor ions; this generates product ions m/z 95. This scan mode enables detection of the presence of alkyl methylphosphonates in an unknown sample.

Neutral Loss Scan

In this mode, both the first quadrupole (Q1 or Q) and the third quadrupole Q3 (QqQ) or TOF (QqTOF) are scanned together, but with a constant mass offset between the two. Thus, for a mass difference a , when ions of mass m go through first quadrupole, detection occurs if these ions have produced fragment ions of mass $(m - a)$ when they leave the collision cell. This is a neutral loss scan, with the neutral having the mass a .

For example, for analysis of ethyl methylphosphonic acid (EMPA):

Precursor ion $[M-H]^-$ m/z 123 produces product ion m/z 95 by losing a $CH_2=CH_2$ alkene (a for this example is equal to 28 u).

Selected Reaction Monitoring (or Multiple Reaction Monitoring)

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For SRM, both the first quadrupole (Q1) and third quadrupole (Q3) are focused on selected masses. The ions selected by the first quadrupole (Q1) are only detected if they produce a given fragment, by a selected reaction (SRM) or by multiple reactions (MRM). The absence of scanning allows one to focus on the precursor and fragment ions over longer times, increasing the sensitivity as for Selected Ion Monitoring (MS mode), but this sensitivity is now associated with a high increase in selectivity. This method is often used to detect a selected compound in an elution timeframe during chromatography. The high selectivity corresponds to a low probability of detecting an interfering compound. This reduces the background noise and eliminates most interference due primarily to the matrix, thereby improving the specificity.

An analysis obtained by SRM or MRM on a triple quadrupole is more sensitive than a MS/MS analysis acquired on an ion trap. This scan mode is particularly dedicated to quantitative analyses or confirmation analyses when the compound was already identified by another technique or technology. It should be mentioned that, at present, the SRM or MRM mode is not accepted for Official OPCW Proficiency Tests dedicated to environmental samples.

The different scan modes above-mentioned are summarised in Figure 1.

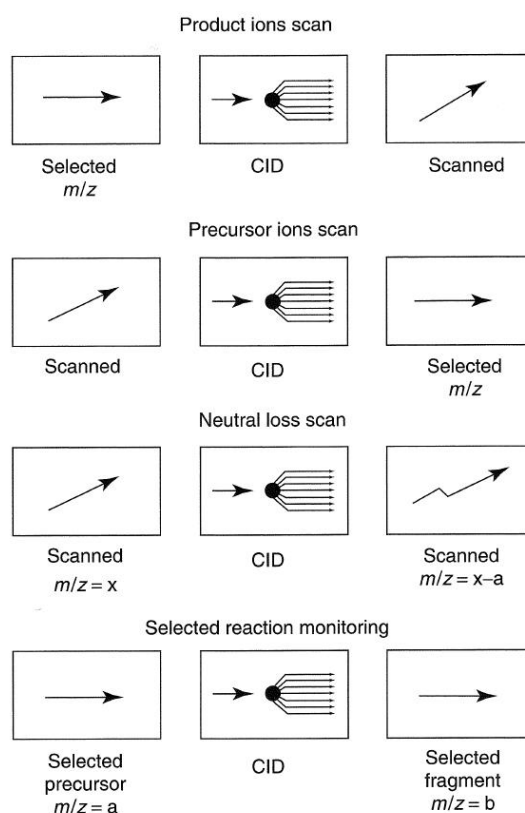


Figure 1. Possible MS/MS scan modes with a spatial tandem mass spectrometer (triple quadrupole).

4.1.2 Tandem mass spectrometry scan modes for temporal instruments

Product Ion Scan (also called Fragment Ion Scan or MS²)

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Firstly, ions of a chosen m/z are selected by ejecting all other ions from the ion trap. Secondly, these chosen ions are excited (supplementary RF voltage applied) and collision induced dissociations (CID) with the He buffer gas inside the ion trap are produced, leading to product ions. Thirdly, product ions are analyzed in the ion trap by rescanning all of the mass range.

Multistage tandem mass spectrometry experiments (MS^n)

Because all of the steps of the MS/MS experiment are performed at the same location, these steps can be run again with product ions obtained by MS/MS. After the MS/MS step, product ions of a chosen m/z are selected by ejecting all other product ions from the ion trap. The product ions at the chosen m/z are excited and fragmented to product ions (MS^3 experiment). This process can be repeated until the sensitivity of the product ion is sufficient, leading to an MS^n spectrum (where n represents the number of isolation-fragmentation cycles). It should be mentioned that, at present, the MS^n experiment is not accepted for Official OPCW Proficiency tests dedicated to environmental samples.

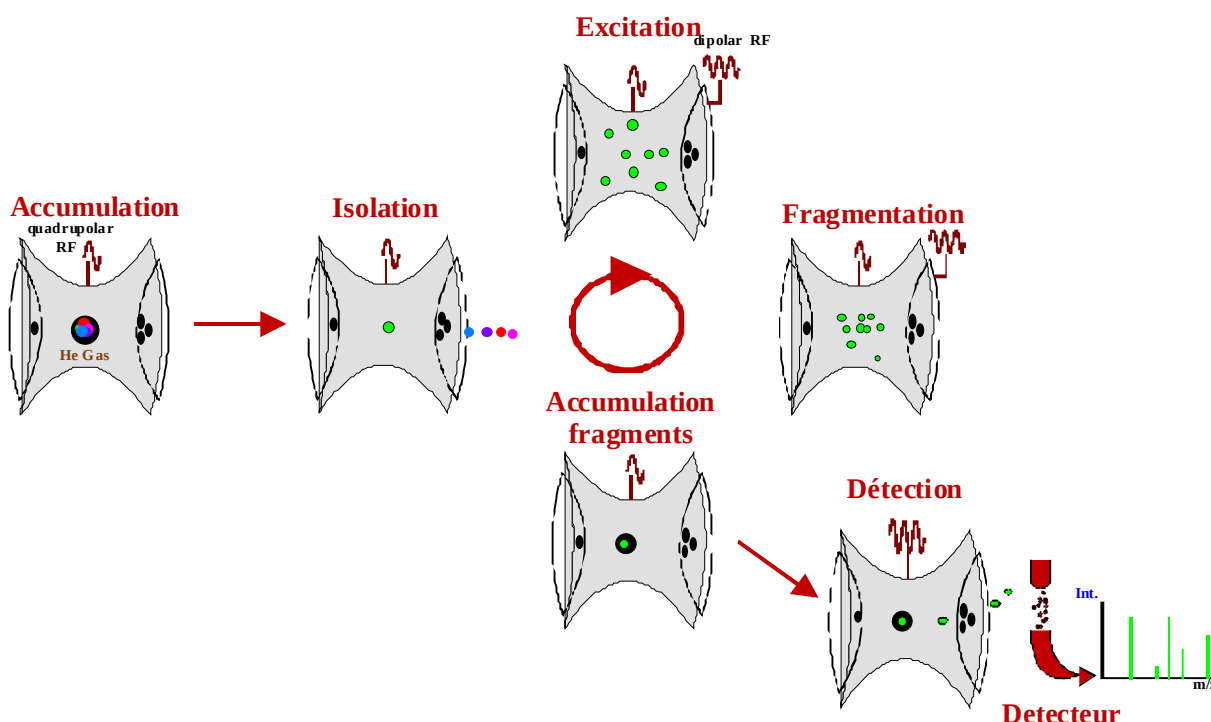


Figure 2. Scheme of a MS^n experiment on a 3D ion trap.³

4.2 LC columns

See Section 3, Part C, Chapter I.

4.3 Reagents

The reagents and chemicals needed for LC–MS/MS analyses are discussed in the following subparagraphs (from 4.3.1 to 4.3.3).

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4.3.1 QC test mixtures

The QC test in MS/MS can be performed on two compounds present in the LC–MS QC test mixture (see chapter II for the LC–MS QC test composition). The two compounds (for example, ethyl methylphosphonate and pinacolyl methylphosphonate) need to be diluted to adapt the concentration to the MS/MS experiment (for example, at 1 µg/ml in deionised water).

4.3.2 Gas

The only supplementary gas required for LC–MS/MS, compared to LC–MS, is the collision gas. In the case of ion traps, the cooling gas is also used as collision gas.

Table 1. Example of gases required for use in LC–QqQ/MS.

Collision Gas	Type of instrument	Purity	Codes for purity*
Helium	Ion trap	≥ 99.9995%	e.g. N55 or 5.5
Argon	Triple quadrupole	≥ 99.9999%	e.g. N60 or 6.0
Nitrogen	Triple quadrupole	≥ 99.9999%	e.g. N60 or 6.0

* Coding of the gas purities varies from country to country

4.3.3 Solvents and buffers

See Section 3, Part C chapters I and II.

5. Experimental

5.1 Recommended LC conditions

See Section 3 Part C chapter I.

5.2 Recommended MS conditions

See Section 3 Part C chapter II, for general conditions. Remember that the MS conditions must be optimized (soft desolvation conditions to avoid in-source fragmentation) in order to have the best intensity of the precursor ion, which will be dissociated by MS/MS experiments.

5.3 Recommended MS/MS conditions

For each compound, several parameters must be optimized to obtain the best response in MS/MS. If possible, this optimization occurs by infusion with an ESI interface. This means that the compound is available to study.

A solution of the compound of interest has to be prepared in a solvent mixture relatively close to the solvent composition at the elution time of that compound; i.e., generally, a mixture deionized water/acetonitrile 50/50 (v/v) with 0.2% formic acid (or acetic acid) or a

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mixture deionized water/methanol with 0.02 M ammonium formate can be used. These solvent proportions correspond to the elution solvent composition of many compounds and provide the best surface tension to obtain a stable spray. The formic acid facilitates ionisation in positive mode.

5.3.1 Triple quadrupole

5.3.1.1 Optimization of MS/MS parameters

The reference chemical must be diluted to obtain a concentration equivalent to 1 µg/ml (or generally between 0.5 and 20 µg/ml).

1. The first step consists of determination of the precursor ion, based on the obtained MS spectra obtained both in positive and negative ion (if applicable) modes (see Section 3 Part C chapter II). Occasionally, the protonated molecule $[M+H]^+$ or the deprotonated molecule $[M-H]^-$ are not detected or are present at very low abundance, so that only a fragment ion (as is the case for thiodiglycol) or an adduct ion, often with sodium, $[M+Na]^+$, is observed. The ionisation mode that gives the best response in terms of intensity is used. The selection of the $[M+Na]^+$ as precursor ion is not recommended because the sodium ion retains the charge and the fragments of the chemical are not detected as neutrals.
2. The solution is then directly infused into the mass spectrometer by the ESI interface to obtain a MS/MS spectrum of the compound under standard conditions (see **Error! Reference source not found.**).
3. The MS/MS experiment could also be used to determine the transitions of interest from precursor ion → product ion for MRM experiments
4. To obtain an optimal MS/MS fragmentation, the collision energy of a given transition for a given capillary value must be optimized. This parameter optimization is done manually (by testing different values and register spectra) or automatically by the software (for example, use of the “MS/MS Breakdown” option), which provides the searched value.
5. Capillary and collision energy values can be more precisely optimized again by injecting the compound directly into the MS source using the diverter valve while the mobile phase is at its elution composition and as it passes at a determined LC flow through the system.
6. Once the other steps are realized, the mass peak width on Q1 and Q3 and the scan time can be optimized to obtain the best MS/MS spectra of the analyzed compound.

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Table 2. Default settings for MS/MS optimization by ESI interface.

MS/MS parameter	Values
Collision Gas pressure	1.5 mTorr
Collision energy	20 eV
Detector EDR	Maximal value given by the last monthly autotune
Scan mode	Centroid
Scan width	0.7 u
Scan time	0.7 s
Mass range	m/z 50 to m/z (precursor ion + 10) on Q3 for product ion scan m/z of (product ion -10) to m/z 500 on Q1 for precursor ion scan m/z of precursor ion on Q1; m/z of product ion on Q3 for MRM
Mass peak width on Q1	Calibrated (0.7 u)
Mass peak width on Q3	Calibrated (0.7 u)

5.3.1.2 MS/MS analyses

MS/MS, SRM or MRM methods must be adapted for each compound studied in accordance with its transitions.

For an identified compound

The precursor ion ($[M+H]^+$ or $[M-H]^-$) is selected on Q1 and, depending on the method (MS/MS or SRM/MRM), Q3 is focused on a mass range or product ion(s). For a product ion scan, the precursor ion must be visible, and should have intensity around 20–33% of the more intense product ion. Optimized capillary and collision energy values, specific to the compound, are defined during the condition optimisation (see paragraph 5.3.1.1).

For an unknown and / or unavailable compound

The precursor ion (typically $[M+H]^+$ or $[M-H]^-$) is selected on Q1 and, depending on the method (MS/MS or SRM / MRM), Q3 is focused on a mass range or product ion(s). For a product ion scan, the precursor ion must be visible, and should have an intensity around 20–33% of the more intense product ion. The capillary value is then defined in the analytical standard method for soft in-source desolvation. Collision energy, normally characteristic of each transition, is first set by default at 20 eV and then manually optimized on the real sample to obtain the best spectra possible in terms of fragmentation and sensitivity.

5.3.2 Ion trap

Two possibilities could be used to perform LC/MS/MS analyses on the ion trap.

For an unknown and/or an unavailable compound

First, automated MS/MS experiments can be utilised with generic collision induced dissociation (CID) conditions on the main precursor ions. This means that during the LC peak elution, MS and MS/MS experiments on the more intensive ion are alternatively performed. It is possible to exclude a list of ions (for example, known interferents) or to

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include a list of preferential ions for MS/MS scan (for example, m/z of certain alkyl methylphosphonates). Several precursor ions during the same LC peak can be isolated and excited for fragmentation depending of their intensity, if option allows exclusion of the MS/MS spectrum registration of an already fragmented ion. The excitation voltage (tickle, if a resonant excitation) should be chosen in the middle of the excitation values usually used. If possible, a ramp of excitation values can be added in order to ensure the fragmentation of main precursor ions; for example, an excitation voltage of $0.6 V_{pp}$ with a ramp (30–200%) means that an averaged MS/MS spectrum is obtained per scan, corresponding to the combined spectra from 0.2 to $1.2 V_{pp}$ excitation voltage.

The generic conditions for automated LC–MS/MS experiment in positive and negative ionisation modes are described in Table 3.

Table 3. Default settings for ion trap MS/MS experiment in positive or negative ionisation mode.

Parameter	Value
Ion Charge control	10,000
Capillary exit offset in positive mode	40 V
Capillary exit offset in negative mode	-40 V
m/z of selected ion	Auto
Isolation window	2 u
Excitation voltage (V_{pp})	$0.6 V_{pp}$
Ramp (30–200%)	On
Active exclusion after 5 spectra	On
Accumulation time	500 ms
Mass range	m/z 50–600
Average	3
Trap drive (Low Mass Cut Off)	25

These two generic LC–MS/MS methods allow structural information on several compounds present in the sample to be obtained rapidly in a single run. The option of active exclusion after several spectra (for example 5) allows MS/MS information to be obtained for co-eluted compounds. The fragmentation conditions are not optimized per compound. This run can be considered as a screening method more than a MS/MS run for identification. After identification of the chemical of interest, the MS/MS conditions for this chemical need to be optimized.

For an identified compound

The reference chemical must be diluted to a concentration equivalent to 1 $\mu\text{g/ml}$ (or generally between 0.5 and 20 $\mu\text{g/ml}$).

The solution is then directly infused into the mass spectrometer by the ESI interface to detect the compound first in Full Scan mode and secondly in MS/MS mode.

1. To obtain an optimal fragmentation in MS/MS, the excitation voltage value must be optimized with a standard isolation window (2 u is a good compromise for unit resolution instrument). This parameter optimization is done manually by testing different values and register spectra. The optimized value corresponds to the value for which the precursor ion is presented in the MS/MS spectrum at 20–33% of the

main product ion. A ramp of excitation voltage can be used, depending on whether an important difference in fragmentation is noted for close values.

2. After optimization of the excitation voltage step, the best isolation window is then defined. In the case of isotopic pattern or co-eluted compounds in the real sample, testing an isolation window at 1 u can be helpful for selecting only the monoisotopic peak of your chemical of interest. This optimization is a compromise between sensitivity (which increases with a large excitation window) and specificity (no selection of ions that do not correspond to the precursor ion fragments).
3. Verify your optimized conditions by analysing the reference chemical by LC–MS/MS.
4. Perform the LC–MS/MS analysis of your system blank and then for your real sample with the optimized fragmentation conditions.

5.4 Recommended tuning and mass calibration of the MS

Use the recommended tuning method of the instrument. Automatic tuning and calibration procedures should be used, if available. Tuning and calibration in MS and MS/MS modes should be carried out once a month, or more often if indicated by the QC test. Printouts of tuning and calibration reports should be archived for further reference.

6. Quality control

6.1 Instrument performance test in LC–MS

The reliable identification of CWC-related compounds requires an operating system. Therefore, running quality control tests continuously is essential to confirm that the instrument is working correctly and the performance of the instrument remains stable during the sample analysis.

The performance of a LC–MS instrument is controlled by carrying out a QC test run (see Section 3, Part C, Chapter II, where two examples of QC test mixture are described), in which the mass spectrometric behaviour of test compounds and the adsorption behaviour of the column are confirmed as acceptable.

Since performance depends on the nature of the analyzed molecules, compounds chosen as references for test solution(s) should be close to those likely to be searched (CWC-related chemicals).

The use of this kind of QC test mixture imposes the absolute need to verify the absence of a QC compound in the system by performing a blank sample run prior to the real sample run.

6.2 Sensitivity test in LC–MS/MS or LC–MRM

For a QC test run in LC–MS/MS or LC–MRM, one or several products of the LC–MS QC test mixture can be used. The paragraphs 6.2.1 and 6.2.2 described the criteria of the LC–MS/MS or LC–MRM QC test run at DGA CBRN Defence, France.

6.2.1 Triple quadrupole

The LC–MS system is assumed to be operating if the LC–MRM data resulting from the test meet the following requirements:

1. Retention time of EMPA and PMPA peaks must be included in the range $t_{\text{Ref}} \pm 0.3$ min, with t_{Ref} the reference retention time established as first run on the new column in the same experimental conditions.
2. Signal-to-noise (S/N) ratio assessed in the specific transition of EMPA (m/z 123→95) and in the specific transition of PMPA (m/z 179→95) must be respectively better than 100:1 and 400:1 (with calibrated mass peak width; i.e., 0.7 u). The evaluation is done manually on the raw data (TIC) according to the formula $S/N = 2H / h$.

6.2.2 Ion trap

The LC–MS system is assumed to be operating if the LC–MS/MS data resulting from the test meet the following requirements:

1. Retention time of EMPA peak must be included in the range $t_{\text{Ref}} \pm 0.3$ min, with t_{Ref} the reference retention time established as first run on the new column in the same experimental conditions.
2. Signal-to-noise (S/N) ratio for EMPA peak must be better than 100:1. The evaluation is done manually on the raw data (TIC) according to the formula $S/N = 2H / h$.
3. The intensity of precursor ion $[M-H]^+$ of EMPA (m/z 123) compared to the intensity of product ion (m/z 95) must lie within the range of 20–40%.
4. No extra ions are present that have m/z greater or less than m/z of $[M-H]^+$ and intensity above 10%.
5. The m/z measured value of the product ion (m/z 95) must be equivalent to the theoretical m/z value ± 0.4 u.

6.3 Blank test

A solvent blank run must be carried out to avoid cross contamination of the equipment (e.g., contamination from the injection needle or a memory effect of the column) and thus to ensure the reliability of the samples analysis by eliminating all risks of false positives. Running a solvent blank is therefore recommended before each sample analysis by injecting 10 μL of the same solvent (or solvent mixture) used in sample preparation. If CWC-related chemicals are found in the blank chromatogram, the solvent blank test should be repeated until the total elimination of the contamination source is confirmed. All contamination coming from glassware, solvents, standard solution (or authentic reference solution), syringe, injector system, or columns must be eliminated before sample analysis is started.

A system blank also needs to be run if a sample preparation (concentration, matrix extraction, etc.) is performed before the LC–MS/MS analysis. In this case, the solvent blank is the system blank.

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7 Analysis of samples

7.1. Sequence of analysis

At the beginning of an analysis batch, a QC test (always) and a blank test are run. When both tests have been accepted, samples can be analysed. Generally, 10 µl of the sample is injected.

7.2 Screening

A list of specific ions for screening of the most important groups of CWC-related chemicals is given in Table 4.

Table 4. Diagnostic ions for identification of CWC degradation products.

Groups of CWC chemicals	Observed product ions (m/z)	
	Pos	Neg
O-alkyl S-[2-(Dimethylamino)ethyl] alkylphosphonothiolates or S-[2-(dimethylamino)ethyl] alkylphosphonothiolates	72	–
Methylphosphonic acid	–	79
Ethylphosphonic acid	–	79
Propylphosphonic acid	–	79
Isopropylphosphonic acid	–	79
Alkyl methylphosphonates or dialkyl methylphosphonates	97	95
Dialkyl dimethylpyrophosphonates	97	–
Alkyl methylphosphonofluoridates	99	–
O-Alkyl S-[2-(diethylamino)ethyl] alkylphosphonothiolates or S-[2-(diethylamino)ethyl] alkylphosphonothiolates	100	–
Hydrolysis products of mustards	105	–
Alkyl ethylphosphonates or dialkyl ethylphosphonates	111	109
Dialkyl diethylpyrophosphonates	111	–
Alkyl propylphosphonates or dialkyl propylphosphonates	125	123
Alkyl isopropylphosphonates or dialkyl isopropylphosphonates	125	123
Dialkyl dipropylpyrophosphonates or dialkyl diisopropylpyrophosphonates or dialkyl isopropylpropylpyrophosphonates	125	–
Alkyl dimethylphosphoramidates	126	124
Dialkyl dimethylphosphoramidates	126	–
O-Alkyl S-[2-(diisopropylamino)ethyl] alkylphosphonothiolates or S-[2-(diisopropylamino)ethyl] alkylphosphonothiolates	128	–
O-Alkyl S-[2-(dipropylamino)ethyl] alkylphosphonothiolates or S-[2-(dipropylamino)ethyl] alkylphosphonothiolates	128	–
Bis[2-(diisopropylamino)ethyl]sulfide or bis[2-(diisopropylamino)ethyl]disulfide	128	–
O-Alkyl N,N-dimethylphosphoroamidocyanidate	135	–

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Groups of CWC chemicals	Observed product ions (m/z)	
	Pos	Neg
Tetramethyl phosphorodiamidic cyanide or hexamethylphosphoro-triamide or alkyl tetramethyl-phosphorodiamidates	135	–
alkyl tetramethylphosphorodiamidates or alkyl phosphoric tetramethylphosphorodiamidic anhydrides or alkyl dimethylphosphoramidic tetramethylphosphoamidic anhydrides or dialkyl phosphoric tetramethylphosphorodiamidic anhydrides	153	–
Bis(ethyl dimethylphosphoramidic) anhydride	154	–

The structures of some characteristic product ions for several compounds are described in Table 5. These specific ions can be obtained after MS/MS experiments as well as during LC–MS analysis if in-source fragmentations are favoured.

Table 5. Product ions observed during analysis of chemical warfare agents by LC–ESI/MS/MS.⁴

m/z	Possible structures	m/z	Possible structures
97	$\begin{array}{c} \text{OH} \\ \\ \text{CH}_3-\text{P}^+-\text{OH} \\ \\ \text{OH} \end{array}$	127	$\begin{array}{c} \text{OH} \\ \\ \text{CH}_3\text{CH}_2\text{O}-\text{P}^+-\text{OH} \\ \\ \text{OH} \end{array}$
99	$\begin{array}{c} \text{OH} \\ \\ \text{HO}-\text{P}^+-\text{OH} \\ \\ \text{OH} \end{array} \quad \text{or} \quad \begin{array}{c} \text{OH} \\ \\ \text{F}-\text{P}^+-\text{OH} \\ \\ \text{CH}_3 \end{array}$	128	$\begin{array}{c} (\text{CH}_3)_2\text{CH}-\text{N}^+-\text{CH}_2 \\ \quad \quad \\ (\text{CH}_3)_2\text{CH}-\text{CH}_2 \end{array}$
101	$\begin{array}{c} \text{OH} \\ \\ \text{F}-\text{P}^+-\text{OH} \\ \\ \text{OH} \end{array}$	135	$\begin{array}{c} \text{O} \\ \\ (\text{CH}_3)_2\text{N}-\text{P}^+ \\ \\ \text{N}(\text{CH}_3)_2 \end{array}$
105	$\begin{array}{c} \text{H} \\ \\ \text{S}^+ \\ / \quad \backslash \\ \text{H}_2\text{C} \quad \text{CH}_2 \\ \quad \quad \\ \text{H}_2\text{C} \quad \text{CH}_2 \\ \backslash \quad / \\ \text{O} \end{array}$	153	$\begin{array}{c} \text{OH} \\ \\ (\text{CH}_3)_2\text{N}-\text{P}^+-\text{OH} \\ \\ \text{N}(\text{CH}_3)_2 \end{array}$
111	$\begin{array}{c} \text{OH} \\ \\ \text{CH}_3\text{CH}_2-\text{P}^+-\text{OH} \\ \\ \text{OH} \end{array}$	154	$\begin{array}{c} \text{OH} \\ \\ \text{CH}_3\text{CH}_2\text{O}-\text{P}^+-\text{OH} \\ \\ \text{N}(\text{CH}_3)_2 \end{array}$
126	$\begin{array}{c} \text{OH} \\ \\ (\text{CH}_3)_2\text{N}-\text{P}^+-\text{OH} \\ \\ \text{OH} \end{array}$	155	$\begin{array}{c} \text{OH} \\ \\ \text{CH}_3\text{CH}_2\text{O}-\text{P}^+-\text{OH} \\ \\ \text{OCH}_2\text{CH}_3 \end{array}$

The loss of an alkene (C_nH_{2n}) and the presence of a product ion at *m/z* 97, 111, 125 or 125 (in positive mode, see

Figure 3) or at *m/z* 95, 109, 123 or 123 (in negative mode, see Figure 4) is characteristic of an alkyl methylphosphonate, an alkyl ethylphosphonate, an alkyl propylphosphonate or alkyl isopropylphosphonate, respectively.

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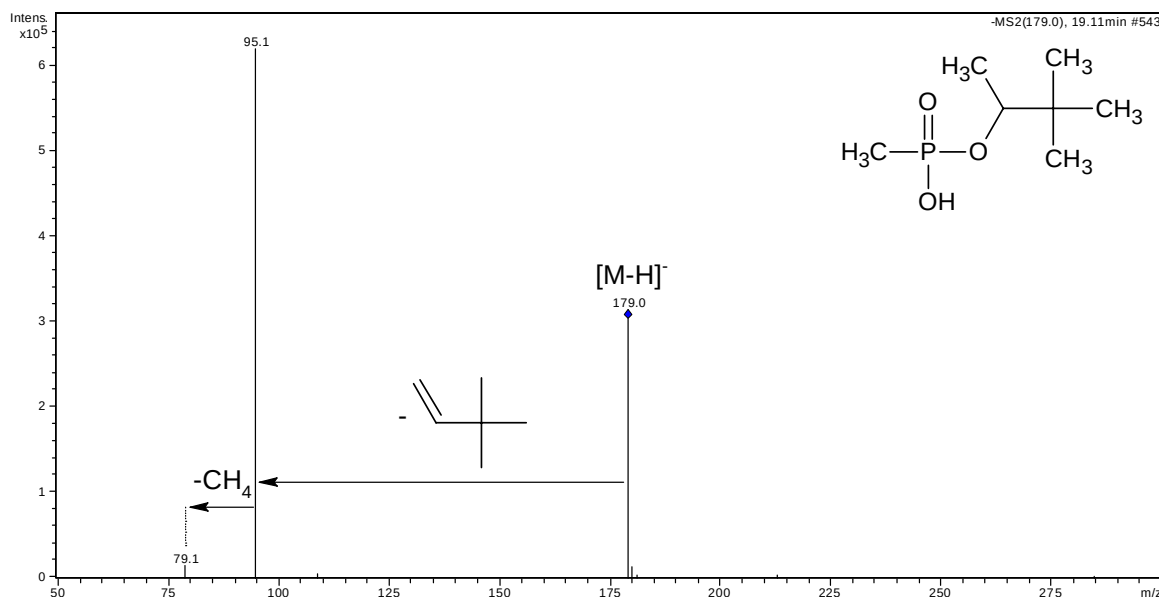


Figure 3. Low energy CID spectrum of: pinacolyl methylphosphonate in a negative ionisation mode

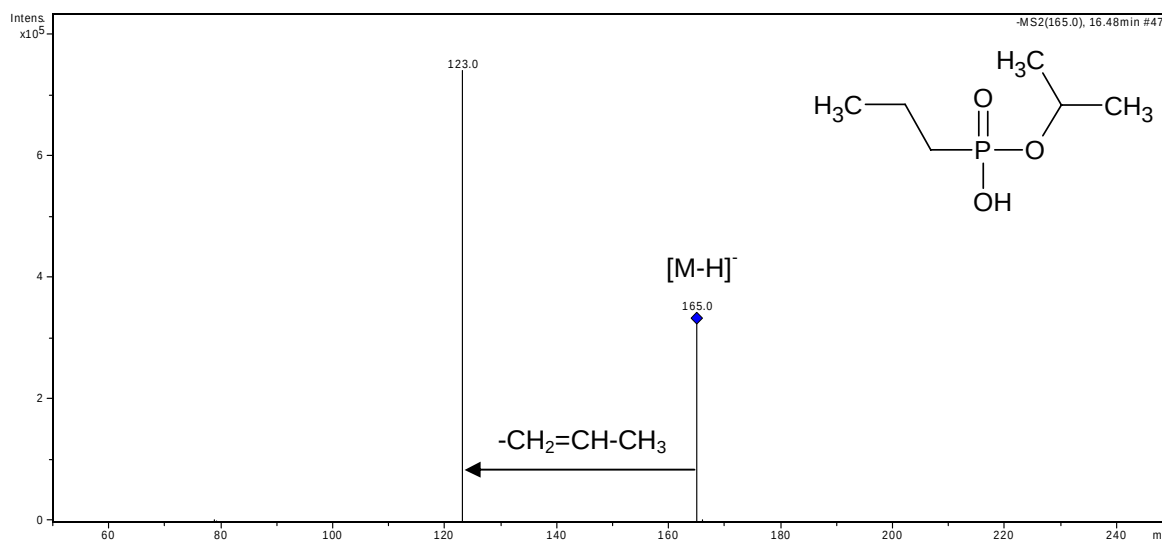


Figure 4. Low energy CID spectrum of: isopropyl propylphosphate in a negative ionisation mode.

In negative ionisation mode, the fragment ion obtained for alkylphosphonic acid is m/z 79 by the loss of an alkane (C_nH_{2n+2}), as presented

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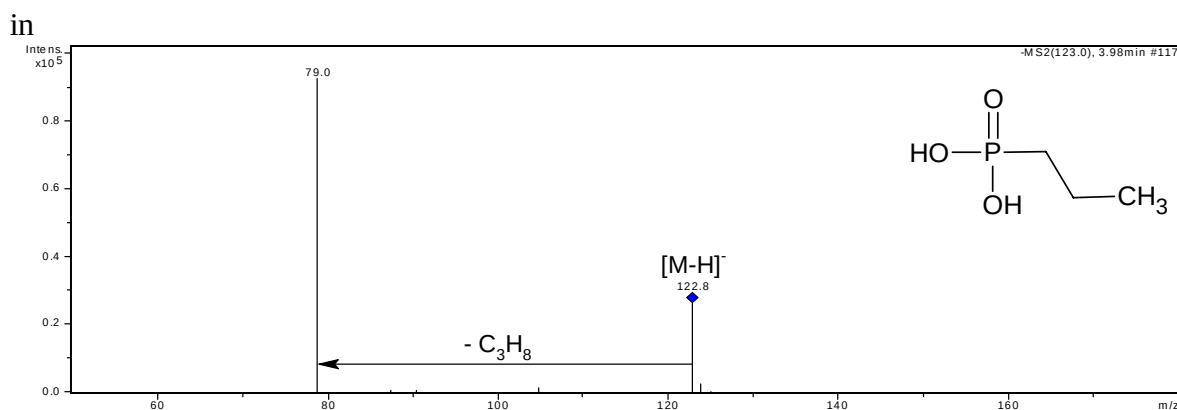


Figure 5. This is also the case for alkyl alkylphosphonates after the first alkene loss related to the O-alkyl chain (see

Figure 3 with the loss of CH_4).

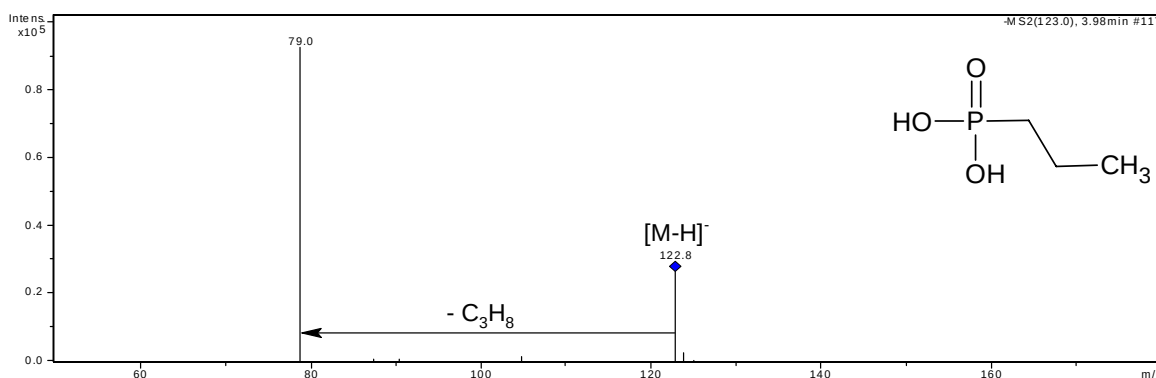


Figure 5. Low energy CID spectrum of: propylphosphonic acid in a negative ionisation mode

In positive ionisation mode, the losses of two alkenes (different or similar) and the presence of a product ion at m/z 97, 111, 125 or 125 is characteristic of, respectively, a dialkyl methylphosphonate, a dialkyl ethylphosphonate, a dialkyl propylphosphonate or dialkyl isopropylphosphonate.

The differentiation between the isopropyl and propyl group linked to the phosphorus (which is a requirement for identification) could be done by MS/MS on triple quadrupole, but not in ion trap.⁵ An MS^3 experiment allows the differentiation on an ion trap.

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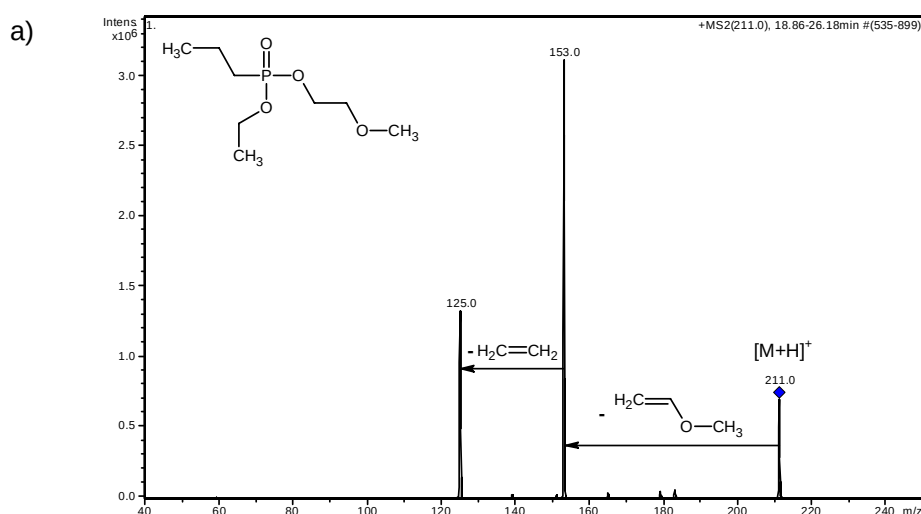
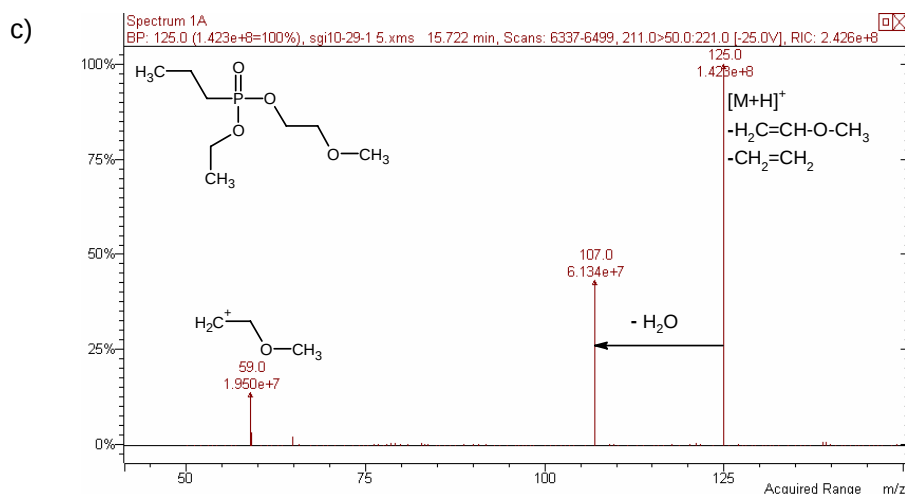
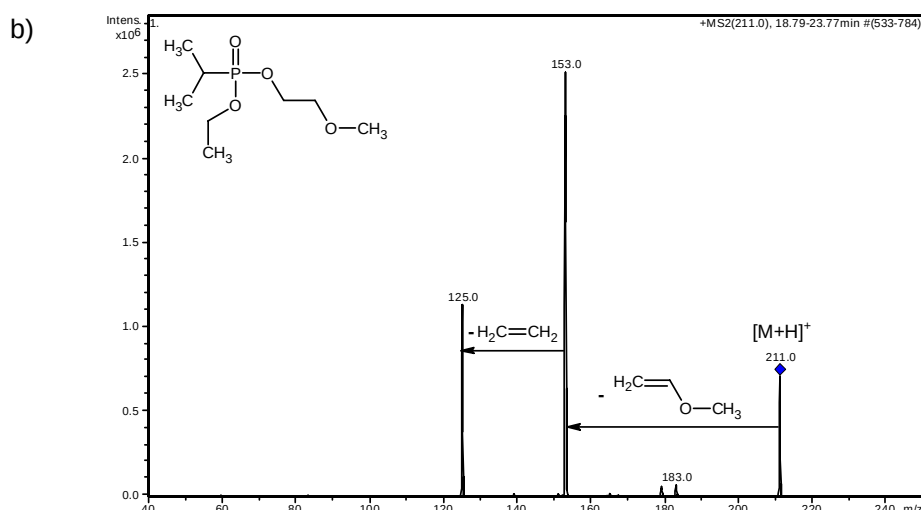


Figure 6. Low energy CID spectra obtained on an ion trap for a) ethyl 2-methoxyethyl isopropylphosphonate and b) ethyl 2-methoxyethyl propylphosphonate and on a triple quadrupole for c) ethyl 2-methoxyethyl isopropylphosphonate and d) ethyl 2-methoxyethyl propylphosphonate. *Figure continues on next page.*



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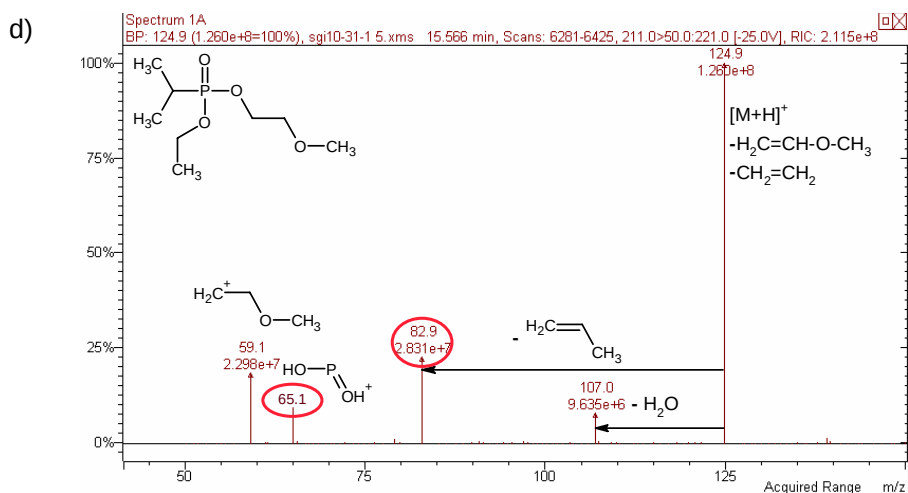


Figure 7. Low energy CID spectra obtained on an ion trap for a) ethyl 2-methoxyethyl isopropylphosphonate and b) ethyl 2-methoxyethyl propylphosphonate and on a triple quadrupole for c) ethyl 2-methoxyethyl isopropylphosphonate and d) ethyl 2-methoxyethyl propylphosphonate. *Figure continued from previous page.*

In the case of an ion trap, some observed fragment ions are obtained after water addition due to the residual water in the ion trap, as illustrated by the example shown in

Figure 8.

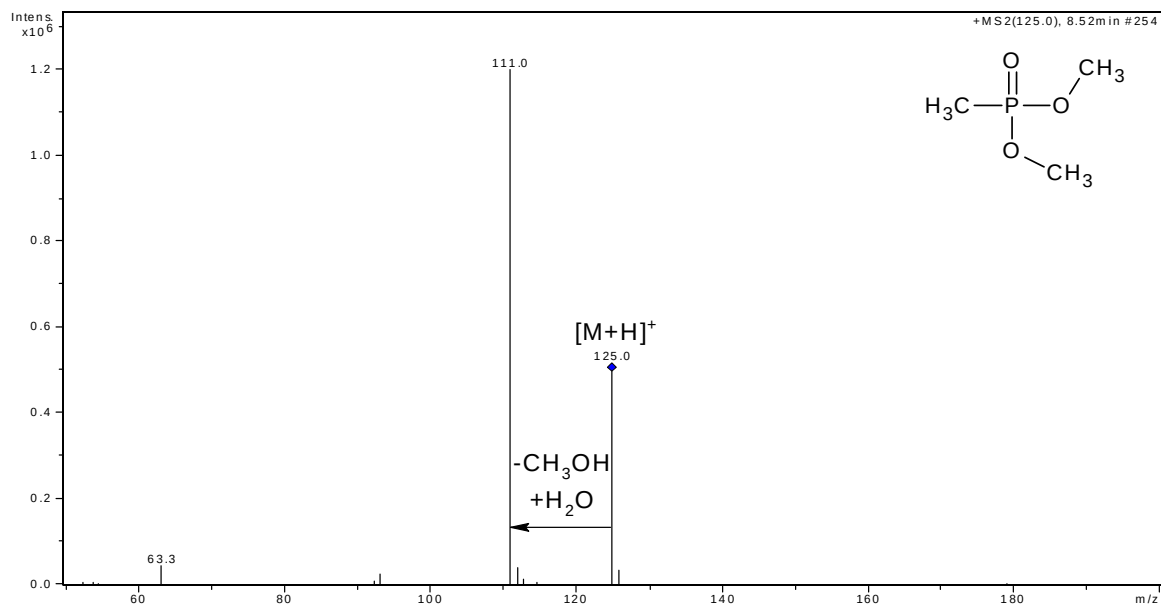


Figure 8. Low energy CID spectrum of: dimethyl methylphosphonate in positive ionisation mode in an ion trap.

Several product ions should be obtained during the fragmentation of the precursor ion.

Table 6 indicates the different observed product ions for some CWC-related compounds. As the relative percentage of the product ions is strongly dependent on the mass spectrometer, this was not mentioned.

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CWC chemicals^{2,6}	[M+H]⁺	Product ions (+)
Ethyl methylphosphonate	125	97
Isopropyl methylphosphonate	139	97
Cyclohexyl methylphosphonate	179	97
Pinacolyl methylphosphonate	181	97
Diethyl methylphosphonate	153	125, 97
Diisopropyl methylphosphonate	181	139, 97
Diethyl dimethylpyrophosphonate	231	203, 175, 97
Isopropyl methylphosphonofluoridate (Sarin, GB)	141	99
Cyclohexyl methylphosphonofluoridate (GF)	181	99
Pinacolyl methylphosphonofluoridate (Soman, GD)	183	99, 85
Bis(2-hydroxyethyl) sulfide (thiodiglycol)	123	105, 87, 77, 61
3,6-Dithiaoctan-1,8-diol	183	165, 137, 105
8-Chloro-6-oxa-3-thiaoctan-1-ol	185	167, 105
6-Oxa-3,9-dithiaundecan-1,11-diol	227	209, 181, 105
3,6,9-Trithiaundecan-1,11-diol	243	225, 197, 165, 137, 105
6-Oxa-3,9,12-trithiatetradecan-1,4-diol	287	269, 241, 209, 165, 137, 105
Ethyl ethylphosphonate	139	111
Isopropyl methyl ethylphosphonate	153	111
Diisopropyl ethylphosphonate	195	153, 111
Ethyl dimethylphosphoramidate	154	126
Diethyl dimethylphosphoramidate	182	154, 126
Ethyl isopropyl dimethylphosphoramidate	196	154, 126
Diisopropyl dimethylphosphoramidate	210	168, 126
S-[2-(Diisopropylamino)ethyl] methylphosphonothiolate	240	198, 162, 128
O-Ethyl S-[2-diisopropylamino)ethyl] methylphosphonothiolate (VX)	268	167, 128
O-Ethyl S-[5-(diisopropylamino)-3-thiapentyl] methylphosphonothiolate	328	227, 199, 167
O-Ethyl S-[8-(diisopropylamino)-3,6-dithiaoctyl] methylphosphonothiolate	388	259, 227, 199, 167, 162, 128
Tetramethyl phosphorodiamidic cyanide	162	135
O-Ethyl N,N-dimethylphosphoroamidocyanide (Tabun, GA)	163	135
Hexamethylphosphorotriamide	180	135
Ethyl tetramethylphosphorodiamidate	181	153, 135, 108
Ethyl phosphoric tetramethylphosphorodiamidic anhydride	261	233, 153, 135
Bis(tetramethylphosphorodiamidic) anhydride	287	242, 153, 135
Ethyl dimethylphosphoramidic tetramethylphosphoamidic anhydride	288	243, 215, 153
Diethyl phosphoric tetramethylphosphorodiamidic anhydride	289	261, 244, 233, 216, 188, 153, 135

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CWC chemicals ^{2,6}	[M+H] ⁺	Product ions (+)
Octamethyltetramidotriphosphoric acid (or isomer)	367	322, 215, 153
Bis(ethyl dimethylphosphoramidic) anhydride	289	261, 244, 233, 216, 188, 154, 126
Diethyl phosphoric ethyl dimethylphosphoramidic anhydride	290	262, 234, 206, 155, 127
Tris(2-chloroethyl)amine (HN3)	204	168, 142, 106, 70
N-Methyl-N,N-diethanolamine	120	102, 58
N-Ethyl-N,N-diethanolamine	134	116, 72
Triethanolamine	150	132, 114, 88, 70
2-(Diisopropylamino)ethyl ethyl sulfide	190	89
Bis[2-(diisopropylamino)ethyl] sulfide	289	188, 160, 145, 128
Bis[2-(diisopropylamino)ethyl] disulfide	321	220, 161, 160, 128

Several examples are presented in Figures 8–11:

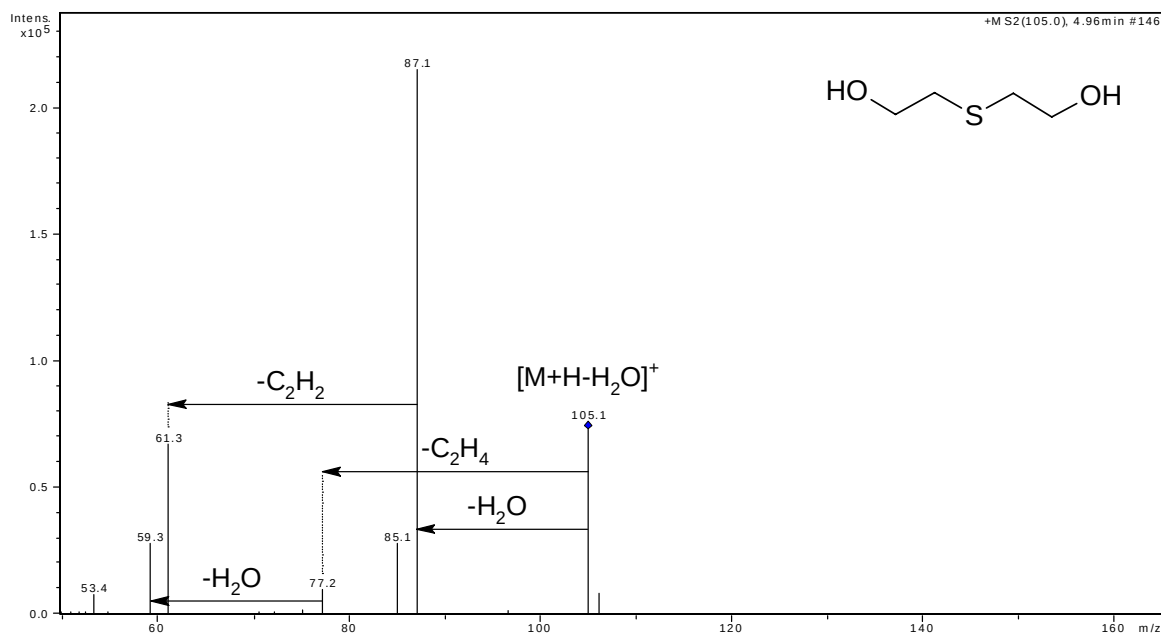


Figure 9. Low energy CID spectrum of thiodiglycol on the [M+H-H₂O]⁺ as precursor ion.

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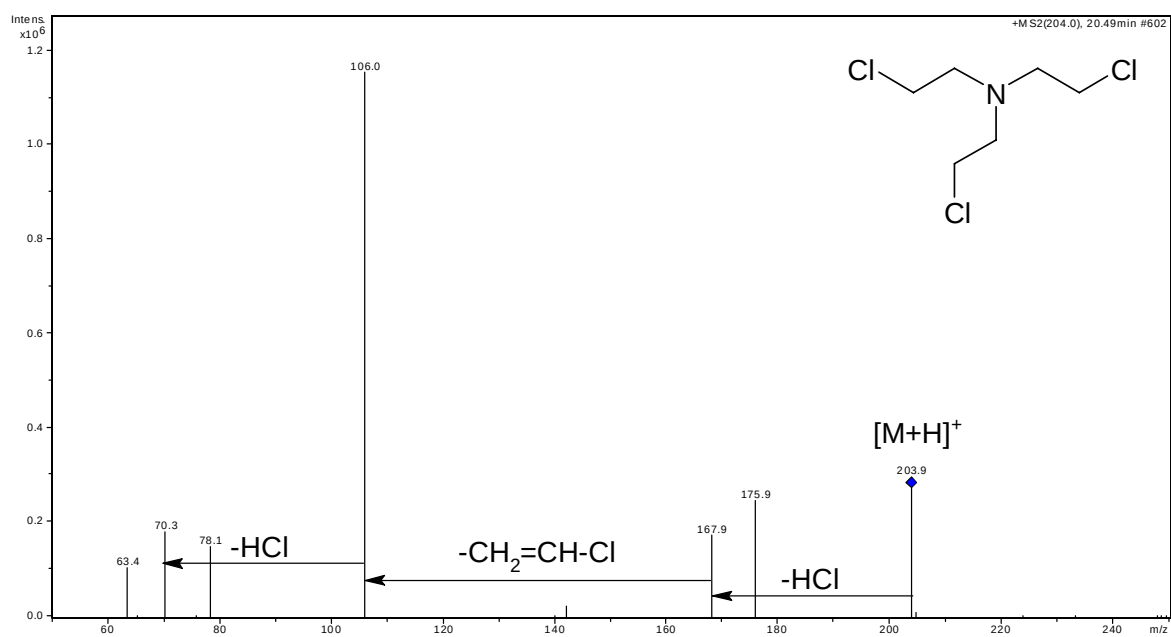


Figure 10. Low energy CID spectrum of tris(2-chloroethyl)amine (HN3) in positive ionisation mode.

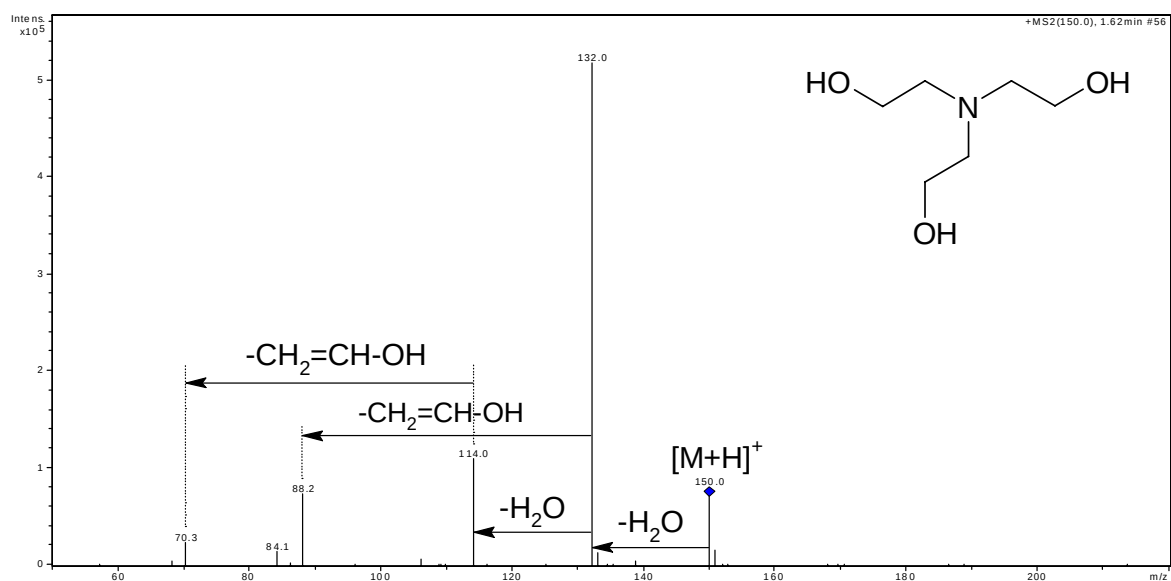


Figure 11. Low energy CID spectrum of triethanolamine in a positive ionisation mode.

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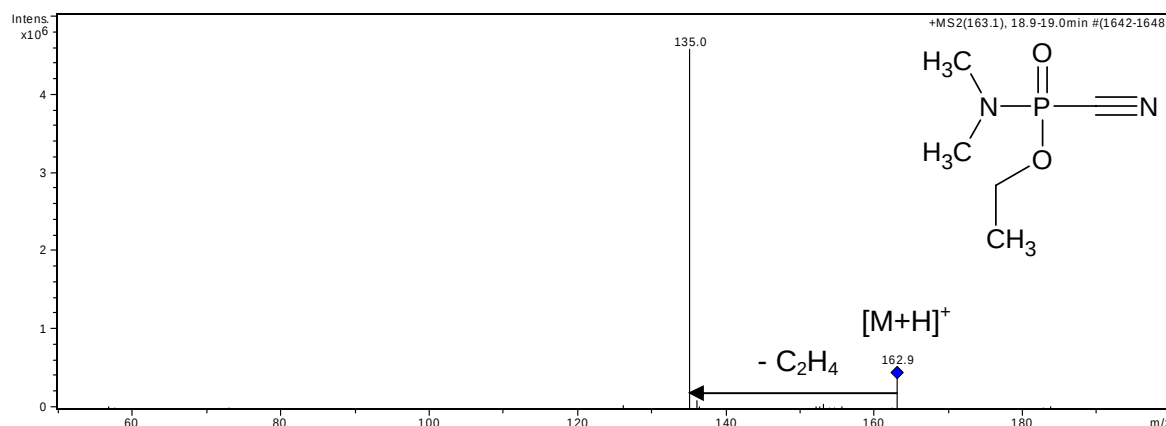


Figure 12. Low energy CID spectrum of tabun in positive ionisation mode.

7.3 Identification

The final identification must always be confirmed by an experienced analyst.

Identification can be based on comparison with authentic reference chemical. If a reference chemical or reference spectra are not available, the identification can be done by spectrum interpretation.

The recorded spectrum (reference and unknown) must be at least ten mass units above the molecular weight of the proposed molecule. No ions having mass to charge ratio above the molecular weight (excluding the isotope ions) must be present. Ions having intensities above 10% in relative scale and not belonging to the reference spectrum must not be present. The signal-to-noise ratio of a recorded TIC peak must have a value greater than five. If identification is based on spectrum interpretation, a spectrum of a closely related chemical together with a spectral interpretation and a fragmentation pathway must be provided.

Note! If extra peaks are present in the spectrum, use extracted ion chromatograms by selecting the impurity ion, together with two ions that belong to the spectrum of authentic reference chemical. If the ion profiles are different, try to extract as pure a spectrum as possible of the chemical of interest. If the target chemical elutes exactly together with a background chemical, a different column should be used to separate the chemicals.

8. Evaluation of methods

8.1 Area of applicability

An LC–MS/MS method can be applied to non-volatile, polar or semi-polar degradation products of chemical warfare agents, and some chemical warfare agents (VX, tabun). The method is suitable for analyzing and identifying any chemical at a level of 10 mg/kg in the full scan mode and at a level of 0.1 mg/kg in the selective ion monitoring mode (SIM mode, target analysis only, MRM mode).

8.2 Advantages/Disadvantages

The advantage of the method is the direct analysis of degradation products of chemical warfare agents. Thus, the sample preparation is simple and rapid due to the absence of a

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derivatization step. Furthermore, the LC–MS/MS method is complementary to GC–MS or NMR techniques.

The disadvantage in identification using LC–MS/MS is the absence of commercial or OCAD databases. In-house LC–MS/MS databases should be used, but the spectra are strongly dependent on the configuration of the instrument used. Few characteristic ions are produced in LC–MS/MS compared to GC–EI/MS.

9. References

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RECOMMENDED OPERATING PROCEDURES FOR CWC-RELATED ANALYSIS

Section 3. Analytical methods

Part C. LC-based techniques

Chapter III. Liquid chromatography–tandem mass spectrometry (LC–MS/MS)

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Chapter I.

Capillary electrophoresis (CE)

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1. Scope

This chapter describes the use of capillary electrophoresis (CE) with indirect ultraviolet (UV) detection for the separation and subsequent analysis of alkylphosphonic acids and related monoesters. Species are separated in CE by their electrophoretic mobilities. This method has many advantages, including low sample consumption and narrow peak widths for excellent analyte separation. Detection (indirect UV) is achieved by using a buffer system containing a background electrolyte (phenylphosphonic acid) and continuously monitoring a specific UV-absorbing background. The loss of signal indicates when non-chromophore-containing analytes have been detected. Generally, the procedure is used for qualitative identification of chemicals above 1 mg/l (ppm). However, the procedure can also be utilised for identifying relevant chemicals down to sub-ppm concentration levels.

2. Summary of the method

Alkylphosphonic acid degradation products of chemical warfare agents can be detected in aqueous samples and extracts. If necessary, aqueous samples are prepared by using a combination of On-Guard sample clean-up cartridges to remove chloride and/or sulfate interferences. Capillary electrophoresis with a reversed electro-osmotic flow (EOF) buffer system and indirect UV detection is used to separate and detect the alkylphosphonic acids. Before each sample batch, a quality control run (QC test, refer to Section 6.1) is carried out to verify the correct performance of the instrument, and a blank run is used to verify the absence of alkylphosphonic acids and related monoesters. Screening for alkylphosphonic acids is carried out by comparing relative migration times of peaks in the electropherograms of unknowns versus the migration times of peaks in a sample made from authentic reference chemicals.

3. Safety aspects

CE analysis does not require any specific precautions concerning safety. Samples to be analysed normally contain only small concentrations of toxic chemicals. Normal laboratory safety measures such as donning appropriate personal protective equipment (PPE) such as gloves, lab coats and safety glasses during sample handling should be used.

RECOMMENDED OPERATING PROCEDURES FOR CWC-RELATED ANALYSIS

Section 3. Analytical methods

Part D. Other separation techniques

Chapter I. Capillary electrophoresis (CE)

4. Instrumentation

4.1 Capillary electrophoresis instrument

Any modern CE instrument equipped with a UV or DAD detector can be used. This method was developed on an Agilent Technologies 3D-CE™.

4.2 CE capillary

An Agilent Technologies (or equivalent) bare fused silica column, with a 50 µm i.d. and 56 cm effective column length, is recommended. Alternatively, Agilent Technologies offers an extended light-path column with a larger detection window in the column. The use of this column can lead to 3–5 times (on average) improvement in the detection limits of alkylphosphonic acids.

4.3 Reagents and chemicals

The reagents and chemicals needed for CE analysis are given in Table 1.

Table 1. Chemicals and reagents needed for CE

Chemical	Purity
Methylphosphonic acid (MPA) (CAS 993-13-5)	≥98.0%
Ethyl methylphosphonic acid (EMPA) (CAS 1832-53-7)	≥98.0%
Isopropyl methylphosphonic acid (IMPA) (CAS 1832-54-8)	≥98.0%
Isobutyl methylphosphonic acid (iBMPA) (CAS 1604-38-2)	≥98.0%
Cyclohexyl methylphosphonic acid (CMPA) (CAS 1932-60-1)	≥98.0%
Pinacolyl methylphosphonic acid (PMPA) (CAS 616-52-4)	≥98.0%
Phenylphosphonic acid (CAS 1571-33-1)	≥98.0%
Boric acid (CAS 10043-35-3)	≥99.99%
Didodecyldimethylammonium bromide (DDAB) (CAS 3282-73-3)	≥98.0%
0.1 N Sodium Hydroxide (NaOH) (CAS 1310-73-2)	p.a. grade or better
Solvents	
High Purity Deionised Water (CAS 7732-18-5)	≥18.0 MΩ·cm
Methanol (MeOH) (CAS 67-56-1)	HPLC grade or better

RECOMMENDED OPERATING PROCEDURES FOR CWC-RELATED ANALYSIS

Section 3. Analytical methods

Part D. Other separation techniques

Chapter I. Capillary electrophoresis (CE)

5. Experimental

Recommended CE conditions are given in Table 2.

Table 2. Recommended CE conditions.

CE parameter	Recommended conditions	Notes
Cassette temperature	25.0 °C	
Buffer solution	200 mM boric acid, 10 mM phenylphosphonic acid, 0.35 mM DDAOH (prepared by passing the DDAB solution through an ion-exchange cartridge)	Adjust the pH to 4.0 by adding 1 M NaOH, if necessary.
Injection mode	50 mbar for 15 s	Applied pressure
Voltage	-25 kV for 5.5 min	Negative polarity
Current	System limit	Averages 5–7 µA
Preconditioning (Column)	1.) 5.0 min with 0.1 N NaOH 2.) 10.0 min with deionised water 3.) 10.0 min with buffer solution	
Preconditioning (Method)	4.0 min with buffer solution	
Response time	0.3 s	
Peak width	>0.03 min	
Signal	254 nm, BW 12 nm; Reference 200 nm, BW 20 nm	
Threshold	2 mAU	
Indirect UV detection	ON	

6. Quality control

6.1 Instrument performance test

The reliable identification of Chemical Weapons Convention (CWC)-related compounds requires that the data produced by the instrument is correct. Therefore, it is essential to continuously run quality control tests to confirm correct and stable instrument performance during the sample analysis.

An instrument performance check is a suggested verification procedure; instrument performance for use with CWC-related compounds is determined by analysing a check standard containing ca. 10.0 µg/ml isopropyl methylphosphonic acid (IMPA), each day the instrument is used to analyse a sample. The instrument is acceptable for use if the data produced from the test compound meets the following requirement: peak area of IMPA (a 10.0 µg/ml solution) shall be no less than 8.0 area counts on the diode array detector set at 254 nm with indirect UV detection.

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Section 3. Analytical methods

Part D. Other separation techniques

Chapter I. Capillary electrophoresis (CE)

6.2 Quality control (QC) standard

Figure 1 shows a typical electropherogram of a QC test sample (containing the reference chemical IMPA) indicating good instrument performance. Figure 2 shows a typical electropherogram of a QC of the alkylphosphonic acids standard (containing the reference chemicals: MPA, EMPA, IMPA, iBMPPA, CMPA and PMPA at 10 ppm).

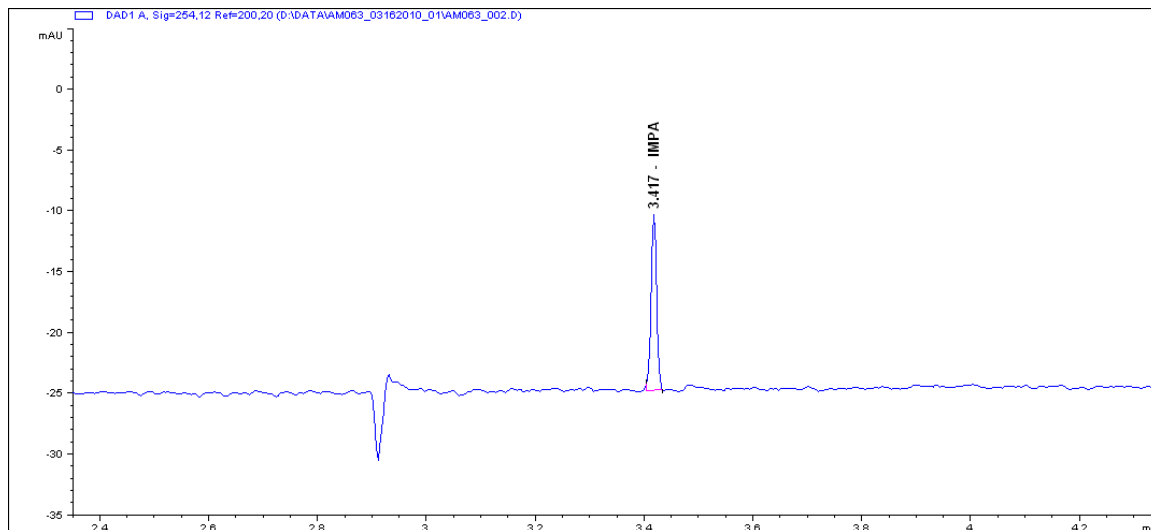


Figure 1. Analysis of the suggested instrument performance standard (10 ppm IMPA) on a 56 cm x 0.50 μ m i.d. bare fused silica column.

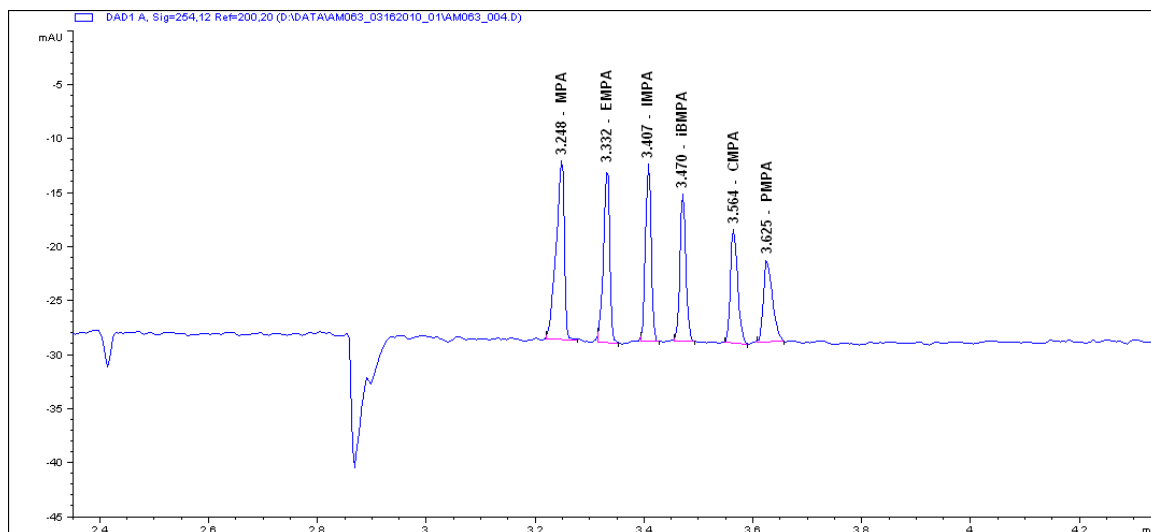


Figure 2. Analysis of an OPCW QC standard containing a 10 μ g/ml mixture of the following reference chemicals: MPA, EMPA, IMPA, iBMPPA, CMPA, and PMPA. Analysis was done on a 56 cm (effective length) x 0.50 μ m i.d. bare fused silica column.

6.3 Blank tests

The running of blanks is highly recommended to check for any possible contamination coming from the sample preparation (method blank) or the instrument (system blank). If the system blank shows contamination, a solvent blank run with the same solvent used in sample preparation must be run on the instrument to eliminate the possibility of contamination due to the equipment (e.g. contamination from the syringe or the memory effect of the equipment). A method blank will test for any contamination that would result from the sample preparation.

If CWC-related chemicals are found in any blank electropherogram, the blank tests are repeated after eliminating the possible contamination source. All contamination coming from glassware, solvents, standard solutions (or authentic reference solutions), syringes, the autosampler system or columns must be eliminated before sample analysis is started.

7. Analysis of samples

7.1. Sequence of analysis

Multiple samples are analysed using the Agilent-CE autosampler with a sequence table containing the respective information for each sample. All samples must have unique identifiers. In the beginning of an analysis batch, a QC test and a blank test are always run. When all tests have been accepted, samples can be analysed. Identification of a compound determined by this method is based on a comparison of the sample migration time to the standard solution migration time. The relative migration time (RMT) of the sample component must be within ± 0.20 RMT units of the RMT of the standard component within each batch (current OPCW criteria limit RTs to ± 0.1 min of the reference standard). The approximate migration times for each component are: MPA, 3.25 min; EMPA, 3.33 min; IMPA, 3.41 min; iBMPA, 3.47 min; CMPA 3.56 min; and PMPA, 3.63 min. In order to track migration time changes, a QC standard should be run at least every 10 samples. It is recommended that batches (sequences) should be kept under 25 total injections, including all standards, samples and controls. A freshly ampouled vial of electrolyte solution needs to be placed into the appropriate vial slot in the autosampler before each batch is run. This will help to ensure that the electrolyte will not be depleted and migration time changes will be kept to a minimum. Note: According to the current OPCW rules, a blank run (method blank) is required before each sample analysis.

7.2 Identification

The final identification must always be approved by an experienced analyst.

Identification is based on RMT and comparison with an authentic reference chemical. The recorded RMT in the electropherograms (reference and unknown) must be within ± 0.1 minutes of each other. The signal-to-noise (S/N) ratio of a recorded peak must be greater than five. The S/N can be calculated by visual inspection or through the use of an S/N calculator in the software.

Note! *If extra peaks are present in the electropherogram within the ± 0.1 min window due to a complex matrix, an electropherogram with standard addition of the reference*

standard must also be provided to confirm the identity of the peak in the electropherogram of the unknown sample.

8. Evaluation of method

8.1 Area of applicability

This CE method can be applied to identification of phosphonic acid degradation products of chemical warfare agents in aqueous samples and extracts. The method is suitable for analysing and identifying alkylphosphonic acids at ≥ 500 ng/ml (ppb) concentration levels.

8.2 Advantages/disadvantages

Advantages of the identification method using CE are reliability and sensitivity. Modern instruments are easy to use, and the identification does not rely on any derivatisation techniques.

A disadvantage is that the technique requires primarily aqueous samples, and the buffer system used in this method is specific for analysing and identifying alkylphosphonic acids and monoesters only. Other buffer systems and methods would need to be developed to expand the method for other CWC-related compounds.

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Chapter I. Capillary electrophoresis (CE)

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Chapter II. Ion chromatography (IC)

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1. Scope

This chapter describes the use of ion-exchange chromatography in the separation and analysis of alkylphosphonic acids using conductivity and/or mass spectrometry detection. Ion-exchange chromatography involves the separation of target analytes based on coulombic interactions of the charged species with an ionic stationary phase. This procedure uses anion exchange chromatography coupled with conductivity detection to identify analytes above a 1 mg/l (ppm) concentration level. The use of a mass spectrometer allows for the detection of analytes at much lower (sub-ppm) concentration levels.

2. Summary of the method

Ion chromatography (IC) is carried out using a modern IC instrument equipped with a conductivity detector (CD) and an optional mass spectrometry (MS) detector. Aqueous samples are analysed with an IC-CD instrument equipped with an analytical and guard column containing an anion-exchange medium. Before each sample batch an instrument performance test and quality control (QC) standard is run to verify the correct performance of the instrument. Additionally, a blank run is performed to verify the absence of alkylphosphonic acids. Identification of analytes: methylphosphonic acid (MPA), ethyl methylphosphonic acid (EMPA), isopropyl methylphosphonic acid (IMPA), isobutyl methylphosphonic acid (iBMPA), and pinacolyl methylphosphonic acid (PMPA) is made by comparison of the retention times in chromatograms of the unknowns to that of authentic reference chemicals. With MS detection, spectra of unknowns and authentic reference chemicals can also be compared.

3. Safety aspects

IC analysis does not require any specific precautions concerning safety. Samples to be analysed normally contain only small concentrations of toxic chemicals. Normal laboratory safety measures such as donning appropriate personal protective equipment (PPE) such as gloves, lab coats, and safety glasses during sample handling should be utilised.

4. Instrumentation

4.1 Ion chromatography instrument

Any modern IC instrument equipped with a conductivity detector, pumps for delivering a gradient program, an autosampler and a suppressor is acceptable. Most modern IC instruments also include eluent generators for delivering the appropriate eluent concentrations during the analysis. For additional MS or MS/MS analysis, an auxiliary pump is also required to provide make-up flow into the electrospray source of the mass spectrometer. This method was developed on a Dionex ICS-3000 system equipped with a conductivity detector coupled to a mass spectrometer. Parameters for detection on an AB Sciex API 3200 triple quadrupole mass spectrometer are also included.

- Dionex ICS-3000 Detector/Chromatography (DC) Module with dual temperature control (Dionex Corporation, Sunnyvale, CA)
- Dionex ICS-3000 Dual Pump-1 (DP-1) Module
- Dionex ICS-3000 Eluent Generator (EG) Module
- Dionex ICS Series Photodiode Array (PDA) Detector
- Dionex Autosampler (AS) with 100-vial tray
- Dionex Conductivity (CD) Detector
- Dionex ASRS 300-2 mm Suppressor [P/N 064555]
- Dionex RFIC Eluent Degasser
- Dionex EluGen® Cartridge EGC II KOH [P/N 058900]
- Dionex CR-ATC Continuously Regenerated – Anion Trap Column [P/N 060477]
- Dionex GM-4 2 mm Gradient Mixer
- Dionex ICS-3000 Eluent Organizer (EO) to contain (1) 4 L water reservoir and (2) 2 L eluent bottles
- API 3200 triple quadrupole mass spectrometer
- Dionex AXP-MS pump

4.2 IC columns

- Dionex Ionpac® AG18 2 x 50 mm guard column
- Dionex Ionpac® AS18 2 x 250 mm analytical column

4.3 Reagents and chemicals

The reagents and chemicals needed for IC analysis are given in Table 1.

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Chapter II. Ion chromatography (IC)

Table 1. Chemicals and reagents needed for IC.

Chemical	Purity
Methylphosphonic acid (MPA) (CAS 993-13-5)	≥98.0%
Ethyl methylphosphonic acid (EMPA) (CAS 1832-53-7)	≥98.0%
Isopropyl methylphosphonic acid (IMPA) (CAS 1832-54-8)	≥98.0%
Isobutyl methylphosphonic acid (iBMPA) (CAS 1604-38-2)	≥98.0%
Pinacolyl methylphosphonic acid (PMPA) (CAS 616-52-4)	≥98.0%
Solvents	
High-purity deionised water (CAS 7732-18-5)	≥18.0 MΩ·cm
Methanol (MeOH) (CAS 67-56-1)	HPLC grade or better
Acetonitrile (ACN) (CAS 75-05-8)	HPLC grade or better

5. Experimental

5.1 Recommended IC conditions

Recommended IC conditions are given in Table 2.

It is important to let the entire system equilibrate before attempting performance verification or calibration. After loading the appropriate program file, let the instrument run for a minimum of twenty minutes under initial conditions. Performing a test injection of a standard is recommended, to ensure that the entire system is working properly.

Table 2. Recommended IC conditions.

IC parameter	Recommended conditions	Notes
Injection volume	5.0 µl	
Flow rate (main pump)	0.350 ml/min.	
Flow rate (AXP-MS pump)	0.150 ml/min.	100% acetonitrile is only necessary if using MS detection.
Column temperature	30 °C	
Compartment temperature	30 °C	
Detector temperature	35 °C	
Suppressor current	18 mA	
Eluent multi-step gradient	0.0 min: 1.0 mM KOH 5.0 min: 1.0 mM KOH 10.0 min: 10.0 mM KOH 19.0 min: 20.0 mM KOH 20.0 min: 1.0 mM KOH	
Stop time	20.0 min	

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5.2 Recommended MS conditions

Recommended MS conditions are given in Table 3. Table 4 contains a list of the relevant reactions to be monitored, as well as the mass-specific parameters (i.e. collision energies and various potentials).

These recommended transitions are for use with a triple quadrupole mass spectrometer. The listed voltages are given for use with an AB Sciex API-3200.

Table 3. Recommended MS conditions.

MS parameter	Recommended conditions	Notes
Ionisation mode	ESI(-)	
IS voltage	-4.5 kV	
Interface heater	On	
Curtain gas	Nitrogen	Set at 35 (arbitrary units)
Nebuliser gas	Nitrogen	Set at 40 (arbitrary units)
Turbo gas	Nitrogen	Set at 40 (arbitrary units)
Turbo gas temperature	600 °C	Equivalent to source temperature
Scan mode	Multiple reaction monitoring (MRM)	
Scan time	200 ms	
Collision gas	Nitrogen	Set at 7 (arbitrary units)
Resolution	Unit, Q1 and Q3	

Table 4. Recommended MRMs and additional MS parameters.

Analyte	Precursor ion (<i>m/z</i>)	Product ion (<i>m/z</i>)	Entrance potential (V)	Declustering potential (V)	Collision energy (V)	Collision cell exit potential (V)
MPA	95	79	-9.5	-30.0	-14.0	-8.0
EMPA	123	78.8	-4.5	-45.0	-26.0	-8.0
	123	94.6	-4.5	-45.0	-8.0	-12.0
IMPA	137	94.8	-9.0	-25.0	-16.0	-8.0
	137	79.1	-9.0	-25.0	-52.0	-10.0
iBMPA	151	95.0	-2.5	-40.0	-12.0	-10.0
	151	77.0	-2.5	-40.0	-20.0	-8.0
PMPA	179	79.1	-4.5	-30.0	-74.0	-8.0
	179	77.1	-4.5	-30.0	-50.0	-50.0

5.3 Recommended tuning of the IC and MS instruments

Use the recommended tuning methods of the instruments. An automatic tuning and calibration procedure should be used, if available. Tuning and calibration should be carried out once a week, or more often if indicated by the QC test. Printouts of tuning and calibration reports should be archived for further reference.

6. Quality control

6.1 Instrument performance test

The reliable identification of Chemical Weapons Convention (CWC)-related compounds requires that the data produced by the instrument is correct. Therefore, it is essential to continuously run QC tests to confirm correct and stable instrument performance during the sample analysis.

An instrument performance check is a suggested verification procedure; instrument performance for use with CWC-related compounds is determined by analysing a check standard containing ca. 10.0 µg/ml isopropyl methylphosphonic acid (IMPA), each day the instrument is used to analyse a sample.

The instrument is accepted for use if the MS and IC data produced from each test compound meet the following requirements:

1. Signal-to-noise (S/N) ratio for each test chemical is >5:1 in the chromatograms. The evaluation is done visually or automatically if available.
2. The m/z values of the major ions of each test chemical are correct, and no extra ions above 5% are present (the mass range to be printed must extend 10 m/z units beyond the molecular mass, so that the whole relevant mass spectrum is registered).

Figure 1 shows a typical conductivity chromatogram of a QC test sample indicating good instrument performance.

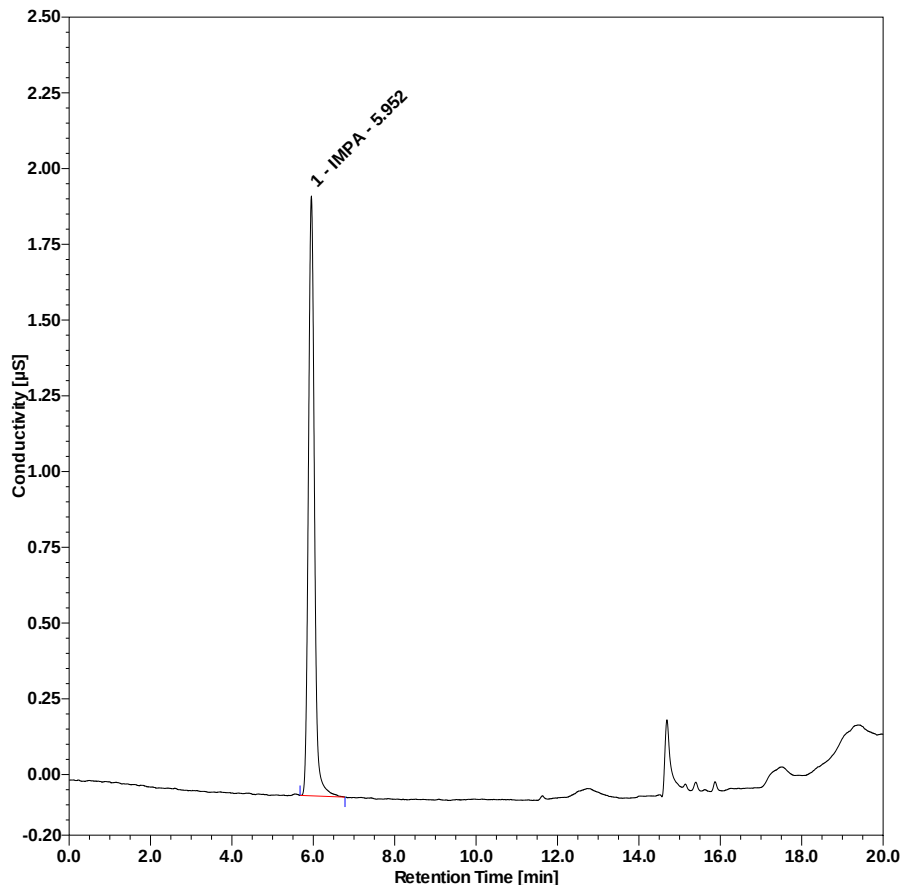


Figure 1. Analysis of the suggested instrument performance standard (10 ppm IMPA) on a Dionex Ionpac® AS18 2 x 250 mm analytical column with conductivity detection.

6.2 Quality control standard

Figure 2 shows a typical conductivity chromatogram of a QC of the alkylphosphonic acids standard (containing the reference chemicals: EMPA, IMPA, iBMPA, PMPA and MPA at 10 ppm). Figure 3 shows a typical extracted ion chromatogram (EIC) of a QC of the alkylphosphonic acids standard (containing the reference chemicals at 10 ppb).

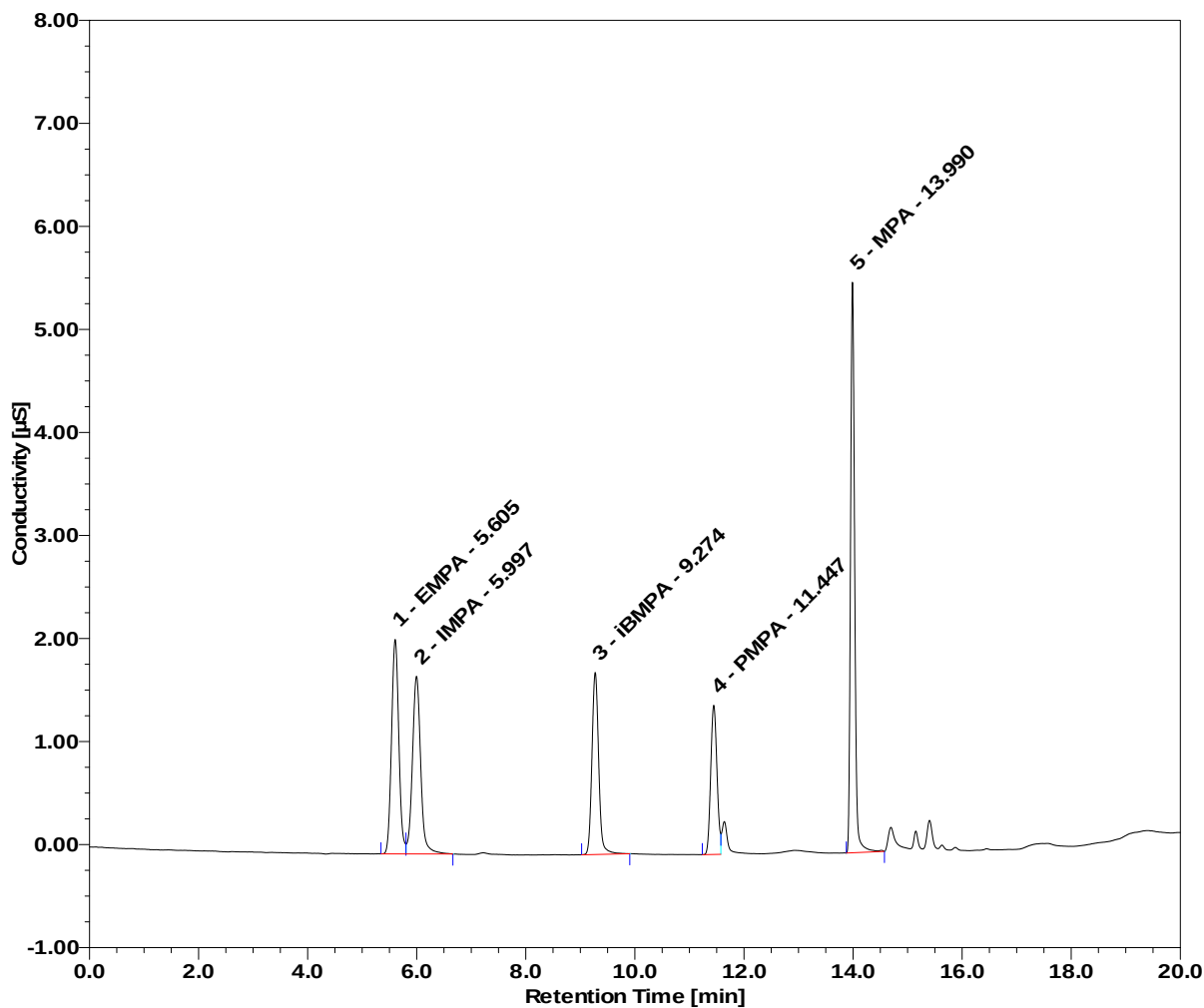


Figure 2. Analysis of an OPCW QC standard containing a 10 µg/ml mixture of the following reference chemicals: EMPA, IMPA, iBMPA, PMPA and MPA. The mixture was separated on a Dionex Ionpac® AS18 2 x 250 mm analytical column, and analytes detected via conductivity detection.

RECOMMENDED OPERATING PROCEDURES FOR CWC-RELATED ANALYSIS

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Chapter II. Ion chromatography (IC)

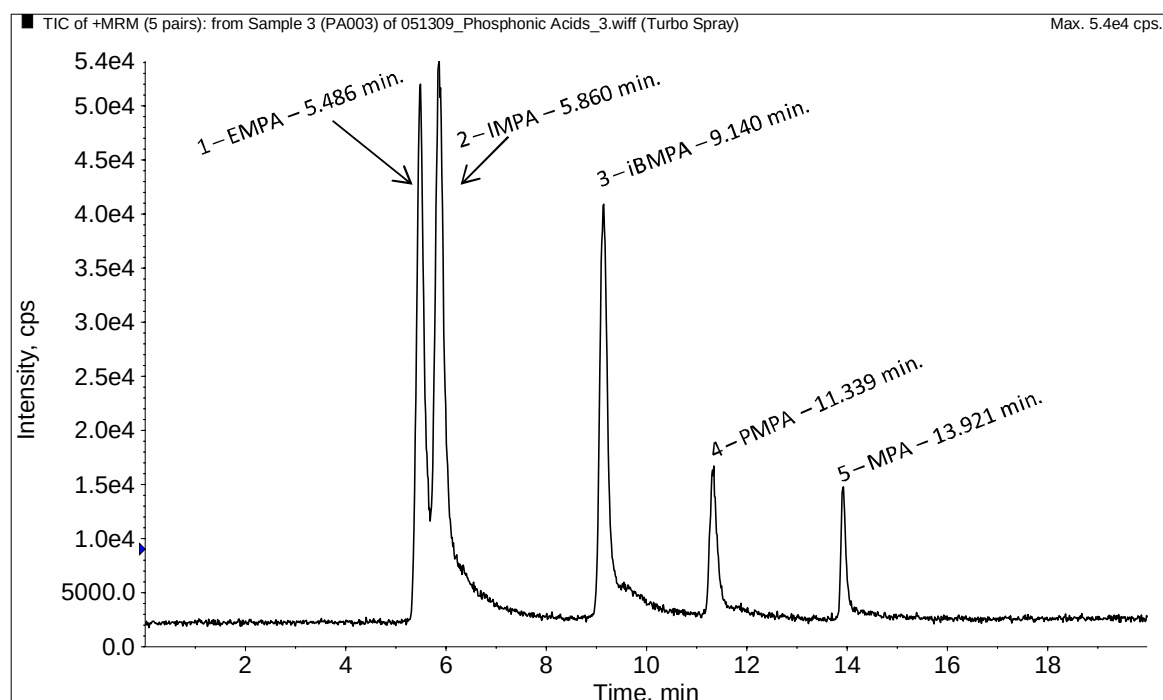


Figure 3. Analysis of an OPCW QC standard containing a 0.01 µg/ml mixture of the following reference chemicals: EMPA, IMPA, iBMPA, PMPA and MPA. The mixture was separated on a Dionex Ionpac® AS18 2 x 250 mm analytical column, and analytes detected through MRMs on a AB Sciex 3200 triple quadrupole mass spectrometer.

6.3 Blank tests

The running of blanks is highly recommended to check for any possible contamination coming from the sample preparation (method blanks) or the instrument (system blanks). If a system blank shows contamination, a solvent blank run with the same solvent used in the sample preparation must be run on the instrument to eliminate the possibility of contamination from the equipment (e.g. contamination from the syringe or the memory effect of the equipment). A method blank will test for any contamination that would result from the sample preparation.

If CWC-related chemicals are found in any blank chromatograms, the blank tests are repeated after eliminating the possible contamination source. All contamination coming from glassware, solvents, standard solutions (or authentic reference solutions), syringes, the autosampler system, or columns must be eliminated before sample analysis is started.

7. Analysis of samples

7.1. Sequence of analysis

Multiple samples are analysed using the Dionex autosampler (AS) with a sequence table containing the respective information for each sample. All samples must have unique identifiers. At the beginning of an analysis batch, a QC standard test and a solvent blank test are run. When all tests have been accepted, samples can be analysed. Identification of a compound determined by this method is based on a comparison of the sample retention time (RT) to the standard-solution RT. The RT of the analyte must be within ±0.2 minutes

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of the RT of the reference standard. The approximate RTs for each component using IC–CD are: EMPA, 5.58 min; IMPA, 5.97 min; iBMPA, 9.23 min; PMPA 11.43 min; and MPA, 13.99 min. In order to track migration time changes, a QC standard should be run at least every 10 samples. It is recommended that batches (sequences) should be kept under 25 total injections, including all standards, samples, and controls. **Note!** *According to the current OPCW rules, a blank run (method blank) is required before each sample analysis.*

7.2 Identification

The final identification must always be approved by an experienced analyst.

Identification is based on comparison with an authentic reference chemical. The recorded RT in the chromatograms (reference and unknown) must be within ± 0.2 minutes of each other. The signal-to-noise ratio of a recorded peak must be greater than 5. The S/N can be calculated by visual inspection or through the use of a S/N calculator in the software. If mass spectrometric detection is used, then the recorded spectra (reference and unknown) must be at least 10 m/z units above the molecular weight of the precursor and be similar in spectral quality.

Note! *If extra peaks are present in the chromatogram within the ± 0.2 min window due to a complex matrix, a chromatogram with standard addition of the reference standard must also be provided to confirm the identity of the peak in the chromatogram of the unknown sample.*

8. Evaluation of method

8.1 Area of applicability

This IC method can be applied to the identification of phosphonic acid degradation products of chemical warfare agents in aqueous samples and extracts. The method is suitable for analysing and identifying alkylphosphonic acids at ≥ 500 ng/ml (ppb) concentration levels via conductivity detection and at ≥ 50 ng/ml (ppb) via mass spectrometric detection.

8.2 Advantages/Disadvantages

The advantages of both the IC–CD and the IC–MS methods are their reliability and sensitivity. Modern instruments are easy to use, and the identification does not rely on any derivatisation techniques. The use of an IC–MS/MS method with MRMs allows for the selectivity of individual ions corresponding to specific phosphonic acids. However, the use of a full-scan MS method may be necessary as a screen to first qualitatively detect the presence of phosphonic acids in the samples. The IC–MS or IC–MS/MS methods can then be used as a positive identification and/or a quantitative analysis technique. Disadvantages in identification using anion-exchange chromatography are that the technique requires primarily aqueous samples, and other common anions, such as fluoride, chloride, bromide, nitrite, nitrate, sulphate, phosphate, carbonate, chlorate, chlorite and perchlorate, may potentially cause interference when using conductivity detection. It may be necessary to perform sample cleanup to remove some of these interferences or to use an MS detection technique. A mass spectrometer is significantly more expensive than an IC system with a conductivity detector.

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Chapter II. Ion chromatography (IC)

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1. Scope

This section describes the use of high-resolution nuclear magnetic resonance (NMR) spectrometry utilising one-dimensional ^1H NMR experiments for the screening and identification of chemical warfare agents, their precursors and degradation/reaction products of chemicals related to the Chemical Weapons Convention (CWC). Generally, the procedure is for identifying chemicals above the 1 $\mu\text{g}/\text{ml}$ concentration level. NMR measurements are carried out with modern NMR instruments typically equipped with 5-mm NMR probe heads.

2. Summary of the method

NMR techniques are commonly used for structural elucidation. With the ^1H NMR spectrum interpretation, a set of information is obtained in the form of NMR parameters (chemical shifts and coupling constants) that converge to a chemical structure. ^1H NMR spectroscopy can also be used as a screening method and give information about chemical families. Identification of a compound is achieved by comparison with reference spectra included in a computer database or spectrum library, or through comparison with a spectrum of the authentic compound. Identification of a chemical by ^1H NMR may also be

achieved by spectrum interpretation. The number of resonances, chemical shifts, coupling constants, and peak areas must be consistent.

3. Safety aspects

The samples can potentially contain a high dosage of toxic chemicals, and sufficient safety measures should be used in handling. Whether to close the tube with a plastic cap or by flame must be decided on the basis of preliminary knowledge of the sample. If the concentration of the agent is high, the tube should be sealed by flame.

4. Instrumentation

4.1 General recommendations

The requirements for spectrometers suitable for the analysis of treaty-related compounds by ^1H NMR are as follows:

1. capability to observe ^1H nucleus,
2. capability to decouple selected second (^{13}C , ^{19}F , or ^{31}P) nucleus,
3. capability to apply signal suppression on selected solvent peak(s),
4. capability for shaping rf pulses and pulsed field gradients,
5. capability for controlling of the sample temperature to an accuracy of within $\pm 0.5\text{ }^\circ\text{C}$,
6. a proton resonance frequency preferably of 400 MHz or more, and
7. a data system adequate for acquisition, processing, and storage of all necessary data.

4.2 Probe head characteristics

The 5-mm room-temperature probe heads generally used in NMR spectroscopy are suitable for the analysis of treaty-related compounds. The coil geometry and number of channels can vary depending on the type and manufacturer of the probe head.

^1H NMR experiments need a probe head with a ^1H channel; optimal sensitivity is achieved with so-called inverse probe heads, where the innermost RF coil is used for proton detection. Using cryogenic and microcoil probe heads may also increase sensitivity (see section 7.5).

The measurement of decoupled experiments (e.g. $^1\text{H}\{^{31}\text{P}\}$ NMR) needs a probe head with two channels tuneable to the observed and decoupled nuclei.

5. Experimental

5.1 Recommended conditions

Temperature: the temperature of the NMR sample must be controlled. Its value is between 20 to 30 $^\circ\text{C}$.

NMR sample: The volume of sample to analyse for a 5-mm NMR tube is 450 to 600 μl . It can be reduced by using a specific tube like shigemi (its price is often prohibitive) or by

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using the sandwiched solvent method (ref. 1). To avoid cross-pollution, a new tube must indeed be used for each sample.

NMR solvent: To lock the frequency of the spectrometer, addition of a miscible deuterated solvent to the liquid sample is required (50 to 150 μl). If possible, using a deuterated solvent similar to the protonated solvent is recommended. To determine the solvent of a liquid sample, a first unlocked ^1H NMR experiment can be performed. If pre-treatment is required to analyse a sample, the used solvent can be fully deuterated.

NMR acquisition parameters: A pulse angle of 30° , an acquisition time of 3.9 s, a relaxation delay of 0.1 s, and a spectral width of 12 ppm, ranging from -2 to 10 ppm, are recommended. These experimental parameters may be optimised for particular samples using the guidelines given in Table 1.

5.2 Selection of parameters

Table 1 provides guidelines for the selection of experimental parameters in ^1H NMR experiments.

Table 1. Guidelines for the selection of parameters in ^1H NMR experiments.

Parameter	Recommendation
operation frequency * (MHz)	≥ 400
pulse sequence	a one-pulse sequence or a sequence with simultaneous solvent/water suppression or a sequence with heteronuclear decoupling
flip angle ($^\circ$)	30–90
pulse duration (μs)	dependent on probe head and electronics, correlated to the flip angle (must be below 20 μs for a 90° pulse)
spectral width (Hz)	large enough to show all resonances of interest (typically 12 ppm)
data points (FID)	enough to reveal the smallest coupling and shift difference of overlapping resonances (resolution 0.2–1.0 Hz)
digital resolution on spectrum	sufficient to give 2–3 datapoints per half linewidth
repetition time(s)	long enough to allow complete relaxation (quantification), or shorter as a compromise between optimum sensitivity and total experiment time
number of pulses	enough to provide desired S/N (generally between 128 and 1024)
temperature ($^\circ\text{C}$)	a known temperature between $+20$ and $+30^\circ\text{C}$. Notice the boiling point of the sample solvent.
filter function	exponential function with 0.1–1 Hz line broadening

*not an adjustable parameter

5.3 Optional modifications

When using solvent suppression by presaturation, increase the relaxation delay to 2 s and reduce the acquisition time. Check the power levels. Decoupled experiments like $^1\text{H}\{^{31}\text{P}\}$ NMR and $^1\text{H}\{^{19}\text{F}\}$ NMR can be used to simplify the appearance of the multiplets and is very useful to screen chemical families while $^1\text{H}\{^{13}\text{C}\}$ NMR can be useful to remove overlapping ^{13}C -satellites.

6. Quality control

6.1 Instrument performance test

The most important quality control operations are the testing of the spectrometer performance and following the identification criteria. The operating procedures included in the Quality Manual for the laboratory should complement the Recommended Operating Procedures (ROP).

Testing of the spectrometer performance to meet specifications requires a variety of test samples, usually provided by the instrument manufacturer. A set of test samples could be as follows:

1. ^1H resolution test sample: 5% of orthodichlorobenzene in acetone- d_6 ,
2. ^1H lineshape and spinning sideband test sample: 1–3% chloroform in acetone- d_6 ,
3. ^1H sensitivity test sample: 0.1% ethylbenzene in chloroform- d_1 ,

Spectrometer performance tests consist of checking of lineshape, sensitivity, and resolution for a probe head for ^1H nucleus. Other tests include checking the radio frequency field strengths of the observation and decoupling channels. The interval between extensive testing should be six to twelve months; otherwise, tests are performed whenever any degradation of performance is detected. Tests are performed according to procedures found in the spectrometer manuals or other literature (e.g. see Recommended Practice for Signal-to-Noise Determination in Fourier Transform NMR in ref. 2).

Note! *The chemical shift is sensitive to changes of the sample temperature; therefore, the temperature of the sample during the NMR experiment should be known and controlled. The calibration of the temperature control unit can be performed with either the ethylene glycol or methanol sample when operating near the room temperature (ref. 3).*

6.2 Solvents

Deuterated solvents should have a high enrichment of deuterium, >99.5%. Chloroform- d_1 , deuterium oxide, acetone- d_6 , dichloromethane- d_2 , acetonitrile- d_3 , benzene- d_6 , and methanol- d_4 are the most common solvents for NMR spectrometry. Chloroform- d_1 and deuterium oxide are considered to be the main solvents, but some others may be preferred for better dissolution or to avoid overlap of resonances. The purity of the solvents should be checked by long-term NMR experiments and, when appropriate, by gas chromatography.

6.3 Internal chemical shift references

Tetramethylsilane [TMS, $\delta(\text{H}) = 0.00$ ppm] should be used in organic solvents as the chemical shift reference for proton NMR spectra. The salts 3-(trimethylsilyl)propane sulfonic acid sodium salt (TSPSA, $\delta(\text{H}) = 0.015$ ppm) or 3-(trimethylsilyl)-3,3,2,2-tetradeuteriopropionic acid sodium salt (TSPA- d_4 , $\delta(\text{H}) = 0.00$ ppm) are commonly used as reference standards in water. They should not be used if reactivity with the sample is suspected. In water, *p*-dioxane [$\delta(\text{H}) = 3.76$ ppm vs. int. TSPSA (0.015 ppm)] could be used instead. In chloroform, the shifts of *p*-dioxane are $\delta(\text{H}) = 3.70$ ppm, vs. int. TMS (0.00 ppm).

6.4 External chemical shift references

If appropriate reference samples or NMR tube inserts are used, internal chemical shift references can also serve as external references.

7. Analysis of samples

7.1. Sequence of analysis

Put the NMR sample tube into the magnet. Wait until the sample temperature has reached the preset value (takes ca. 10 min.). Meanwhile, lock the field to the ^2H resonance of the solvent (if no deuterated solvent is present, skip this step) and set the appropriate experimental parameters. Tune the observation and, if necessary, the decoupling channel. Adjust the magnetic field homogeneity. Adjust the receiver gain and start the data acquisition. After the NMR experiment is completed, store the original data (FID) on suitable media (e.g. hard disk and/or external media), and include all necessary information about the sample and recording conditions. Calculate the Fourier-transformed NMR spectrum, using a filter function if necessary, and perform the phase correction needed to obtain the real spectrum in pure absorption lineshape. The matched filter gives best signal-to-noise ratio, but if better resolution is desired, use appropriate weighting (e.g. less line broadening). Always use the same or comparable conditions for the Fourier transformation. Use an internal or external reference to calibrate the spectrum.

Plot the whole spectrum. Print the parameter list of experimental conditions. Record all necessary expansions on a plotter, and print out the peak positions. If useful, integrate the spectrum and plot expansions with integrals. Formats other than paper are possible (e.g. pdf file format).

Compared with the sample matrix, the experimental parameters have a minor effect on the spectral parameters. Experimental parameters such as spectral width, flip angle, repetition time, number of points in the FID and in the real spectrum, number of pulses (scans), and processing parameters need to be comparable with those used for the acquisition of the database spectrum or spectrum of the authentic reference compound.

Note! *It is not always possible or feasible to use the same experimental parameters like the number of pulses (scans) for the test sample as for the reference standard due to, e.g., different concentration of the compound.*

7.2 Screening

^1H NMR experiments can be used for the screening of hydrogen-containing organic compounds, in which case ^1H NMR may provide a high level of proof for the presence of a particular compound. However, the usefulness of the ^1H NMR experiment may be limited if the sample contains a strong proton signal background due to the solvent or sample matrix.

7.3 Identification

First, the spectrum is examined for resonance patterns, their positions, number of resonances, and their intensities or areas (Figure 1). This gives an idea of the functional

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groups of the compounds of interest. An immediate identification of a compound on the basis of familiar features may also result. Information obtained from each NMR experiment is consolidated. Comparisons are made with spectra in a computer database or spectral library, or with the spectrum of an authentic compound (Figures 2–3). All available information, including that obtained by other techniques, is used to construct a complete picture.

To obtain a positive identification, the NMR spectral parameters for a compound must agree within certain limits with those of the database or the authentic compound. If the sample of the compound to be identified has been prepared so as to correspond in solvent, concentration, pH, and chemical shift references to the compound in the database (or authentic compound), the values of $\delta(\text{H})$ should agree with the corresponding values in the database (or authentic compound) within limits of ± 0.05 ppm. The values of coupling constants $J(\text{HH})$ should agree within limits of ± 0.2 Hz. In addition, the resonance patterns and their intensities must match. An identification made by spiking a sample with the suspected chemical requires exact overlapping of the resonances. The amount of compound spiked should increase the signals of the compound by at least a factor of 2 and a maximum factor of 10. Identification of a chemical by ^1H NMR may also be achieved by spectrum interpretation.

Note! It is not always possible to obtain the same conditions for the reference standard as for the test sample (e.g. matrix effect), so that differences in the NMR spectral parameters may be considerable. The identification criteria may then be expanded within reasonable limits (e.g. doubled). In this case, the identification by adding the authentic compound in the sample is recommended.

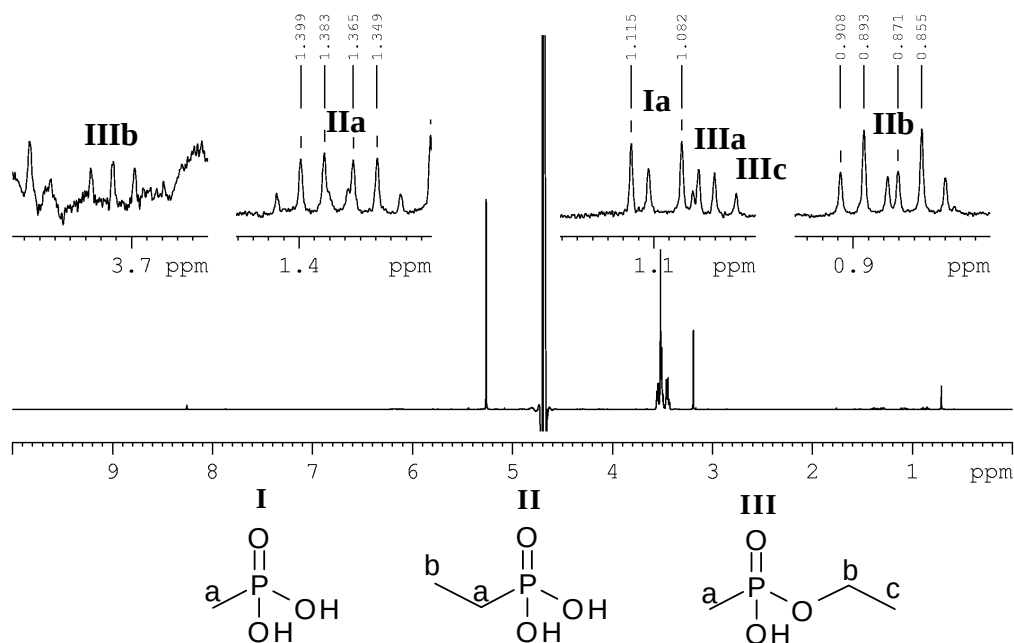


Figure 1. 500 MHz ^1H NMR spectrum with presaturation of sample W from the 22nd Proficiency Test acquired with a 5 mm broadband inverse probe head after an experimental time of 48 min. Analyte concentrations 2–10 $\mu\text{g}/\text{ml}$, solvent water, 100 μl D_2O added to 500 μl sample W. Methylphosphonic acid (I): $\delta_{\text{Ha}} = 1.10$, $J(\text{HaP}) = 16.5$ Hz. Ethylphosphonic acid: $\delta_{\text{Ha}} = 1.37$, $\delta_{\text{Hb}} = 0.87$, $J(\text{HaP}) = 17.2$ Hz, $J(\text{HbP}) = 18.8$ Hz, $J(\text{HaHb}) = 7.7$ Hz. Ethyl methylphosphonic acid: $\delta_{\text{Ha}} = 1.09$, $\delta_{\text{Hb}} = 3.71$, $\delta_{\text{Hc}} = 1.06$, $J(\text{HaP}) = 16.4$ Hz, $J(\text{HaHb}) = 7.1$ Hz.

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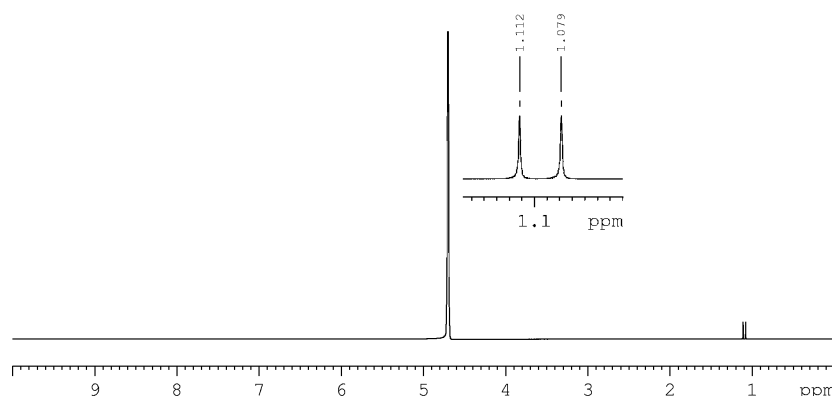


Figure 2. 500 MHz ^1H NMR spectrum of sample from authentic reference compound methylphosphonic acid acquired with a 5 mm broadband inverse probe head after an experimental time of 1 min. Analyte concentration 1.6 mg/ml, solvent water/ D_2O . Methylphosphonic acid: $\delta_{\text{Ha}} = 1.10$, $J(\text{HaP}) = 16.5$ Hz.

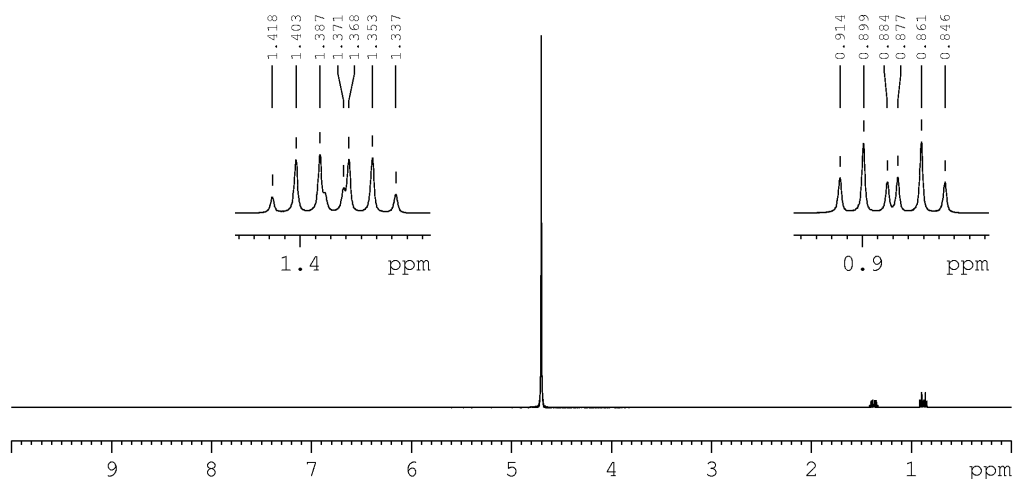


Figure 3. 500 MHz ^1H NMR spectrum of sample from authentic reference compound ethylphosphonic acid acquired with a 5 mm broadband inverse probe head after an experimental time of 1 min. Analyte concentration 1.6 mg/ml, solvent water/ D_2O . Ethylphosphonic acid: $\delta_{\text{Ha}} = 1.38$, $\delta_{\text{Hb}} = 0.87$, $J(\text{HaP}) = 17.2$ Hz, $J(\text{HbP}) = 18.8$ Hz, $J(\text{HaHb}) = 7.7$ Hz.

7.4 Quantification

NMR spectrometry is well-suited for quantification because, under certain experimental conditions, the signal intensity is directly proportional to the amount of nuclei present. Quantification accuracy of ca. 5–10% can be achieved. Quantification is performed most frequently using an internal standard, although other approaches may also be applied (for example, using the ERETIC method, which utilises a reference signal synthesised by an electronic device to determine absolute concentrations, see ref. 4; or another example is the PULCON method that correlates the absolute intensities of two spectra measured in different solution conditions, see ref. 5). A pulse angle of 90° and a repetition time (sum of the acquisition time and the relaxation delay) that equals five times the longest T_1 are recommended, using a pulse program that does not build up NOE. Signal-to-noise of the quantified compound and the internal reference must be comparable and above 10:1.

7.5 Trace analysis

The detection limit for a compound depends on the magnetic field strength of the spectrometer, the probe head, the number of pulses, the flip angle, the observed nucleus and its relaxation time, the NMR tube diameter, the sample matrix, and the homogeneity of the magnetic field. With modern NMR spectrometers having a magnetic field of ca. 12 Tesla and a room-temperature probe head, a limit of detection of 1 $\mu\text{g}/\text{ml}$ can be obtained. With special probe heads (ref. 6, ref. 7), the sensitivity and, hence, the detection limit can be further improved, and the measurement time can be reduced (Figure 4).

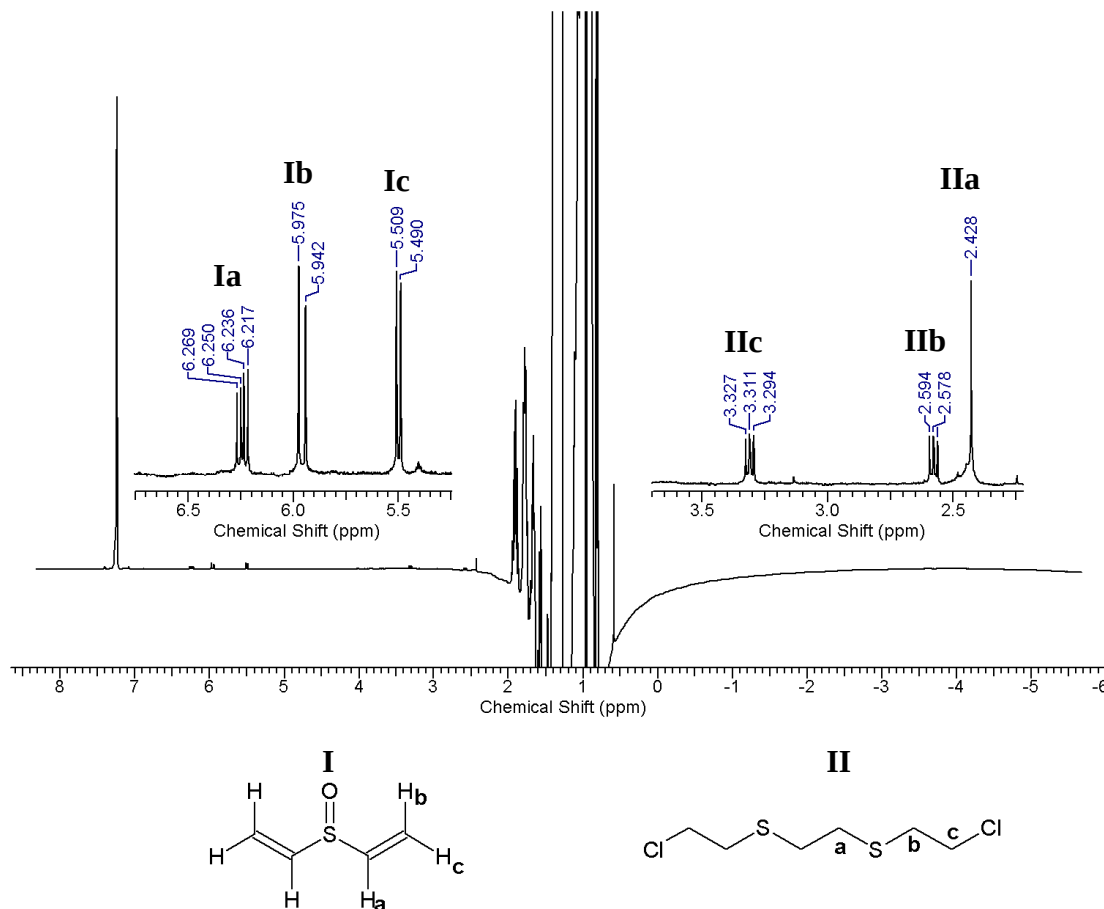


Figure 4. 500 MHz ^1H NMR spectrum with presaturation of sample O from the 25th Proficiency Test acquired with a 5 mm quadruple-nucleus (^1H , ^{13}C , ^{31}P , ^{19}F) cryoprobe after an experimental time of 17 min. Analyte concentration is 20 $\mu\text{g}/\text{ml}$, solvent *n*-hexane with 5% (v/v) *n*-nonane, 150 μl C_6D_6 added to 500 μl O. Divinyl sulfoxide (I): $J(\text{H}_a\text{H}_b) = 16.5$ Hz, $J(\text{H}_a\text{H}_c) = 9.6$ Hz. 1,2-Bis(2-chloroethylthio)ethane (II): $J(\text{H}_b\text{H}_c) = 8.1$ Hz.

7.6 Remarks on analysis

Non-deuterated solvents can be used if a small amount of the corresponding deuterated solvent is added to the sample to provide the lock conditions, if the experiment is performed unlocked, or if a lock nucleus other than deuterium is used. For strongly salty samples, cryogenic probe heads are disadvantageous, but it could be beneficial using smaller diameter tubes (e.g. 3-mm tube and spinner on a 5-mm probe head) or microcoil probe heads. The 90° pulse width should be recalibrated. Paramagnetic salts in aqueous

samples can enhance relaxation so that the resonances became too broad to be observable. In this case, a pre-treatment must be performed (e.g. cation exchange, application of chelating agents).

8. Evaluation of method

8.1 Area of applicability

The method is applicable to all compounds containing hydrogen atoms other than exchangeable and dissolving to a sufficient degree in a solvent. The analysed liquid sample must be homogeneous and without paramagnetic properties.

8.2 Advantages/Disadvantages

The main advantage of ^1H NMR spectroscopy is that the spectrum of a compound is in most cases specific and the measurement does not degrade the sample. The disadvantages are the non-selectivity causing overlap of signals and the modest sensitivity (several orders of magnitude compared to GC–MS if absolute amounts of compounds are considered).

When unknown compounds (without reference data) are present, ^1H NMR spectrometry may be the best source of structural information, with other techniques providing complementary information. Although they are not described here, other one-dimensional (e.g. techniques utilising polarisation transfer) or two-dimensional homonuclear and heteronuclear correlation techniques can be used as well.

9. References

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Chapter II. Basic ^{31}P NMR techniques

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1. Scope

This section describes the use of high-resolution nuclear magnetic resonance (NMR) spectrometry utilising one-dimensional ^{31}P NMR experiments for the screening and identification of chemical warfare agents, their precursors, and degradation/reaction products of chemicals related to the Chemical Weapons Convention (CWC). Generally, the procedure is for identifying chemicals above the 1 to 5 $\mu\text{g/ml}$ concentration level. NMR measurements are carried out with modern NMR instruments typically equipped with 5-mm NMR probe heads.

2. Summary of the method

NMR techniques are commonly used for structural elucidation. With ^{31}P NMR spectrum interpretation, a set of information is obtained in the form of NMR parameters (chemical shifts and coupling constants) that converge to a chemical structure. ^{31}P NMR spectroscopy is used as a screening method and to give information about chemical families. Identification of a compound is achieved by comparison with reference spectra included in a computer database or spectrum library, through comparison with a spectrum of the authentic compound, or with standard addition of the authentic compound. Number of resonances, chemical shifts, and coupling constants must be consistent.

3. Safety aspects

The samples can potentially contain a high dosage of toxic chemicals, and sufficient safety measures should be used in handling. Whether to close the tube with a plastic cap or by flame must be decided on the basis of preliminary knowledge of the sample. If the concentration of the agent is high, the tube should be sealed by flame.

4. Instrumentation

4.1 General recommendations

The requirements for spectrometers suitable for the analysis of treaty-related compounds by ^{31}P NMR are as follows:

1. capability to observe ^{31}P nucleus,
2. capability to decouple ^1H nucleus,
3. capability for controlling of the sample temperature to an accuracy of within $\pm 0.5\text{ }^\circ\text{C}$,
4. a phosphorus resonance frequency preferably of 161 MHz or more, and
5. a data system adequate for acquisition, processing, and storage of all necessary data.

4.2 Probe head characteristics

The 5-mm room-temperature probe heads generally used in NMR spectroscopy are suitable for the analysis of treaty-related compounds. The coil geometry and number of channels can vary depending on the type and manufacturer of the probe head.

^{31}P NMR experiments need a probe head with a ^{31}P channel; optimal sensitivity is achieved with so-called direct probe heads, where the innermost RF coil is used for phosphorus detection. Using cryogenic and microcoil probe heads may also increase sensitivity (see section 7.5).

The measurement of $^{31}\text{P}\{^1\text{H}\}$ NMR experiments needs a probe head with two channels tuneable to the ^{31}P and ^1H nuclei.

5. Experimental

5.1 Recommended conditions

Temperature: the temperature of the NMR sample must be controlled. Its value is between 20 to 30 $^\circ\text{C}$.

NMR sample: The volume of the sample to analyse for a 5-mm NMR tube is 450 to 600 μl . It can be reduced by using a specific tube like shigemi, but its price is often prohibitive, or by using the sandwiched solvent method (ref. 1). To avoid cross-pollution, a new tube must indeed be used for each sample.

NMR solvent: To lock the frequency of the spectrometer, addition of a miscible deuterated solvent to the liquid sample is required (50 to 150 μl). If possible, using a deuterated solvent similar to the protonated solvent is recommended. To determine the solvent of a

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liquid sample, a first unlocked ^1H NMR experiment can be performed. If a pre-treatment is required to analyse a sample, the used solvent can be fully deuterated.

NMR acquisition parameters: Optimised sensitivity for an interscan delay T_c and longitudinal relaxation time T_1 is obtained by the application of an excitation angle α_{ernst} (Ernst angle) given by the following relation: $\cos(\alpha_{\text{ernst}}) = \exp(-T_c/T_1)$, (ref. 2). Current recommended parameters corresponding to a longitudinal relaxation time T_1 closed to 4 s are as follows: a pulse angle of 45° , an acquisition time of 1 s, a relaxation delay of 0.3 s, and a spectral width of 200 ppm, ranging from 170 to -30 ppm. These experimental parameters may be optimised for particular samples using the guidelines given in Table 1.

5.2 Selection of parameters

Table 1 provides guidelines for the selection of experimental parameters in ^{31}P NMR experiments.

Table 1. Guidelines for the selection of parameters in ^{31}P NMR experiments.

Parameter	Recommendation
operation frequency * (MHz)	≥ 161
pulse sequence	a one-pulse sequence or a sequence with proton decoupling
flip angle ($^\circ$)	30–90
pulse duration (μs)	dependent on probe head and electronics, correlated to the flip angle (must be below 20 μs for the 90° pulse)
spectral width (Hz)	large enough to show all resonances of interest (standard 200 ppm)
data points (FID)	enough to reveal the smallest coupling and shift difference of overlapping resonances (resolution 0.5–1.5 Hz)
digital resolution on spectrum	sufficient to give 2–3 datapoints per half linewidth
repetition time(s)	long enough to allow complete relaxation (quantification), or shorter as a compromise between optimum sensitivity and total experiment time
number of pulses	enough to provide desired S/N, must be above the limit of detection (3)
temperature ($^\circ\text{C}$)	a known temperature between +20 and +30 $^\circ\text{C}$. Notice the boiling point of the sample solvent
filter function	Exponential function with 1–3 Hz line broadening

*not an adjustable parameter

5.3 Optional modifications

The proton-decoupled ^{31}P NMR experiment, (i.e. $^{31}\text{P}\{^1\text{H}\}$ NMR) is more common due to a better signal-to-noise ratio, simplified appearance of the multiplets and its usefulness to screen chemical families of organophosphorus compounds. The proton decoupling is achieved by standard composite pulse decoupling (CPD) like WALTZ or MLEV.

6. Quality control

6.1 Instrument performance test

The most important quality control operations are the testing of the spectrometer performance and following the identification criteria. The operating procedures included in the Quality Manual for the laboratory should complement the Recommended Operating Procedures (ROP).

Testing of the spectrometer performance to meet specifications requires a variety of test samples usually provided by the instrument manufacturer. A set of test samples could be as follows:

1. ^{31}P sensitivity test sample: 0.0485 M triphenyl phosphate in acetone- d_6 ,
2. A suitable test sample for calibration of the proton pulse width and power on decoupling channel

Spectrometer performance tests consist of checking of lineshape and sensitivity for a probe head for the ^{31}P nucleus. Other tests include checking the radio frequency field strengths of the observation and decoupling channels. The interval between extensive testing should be six to twelve months; otherwise, tests are performed whenever any degradation of performance is detected. Tests are performed according to procedures found in the spectrometer manuals or other literature (e.g. see Recommended Practice for Signal-to-Noise Determination in Fourier Transform NMR in ref. 3).

Note! *The chemical shift is sensitive to changes of the sample temperature; therefore, the temperature of the sample during the NMR experiment should be known. The calibration of the temperature control unit can be performed with either the ethylene glycol or methanol sample when operating near the room temperature (ref. 4).*

6.2 Solvents

Deuterated solvents should have a high enrichment of deuterium, >99.5%. Chloroform- d_1 , deuterium oxide, acetone- d_6 , dichloromethane- d_2 , acetonitrile- d_3 , benzene- d_6 , and methanol- d_4 are the most common solvents for NMR spectrometry. Chloroform- d_1 and deuterium oxide are considered to be the main solvents, but some others may be preferred for better dissolution. The purity of the solvents should be checked by long-term NMR experiments and, when appropriate, by gas chromatography.

6.3 Internal chemical shift references

Triphenyl phosphate (TPP) dissolves well in organic solvents and could serve as an internal chemical shift reference for phosphorus. The chemical shift of TPP is -17.80 ppm. Note that TPP gives resonances in the aromatic region of the ^1H NMR spectrum.

6.4 External chemical shift references

The most common external chemical shift reference in ^{31}P NMR is 85% phosphoric acid. In aqueous samples, $\delta(\text{P}) = 0.00$ ppm is measured in an ideal spherical container inside the solution (see ref. 5). In practice, however, it is more convenient to use a cylindrical

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capillary tube to hold the standard, and a correction should then be applied to take into account the differences in diamagnetic susceptibility. To correct for diamagnetic susceptibility differences in aqueous samples, the signal of 85% H_3PO_4 in a cylindrical capillary in superconducting solenoid spectrometers should be set at -0.73 ppm. If a narrower line is desired, a 0.2 M solution of crystalline phosphoric acid in 14% aqueous perchloric acid (tetrahydroxyphosphonium perchlorate, $\text{P}(\text{OH})_4\text{ClO}_4$, see ref. 6) in a cylindrical capillary could be chosen, and the signal should then be set at $\delta(\text{P}) = 0.09$ ppm in superconducting solenoid spectrometers. For other solvents, corrections to the phosphorus chemical shifts of H_3PO_4 and tetrahydroxyphosphonium perchlorate would be different.

For reasons of comparability, the use of $\delta(\text{P}) = 0.00$ ppm for 85% H_3PO_4 (in a cylindrical capillary tube) without corrections for diamagnetic susceptibility differences is recommended as a primary chemical shift reference. Because of the number of secondary standards, the referencing method should always be reported.

7. Analysis of samples

7.1 Sequence of analysis

Put the NMR sample tube into the magnet. Wait until the sample temperature has reached the preset value (takes ca. 10 min.). Meanwhile, lock the field to the ^2H resonance of the solvent (if no deuterated solvent is present, skip this step) and set the appropriate experimental parameters. Tune the observation channel to ^{31}P and, if necessary, the decoupling channel. Adjust the magnetic field homogeneity. Adjust the receiver gain and start the data acquisition. After the NMR experiment is completed, store the original data (FID) on suitable media (e.g. hard disk and/or external media), and include all necessary information about the sample and recording conditions. Calculate the Fourier-transformed NMR spectrum, using a filter function if necessary, and perform the phase correction needed to obtain the real spectrum. The matched filter gives the best signal-to-noise ratio, but if better resolution is desired, use appropriate weighting (e.g. less line broadening). Always use the same or comparable conditions for the Fourier transformation. Use an internal or external reference to calibrate the spectrum.

Plot the whole spectrum. Print the parameter list of experimental conditions. Record all necessary expansions on a plotter, and print out the peak positions. If useful, integrate the spectrum and plot expansions with integrals. Formats other than paper are possible (e.g. pdf file format).

Compared with the sample matrix, the experimental parameters have a minor effect on the spectral parameters. Experimental parameters such as spectral width, flip angle, repetition time, number of points in the FID and in the real spectrum, number of pulses (scans), and processing parameters need to be comparable with those used for the acquisition of the database spectrum or spectrum of the authentic reference compound.

Note! *It is not always possible or feasible to use the same experimental parameters like number of pulses (scans) for the test sample as for the reference standard due to, e.g., the different concentration of the compound.*

7.2 Screening

^{31}P NMR experiments can be used for screening of organophosphorus compounds, in which case ^{31}P NMR may provide a high level of proof for the presence of a particular compound. The chemical shift range of phosphorus is broad in comparison with proton (from -50 to 250 ppm with CWC-related compounds), and the phosphorus chemical shift is generally characteristic of the phosphorus chemical family (i.e. number of valence, functional groups around phosphorus).

7.3 Identification

First, the spectrum is examined for resonances, their chemical shifts, number of resonances, and their intensities or areas (Figure 1A). This gives an idea of the chemical families (e.g. phosphonofluoridate, phosphonothiolate, phosphoramidocyanidate) of the compounds of interest. Information obtained from each NMR experiment is consolidated. Comparisons are made with spectra in a computer database or spectral library, or with the spectrum of an authentic compound. However, derivation of the complete chemical structure solely with ^{31}P NMR experiments may not be possible; therefore, all available information, including that obtained by other techniques, is used to construct a complete picture.

To obtain a positive identification, the NMR spectral parameters for a compound must agree within certain limits with those of the database or the authentic compound. If the sample of the compound to be identified has been prepared so as to correspond in solvent, concentration, pH, and chemical shift references to the compound in the database (or authentic compound), the values of $\delta(\text{P})$ should agree with the corresponding values in the database (or authentic compound) within limits of ± 0.2 ppm. The values of coupling constants should agree within certain limits, e.g. $J(\text{PH})$ should agree within limits of ± 0.2 Hz, (except for $^1J(\text{PH}) \pm 2$ Hz) and $J(\text{PF})$ should agree within limits of ± 2 Hz. In addition, the resonance patterns and their intensities must match. An identification made by spiking a sample with the suspected chemical (standard addition) requires exact overlapping of the resonances (Figure 1B). The amount of compound spiked should increase the signals of the compound by at least a factor of 2 and a maximum factor of 10.

Note! *It is not always possible to obtain the same conditions for the reference standard as for the test sample (e.g. matrix effect), so that differences in the NMR spectral parameters may be considerable. The identification criteria may then be expanded within reasonable limits (e.g. doubled). In this case, the identification by standard addition is recommended.*

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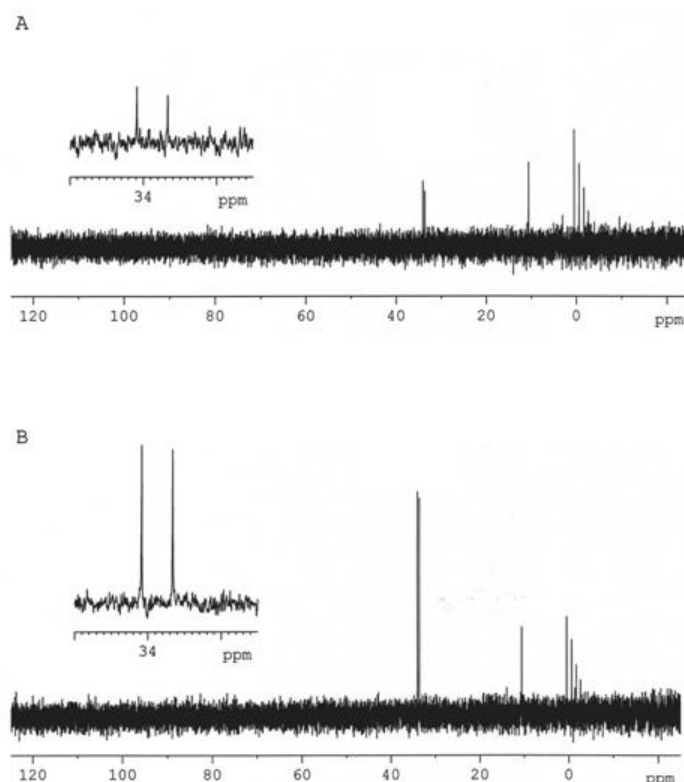


Figure 1. (A) 400 MHz $^{31}\text{P}\{^1\text{H}\}$ NMR spectrum acquired with a 5 mm quadruple-nucleus (^1H , ^{13}C , ^{31}P , ^{19}F) probe head of sample W from the 18th Proficiency Test (concentration 10 $\mu\text{g}/\text{ml}$, solvent water, 100 μl D_2O added to 500 μl sample W). Dipinacolyl methylphosphonate: $\delta_{\text{P(dia stereomer A)}} = 34.1$, $\delta_{\text{P(dia stereomer B)}} = 33.7$). (B) 400 MHz ^{31}P NMR with proton decoupling of sample W from the 18th Proficiency Test after standard addition of 10 μg authentic reference dipinacolyl methylphosphonate ($\delta_{\text{P(dia stereomer A)}} = 34.1$, $\delta_{\text{P(dia stereomer B)}} = 33.7$). The experimental time was 5 h 45 min in both of the spectra.

7.4 Quantification

NMR spectrometry is well-suited for quantification because, under certain experimental conditions, the signal intensity is directly proportional to the amount of nuclei present in the sample. Quantification accuracy of ca. 5–10% can be achieved. Quantification is performed most frequently using an internal standard, although other approaches may also be applied (for example, using the ERETIC method, which utilises a reference signal synthesised by an electronic device to determine absolute concentrations, see ref. 7, or another example is the PULCON method that correlates the absolute intensities of two spectra measured in different solution conditions, see ref. 8). A pulse angle of 90° and a repetition time (sum of the acquisition time and the relaxation delay) that equals five times the longest T_1 are recommended, using a pulse program that does not build up NOE. Signal-to-noise of the quantified compound and the internal reference must be comparable and above 10:1.

7.5 Trace analysis

The detection limit for a compound depends on the magnetic field strength of the spectrometer, the probe head, the number of pulses, the flip angle, the observed nucleus and its relaxation time, the NMR tube diameter, the sample matrix, and the homogeneity of the magnetic field. With modern NMR spectrometers having a magnetic field of ca. 12 Tesla and a room-temperature probe head, a limit of detection of 1 to 5 $\mu\text{g/ml}$ can be obtained. With special probe heads (see ref. 9 and ref. 10), the sensitivity and, hence, the detection limit can be further improved, and the measurement time can be reduced (Figures 2 and 3).

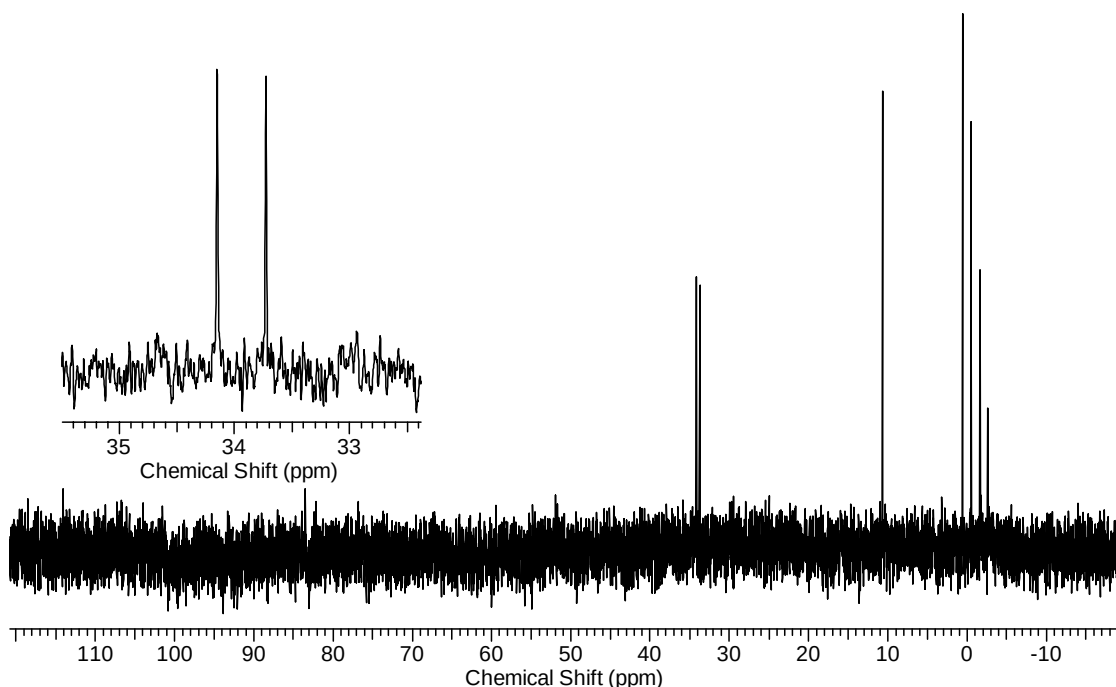


Figure 2. 500 MHz $^{31}\text{P}\{^1\text{H}\}$ NMR spectrum of sample W from the 18th Proficiency Test acquired with a 5 mm quadruple-nucleus (^1H , ^{13}C , ^{31}P , ^{19}F) cryoprobe after an experimental time of 30 min. Analyte concentration 10 $\mu\text{g/ml}$, solvent water, 100 μl D_2O added to 500 μl sample W. Dipinacolyl methylphosphonate: $\delta_{\text{P(dia stereomer A)}}$ = 34.1, $\delta_{\text{P(dia stereomer B)}}$ = 33.7.

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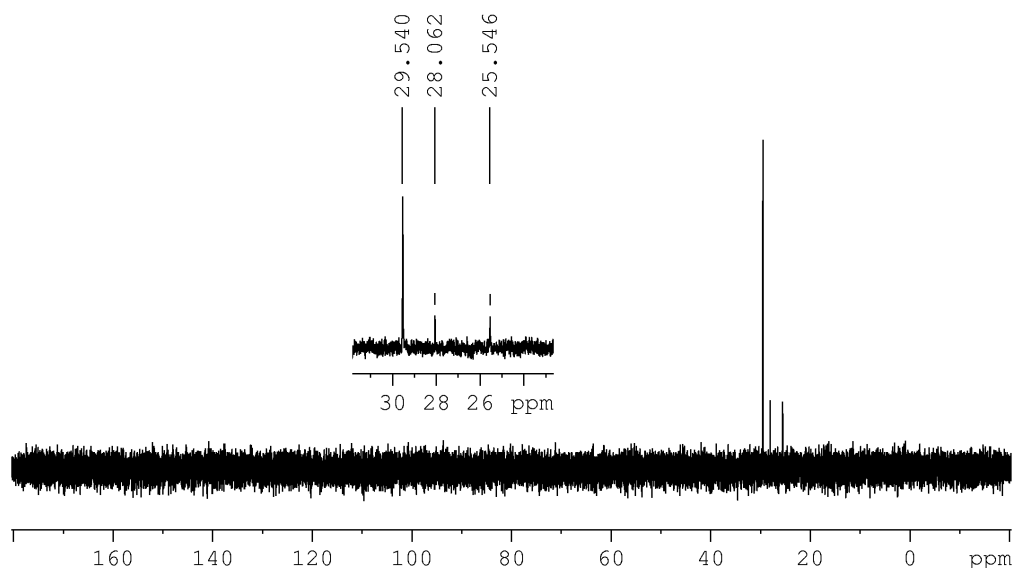


Figure 3. 500 MHz $^{31}\text{P}\{^1\text{H}\}$ NMR spectrum of sample W from the 22nd Proficiency Test acquired with a 1.7 mm triple-resonance (^1H , ^{13}C , ^{31}P) microcoil probe head after an experimental time of 21 min. Analyte concentrations 2–10 $\mu\text{g}/\text{ml}$, solvent water, sample enrichment from 1 ml W to 20 μl , 10 μl D_2O added. Methylphosphonic acid: $\delta_{\text{P}} = 25.5$, ethylphosphonic acid: $\delta_{\text{P}} = 29.5$, ethyl methylphosphonic acid: $\delta_{\text{P}} = 28.1$.

7.6 Remarks on analysis

Non-deuterated solvents can be used if a small amount of the corresponding deuterated solvent is added to the sample to provide the lock conditions. Due to long total measurement times, acquisition of ^{31}P NMR spectra unlocked is not recommended. For strongly salty samples, cryogenic probe heads are disadvantageous, but it could be beneficial using smaller diameter tubes (e.g. 3-mm tube and spinner on a 5-mm probe head) or microcoil probe heads. The 90° pulse width should be recalibrated. Paramagnetic salts in aqueous samples can enhance relaxation so that the resonances became too broad to be observable. In this case, a pretreatment must be performed (e.g. cation exchange, application of chelating agents).

8. Evaluation of method

8.1 Area of applicability

The method is applicable to all organophosphorus compounds dissolving in sufficient amount (concentration above the limit of detection) in a solvent. The analysed liquid sample must be homogeneous and without paramagnetic properties.

8.2 Advantages/Disadvantages

The main advantage of ^{31}P NMR spectroscopy is its selectivity, since it reveals the presence of organophosphorus compounds in the sample, and the measurement does not degrade the sample. The disadvantages are the low information content of the spectrum in

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comparison with information obtained in ^1H NMR spectrum, and the modest sensitivity (several orders of magnitude compared to GC–MS if absolute amounts of compounds are considered).

Although they are not described here, other one-dimensional (e.g. techniques utilising polarisation transfer) or two-dimensional heteronuclear correlation techniques can be used as well. Generally, to determine a complete picture of an organophosphorus compound, complementary NMR techniques should be used.

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Chapter III. Basic ^{19}F NMR techniques

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1. Scope

This section describes the use of high-resolution nuclear magnetic resonance (NMR) spectrometry utilising one-dimensional ^{19}F NMR experiments for the screening and identification of chemical warfare agents, their precursors, and degradation/reaction products of chemicals related to the Chemical Weapons Convention (CWC). Generally, the procedure is for identifying chemicals above a 1 to 5 $\mu\text{g/ml}$ concentration level. NMR measurements are carried out with modern NMR instruments equipped with 5-mm NMR probe heads.

2. Summary of the method

NMR techniques are commonly used for structural elucidation. With ^{19}F NMR spectrum interpretation, a set of information is obtained in the form of NMR parameters (chemical shifts and coupling constants) that converge to a chemical structure. ^{19}F NMR spectroscopy is used as a screening method and to give information about chemical families. Identification of a compound is achieved by comparison with reference spectra included in a computer database or spectrum library, through comparison with a spectrum of the authentic compound, or with standard addition of the authentic compound. The number of resonances, chemical shifts, and coupling constants must be consistent.

3. Safety aspects

The samples can potentially contain a high dosage of toxic chemicals, and sufficient safety measures should be used in handling. Whether to close the tube with a plastic cap or by flame must be decided on the basis of preliminary knowledge of the sample. If the concentration of the agent is high, the tube should be sealed by flame.

4. Instrumentation

4.1 General recommendations

The requirements for spectrometers suitable for the analysis of treaty-related compounds by ^{19}F NMR are as follows:

1. capability to observe ^{19}F nucleus,
2. capability to decouple ^1H nucleus,
3. capability for controlling of the sample temperature to an accuracy of within $\pm 0.5^\circ\text{C}$,
4. a fluorine resonance frequency preferably of 376 MHz or more, and
5. a data system adequate for acquisition, processing, and storage of all necessary data.

4.2 Probe head characteristics

The 5-mm room-temperature probe heads generally used in NMR spectroscopy are suitable for the analysis of treaty-related compounds. The coil geometry and number of channels can vary depending on the type and manufacturer of the probe head.

^{19}F NMR experiments need a probe head with a ^{19}F channel; optimal sensitivity is achieved with so-called direct probe heads, where the innermost RF coil is used for fluorine detection. Using cryogenic and microcoil probe heads may also increase sensitivity.

The measurement of $^{19}\text{F}\{^1\text{H}\}$ NMR experiments needs a probe head with two channels tuneable to the ^{19}F and ^1H nuclei.

5. Experimental

5.1 Recommended conditions

Temperature: the temperature of the NMR sample must be controlled. Its value is between 20 to 30 $^\circ\text{C}$.

NMR sample: The volume of sample to analyse for a 5-mm NMR tube is 450 to 600 μl . It can be reduced by using a specific tube like a shigemi tube, but its price is often prohibitive or by using the sandwiched solvent method (ref. 1). To avoid cross-pollution, a new tube must indeed be used for each sample.

NMR solvent: To lock the frequency of the spectrometer, addition of miscible deuterated solvent in the liquid sample is required (50 to 150 μl). If possible, using the deuterated solvent similar to the protonated solvent is recommended. To determine the solvent of a liquid sample, a first unlocked ^1H experiment can be performed. If a pre-treatment is required to analyse a sample, the used solvent can be fully deuterated.

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NMR acquisition parameters: Optimised sensitivity for an interscan delay T_c and longitudinal relaxation time T_1 is obtained by the application of an excitation angle α_{ernst} (Ernst angle) given by the following relation: $\cos(\alpha_{\text{ernst}}) = \exp(-T_c/T_1)$, (ref. 2). Current recommended parameters corresponding to a longitudinal relaxation time T_1 close to 3 s are as follows: a pulse angle of 45° , an acquisition time of 0.5 s, a relaxation delay of 0.5 s and a spectral width of 230 ppm, ranging from 50 to -180 ppm. These experimental parameters may be optimised for particular samples using the guidelines given in Table 1.

5.2 Selection of parameters

Table 1 provides guidelines for the selection of experimental parameters in ^{19}F NMR experiments.

Table 1. Guidelines for the selection of parameters in ^{19}F NMR experiments.

Parameter	Recommendation
operation frequency * (MHz)	≥ 376
pulse sequence	a one-pulse sequence or a sequence with proton decoupling
flip angle ($^\circ$)	30–90
pulse duration (μs)	dependent on probe head and electronics (must be below 20 μs for a 90° pulse)
spectral width (Hz)	large enough to show all resonances of interest (standard spectral width is commonly between 50 to -180 ppm)
data points (FID)	enough to reveal the smallest coupling and shift difference of overlapping resonances (resolution 0.5–1.5 Hz)
digital resolution on spectrum	sufficient to give 2–3 datapoints per half linewidth
repetition time(s)	long enough to allow complete relaxation (quantification), or shorter as a compromise between optimum sensitivity and total experiment time
number of pulses	enough to provide desired S/N, must be above the limit of detection (ref. 3)
temperature ($^\circ\text{C}$)	a known temperature between +20 and +30 $^\circ\text{C}$. Notice the boiling point of the sample solvent.
filter function	exponential function with 0.1–2 Hz line broadening

*not an adjustable parameter

5.3 Optional modifications

The proton-decoupled ^{19}F NMR experiment (i.e. $^{19}\text{F}\{^1\text{H}\}$ NMR) can be used to simplify the appearance of the multiplets and to increase the sensitivity of the experiment. The proton decoupling is achieved by standard composite pulse decoupling (CPD) like WALTZ or MLEV.

6. Quality control

6.1 Instrument performance test

The most important quality control operations are the testing of the spectrometer performance and following the identification criteria. The operating procedures included in the Quality Manual for the laboratory should complement the Recommended Operating Procedures (ROP).

Testing of the spectrometer performance to meet specifications requires a variety of test samples, usually provided by the instrument manufacturer. A set of test samples could be as follows:

1. ^{19}F sensitivity test sample: 0.05% trifluorotoluene in chloroform- d_1 ,
2. A suitable sample for the calibration of the proton pulse width and the power on the decoupling channel.

Spectrometer performance tests consist of checking of lineshape and sensitivity for a probe head for the ^{19}F nucleus. Other tests include checking the radio frequency field strengths of the observation and decoupling channels. The interval between extensive testing should be six to twelve months; otherwise, tests are performed whenever any degradation of performance is detected. Tests are performed according to procedures found in the spectrometer manuals or other literature (e.g. see Recommended Practice for Signal-to-Noise Determination in Fourier Transform NMR in ref. 3).

Note! *The chemical shift is sensitive to changes of the sample temperature; therefore, the temperature of the sample during the NMR experiment should be known and controlled. The calibration of the temperature control unit can be performed with either the ethylene glycol or methanol sample when operating near the room temperature (ref. 4).*

6.2 Solvents

Deuterated solvents should have a high enrichment of deuterium, >99.5%. Chloroform- d_1 , deuterium oxide, acetone- d_6 , dichloromethane- d_2 , acetonitrile- d_3 and methanol- d_4 are the most common solvents for NMR spectrometry. Chloroform- d_1 and deuterium oxide are considered to be the main solvents, but some others may be preferred for better dissolution. The purity of the solvents should be checked by long-term NMR experiments and, when appropriate, by gas chromatography.

6.3 Internal chemical shift references

Trichlorofluoromethane [$\text{Cl}^{35}\text{Cl}_2^{37}\text{CF}$, $\delta(\text{F}) = 0.00$ ppm] should be used in organic solvents as the chemical shift reference for fluorine spectra. The fine structure of the ^{19}F signal of CCl_3F results from the different isotopomers of the chlorine. Alternatively, fluorobenzene ($\text{C}_6\text{H}_5\text{F}$, $\delta(\text{F}) = -113.5$ ppm) or hexafluorobenzene (C_6F_6 , $\delta(\text{F}) = -163$ ppm) (ref. 5) may be used. To calibrate the chemical shift scale in water, trifluoroacetic acid can be used (CF_3COOH , $\delta(\text{F}) = -74.225$ ppm vs. external CCl_3F) (ref. 6).

6.4 External chemical shift references

If appropriate reference samples or NMR tube inserts are used, internal chemical shift references can also serve as external references.

7. Analysis of samples

7.1 Sequence of analysis

Put the NMR sample tube into the magnet. Wait until the sample temperature has reached the preset value (takes ca. 10 min.). Meanwhile, lock the field to the ^2H resonance of the solvent (if no deuterated solvent is present, skip this step) and set the appropriate experimental parameters. Tune the observation channel to ^{19}F and, if necessary, the decoupling channel. Adjust the magnetic field homogeneity. Adjust the receiver gain and start the data acquisition. After the NMR experiment is completed, store the original data (FID) on suitable media (e.g. hard disk and/or CD/DVD), and include all necessary information about the sample and recording conditions. Calculate the Fourier-transformed NMR spectrum, using a filter function if necessary, and perform the phase correction needed to obtain the real spectrum. The matched filter gives the best signal-to-noise ratio, but if better resolution is desired, use appropriate weighting (e.g. less line broadening). Always use the same or comparable conditions for the Fourier transformation. Use an internal or external reference to calibrate the spectrum.

Plot the whole spectrum. Print the parameter list of experimental conditions. Record all necessary expansions on a plotter, and print out the peak positions. If useful, integrate the spectrum and plot expansions with integrals. Formats other than paper are possible (e.g. pdf file format).

Compared with the sample matrix, the experimental parameters have a minor effect on the spectral parameters. Experimental parameters such as spectral width, flip angle, repetition time, number of points in the FID and in the real spectrum, number of pulses (scans), and processing parameters need to be comparable with those used for the acquisition of the database spectrum or spectrum of the authentic reference compound.

Note! *It is not always possible or feasible to use the same experimental parameters like number of pulses (scans) for the test sample as for the reference standard due to e.g. different concentration of the compound.*

7.2 Screening

^{19}F NMR experiments can be used for screening of fluorine-containing organic compounds, in which case ^{19}F NMR may provide a high level of proof for the presence of a particular compound. The fluorine chemical shift range is broad in comparison with protons (from -180 to 50 ppm for CWC-related compounds), and the fluorine chemical shift is generally characteristic of the fluorine chemical family (i.e. functional groups connected to the fluorine).

7.3 Identification

First, the spectrum is examined for resonance patterns, their positions, number of resonances, and their intensities or areas (Figure 1). This gives an idea of the chemical families (e.g. phosphonofluoridate) of the compounds of interest. Information obtained from each NMR experiment is consolidated. Comparisons are made with spectra in a computer database or spectral library, or with the spectrum of an authentic compound. However, derivation of the complete chemical structure solely with ^{19}F NMR experiments may not be possible; therefore, all available information, including that obtained by other techniques, is used to construct a complete picture.

To obtain a positive identification, the NMR spectral parameters for a compound must agree within certain limits with those of the database or the authentic compound. If the sample of the compound to be identified has been prepared so as to correspond in solvent, concentration, pH, and chemical shift references to the compound in the database (or authentic compound), the values of $\delta(\text{F})$ should agree with the corresponding values in the database (or authentic compound) within limits of ± 0.2 ppm. The values of coupling constants $^2\text{-}^3\text{J}(\text{FH})$ should agree within limits of ± 0.2 Hz and $^1\text{J}(\text{FP})$ should agree within 2 Hz. Also, the resonance patterns and their intensities must match. An identification made by spiking a sample with the suspected chemical requires exact overlapping of the resonances. The amount of compound spiked should increase the signals of the compound by at least a factor of 2 and a maximum factor of 10.

Note! It is not always possible to obtain the same conditions for the reference standard as for the test sample (e.g. matrix effect), so that differences in the NMR spectral parameters may be considerable. The identification criteria may then be expanded within reasonable limits (e.g. doubled). In this case, the identification by standard addition is recommended.

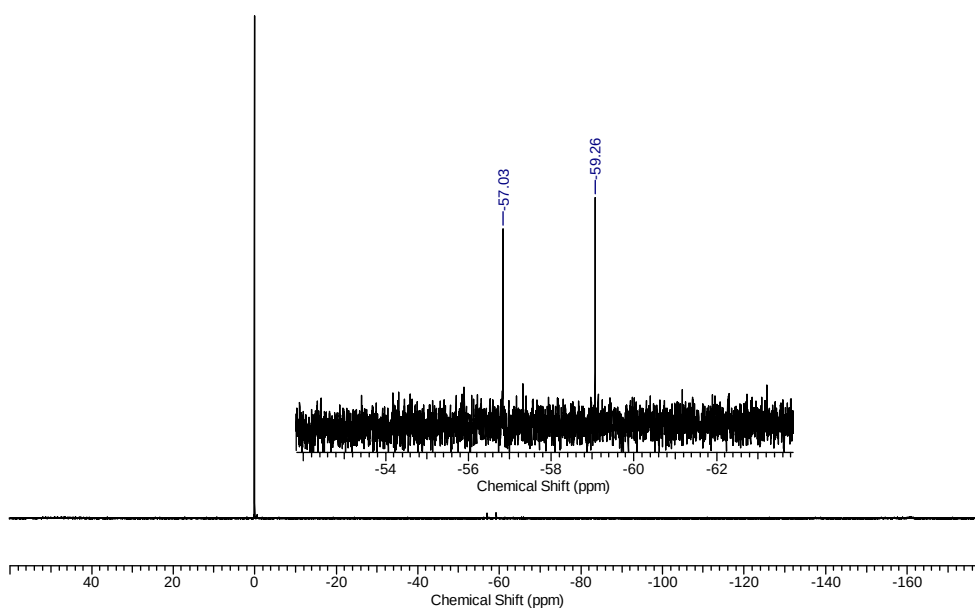


Figure 1. 470 MHz $^{19}\text{F}\{^1\text{H}\}$ NMR of sample O219 from the 19th Proficiency Test acquired with a 5 mm quadruple-nucleus (^1H , ^{13}C , ^{31}P , ^{19}F) after an experimental time of 96 min (concentration 8 $\mu\text{g}/\text{ml}$, solvent tetrachloroethylene, 150 μl benzene- d_6 added to 500 μl O219). (4-Methylpentyl methylphosphonofluoridate): $\text{J}(\text{PF}) = 1048$ Hz.

7.4 Quantification

NMR spectrometry is well-suited for quantification because, under certain experimental conditions, the signal intensity is directly proportional to the amount of nuclei present. Quantification accuracy of ca. 5–10% can be achieved. Quantification is performed most frequently using an internal standard, although other approaches may also be applied (for example, using the ERETIC method, which utilises a reference signal synthesised by an electronic device to determine absolute concentrations, see ref. 7, or another example is the PULCON method that correlates the absolute intensities of two spectra measured in different solution conditions, see ref. 8). A pulse angle of 90° and a repetition time (sum of the acquisition time and the relaxation delay) that equals five times the longest T_1 are recommended, using a pulse program that does not build up NOE. Signal-to-noise of the quantified compound and the internal reference must be comparable and above 10:1.

7.5 Trace analysis

The detection limit for a compound depends on the magnetic field strength of the spectrometer, the probe, the number of pulses, the flip angle, the observed nucleus and its relaxation time, the NMR tube diameter, the sample matrix, and the homogeneity of the magnetic field. With modern NMR spectrometers having a magnetic field of ca. 12 Tesla and a room-temperature probe head, a limit of detection of 1 to 5 $\mu\text{g/ml}$ can be obtained. With special probe heads (ref. 9, ref. 10), the sensitivity and, hence, the detection limit can be further improved, and the measurement time can be reduced (cf. Figure 2).

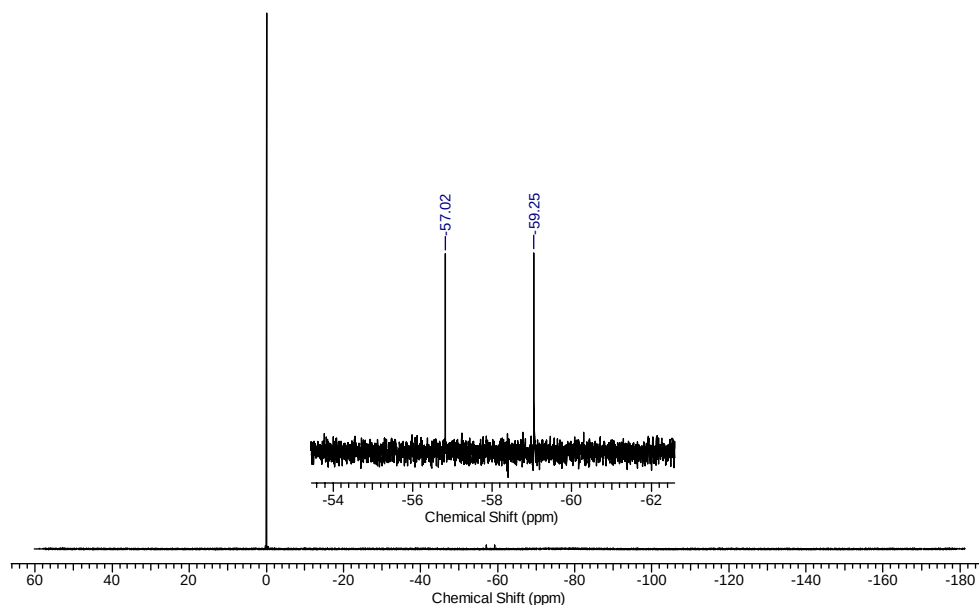


Figure 2. 470 MHz $^{19}\text{F}\{^1\text{H}\}$ NMR of sample O219 from the 19th Proficiency Test acquired with a 5 mm quadruple-nucleus (^1H , ^{13}C , ^{31}P , ^{19}F) cryoprobe after an experimental time of 11 min. (concentration 8 $\mu\text{g/ml}$, solvent tetrachloroethylene), 150 μl benzene- d_6 added to 500 μl O219. (4-Methylpentyl ,methylphosphonofluoridate): J(PF) = 1048 Hz.

7.6 Remarks on analysis

Non-deuterated solvents can be used if a small amount of the corresponding deuterated solvent is added to the sample to provide the lock conditions, if the experiment is performed unlocked, or if a lock nucleus other than deuterium is used. For strongly salty samples, cryoprobes are disadvantageous, but it could be beneficial using smaller diameter tubes (e.g. 3-mm tube and spinner on a 5-mm probe head) or microprobes. The 90° pulse width should be recalibrated. Paramagnetic salts in aqueous samples can enhance relaxation so that the resonances became too broad to be observable. In this case, a pretreatment must be performed (e.g. cation exchange, application of chelating agents).

Note! *Most treaty-related chemicals with fluorine hydrolyse rapidly in aqueous conditions, hence losing the fluorine atom. Observation of ^{19}F NMR signal from these compounds is, thus, not feasible.*

8. Evaluation of method

8.1 Area of applicability

The method is applicable to all compounds containing fluorine atoms dissolving to a sufficient degree in a solvent (concentration upper to the limit of detection). The analysed liquid sample must be homogeneous and without paramagnetic properties.

8.2 Advantages/Disadvantages

The main advantage of ^{19}F NMR spectroscopy is its selectivity, since it reveals the presence of organofluorine compounds in the sample, and the measurement does not degrade the sample. The disadvantages are the low information content of the spectrum in comparison with the information obtained in a ^1H NMR spectrum and the modest sensitivity (several orders of magnitude compared to GC–MS if absolute amounts of compounds are considered).

Although they are not described here, other one-dimensional (e.g. techniques utilising polarisation transfer) or two-dimensional heteronuclear correlation techniques can be used as well. Generally, to determine a complete picture of an organofluorine compound, complementary NMR techniques should be used.

9. References

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Chapter IV. Nonselective 1D ^1H - ^{31}P HSQC technique

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1. Scope

This section describes the use of high-resolution nuclear magnetic resonance (NMR) spectrometry utilising nonselective 1D ^1H - ^{31}P HSQC experiments (ref. 1) for the screening of samples for the presence of organophosphorus chemicals related to the Chemical Weapons Convention (CWC). Generally, the procedure is suitable for screening organophosphorus chemicals above a 1–5 $\mu\text{g/ml}$ concentration level. NMR measurements are carried out with modern NMR instruments typically equipped with 5-mm NMR probe heads.

2. Summary of the method

NMR techniques are commonly used for structural elucidation. With the nonselective 1D ^1H - ^{31}P HSQC spectrum interpretation, a set of information is obtained in the form of NMR parameters (chemical shifts and coupling constants) that converge to a chemical structure. The main information obtained by the nonselective 1D ^1H - ^{31}P HSQC experiment is the structure of the alkyl group (e.g. methyl, ethyl, propyl or isopropyl) directly bonded to the phosphorus atom and the number of different organophosphorus compounds present in a sample. Usually, a subset of all proton signals of a given organophosphorus compound is observed, and the experiment is mostly used for screening and rarely for identification of

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organophosphorus compounds (Fig. 1b). If used for identification, identification by addition of an authentic reference compound is recommended.

3. Safety aspects

The samples can potentially contain a high dosage of toxic chemicals, and sufficient safety measures should be used in handling. Whether to close the tube with a plastic cap or by flame must be decided on the basis of preliminary knowledge of the sample. If the concentration of the agent is high, the tube should be sealed by flame.

4. Instrumentation

4.1 General recommendations

The requirements for spectrometers suitable for the analysis of treaty-related compounds by the nonselective 1D ^1H - ^{31}P HSQC are as follows:

1. capability to observe ^1H nucleus,
2. capability to decouple ^{31}P nucleus,
3. capability to apply signal suppression on selected solvent peak(s),
4. capability for shaping rf pulses and pulsed field gradients,
5. capability for controlling of the sample temperature to an accuracy of within $\pm 0.5^\circ\text{C}$,
6. a proton resonance frequency preferably of 400 MHz or more, and
7. a data system adequate for acquisition, processing, and storage of all necessary data.

4.2 Probe head characteristics

The 5-mm room-temperature probe heads generally used in NMR spectroscopy are suitable for the analysis of treaty-related compounds. The coil geometry and number of channels can vary depending on the type and manufacturer of the probe head.

The 1D ^1H - ^{31}P HSQC experiment needs a probe head with two channels tuneable to ^1H and ^{31}P . Optimal sensitivity is achieved with so-called inverse probe heads, where the innermost RF coil is used for proton detection and the outer coil for ^{31}P pulses. Using cryogenic and microcoil probe heads may also increase sensitivity.

5. Experimental

5.1 Recommended conditions

Temperature: the temperature of the NMR sample must be controlled. Its value is between 20 to 30 $^\circ\text{C}$.

NMR sample: The volume of sample to analyse for a 5-mm NMR tube is 450 to 600 μl . It can be reduced by using a specific tube like shigemi (its price is often prohibitive) or by using the sandwiched solvent method (ref 2). To avoid cross-pollution, a new tube must indeed be used for each sample.

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NMR solvent: To lock the frequency of the spectrometer, addition of a miscible deuterated solvent to the liquid sample is required (50 to 150 μl). If possible, using a deuterated solvent similar to the protonated solvent is recommended. To determine the solvent of a liquid sample, a first unlocked ^1H NMR experiment can be performed. If a pre-treatment is required to analyse a sample, the used solvent can be fully deuterated.

NMR acquisition parameters: ^1H and ^{31}P pulse widths corresponding to angles of 90° and 180° , a coupling constant of 20 Hz, a gradient width of 1 ms, gradient strengths of -40%, 40% and 32.36%, an acquisition time of 1 s, a relaxation delay of 2.5 s, an ^1H spectral width of 12 ppm, ranging from -2 to 10 ppm and setting the ^{31}P transmitter to 30 ppm are recommended. These experimental parameters may be optimised for particular samples using the guidelines given in Table 1.

5.2 Selection of parameters

Table 1 provides guidelines for the selection of experimental parameters for nonselective 1D ^1H - ^{31}P HSQC experiments.

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Table 1. Guidelines for the selection of parameters in 1D ^1H - ^{31}P HSQC experiments.

Parameter	Recommendation
operation frequency * (MHz)	≥ 400 for ^1H
pulse sequence [#]	
flip angles (°)	90° (narrow rectangles in the pulse sequence), 180° (wide rectangles in the pulse sequence)
pulse duration (μs)	dependent on probe head and electronics, correlated to the flip angle (for a 90° pulse $\leq 20 \mu\text{s}$)
^1H spectral width (Hz)	large enough to show all resonances of interest (standard 12 ppm)
data points (FID)	enough to reveal the smallest coupling and shift difference of overlapping resonances (resolution 0.5–1.5 Hz)
digital resolution on spectrum	sufficient to give 2–3 datapoints per half linewidth
^{31}P transmitter	set to region of interest
polarisation transfer delay (Δ)	$\Delta = 1/2J_{^1\text{H}-^{31}\text{P}} = 25 \text{ ms}$ to 28 ms optimised for alkyl group ($J = 18$ to 20 Hz) and 40 to 50 ms optimised for alkyl ester or thioester group ($J = 10$ to 12.5 Hz) <i>For information: alkyl group bonded to P, $^2J(\text{PH})$, $^3J(\text{PH})$: 15–30 Hz</i> <i>alkyl ester or thioester group, $^3J(\text{PH})$: 7–12 Hz</i> <i>N,N-dialkyl phosphor/phosphonoamidates, $^3J(\text{PH})$: 10–30 Hz</i> <i>one bond $^1J(\text{PH})$: $\sim 700 \text{ Hz}$</i>
gradient width	1 ms
gradient strength	$G_1, -G_1, G_2 = \gamma_{^{31}\text{P}} / \gamma_{^1\text{H}} * 2 * G_1, G_2 = 0.81 * G_1$
number of pulses	enough to provide desired S/N (generally between 128 and 1024)
temperature (°C)	a known temperature between +20 and +30 °C. Notice the boiling point of the sample solvent.
filter function	exponential function with 0.5–1 Hz line broadening

*not an adjustable parameter, [#]different implementations are possible (e.g. ^{31}P decoupling during acquisition)

5.3 Optional modifications

^{31}P decoupling during acquisition can be used to simplify the appearance of the multiplets and to increase the sensitivity of the experiment. To increase the ^{31}P sampling range, the hard ^{31}P 180° pulses may be replaced by adiabatic pulses.

6. Quality control

6.1 Instrument performance test

The most important quality control operations are the testing of the spectrometer performance and following the identification criteria. The operating procedures included in the Quality Manual for the laboratory should complement the Recommended Operating Procedures (ROP).

Testing of the spectrometer performance to meet specifications requires a variety of test samples, usually provided by the instrument manufacturer. A set of test samples could be as follows:

1. ^1H resolution test sample: 5% of orthodichlorobenzene in acetone- d_6 .
2. ^1H lineshape and spinning sideband test sample: 1–3% chloroform in acetone- d_6 .
3. ^1H sensitivity test sample: 0.1% ethylbenzene in chloroform- d_1 .
4. ^{31}P sensitivity test sample: 0.0485 M triphenyl phosphate in acetone- d_6 .

Spectrometer performance tests consist of checking of lineshape, sensitivity, and resolution of a probe head for ^1H nucleus and the sensitivity of the ^{31}P nucleus. Other tests include checking the radio frequency field strengths of the observation and decoupling channels. Extensive testing should be six to twelve months; otherwise, tests are performed whenever any degradation of performance is detected. Tests are performed according to procedures found in the spectrometer manuals or other literature (e.g. see Recommended Practice for Signal-to-Noise Determination in Fourier Transform NMR in ref. 3).

Note! *The chemical shift is sensitive to changes of the sample temperature, and therefore the temperature of the sample during the NMR experiment should be known and controlled. The calibration of the temperature control unit can be performed with either the ethylene glycol or methanol sample when operating near the room temperature (ref. 4).*

6.2 Solvents

Deuterated solvents should have a high enrichment of deuterium, >99.5%. Chloroform- d_1 , deuterium oxide, acetone- d_6 , dichloromethane- d_2 , acetonitrile- d_3 , and methanol- d_4 are the most common solvents for NMR spectrometry. Chloroform- d_1 and deuterium oxide are considered to be the main solvents, but some others may be preferred for better dissolution. The purity of the solvents should be checked by long-term NMR experiments and, when appropriate, by gas chromatography.

6.3 Internal chemical shift references

The same chemical shift references as for the basic ^1H NMR experiments should be used:

Tetramethylsilane [TMS, $\delta(\text{H}) = 0.00$ ppm] should be used in organic solvents as the chemical shift reference for proton NMR spectra. The salts 3-(trimethylsilyl)propane sulfonic acid sodium salt (TSPSA, $\delta(\text{H}) = 0.015$ ppm) or 3-(trimethylsilyl)-3,3,2,2-tetradeuteriopropionic acid sodium salt (TSPA- d_4 , $\delta(\text{H}) = 0.00$ ppm) are commonly used as reference standards in water. They should not be used if reactivity with the sample is suspected. In water, *p*-dioxane [$\delta(\text{H}) = 3.76$ ppm vs. int. TSPSA (0.015 ppm) could be used instead. In chloroform, the shifts of *p*-dioxane are $\delta(\text{H}) = 3.70$ ppm, vs. int. TMS (0.00 ppm).

As these chemical shift references do not display a signal in a 1D ^1H - ^{31}P HSQC spectrum, a direct proton chemical shift calibration is not possible. Instead, a proton spectrum should be acquired first, followed by a transfer of the relevant parameters to the 1D ^1H - ^{31}P HSQC experiment.

6.4 External chemical shift references

If appropriate reference samples or NMR tube inserts are used, internal chemical shift references can also serve as external references. ^1H NMR spectra are acquired from the external chemical shift reference, followed by a transfer of the relevant parameters to the 1D ^1H - ^{31}P HSQC experiment.

7. Analysis of samples

7.1. Sequence of analysis

Put the NMR sample tube into the magnet. Wait until the sample temperature has reached the preset value (takes ca. 10 min.). Meanwhile, lock the field to the ^2H resonance of the solvent (if no deuterated solvent is present, skip this step) and set the appropriate experimental parameters. Tune the observation channel to ^1H and the decoupling channel to ^{31}P . Adjust the magnetic field homogeneity. Adjust the receiver gain and start the data acquisition. After the NMR experiment is completed, store the original data (FID) on suitable media (e.g. hard disk and/or external media), and include all necessary information about the sample and recording conditions. Calculate the Fourier-transformed NMR spectrum, using a filter function if necessary, and perform the phase correction needed to obtain the real spectrum. The matched filter gives the best signal-to-noise ratio, but if better resolution is desired, use appropriate weighting (e.g. less line broadening). Always use the same or comparable conditions for the Fourier transformation. Reference the spectrum.

Plot the whole spectrum. Print the parameter list of experimental conditions. Record all necessary expansions on a plotter, and print out the peak positions. If useful, integrate the spectrum and plot expansions with integrals. Formats other than paper are possible (pdf file format, for example).

Compared with the sample matrix, the experimental parameters have a minor effect on the spectral parameters. Experimental parameters such as spectral width, flip angle, repetition time, number of points in the FID and in the real spectrum, number of pulses (scans), and

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processing parameters need to be comparable with those used for the acquisition of the database spectrum or spectrum of the authentic reference compound.

Note! *It is not always possible or feasible to use the same experimental parameters like number of pulses (scans) for the test sample as for the reference standard due to, e.g., the different concentration of the compound.*

7.2 Screening

Screening solutions for the presence of organophosphorus chemicals relevant to the CWC is the main application of the nonselective 1D ^1H - ^{31}P HSQC experiment. In this case, the 1D ^1H - ^{31}P HSQC experiment may provide a high level of proof for the presence of a particular organophosphorus chemical.

7.3 Identification

First, the spectrum is examined for resonance patterns, their positions, number of resonances, and their intensities or areas. This gives an idea of the functional groups of the compounds of interest. Information obtained from each NMR experiment is consolidated. Comparisons are made with spectra in a computer database or spectral library, or with the spectrum of an authentic compound. All available information, including that obtained by other techniques, is used to construct a complete picture.

To obtain a positive identification, the NMR spectral parameters for a compound must agree within certain limits with those of the database or the authentic compound. If the sample of the compound to be identified has been prepared so as to correspond in solvent, concentration, pH, and chemical shift references to the compound in the database (or authentic compound), the values of $\delta(\text{H})$ should agree with the corresponding values in the database (or authentic compound) within limits of ± 0.05 ppm. The values of coupling constants $J(\text{HH})$ and $J(\text{PH})$ should agree within limits of ± 0.2 Hz, (except for $^1J(\text{PH}) \pm 2$ Hz). In addition, the resonance patterns and their intensities must match. An identification made by spiking a sample with the suspected chemical requires exact overlapping of the resonances and is recommended for the identification of compounds by the nonselective 1D ^1H - ^{31}P HSQC experiment. The amount of compound spiked should increase the signals of the compound by at least a factor of 2 and a maximum factor of 10.

Note! *It is not always possible to obtain the same conditions for the reference standard as for the test sample (e.g. matrix effect), so that differences in the NMR spectral parameters may be considerable. The identification criteria may then be expanded within reasonable limits (e.g. doubled). In this case, the identification by adding the authentic compound in the sample is recommended.*

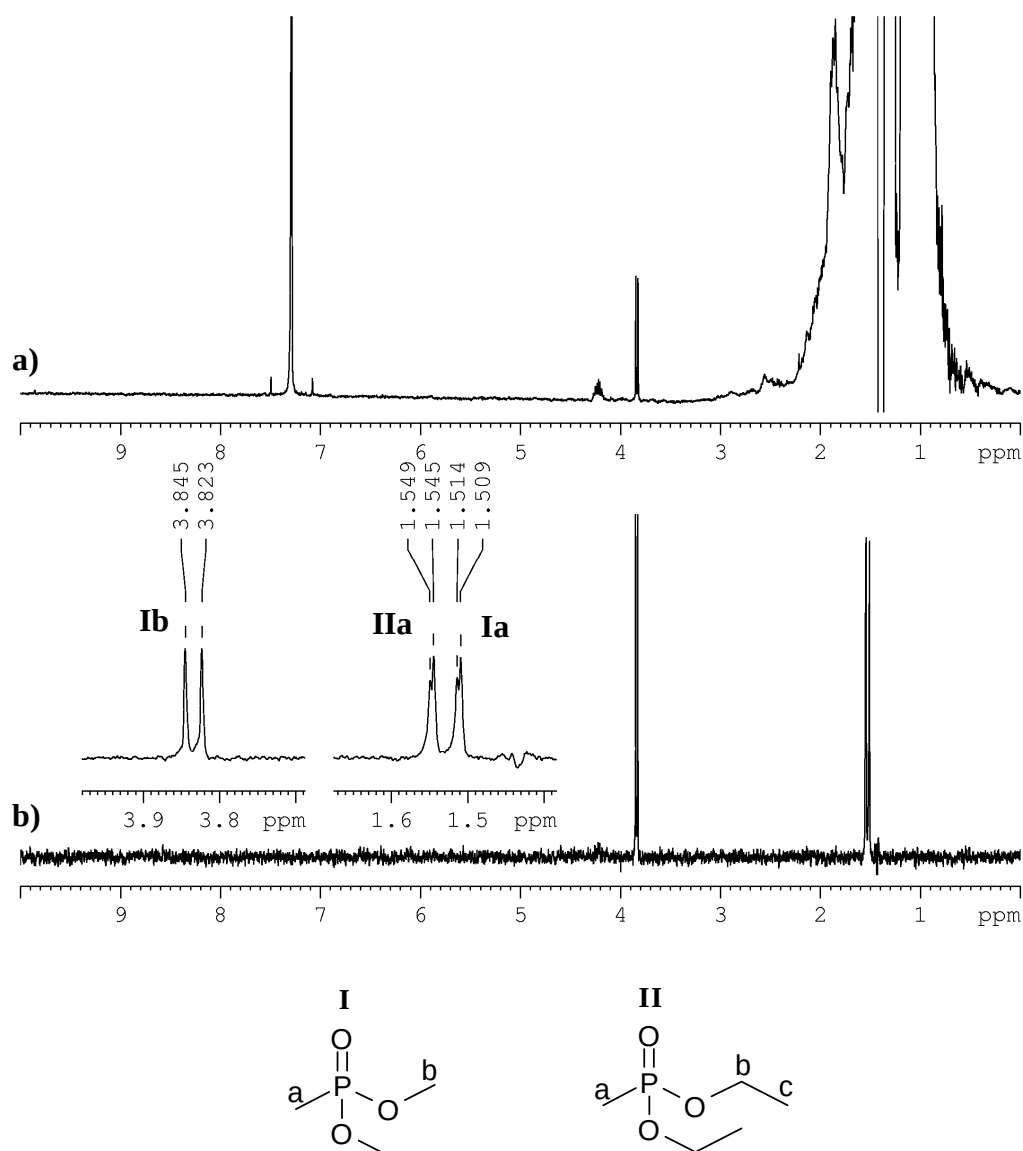


Figure 1. Sample OX from the 23rd Proficiency Test (concentration 10 $\mu\text{g}/\text{ml}$, solvent n-decane, 100 μl chloroform- d_1 added to 500 μl OX). Dimethyl methylphosphonate (I); $J(\text{aP}) = 17.7$ Hz, $J(\text{bP}) = 10.9$ Hz. Diethyl methylphosphonate (II): $J(\text{aP}) = 17.7$ Hz. a) 500 MHz ^1H NMR spectrum (256 scans, 23 min), b) 500 MHz 1D ^1H - ^{31}P HSQC spectrum (368 scans, 28 min) acquired with a 5-mm inverse broadband probe head.

7.4 Quantification

NMR spectrometry is well-suited for quantification because, under certain experimental conditions, the signal intensity is directly proportional to the amount of nuclei present. Quantification accuracy of ca. 5–10% can be achieved. In the nonselective 1D ^1H - ^{31}P HSQC experiment, signal intensity is strongly dependent on the $J(\text{PH})$ of the protons and the used polarisation transfer delay, so the relative number of protons contributing to a signal cannot be deduced like in a ^1H NMR spectrum. However, when taking these variations into account, the nonselective 1D ^1H - ^{31}P HSQC experiment can be used for the quantification of organophosphorus analytes using an external calibration (ref. 5).

7.5 Trace analysis

The detection limit for a compound depends on the magnetic field strength of the spectrometer, the probe, the number of pulses, the flip angles, the observed nucleus and its relaxation time, the NMR tube diameter, the sample matrix, and the homogeneity of the magnetic field. With modern NMR spectrometers having a magnetic field of ca. 12 Tesla and a room-temperature probe head, a limit of detection of 1 $\mu\text{g/ml}$ can be obtained. With special probe heads (ref. 6, ref. 7), the sensitivity and, hence, the detection limit can be further improved, and the measurement time can be reduced (Figures 2 and 3).

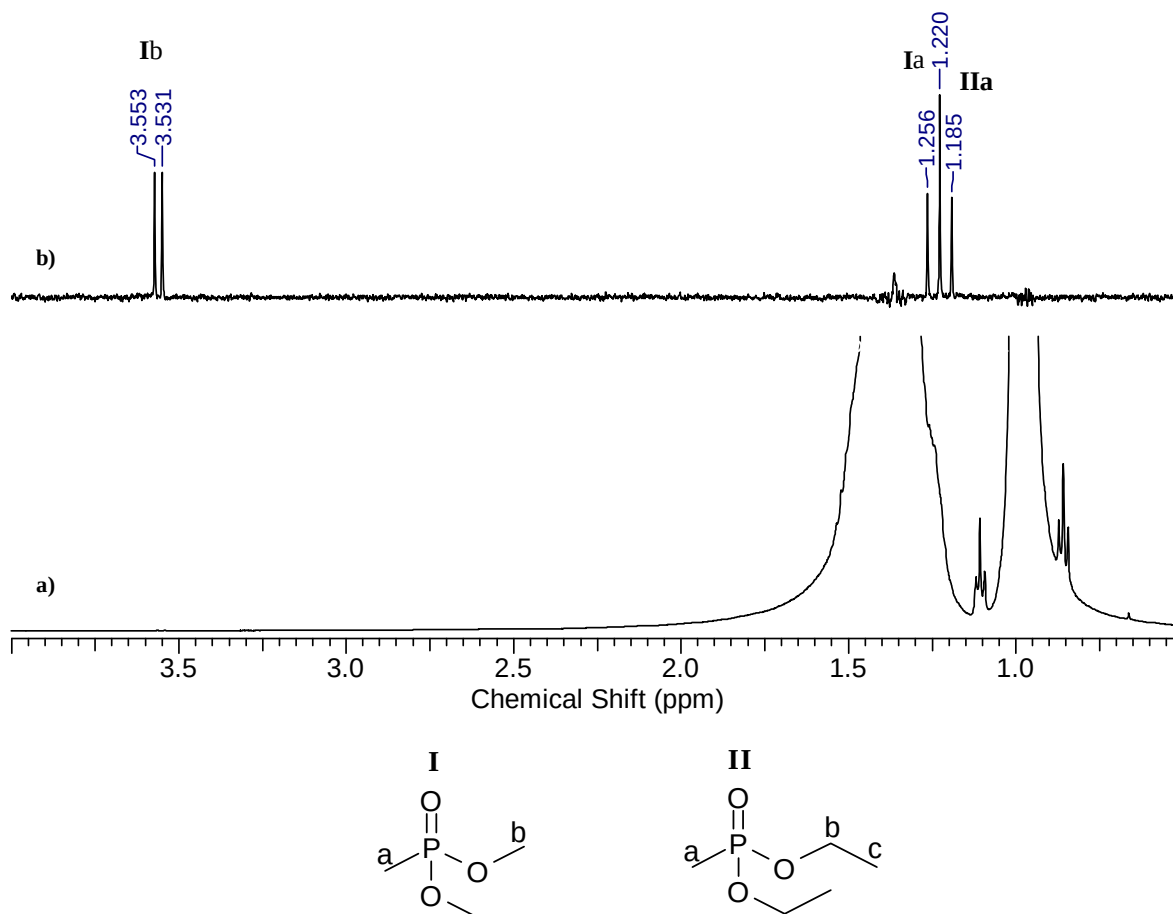


Figure 2. Sample OX from the 23rd Proficiency Test (concentration 10 $\mu\text{g/ml}$, solvent n -decane, 150 μl benzene- d_6 added to 500 μl OX. Dimethyl methylphosphonate: $J(\text{aP}) = 17.7$ Hz, $J(\text{bP}) = 10.9$ Hz. Diethyl methylphosphonate: $J(\text{aP}) = 17.7$ Hz. a) 500 MHz ^1H NMR spectrum (64 scans, 5 min), b) 500 MHz 1D ^1H - ^{31}P HSQC spectrum (64 scans, 5 min) acquired with a 5-mm quadruple-nucleus (^1H , ^{13}C , ^{31}P , ^{19}F) cryoprobe.

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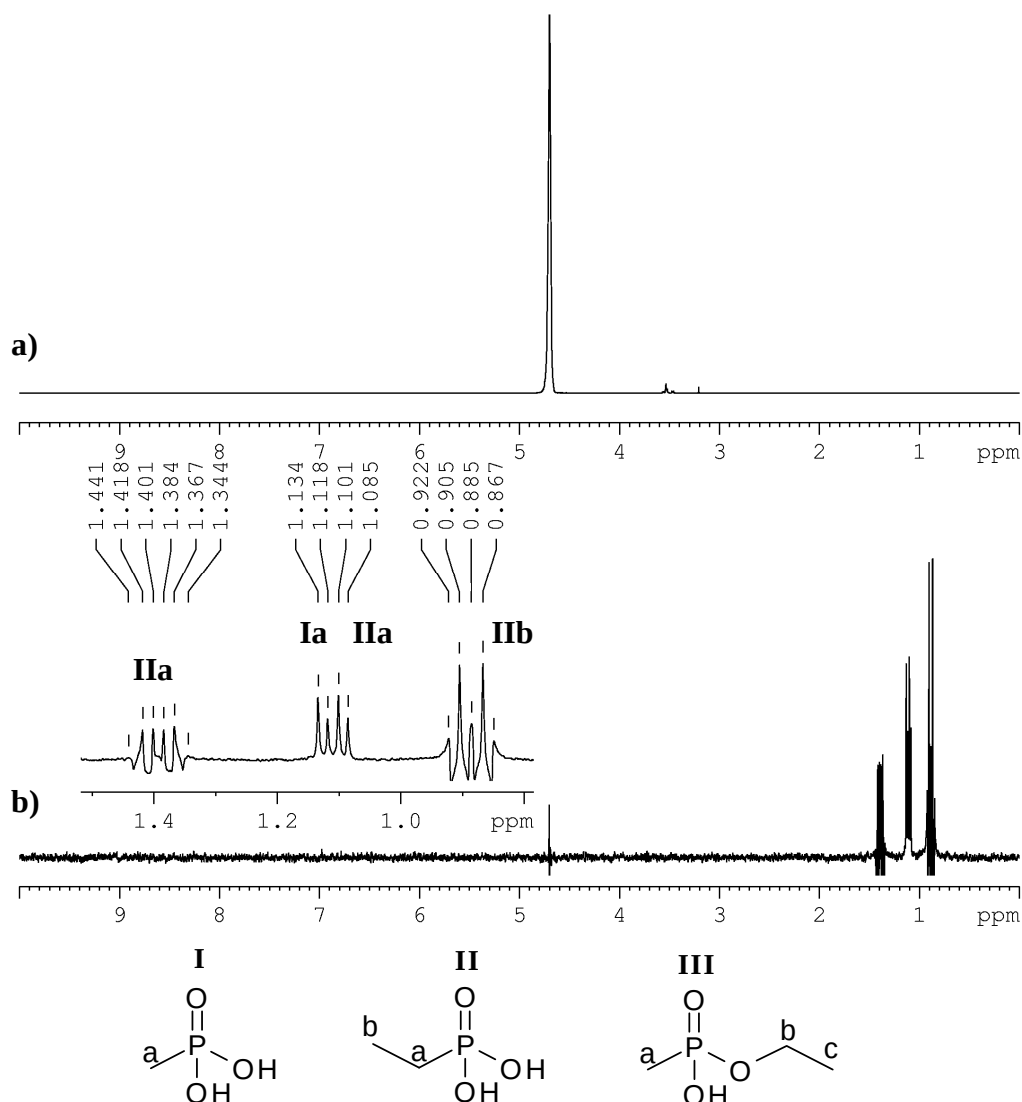


Figure 3. Sample W from the 22nd Proficiency Test (concentration 2–10 $\mu\text{g}/\text{ml}$, solvent water, sample enrichment of 1 ml sample W to 20 μl , 10 μl D_2O added. Methylphosphonic acid (I), ethylphosphonic acid (II), ethyl methylphosphonic acid (III). a) 500 MHz ^1H NMR spectrum (1 scan), b) 500 MHz 1D ^1H - ^{31}P HSQC spectrum (128 scans, 9 min) acquired with a 1.7-mm triple-resonance (^1H , ^{13}C , ^{31}P) microcoil probe head.

7.6 Remarks on analysis

Non-deuterated solvents can be used if a small amount of the corresponding deuterated solvent is added to the sample to provide the lock conditions, if the experiment is performed unlocked, or if a lock nucleus other than deuterium is used. For strongly salty samples, cryoprobes are disadvantageous, but it could be beneficial using smaller diameter tubes (e.g. 3-mm tube and spinner on a 5-mm probe head) or microprobes. The 90° pulse widths should be recalibrated. Paramagnetic salts in aqueous samples can enhance relaxation so that the resonances became too broad to be observable. In this case, a pretreatment must be performed (e.g. cation exchange, application of chelating agents).

8. Evaluation of method

8.1 Area of applicability

The method is applicable to all compounds containing hydrogen atoms other than exchangeable with a scalar coupling to a phosphorus atom and dissolving in sufficient amount in a solvent. The analysed liquid sample must be homogeneous and without paramagnetic properties.

8.2 Advantages/Disadvantages

The main advantage of the 1D ^1H - ^{31}P HSQC experiment is its selectivity: signals exclusively from protons with a scalar coupling to ^{31}P nuclei are recorded. Signals from the sample background obscuring signals from organophosphorus compounds in ^1H NMR spectra are efficiently removed (Fig. 1a and b). Compared to the $^{31}\text{P}\{^1\text{H}\}$ experiment, the sensitivity is considerably enhanced. The disadvantage is the modest sensitivity compared to mass spectrometric techniques (several orders of magnitude compared to GC-MS if absolute amounts of compounds are considered). Another disadvantage is that not all proton signals of a compound are present in 1D ^1H - ^{31}P HSQC spectra – only protons with a scalar coupling with phosphorous are detectable. To obtain a complete picture of a compound (with all spin-systems assigned), other NMR techniques can be recorded.

9. References

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Chapter V. 2D ^1H - ^{31}P correlation techniques

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1. Scope

This section describes the use of high-resolution nuclear magnetic resonance (NMR) spectrometry utilising 2D ^1H - ^{31}P correlation techniques like 2D ^1H - ^{31}P HSQC (ref. 1) and 2D ^1H - ^{31}P HMQC (ref. 2) for the screening and identification of organophosphorus chemicals related to the Chemical Weapons Convention (CWC). Generally, the procedure is suitable for identification of organophosphorus chemicals above the 1 $\mu\text{g/ml}$ concentration level. NMR measurements are carried out with modern NMR instruments typically equipped with 5-mm NMR probe heads.

2. Summary of the method

NMR techniques are commonly used for structural elucidation. With the 2D ^1H - ^{31}P HSQC/HMQC spectrum interpretation, a set of information is obtained in the form of NMR parameters (chemical shifts and coupling constants) that converge to a chemical structure. The information obtained with these two experiments is virtually identical; only the appearance of the correlation peaks may vary due to different $J(\text{HH})$ evolution characteristics during the pulse sequences. The main information obtained by the 2D ^1H - ^{31}P HSQC/HMQC experiments is the information about organophosphorus chemical family, the structure of the alkyl group (e.g. methyl, ethyl, propyl or isopropyl) directly

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bonded to the phosphorus atom, and the number of different organophosphorus compounds present in a sample. Usually, a subset of all proton signals with a scalar coupling to phosphorus of organophosphorus compounds is observed. Identification of a compound is achieved by comparison with a reference spectrum of the authentic compound. The number of resonances, chemical shifts, and coupling constants must be consistent.

3. Safety aspects

The samples can potentially contain a high dosage of toxic chemicals, and sufficient safety measures should be used in handling. Whether to close the tube with a plastic cap or by flame must be decided on the basis of preliminary knowledge of the sample. If the concentration of the agent is high, the tube should be sealed by flame.

4. Instrumentation

4.1 General recommendations

The requirements for spectrometers suitable for the analysis of treaty-related compounds by the 2D ^1H – ^{31}P correlation experiments are as follows:

1. capability to observe ^1H nucleus,
2. capability to decouple ^{31}P nucleus,
3. capability to apply signal suppression on selected solvent peak(s),
4. capability for shaping rf pulses and pulsed field gradients,
5. capability for controlling of the sample temperature to an accuracy of within $\pm 0.5\text{ }^\circ\text{C}$,
6. a proton resonance frequency preferably of 400 MHz or more, and
7. a data system adequate for acquisition, processing, and storage of all necessary data.

4.2 Probe head characteristics

The 5-mm room-temperature probe heads generally used in NMR spectroscopy are suitable for the analysis of treaty-related compounds. The coil geometry and number of channels can vary depending on the type and manufacturer of the probe head.

The 2D ^1H – ^{31}P HSQC/HMQC experiments need a probe head with two channels tuneable to ^1H and ^{31}P . Optimal sensitivity is achieved with so-called inverse probe heads, where the innermost RF coil is used for proton detection and the outer coil for ^{31}P pulses. Using cryogenic and microcoil probe heads may also increase sensitivity (see section 7.5).

5. Experimental

5.1 Recommended conditions

Temperature: the temperature of the NMR sample must be controlled. Its value is between 20 and 30 $^\circ\text{C}$. Notice the boiling point of the sample solvent.

NMR sample: The volume of sample to analyse for a 5-mm NMR tube is 450 to 600 μl . It can be reduced by using a specific tube-like shigemi (its price is often prohibitive) or by

using the sandwiched solvent method (ref. 3). To avoid cross-pollution, a new tube must indeed be used for each sample.

NMR solvent: To lock the frequency of the spectrometer, addition of miscible deuterated solvent in the liquid sample is required (50 to 150 μl). If possible, using a deuterated solvent similar to the protonated solvent is recommended. To determine the solvent of a liquid sample, first an unlocked ^1H NMR experiment should be performed. If a pretreatment is required to analyse a sample, solvent exchange to the compatible deuterated solvent is recommended.

NMR acquisition parameters: A ^1H spectral width of 12 ppm ranging from -2 to 10 ppm and a ^{31}P spectral width of 200 ppm, ranging from -30 to 170 ppm, are recommended. The experimental parameters may be optimised for particular samples using the guidelines given in Table 1 and Table 2.

5.2 Selection of parameters

Table 1 and Table 2 provide guidelines for the selection of acquisition and processing parameters for 2D ^1H - ^{31}P HSQC/HMQC experiments.

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Table 1. Guidelines for the selection of parameters in 2D ^1H - ^{31}P HSQC experiment (see ref. 1 for details).

Parameter	Recommendation
operation frequency* (MHz)	≥ 400 for ^1H
pulse sequence [#]	
flip angles (°)	90° (narrow rectangles in the pulse sequence), 180° (wide rectangles in the pulse sequence)
pulse duration (μs)	dependent on probe head and electronics, correlated to the flip angle (for a 90° pulse ≤ 20 μs)
^1H spectral width (ppm)	large enough to show all resonances of interest (typically 12 ppm)
^{31}P spectral width (ppm)	large enough to show all resonances of interest (typically 200 ppm)
data points (FID)	enough to provide sufficient spectral resolution (< 1 Hz per point)
number of increments	enough to provide sufficient spectral resolution (< 100 Hz per point) on ^{31}P dimension (typically 64 to 512)
polarisation transfer delay (Δ)	27.8 ms (= $1/(2 \times 18 \text{ Hz})$) <i>For information: alkyl group bonded to P, $^2J(\text{PH})$, $^3J(\text{PH})$: 15–30 Hz</i> <i>alkyl ester or thioester group, $^3J(\text{PH})$: 7–12 Hz</i> <i>N,N-dialkyl phosphor/phosphonoamidates, $^3J(\text{PH})$: 10–30 Hz</i> <i>one bond $^1J(\text{PH})$: ~700 Hz</i>
repetition time	2 to 3 s (sum of acquisition time and relaxation delay)
gradient width (δ)	1 ms
gradient strength	$g_2 = \gamma_{^{31}\text{P}} / \gamma_{^1\text{H}} \times g_1 = 0.405 \times g_1$
number of scans	enough to provide desired S/N (generally between 2 and 64) per increment
digital resolution on spectrum	< 1 Hz per point on ^1H dimension, < 100 Hz per point on ^{31}P dimension
filter function	Squared sine with 45° to 90° shifted sine bell on both dimensions

*not an adjustable parameter, [#]different implementations are possible (e.g. ^{31}P decoupling during acquisition)

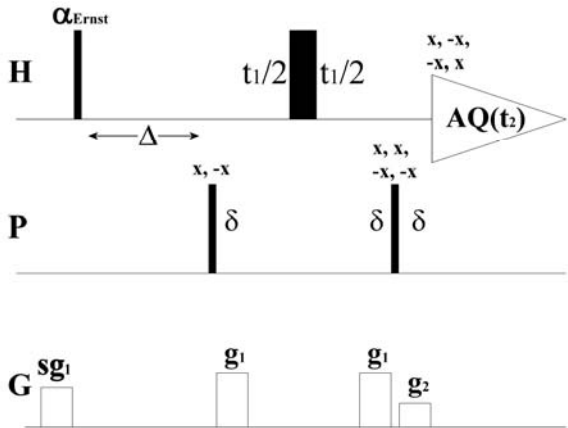
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Table 2. Guidelines for the selection of parameters in 2D ^1H - ^{31}P HMQC experiment (see ref. 2 for details).

Parameter	Recommendation
operation frequency * (MHz)	≥ 400 for ^1H
pulse sequence [#]	
flip angles (°)	128.5° (Ernst angle pulse in the pulse sequence), 90° (narrow rectangles in the pulse sequence), 180° (wide rectangles in the pulse sequence)
pulse duration (μs)	dependent on probe head and electronics, correlated to the flip angle (for a 90° pulse ≤ 20 μs)
^1H spectral width (ppm)	large enough to show all resonances of interest (typically 12 ppm)
^{31}P spectral width (ppm)	large enough to show all resonances of interest (typically 200 ppm)
data points (FID)	enough to provide sufficient spectral resolution (< 1 Hz per point)
number of increments	enough to provide sufficient spectral resolution (< 100 Hz per point) on ^{31}P dimension (typically 64 to 512)
polarisation transfer delay (Δ)	20.8 ms (= $1/(2 \times 24 \text{ Hz})$) For information: alkyl group bonded to P, $^2J(\text{PH})$, $^3J(\text{PH})$: 15–30 Hz alkyl ester or thioester group, $^3J(\text{PH})$: 7–12 Hz <i>N,N</i> -dialkyl phosphor/phosphonoamidates, $^3J(\text{PH})$: 10–30 Hz one bond $^1J(\text{PH})$: ca. 700 Hz
repetition time	1 s (sum of acquisition time and relaxation delay)
gradient width (δ)	1 ms
gradient strength	$g_2 = \gamma_{^{31}\text{P}} / \gamma_{^1\text{H}} \times 2 \times g_1 = 0.81 \times g_1$, $sg_1 = 0.66 \times g_1$
number of scans	enough to provide desired S/N (generally between 8 and 100) per increment
digital resolution on spectrum	< 1 Hz per point on ^1H dimension, < 100 Hz per point on ^{31}P dimension
filter function	squared sine with 45° to 90° shifted sine bell on both dimensions

*not an adjustable parameter, [#]different implementations are possible

5.3 Optional modifications

The 2D ^1H – ^{31}P HSQC experiment can be conducted in a phase-sensitive mode, which will provide narrower peaks in the spectrum. A ^{31}P decoupling during acquisition can be used to simplify the appearance of the multiplets and to increase the sensitivity of the experiment. To increase the ^{31}P excitation range, the hard ^{31}P 180° pulses may be replaced by adiabatic pulses.

The 2D ^1H – ^{31}P HMQC experiment is not phase-sensitive and must be processed in magnitude mode. A better resolution can be achieved using squared sine filtering with 128° and 147° shifted sine bell on proton and phosphorus dimensions, respectively. The experiment is accomplished using the Ernst angle excitation, which will provide the optimised sensitivity. The HMQC is less sensitive to the ^{31}P 90° pulse value deviation compared to the HSQC sequence.

6. Quality control

6.1 Instrument performance test

The most important quality control operations are the testing of the spectrometer performance and following the identification criteria. The operating procedures included in the Quality Manual for the laboratory should complement the Recommended Operating Procedures (ROP).

Testing of the spectrometer performance to meet specifications requires a variety of test samples, usually provided by the instrument manufacturer. A set of test samples could be as follows:

1. ^1H resolution test sample: 5% of orthodichlorobenzene in acetone- d_6 .
2. ^1H lineshape and spinning sideband test sample: 1–3% chloroform in acetone- d_6 .
3. ^1H sensitivity test sample: 0.1% ethylbenzene in chloroform- d_1 .
4. ^{31}P sensitivity test sample: 0.0485 M triphenylphosphate in acetone- d_6 .

Spectrometer performance tests consist of checking of lineshape, sensitivity, and resolution of a probe head for ^1H nucleus and the sensitivity for the ^{31}P nucleus. Other tests include checking the radio frequency field strengths of the observation and decoupling channels. The interval between extensive testing should be six to twelve months; otherwise, tests are performed whenever any degradation of performance is detected. Tests are performed according to procedures found in the spectrometer manuals or other literature (e.g. see Recommended Practice for Signal-to-Noise Determination in Fourier Transform NMR in ref. 5).

Note! The chemical shift is sensitive to changes of the sample temperature; therefore, the temperature of the sample during the NMR experiment should be known. The calibration of the temperature control unit can be performed with either the ethylene glycol or methanol sample when operating near room temperature (ref. 4).

6.2 Solvents

Deuterated solvents should have a high enrichment of deuterium, > 99.5%. Chloroform- d_1 , deuterium oxide, acetone- d_6 , dichloromethane- d_2 , acetonitrile- d_3 , and methanol- d_4 are the

most common solvents for NMR spectrometry. Chloroform- d_1 and deuterium oxide are considered to be the main solvents, but others may be preferred for better dissolution. The purity of the solvents should be checked by long-term NMR experiments and, when appropriate, by gas chromatography.

6.3 Internal chemical shift references

The same chemical shift references as for basic ^1H NMR experiments and basic ^{31}P NMR experiments should be used for the corresponding dimension:

Tetramethylsilane [TMS, $\delta(\text{H}) = 0.00$ ppm] should be used in organic solvents as the chemical shift reference for proton NMR spectra. The salts 3-(trimethylsilyl)propane sulfonic acid sodium salt (TSPSA, $\delta(\text{H}) = 0.015$ ppm) or 3-(trimethylsilyl)3,3,2,2-tetradeuteriopropionic acid sodium salt (TSPA- d_4 , $\delta(\text{H}) = 0.00$ ppm) are commonly used as reference standards in water. They should not be used if reactivity with the sample is suspected. In water, *p*-dioxane [$\delta(\text{H}) = 3.76$ ppm vs. int. TSPSA (0.015 ppm)] could be used instead. In chloroform, the shifts of *p*-dioxane are $\delta(\text{H}) = 3.70$ ppm vs. int. TMS (0.00 ppm).

Triphenyl phosphate (TPP) dissolves well in organic solvents and could serve as an internal chemical shift reference for phosphorus. The chemical shift of TPP is -17.80 ppm.

As these chemical shift references do not display a signal in a 2D ^1H - ^{31}P HSQC/HMQC spectrum, a direct proton/phosphorus chemical shift calibration is not possible. Instead, ^1H NMR and ^{31}P NMR spectra should be acquired first, followed by a transfer of the relevant parameters to the 2D ^1H - ^{31}P HSQC/HMQC experiments.

6.4 External chemical shift references

If appropriate reference samples or NMR tube inserts are used, internal ^1H chemical shift references can also serve as external references.

The most common external chemical shift reference in ^{31}P NMR is 85% phosphoric acid. In aqueous samples, $\delta(\text{P}) = 0.00$ ppm is measured in an ideal spherical container inside the solution (see ref. 2). In practice, however, it is more convenient to use a cylindrical capillary tube to hold the standard, and a correction should then be applied to take into account the differences in diamagnetic susceptibility. To correct for diamagnetic susceptibility differences in aqueous samples, the signal of 85% H_3PO_4 in a cylindrical capillary in superconducting solenoid spectrometers should be set at -0.73 ppm. If a narrower line is desired, a 0.2 M solution of crystalline phosphoric acid in 14% aqueous perchloric acid (tetrahydroxyphosphonium perchlorate, $\text{P}(\text{OH})_4\text{ClO}_4$, see ref. 3) in a cylindrical capillary could be chosen, and the signal should then be set at $\delta(\text{P}) = 0.09$ ppm in superconducting solenoid spectrometers. For other solvents, corrections to the phosphorus chemical shifts of H_3PO_4 and tetrahydroxyphosphonium perchlorate would be different.

For reasons of comparability, the use of $\delta(\text{P}) = 0.00$ ppm for 85% H_3PO_4 (in a cylindrical capillary tube) without corrections for diamagnetic susceptibility differences is recommended as a primary chemical shift reference. Because of the number of secondary standards, the referencing method should always be reported.

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^1H NMR and ^{31}P NMR spectra are acquired from the external chemical shift reference, followed by a transfer of the relevant parameters to the 2D ^1H - ^{31}P HSQC/HMQC experiment.

7. Analysis of samples

7.1. Sequence of analysis

Put the NMR sample tube into the magnet. Wait until the sample temperature has reached the preset value (takes ca. 10 min.). Meanwhile, lock the field to the ^2H resonance of the solvent (if no deuterated solvent is present, skip this step) and set the appropriate experimental parameters. Tune the observation channel to ^1H and the decoupling channel to ^{31}P . Adjust the magnetic field homogeneity. Adjust the receiver gain and start the data acquisition. After the NMR experiment is completed, store the original data (FID) on suitable media (e.g. hard disk and/or external media), and include all necessary information about the sample and recording conditions. Calculate the Fourier-transformed NMR spectrum, using a suitable filter function, and perform the phase correction if the spectrum has been acquired in phase-sensitive mode. Always use the same or comparable conditions for the Fourier transformation. Record ^1H NMR and ^{31}P NMR spectra for projections, if necessary. Reference the spectrum.

Plot the whole spectrum. Print the parameter list of experimental conditions. Record all necessary expansions on a plotter. Formats other than paper are possible (pdf file format, for example).

Compared with the sample matrix, the experimental parameters have a minor effect on the spectral parameters. Experimental parameters such as spectral width, flip angle, repetition time, number of points in the FID and in the real spectrum, number of pulses (scans), and processing parameters need to be comparable with those used for the acquisition of the spectrum of the authentic reference compound.

Note! *It is not always possible or feasible to use the same experimental parameters like number of pulses (scans) for the test sample as for the reference standard due to, e.g., the different concentration of the compound.*

7.2 Screening

The 2D ^1H - ^{31}P HSQC/HMQC experiment can be used for screening solutions for the presence of organophosphorus chemicals relevant to the CWC. In this case, the 2D ^1H - ^{31}P HSQC/HMQC experiment may provide a high level of proof for the presence of a particular organophosphorus chemical.

7.3 Identification

First, the spectrum is examined for resonance patterns, their positions, number of resonances, and their intensities or areas (Figure 1). This gives an idea of the organophosphorus chemical family, and the structure of the alkyl group of the compounds of interest. An immediate identification of a compound on the basis of familiar features may also result. Information obtained from each NMR experiment is consolidated. Comparisons are made with spectra in a computer database or with the spectrum of an

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authentic compound. All available information, including that obtained by other techniques, is used to construct a complete picture.

To obtain a positive identification, the NMR spectral parameters for a compound must agree within certain limits with those of the database or the authentic compound. If the sample of the compound to be identified has been prepared so as to correspond in solvent, concentration, pH, and chemical shift references to the compound in the database (or authentic compound), the values of $\delta(\text{H})$ and $\delta(\text{P})$ should agree with the corresponding values in the database (or authentic compound) within limits of ± 0.05 ppm and ± 0.2 ppm, respectively. The values of coupling constants $J(\text{HH})$ and $J(\text{PH})$ should agree within limits of ± 0.2 Hz (except for $^1J(\text{PH}) \pm 2$ Hz). In addition, the resonance patterns and their intensities must match. An identification made by spiking a sample with the suspected chemical requires exact overlapping of the resonances. The amount of compound spiked should increase the peak integral of the compound by at least a factor of 2 and a maximum factor of 10.

Note. *It is not always possible to obtain the same conditions for the reference standard as for the test sample (e.g. matrix effect), so that differences in the NMR spectral parameters may be considerable. The identification criteria may then be expanded within reasonable limits (e.g. doubled). In this case, the identification by adding the authentic compound in the sample is recommended.*

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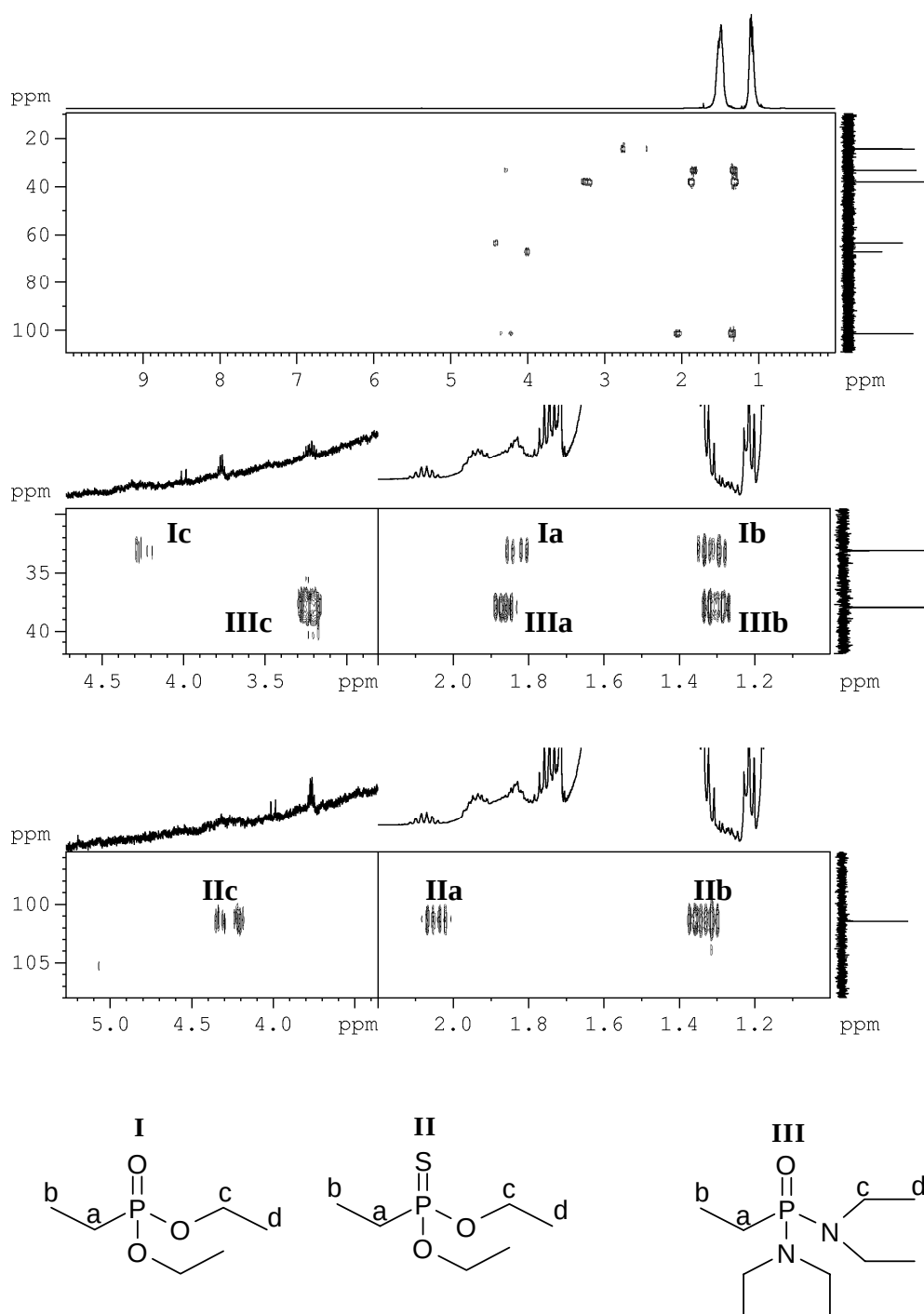


Figure 1. Sample O from the 22nd Proficiency Test (concentrations 10–20 $\mu\text{g}/\text{ml}$), solvent hexane, 100 μl dichloromethane- d_2 added to 500 μl O. Diethyl ethylphosphonate (I), O,O-diethyl ethylphosphonothionate (II), Bis(N,N-diethyl) ethylphosphonodiamide (III). 500 MHz 2D ^1H - ^{31}P HMQC (100 scans per increment, 200 increments, 5 h 46 min), 500 MHz ^1H NMR (256 scans, 24 min), 202 MHz $^{31}\text{P}\{^1\text{H}\}$ NMR (16384 scans, 5 h 28 min) acquired with a 5 mm inverse broadband probe head.

7.4 Quantification

The peak volumes in the 2D ^1H - ^{31}P HSQC/HMQC spectra are dependent on a multitude of parameters. Therefore, these experiments are not recommended for routine quantification of sample compounds. However, the observed cross-peak volume can be regarded as linear with respect to the concentration of the analyte, and peak volume integration can be used in standard addition for identification of particular compound.

7.5 Trace analysis

The detection limit for a compound depends on the magnetic field strength of the spectrometer, the probe head, the number of pulses, the flip angles, the observed nucleus and its relaxation time, the NMR tube diameter, the sample matrix, and the homogeneity of the magnetic field. With modern NMR spectrometers having a magnetic field of ca. 12 Tesla and a room-temperature probe head, a limit of detection of 1 $\mu\text{g/ml}$ can be obtained. With special probe heads (ref. 5, ref. 6), the sensitivity and, hence, the detection limit can be further improved, and the measurement time can be reduced (Figures 2 and 3).

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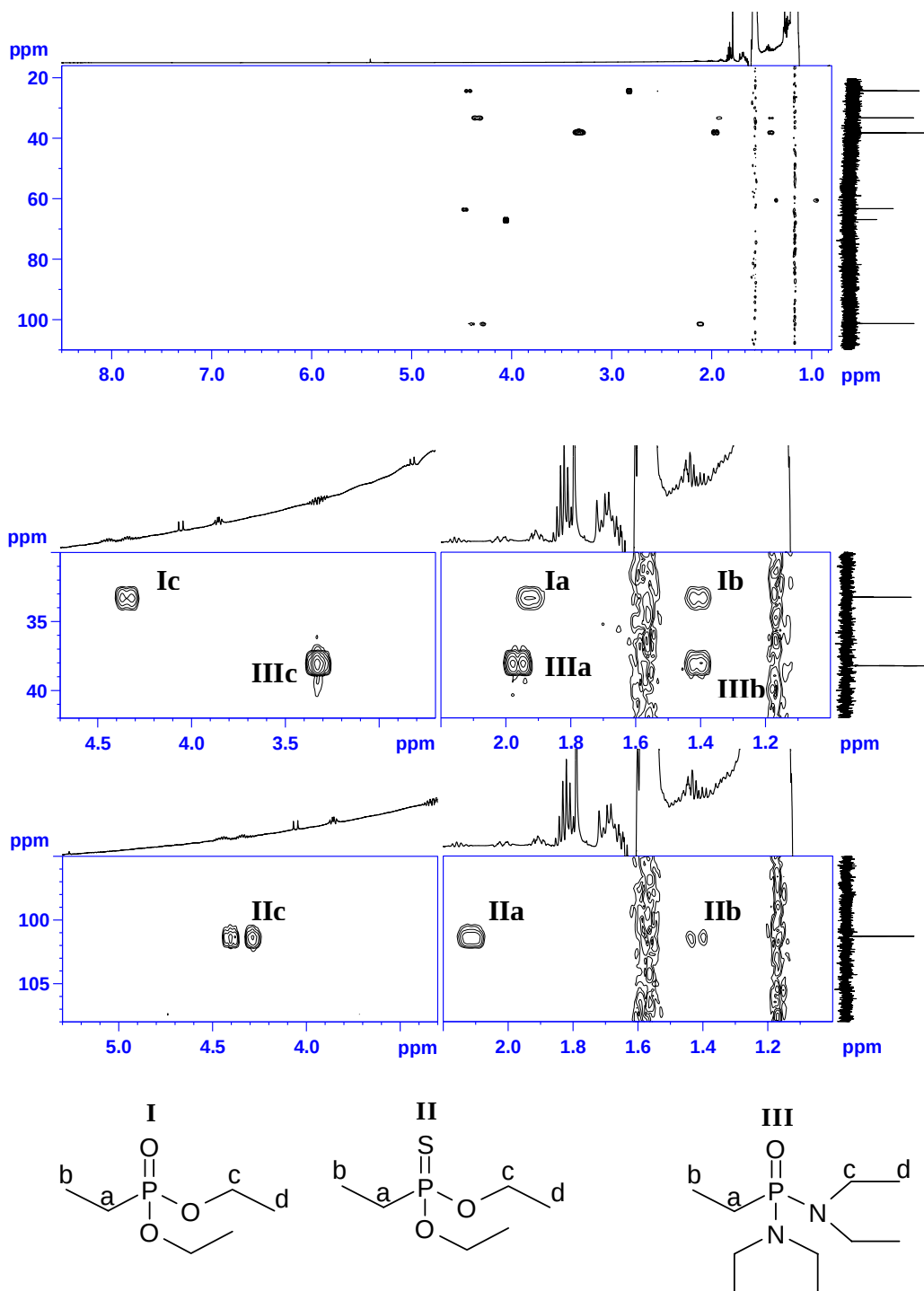


Figure 2. Sample O from the 22nd Proficiency Test (concentrations 10–20 $\mu\text{g}/\text{ml}$), solvent hexane, 50 μl chloroform- d_1 added to 450 μl of sample O. Diethyl ethylphosphonate (I), O,O-diethyl ethylphosphonothionate (II), bis(N,N-diethyl) ethylphosphonodiamide (III). 600 MHz 2D ^1H - ^{31}P HMQC (8 scans per increment, 200 increments, 40 min), 600 MHz ^1H NMR (128 scans, 10 min), 243 MHz $^{31}\text{P}\{^1\text{H}\}$ NMR (4096 scans, 2 h 21 min) acquired with a 5 mm inverse triple-resonance (^1H , ^{31}P , ^{13}C) cryoprobe.

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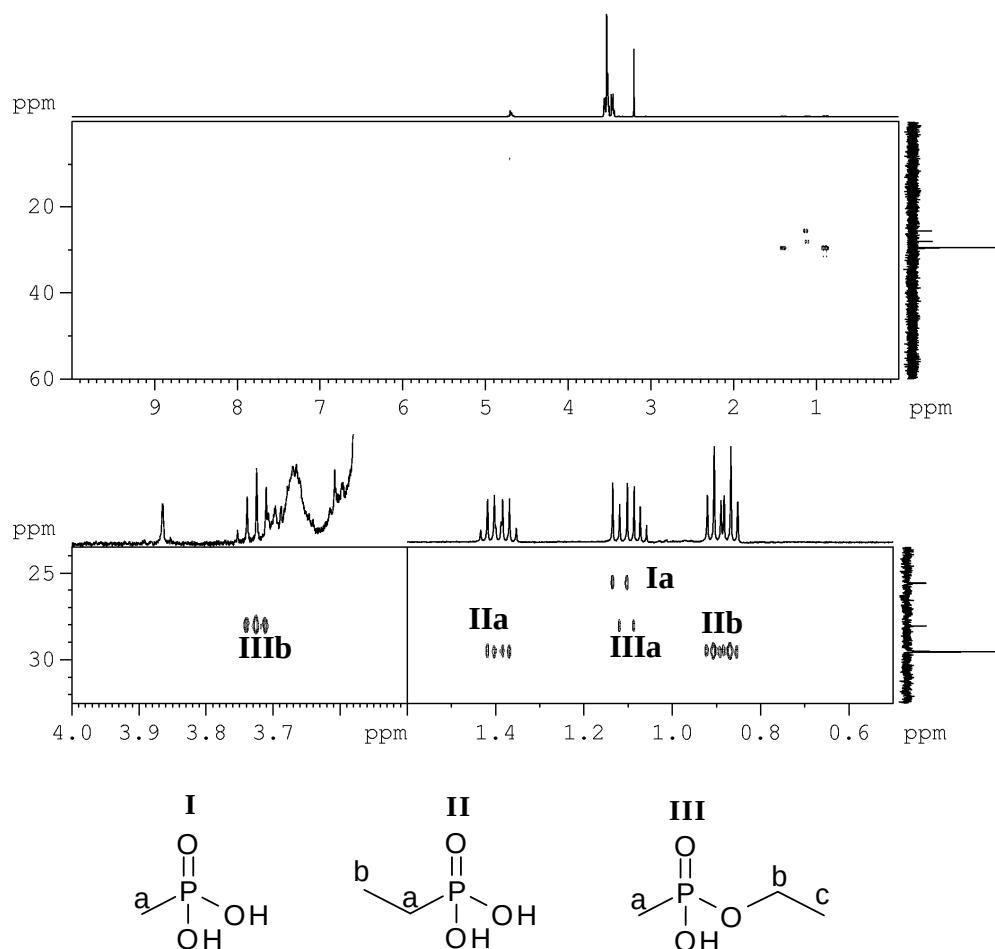


Figure 3. Sample W from the 22nd Proficiency Test. Analyte concentrations 2–10 $\mu\text{g/ml}$, solvent water, sample enrichment from 1 ml sample W to 20 μl , 10 μl D_2O added. Methylphosphonic acid (**I**), ethylphosphonic acid (**II**), ethyl methylphosphonic acid (**III**). 500 MHz 2D ^1H - ^{31}P HMQC (16 scans per increment, 100 increments, 29 min), 500 MHz ^1H NMR (64 scans, 9 min), 202 MHz $^{31}\text{P}\{^1\text{H}\}$ NMR (1024 scans, 21 min) acquired with a 1.7 mm triple-resonance (^1H , ^{13}C , ^{31}P) microcoil probe head.

7.6 Remarks on analysis

Non-deuterated solvents can be used if a small amount of the corresponding deuterated solvent is added to the sample to provide the lock conditions, if the experiment is performed unlocked, or if a lock nucleus other than deuterium is used. In this case, the intense solvent signal must be eliminated with presaturation or pulsed field gradients method (ref. 4). For strongly salty samples, cryoprobes are disadvantageous, but it could be beneficial using smaller diameter tubes (e.g. 3-mm tube and spinner on a 5-mm probe head) or microprobes. The 90° pulse widths should be recalibrated. Paramagnetic salts in aqueous samples can enhance relaxation so that the resonances became too broad to be observable. In this case, a pretreatment must be performed (e.g. cation exchange, application of chelating agents).

8. Evaluation of method

8.1 Area of applicability

The method is applicable to all compounds containing hydrogen atoms other than exchangeable with a scalar coupling to a phosphorus atom and dissolving to a sufficient degree in a solvent. The analysed liquid sample must be homogeneous and without paramagnetic properties.

8.2 Advantages/Disadvantages

The main advantage of the 2D ^1H – ^{31}P HSQC/HMQC experiment is its selectivity: signals exclusively from protons with a scalar coupling to ^{31}P nuclei are recorded. Signals from the sample background obscuring signals from organophosphorus compounds in ^1H NMR spectra are efficiently removed (Fig. 1). The experiment has better sensitivity compared to the $^{31}\text{P}\{^1\text{H}\}$ experiment with additional structural information about the structure of the alkyl group and partial characterisation of the ester side chain. The disadvantage is the modest sensitivity compared to mass spectrometric techniques (several orders of magnitude compared to GC–MS if absolute amounts of compounds are considered). Another disadvantage is that all proton signals of a compound are not present in 2D ^1H – ^{31}P HSQC/HMQC spectra – only proton signals with a scalar coupling with phosphorous are detectable. So to obtain a complete picture of a compound (with all spin-system assigned), other NMR techniques can be recorded.

9. References

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Chapter I. Analysis of saxitoxin: LC–MS/MS

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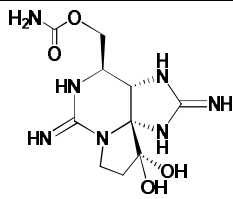
1. Scope

This chapter describes the identification of saxitoxin (STX) using liquid chromatography (LC) combined with mass spectrometry (MS) and tandem mass spectrometry (MS/MS). STX is a neurotoxin that belongs to the group of paralytic shellfish poisoning (PSP) toxins. It is listed in Schedule 1 of the Chemical Weapons Convention (CWC).

The most important technique for analysing chemical warfare agents (CWA) remains the GC–MS technique, but LC–MS has evolved over the last few years to become a powerful and important method in the field of the analysis of CWA and related chemicals. The LC–MS method allows the analysis of nonvolatile or low volatility chemicals with very little sample preparation. Typically, during CWA verification analysis of unknown samples, LC–MS will be used to screen for degradation products of CWA in aqueous matrices.

Due to the high polarity and cationic nature of STX, LC–MS is one of the spectrometric methods of choice for the identification analysis of this compound. Atmospheric pressure ionisation (API) techniques are very soft ionisation methods and therefore STX can be detected as protonated molecule with a low degree of fragmentation by LC–MS. Although several structurally related toxins exist in the group that comprises the PSP toxins, this chapter focuses only on the application of LC–MS for the identification of STX. The properties of STX are given in Table 1.

RECOMMENDED OPERATING PROCEDURES FOR CWC-RELATED ANALYSIS**Section 3.** Analytical Methods**Part F.** Other Analysis**Chapter I** Analysis of saxitoxin: LC–MS/MS**Table 1.** Properties of STX.

Properties	Value
Structure of STX dehydrate	
CAS-Number	35523-89-8
Chemical formula	C ₁₀ H ₁₇ N ₇ O ₄
Average M _w	299.3 Da
Ionic character	Basic, polar detected with API–MS as [M+H] ⁺
Polarity	High
Toxicity	LD50 comparable to VX, inhalation <2 µg/kg bw (hamster)

2. Summary of the method

LC–MS and LC–MS/MS measurements of STX are carried out with modern LC–MS instruments operating in either full scan MS/MS or selected reaction monitoring (SRM) mode. Due to the polar characteristics of STX, sample analysis ideally should be undertaken using hydrophilic interaction chromatography (HILIC) and electrospray ionisation (ESI) as the preferred ionisation method. Reversed-phase LC (RPLC) might also be used, but the retention is low (without the use of, for example, ion pairing agents) and therefore signal suppression and other chromatographic interferences can occur. HILIC remains the better choice as an efficient technique for the separation and identification of STX.^{1,2}

Prior to measurement, a quality control run (QC test) is carried out to verify the correct performance of the instrument, and a blank run is conducted to verify the absence of STX traces. Screening of STX is carried out by selecting characteristic product ions and using two selected reactions. Identification of STX is made by comparison of the measured MS/MS spectrum to that of the known reference chemical.

Table 2. LC–MS-methods for the analysis of STX

Method	Information
LC–MS: Full scan	Molecular weight, isotopic distribution, retention time, elemental composition, isotopic distribution (HR instruments)
LC–MS/MS: Product ion scan	Fragment ion spectrum, structural information, retention time
LC–MS/MS: SRM/MRM	Qualifier and quantifier, signal ratio, retention time

3. Safety aspects

In general, LC–MS analysis does not require specific safety precautions. However, STX is one of the most potent natural toxins known. The LD50 is similar to that of VX nerve agent. In addition, the mobile phase of the LC system usually contains toxic solvents and chemicals. Therefore, contamination of the laboratory with the exhaust of the ionisation source must be avoided and personal protection measures are required when cleaning the LC–MS system. Hot parts of the instrument (e.g., the ionisation source) must also be taken into account when operating or maintaining the system.

4. Instrumentation

4.1 LC–ESI/MS instrument

Any modern LC–MS instrument that covers a mass range of m/z 50–400 can be used. In general, any modern HPLC system is sufficient. Single quadrupole instruments allow the detection of the protonated molecule ion. Combination with immunoassays or other analytical techniques is necessary for an unambiguous identification of STX in unknown samples.

Due to the high polarity of STX, the mass spectrometer should be operated using an ESI source. For a more sensitive and selective screening, the mass spectrometer system should offer SRM or multiple SRM (MRM) or full scan MS/MS mode (e.g., triple quadrupole, linear ion trap). With today's high resolution mass spectrometry (HRMS) instruments, such as Q-TOFs or Orbitraps, additional screening strategies are feasible.

4.2 LC columns

Reversed phase LC (RPLC) systems are often used for the screening of CWC-related chemicals. However, the polarity of STX results in low retention and it elutes close to the solvent front at t_0 . In this region, chromatographic interferences such as signal suppression have to be expected. If the concentration of STX is at trace level, its signal could be completely suppressed. For this reason, HILIC is a better choice because it is an efficient technique for the separation and identification of polar and hydrophilic compounds. A 150 mm long HILIC column (4.6 mm i.d.) packed with 3 μ m particles is recommended. The choice of the actual HILIC stationary phase is left to personal preference.

4.3. Reagents and chemicals

A reference chemical of STX hydrochloride salt is commercially available from the following address:

“CRM-STX-e”

National Research Council of Canada

NRC Communications & Corporate Relations

1200 Montreal Road, Bldg. M-58

Ottawa, Ontario, Canada K1A 0R6

E-mail: info@nrc-cnrc.gc.ca

<http://www.nrc-cnrc.gc.ca/eng/programs/imb/crmp/shellfish/crm/stx-e.html>

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Additional chemicals are given in Table 3.

Table 3. Chemicals and reagents needed for STX analysis by HILIC or RPLC–MS.

Chemical	Purity
Water (HILIC, RP)	18 MΩ or LC–MS grade
Acetonitrile (HILIC)	≥99.9% LC–MS grade
Ammonium formate (HILIC)	≥99% MS grade
Formic acid (HILIC)	LC–MS grade, puriss p.a.
Ammonium acetate (RP)	LC–MS grade, puriss p.a.
Methanol (RP)	HPLC grade
QC Test	
Saxitoxin dihydrochloride (CAS 35554-08-6)	

5. Experimental

5.1 Recommended HILIC–MS and MS/MS method (proposed by VERIFIN)

HILIC–MS is an efficient technique for separation and identification of polar and hydrophilic STX. HILIC–MS/MS analysis gives good retention, an identifiable full scan product ion spectrum and very low detection limits (even below 100 ppb) without using any ion-pair reagents or derivatisation. The guideline conditions for HILIC and MS/MS analyses are given in Table 4.

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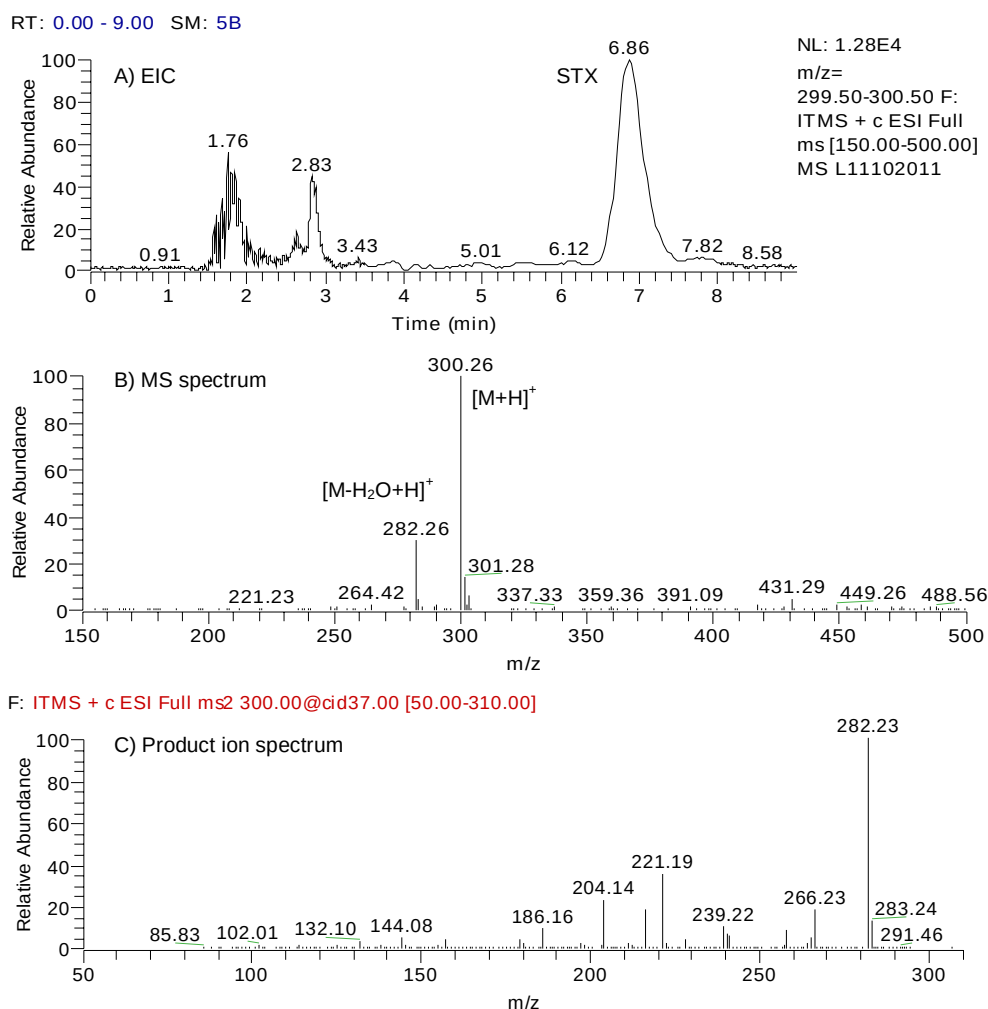


Figure 1. Analysis of 0.2 ppm STX: A) Extracted positive ion chromatogram, B) MS spectrum and C) product ion spectrum of m/z 300 (precursor ion) by LXQ (linear ion trap) Thermo Fisher. Column: Tosoh Amide-80, 3 μ , 150 x 4.6 mm. Eluent: A: H₂O, 4 mM NH₄HCO₂, pH 3.5, B: ACN, 4 mM NH₄HCO₂, isocratic elution.

5.2 Recommended RPLC-MS and MS/MS method (proposed by SPIEZ LABORATORY)

Although HILIC is the preferred separation technique for STX, a reversed phase column such as C18 can also be used. This is especially useful for the screening of CWC-related chemicals as no column switching is necessary for STX screening. The guideline conditions for RPLC are given in Table 4. Due to its highly polar character, STX will elute on an RP column close to the void volume, where signal suppression and/or high background signals have to be expected. Nevertheless, the protonated molecule and the intense fragment ion at m/z 282 (loss of H₂O) of STX can be detected at concentrations even below 1 ppm.

The low energy collision induced dissociation (CID) spectrum of STX reveals a multitude of fragment ions with either LC method, as shown in Figures 1 or 3.

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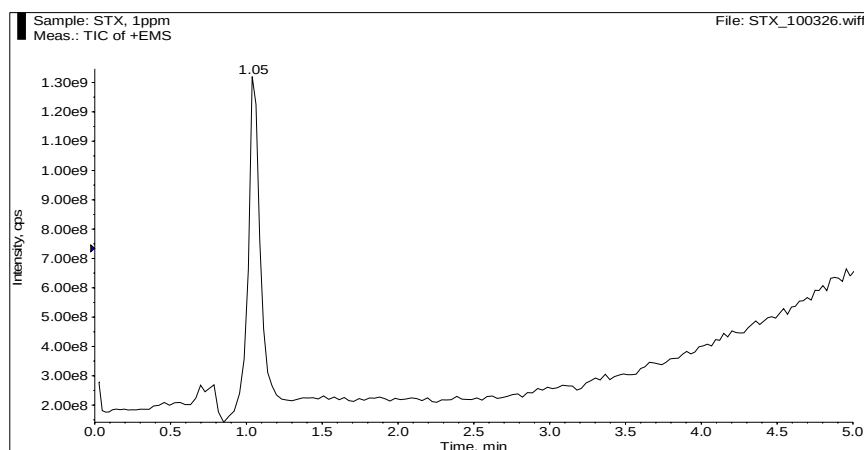


Figure 2. TIC of 10 ppm STX in positive ion mode. MS: 3200QTrap, Applied Biosystems/MDS Sciex. Column: Supelco Discovery HS C18, 5 μ , 150 x 2.1 mm. Eluent: A: H₂O, 5 mM NH₄Ac, B: MeOH, 5 mM NH₄Ac. Gradient elution.

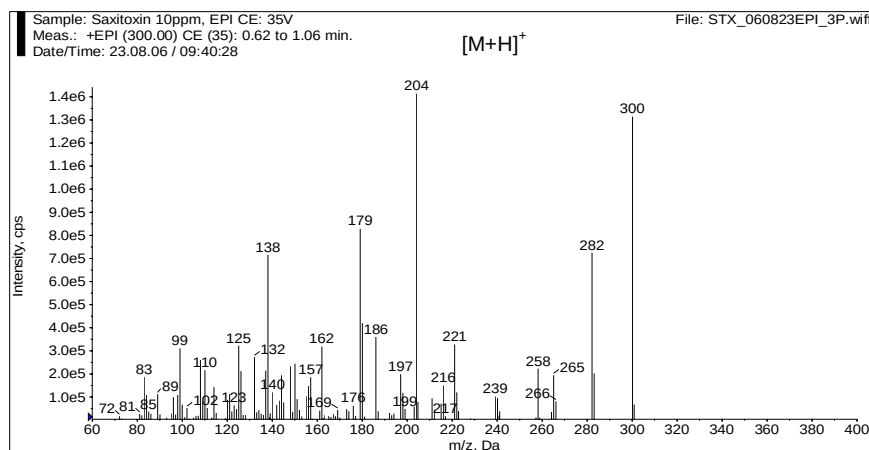


Figure 3. 10 ppm STX: Product ion spectrum of m/z 300 (precursor ion). Collision energy: 35 eV. Instrument: Applied Biosystems 3200QTrap.

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5.3. Guidelines for LC and MS/MS conditions

Guidelines for HPLC and MS conditions are given in Tables 4 and 5, respectively.

Table 4. Guidelines for HILIC and Reversed phase LC conditions. (These parameters are guidelines and may differ for other instrument types and/or manufacturers.)

LC parameter	Guideline conditions	
	HILIC	Reversed Phase
Eluents	A = 4 mM ammonium formate in water; pH 3.5 (adjusted with formic acid) B = 4 mM ammonium formate in ACN (95:5, v/v); pH 3.5	A = 5 mM ammonium acetate in water B = 5 mM ammonium acetate in methanol
Injection mode	No waste	No waste
Injection volume	20 µl	25 µl
Flow type	Isocratic	Gradient
Flow rate	1 ml/min, split into 1:20 (0.05 ml/min)	0.6 ml/min
Column	TSK-gel Amide-80® (3 µm, 150 mm x 4.6 mm)	Supelco Discovery HS C18 (5µ, 150 mm x 2.1 mm)
Column temperature	Room temperature	40 °C

Table 5. Guidelines for HILIC–MS and MS/MS conditions. (These parameters are guidelines and may differ for other instrument types and/or manufacturers.)

MS parameter*	Condition
Probe	ESI
Spray voltage	5 kV
Ionisation mode	Positive
Sheath gas (N ₂)	40 (arb)
Aux Gas	15 (arb)
Sweep Gas	5 (arb)
Capillary temperature	300 °C
Capillary voltage	40 V
Scan range	(ESI ⁺ /MS) <i>m/z</i> 150–500; (ESI ⁺ /MS/MS) <i>m/z</i> 50–310
Collision gas pressure	1.5 mTorr
Isolation width	1.0 amu
Collision energy	37%

*LXQ ThermoFisher

5.4. Quality control samples

The performance of the instrument is controlled by using a diluted solution of the saxitoxin reference chemical (see chapter 4.3) in the appropriate solvent (e.g., LC eluent).

6. Sample analysis

6.1. Screening and identification of STX in MS/MS mode

Due to its richness, the fragment ion spectrum can be used for an unambiguous **identification** of STX when compared to a reference chemical spectrum. The following MS/MS-fragment ions can be detected by triple quadrupole and linear ion trap instruments.

Table 6. Product ions of STX (precursor ion at m/z 300).

Fragment ion (m/z)	Loss of ^{1,2}
282	-H ₂ O
221	-H ₂ O-NH ₃ -CO ₂
204	-2H ₂ O-NH ₃ -NHCO
186	-3H ₂ O-NH ₃ -NHCO

6.2 Screening and identification of STX in the SRM-mode

The SRM/MRM method on triple quadrupoles and hybrid instruments is widely used for the detection and quantification of toxins, metabolites, drugs, dopes and environmental pollutants. As shown in Figures 1 and 3, the MS/MS spectrum offers several fragment ions that can be exploited to develop a selective and sensitive SRM/MRM-method, independent of the instrument used.

Based on the MS/MS spectrum, specific fragment masses for Q and q are chosen. Usually Q will be set to m/z -value of the protonated molecule. For q, the most intense fragment ion is chosen as a quantifier. Preferentially, more than one transition Q→q should be monitored for unambiguous identification (quantifier and qualifier). The ratio of the relative areas of the respective SRM signals between the unknown and a reference chemical should then be calculated (see, e.g., WADA criteria³). For STX, the examples of recommended diagnostic transitions are given in Table 7. The intensities of the fragment ions can vary depending on the instrument type and the setting of the experimental parameters. Thus, other m/z values could also be used for q.

Table 7. Examples of diagnostic SRM/MRM-transitions for STX.

Transition	Q	q
Quantifier	300	221
Qualifier	300	204

7. OPCW-criteria for the unambiguous identification of STX

A proposed set of criteria is given in Table 8. Taking into account the different information content and the lack of ambiguity of these methods, the given combinations of methods A and B are required for an unequivocal identification of STX.

Table 8. Overview of proposed combinations of analytical/biochemical methods A and B and the respective criteria for the identification of STX, effective as of November 2010 (meeting of the TWG on Sampling & Analysis).

	Method A LC/ESI–MS/MS	Method B LC/ESI–MS/MS
Lateral Flow Assay	: Positive response : Retention time $\pm$ 0.2min	: Positive response : Retention time $\pm$ 0.2min. Ratio of 2 SRM-transitions) must match within $\pm$ 20% of the reference value. Resolution $\leq$ Unit
ELISA	: Positive response : Retention time $\pm$ 0.2min	: Positive response : Retention time $\pm$ 0.2min. Ratio of 2 SRM-transitions) must match within $\pm$ 20% of the reference value. Resolution $\leq$ Unit
LC/Fluorescence	: Retention time $\pm$ 0.2min, S/N>5 : Retention time $\pm$ 0.2min	: Retention time $\pm$ 0.2min, S/N>5 : Retention time $\pm$ 0.2min. Ratio of 2 SRM-transitions) must match within $\pm$ 20% of the reference value. Resolution $\leq$ Unit

8. Conclusion

Due to its basicity and polarity, the HILIC–MS/MS method is suitable for identification and verification of STX. The identification is based on either a full scan product ion (MS/MS) spectrum and diagnostic product ions or (e.g., two) diagnostic SRM reactions. For an unambiguous identification, LC–MS/MS results have to be combined with a second technique, which is either LC/fluorescence or a biochemical method. The respective criteria can be found in the OPCW Work Instructions.

9. References

1. Carmela Dell'Aversano, Philipp Hess, Michael A. Quilliam. Hydrophilic interaction chromatography-mass spectrometry for the analysis of paralytic shellfish poisoning (PSP) toxins. *Journal of Chromatography A*. 1081, 2005, pp. 190-201.
2. Mia Halme, Marja-Leena Rapinoja, Maaret Karjalainen, Paula Vanninen. Verification and quantification of saxitoxin from algal samples using fast and validated hydrophilic interaction liquid chromatography–tandem mass spectrometry method. *Journal of Chromatography B*. 2011, in press. (doi:10.1016/j.jchromb.2011.11.015)

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3. World Anti-Doping Agency (2010) WADA Technical Document TD2010IDCR. World Anti-Doping Agency, Montreal, Canada.

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Chapter IV

Analysis of ricin: ELISA

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1. Scope

This chapter describes the identification of ricin by the use of immunoassay. Chemiluminescence enzyme-linked immunosorbent assay (Chemiluminescence ELISA) is based on the interactions of capturing antibody (Ab I) immobilised on solid material and its antigen (ricin). Here this complex is detected with a specific detection antibody (Ab II) labelled with chemiluminescent horseradish peroxidase (HPR).

2. Summary of the method

Here a sandwich format of chemiluminescence ELISA, a consistent alternative to colorimetric detection, is recommended to identify and further quantify ricin toxin both at a low concentration level (typically, 0.05 to 5 µg/l) and in complicated biosamples.

The sandwich ELISA measures the amount of antigen between two layers of antibodies (i.e. the capture and detection antibody). Generally, the antigen to be measured should contain at least two antigen sites capable of binding to an antibody. Either monoclonal or polyclonal antibodies can be used as the capture and detection antibodies.

The chemiluminescent substrates differ from other substrates in that the light detected is a transient reaction product that is only present while the enzyme-substrate reaction is occurring.

3. Safety aspects

Since ricin is a ribosome-inactivating protein type II with high toxicity, all related experiments should be performed according to the operating biosafety guidelines. All contacts with the active toxin should be avoided and during sample treatment and analysis, protective gloves and goggles should be worn.

After analysis, the samples containing ricin should be totally decontaminated. A recommended procedure is to heat all ricin-distributed solutions for an hour at 100 °C and then treat them with, e.g. alkali for the total denaturation and detoxication of the ricin.

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4. Handling and storing of samples

All ricin samples should be handled with care to avoid contamination with foreign materials which could give false-positive reactions and to prevent cross-contamination between positive and negative samples. Ricin samples should be kept cold during handling and stored in a freezer as soon as possible. Avoid freeze/thaw cycles in all samples containing protein.

5. Instrumentation

5.1 Chemiluminescence ELISA instrument

Any modern microplate reader with chemiluminescence mode is applicable.

5.2 Chemicals, reagents and materials

The chemicals, reagents and materials needed for chemiluminescence ELISA are given in Table 1.

Table 1. Chemicals, reagents and materials needed for chemiluminescence ELISA.

Chemical	Concentration	Purity
bovine serum albumin (BSA, CAS 9048-46-8)	10 g/l	≥98% (agarose gel electrophoresis)
primary mAb I	5 mg/l	electrophoresis grade, home-produced or commercially produced
HRP-labelled primary mAb II	2.5 g/l	electrophoresis grade, home-produced or commercially produced
ultrapure water	-	18.2 MΩ·cm, pyrogen-free
Reagents and solution		
bicarbonate buffer, pH 9.6	50 mM	p.a. grade or better
chemiluminescent horseradish peroxidase (HRP) substrate	-	
Material		
microtiter plates (black or white opaque)	-	8×12 wells per plate

6. Experimental

6.1 Recommended chemiluminescence ELISA conditions

Recommended chemiluminescence ELISA conditions are given in Table 2. Either of the different antigen capture ELISA formats, e.g. direct ELISA, indirect ELISA, sandwich ELISA and competitive ELISA, can be used. Here we describe a direct sandwich type,^{1, 2} in which two antibodies recognise different epitopes in ricin (here as an antigen).

Here the primary antibody I, called the “capture antibody”, is passively adsorbed onto the solid phase of the plate to capture ricin. The primary antibody II, called the “detection antibody”, is an HRP-labelled conjugate used to bind ricin to complete the sandwich format. Either monoclonal or polyclonal antibodies can be used as the capture and detection antibodies. An inherent monospecificity toward a single epitope in a monoclonal antibody allows for the fine detection and quantitation of small differences in the antigen. A polyclonal antibody is often used as the capture antibody to pull down as much of the antigen as possible.

The experimental conditions in chemiluminescence ELISA may be modified according to the type of sample. The typical volumes used for general reagents and samples are 50–300 µl per well. In direct sandwich ELISA, the samples do not have to be purified before analysis is performed.

Note! *In any variable formats of chemiluminescence ELISA, the main reagents include the immobilised antigen/antibody, enzyme-linked antigen/antibody and the substrate, with the quality (including the binding affinity and specificity and purity) of the antibody is the key factor on such an assay. In the general case, the cross-reactivity of anti-ricin antibodies to RCA120 has to be investigated, too.*

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Table 2. Recommended chemiluminescence ELISA conditions in a direct sandwich type.

Procedures	Recommended conditions	Notes
Coating of wells with Ab I	Add primary Ab I (capture Ab) at its optimum dilution in wells, incubate overnight at 4 °C	Dissolve capture Ab in bicarbonate buffer, pH 9.6 Cover the plate with an adhesive plastic to prevent evaporation
Washing	Wash wells twice with PBST buffer	PBST buffer: phosphate-buffered saline (PBS) containing 0.05% Tween 20
Blocking	Add PBS/BSA (bovine serum albumin, 10 g/l in PBA) in wells, incubate 1–2 h at RT, or overnight at 4 °C	Here an HRP-labelled antibody is used for detection Do not use sodium azide in buffers or washing solutions because it is an HRB inhibitor
Washing	Wash wells 3 times with PBST buffer	
Adding samples	Add diluted samples in wells, incubate for 1.5 h at 37 °C	A series of samples with different dilutions should be done in the blocking buffer
Washing	Wash wells 3 times with PBST buffer	
Adding detection Ab	Add HRP-labelled primary Ab II (detection Ab) at its optimum dilution, incubate 40 min at 37 °C	Alkaline phosphatase-labelled detection Ab is also suitable with the proper substrate pair
Washing	Wash wells 5 times with PBST buffer	
Chemiluminescence generation	Add chemiluminescent HRP substrate mixture in wells, incubate 5 min at RT in the dark	Chemiluminescence is sensitive to light; the plate should be protected against light during this incubation
Measurement	Perform measurement on a modern microplate reader equipped with chemiluminescence mode	

7. Quality control

In developing any chemiluminescence ELISA assays, it is recommended that the performance be monitored with a series of quality control samples. The quality control (QC) plate should contain a positive control (ricin in a defined concentration spiked in a certain matrix), a negative control (a matrix without ricin), a blank control (wells containing PBST diluents only), a detection conjugate control (a non-capture antibody, which represents nonspecific binding of the HRP-conjugated Ab to the well surfaces), and a substrate control (a well without HRP-Ab, which represents non-enzymatic production of chemiluminescence). All control wells in the QC plate should be triplicated. The full control plate should be measured after every fifth sample plates.

Each sample plate should contain control wells: 4 wells for the negative control and 4 wells for the blank sample. Standard samples should also be included in sample plates if quantitative results are calculated. Control and standard measurements have to be accepted before the sample measurements are reported.

When developing a new ELISA make sure that microtiter plates with good quality are chosen for the measurements. The performance of the plates should also be tested when different batches of plates are used.

8. Analysis of samples

8.1 Procedure of analysis

The passive binding of antibodies and proteins, generally in 96-well polystyrene plates, and the reactants of ELISAs immobilised to microplate surface make it easy to separate bound and unbound material during the assay. The ability to wash away nonspecifically bound materials makes ELISA a powerful tool for measuring specific analytes. A number of modifications to the basic procedure of ELISA can be performed.

A highly specific interaction between antigen and antibody is the most crucial point in the detection strategy of ELISA. In the direct sandwich chemiluminescence ELISA, antibody reagents are standardised, i.e. the optimal dilutions of the capture antibody and the HRP-labelled antibody (i.e. the detection conjugate) are determined by checkerboard titration. Once optimum working dilutions are established, they are used for sequential assays of all quality control and sample analyses.

The samples homogenised in buffer are added into respective wells (duplicates or triplicates). Twofold dilution range of each sample is recommended. For quantitative analyses, standard samples (duplicates or triplicates) as well as blank samples (duplicates or triplicates) are run in each plate to ensure accuracy. The concentrations of the standard samples should cover the most dynamic range.

8.2 Identification

Identification is established by comparing the produced chemiluminescent intensity of the samples with the positive threshold. Generally, two standard deviations above the mean value of the negative control can be used to establish the positive threshold. The mean value of the replicate sample wells is compared to the threshold value to determine the positive or negative status of test samples.

A further quantification can be completed in this assay. The method validation should be developed before the assay of test samples. Prepare a standard curve from the data produced from the serial dilutions with concentration on the x-axis (log. scale) versus chemiluminescent intensity on the y-axis (log. scale). Interpolate the concentration of the sample from this standard curve.

9. Evaluation of method

9.1 Area of applicability

The analytical and systematic characteristics of chemiluminescence ELISA are quite suitable to examine unknown substances at a screening level, especially when large scales of samples are analysed. The samples suspected to contain ricin can be environmental samples, food, drinks and samples of biological origin such as plasma, urine or tissue.

9.2 Advantages/Disadvantages

Chemiluminescence ELISA is one of the most sensitive ELISA, a combination of the specificity of antibodies and the sensitivity of enzymatic chemiluminescence assays. The advantages of chemiluminescence ELISA include versatility, simplicity, high-throughput and the sensitivity of the analyses. Quantification of the samples is also possible. Many systems using different combinations of reagents can be developed in chemiluminescence ELISA and more than one system can be used to measure the same targets. With an enzyme catalysed amplification of chemiluminescent signal, chemiluminescence ELISA can provide a useful measurement of ricin at a concentration lowered to the sub µg/l level.

The disadvantages of chemiluminescence ELISA depend upon its formats. For a direct sandwich chemiluminescence ELISA, the primary Ab II should be enzymatically labelled and ricin (as an antigen) must have at least two different epitopes for both capture and detection antibodies, respectively. Another common disadvantage is that the chemiluminescence ELISA method cannot distinguish between the biologically active and inactive form of ricin (i.e. the loss of toxicity).

10. References

1. Guo J, Shen B, Feng J, Sun Y, Yu M, Hu M., “A novel neutralizing monoclonal antibody against both ricin toxin A and ricin toxin B, and application of a rapid sandwich enzyme-linked immunosorbent assay”, *Hybridoma* **25 (2006)** 225–229.
2. “Systems in ELISA” in Crowther JR (ed.) “Methods in Molecular Biology, The ELISA Guidebook, Vol. 516”, Humana Press, **2009**, pp. 16–19.

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Chapter V

Analysis of ricin: immunocapture

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1. Scope

This chapter describes the enrichment of ricin through the use of immunocapture. Magnetic beads (MB)-based immunocapture of ricin presented here is based on the efficiency of the monoclonal antibody (mAb) directed against the B-chain of ricin to interact specifically with its epitope. The streptavidin-coated magnetic beads can specifically capture the ricin complex and its biotinylated anti-ricin mAb. Enriched ricin can be detected with different detection methods.

2. Summary of the method

Here an MB-based technique is recommended to immunocapture ricin using a monoclonal antibody (mAb) directed against the B-chain of ricin via streptavidin-biotin interaction. This preconcentration and cleanup technique is a good fit to enrich the target protein and remove the unrelated but highly abundant proteins in the matrix.

Generally, in immunopurification, a specific antibody is conjugated to an insoluble substrate (e.g. agarose beads or magnetic beads). This is achieved through interaction with either Protein A or G or through covalent linkage with the substrate surface. Once the antibody is bound, the beads are exposed to a complex mixture of proteins containing the target protein (the antigen). Through the specific antibody, the target protein will bind to the immunomagnetic structure. Following this binding, the antigen-coupled surface may be washed and after that, the target protein can be eluted from the magnetic beads. Variability from this basic principle of immunopurification may be seen in the type of surface used as well as the antibody coupling method.

3. Safety aspects

Since ricin is a ribosome-inactivating protein type II with high toxicity, all related experiments should be performed according to the operating biosafety guidelines. All contact with the active toxins should be avoided and protective gloves and goggles should be worn during sample treatment and analysis.

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After analysis, the samples containing ricin should be totally decontaminated. A recommended procedure is to heat all ricin-distributed solutions for an hour at 100 °C and then treat with, e.g. alkali lye to make sure all ricin is fully denatured and detoxicated.

4. Handling and storing of samples

All ricin samples should be handled with care to avoid contamination with foreign materials which could give false-positive reactions and to prevent cross-contamination between positive and negative samples. Ricin samples should be kept cold during handling and stored in the freezer as soon as possible. Avoid freeze/thaw cycles in all samples containing protein.

5. Chemicals, solutions and materials

The chemicals, solutions and materials needed for the immunocapture of ricin are given in Table 1.

Table 1. Chemicals, solutions and materials needed for immunocapture of ricin.

Chemicals	Concentration	Purity
ammonium chloride (CAS 12125-02-9)	0.1 M	p.a. grade or better
biotinylated N-hydroxysuccinimide ester (bio-NHS, CAS 35013-72-0)	1 g/l	p.a. grade or better
bovine serum albumin (BSA, CAS 9048-46-8)	1 g/l	≥98% (agarose gel electrophoresis)
N,N-dimethylformamide (DMF, CAS 68-12-2)	5%(v/v)	p.a. grade or better
formic acid (FA, CAS 64-18-6)	5%(v/v)	p.a. grade or better
mAb specific to ricin	3 g/l	electrophoresis grade, home-produced or commercially produced
trifluoroacetic acid (TFA, CAS 76-05-1)	0.1% (v/v)	p.a. grade or better
ultrapure water		18.2 MΩ·cm, pyrogen-free
Solutions		
phosphate-buffered saline (PBS)	10 mM phosphate 140 mM NaCl pH 7.4	p.a. grade or better
Material		
magnetic particle separator (Magnet)	-	
streptavidin coated magnetic beads	1 g/l	1 µm diameter
ultrafilter with a molecular weight cutoff of 30 kDa	-	

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6. Experimental

Recommended immunocapture conditions are given in Table 2. Streptavidin-coated MB coupled with mAb against a certain epitope of ricin is the main material used in this technique.

Note! *Different streptavidin-coated MBs have different binding capacities with the biotinylated molecules, depending on the size, the manufacturing approach, etc. Before conjugation with biotinylated antibodies, optimise and calculate the required amount of beads based on their binding capacity.*

Table 2. Recommended immunocapture conditions.

Procedures	Recommended conditions	Notes
Biotinylation of anti-ricin mAb	Add Bio-NHS into the anti-ricin mAb solution and incubate for 4 hours at room temperature (RT) Terminate the reaction using 0.1 M ammonium chloride, incubate for 10 min at RT Ultrafilter the biotinylated mAbs from the excess of bio-NHS molecules	Ratio of Bio-NHS and mAb is 1:10 (w/w); Use freshly prepared Bio-NHS
Preparation of streptavidin-coated magnetic beads (MBs)	Resuspend the MBs in the original vial by vortexing Transfer the desired amounts of MBs to a new tube Wash MBs 3 times with PBS	In washing procedure preservatives, i.e. NaN_3 is removed The separation of the MBs from the solutions, e.g. the removal of supernatant and washing and changing solution procedures, are all done with the help of a magnet
Immobilisation of anti-ricin mAbs	Remove the washing solution (PBS) Add biotinylated anti-ricin mAb onto washed MBs Incubate for 4 hours at RT with gentle rotation of the tube Discard the supernatant Wash the MBs twice with PBS and resuspend the MBs in PBS/BSA in a desired concentration	Biotinylated Anti-ricin mAb can be suspended in PBS buffer containing 1 g/l BSA (PBS/BSA) to keep its bioactivity Incubation time depends on the concentrations of biotinylated mAbs utilised
Immunocapture of ricin from sample	Add sample onto biotinylated mAb-coated MBs Incubate for 4 hours at RT with gentle rotation Remove supernatant and wash the MBs twice with PBS	Gentle rotation can prevent MBs' aggregation
Release of ricin	Remove the washing solution (PBS) Add 0.1% TFA or 5% FA in H_2O and incubate for 5 min Transfer the ricin containing supernatant to a new tube	After elution, ricin-containing solutions should be analysed quickly

7. Quality control

In the quality control (QC) test for such an immunocapture technique, pure ricin with known concentration in PBS can be used as a positive control.

Since BSA-containing PBS solution (PBS/BSA) has been recommended to maintain the bioactivity of streptavidin and anti-ricin antibodies, and to reduce nonspecific bindings, the enrichment efficiency depends mainly upon the specificity and binding capacity of the antibody, while the potential interference of unrelated proteins that occurred in biomatrix is avoided.

The enrichment efficiency is heavily dependent on the performance of the biotinylation of antibodies and the binding capacity of streptavidin with biotinylated antibodies.

Initially, the optimisation of the biotinylation of the antibodies should be performed. The titration experiment to determine the required amount of biotinylated antibodies versus the required amount of streptavidin-MBs should also be performed before the immunocapture of ricin from samples. The preparation of parallel samples is recommended. Disposable plastic tubes should be used to avoid cross-contamination.

8. Evaluation of method

8.1 Area of applicability

This immunocapture technique can be applied to any kind of complicated samples, such as food, drinks, soil, water, plasma, serum, urine and tissues. Depending on the different binding capacities of streptavidin-coated MB towards biotinylated antibodies, and the different binding affinity and specificity of the anti-ricin antibody and ricin, the largest enrichment ratio is varied but is usually higher than tenfold. As well as providing an effective enrichment of the target, the technique can also be used to remove the background which may cause potential interference.

8.2 Advantages/Disadvantages

The advantages of this immunocapture technique are its specificity, reliability and simplicity. The technique is quite suitable to be employed before mass spectrometry-based identification methods are enacted, especially for the effective extraction of ricin from complicated samples.

The main disadvantage of this immunocapture technique is that time is still needed for the optimisation of several of the involved procedures to ensure the highest level of efficiency. At the moment, only common recommendations can be followed.

9. References

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1. Scope

This chapter describes the role, the purpose and the history of Proficiency Tests (PT) organised by the Technical Secretariat of the Organisation for Prohibition of Chemical Weapons (OPCW), as well as general provisions of sampling and analysis under the Chemical Weapons Convention (CWC).

2. Sampling and Analysis under the CWC

The CWC calls for using Sampling and Analysis (S&A) as one inspection tool to support verification activities. Samples can be analysed on-site by the inspection team (IT) or transferred off-site to Designated Laboratories (DL). The objective of S&A can be different depending on the type of the particular inspection. At chemical weapons related facilities, the objective of S&A is to confirm the type of agent being destroyed (inspections under article IV and V of the CWC), and at industrial facilities the objective is to confirm the absence of Schedule 1 chemicals or any other undeclared Scheduled chemicals (inspections under Article VI of the CWC). See chapter “*Chemicals*” (1-A-III) for the discussion on the Schedules of chemicals.

The provisions for S&A under the CWC can be summarised as follows:

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- Samples are generally collected by Inspected State Party (ISP) site representatives, or alternatively, by the inspection IT, if agreed upon in advance.
- Where possible, the analysis will be done on-site. Samples can be analysed by the IT using its approved equipment, or alternatively, by the ISP in the presence of the IT, if so requested by the IT.
- If the IT deems it necessary, it shall transfer samples for analysis off-site to laboratories designated by the OPCW. However, during Schedule 3 and Other Chemical Production Facilities (OCPF) inspections, it can do so only in case of unresolved ambiguities, and also only subject to ISP agreement.

The provisions in the CWC pertaining to S&A can be found under the following references:¹

- VA.II.52–58: General rules for S&A
- VA.IV (A).49b, 66, 70: Storage facilities (CWSF), destruction facilities (CWDF)
- VA.V.49 (iii): Production facilities (CWPF)
- VA.VII.27: Schedule 2 inspections
- VA.VIII.22: Schedule 3 inspections
- VA.IX.19: Other chemical production facilities (Discrete Organic Chemicals, DOCs)
- VA.X.27c, 36b, 47, 48: Challenge inspections
- VA.XI.16–18: Investigation of alleged use
- CA.16: Facility agreement, taking of samples and their analysis
- VA.II.11 (d), 12: Inviolability of samples

3. OPCW Proficiency Tests and Designated Laboratories

3.1 Introduction

The first international interlaboratory comparison (round-robin) test for the verification of chemical disarmament took place in 1989 on Finnish initiative. This test and three subsequent tests were coordinated by Finland.[1–4] The Provisional Technical Secretariat (PTS) of the Preparatory Commission of the OPCW coordinated the first international OPCW/PTS interlaboratory comparison test in 1994.[5] Two further tests—so called ‘trial proficiency tests’—were arranged by the PTS in 1995.

The OPCW has been conducting Official PTs since 1996 to designate laboratories for the analysis of authentic samples. In 1996 and 1997 one test was conducted each year, and since 1998 two annual tests are conducted. The OPCW Designated Laboratory network is an important element of the verification regime of the CWC. The existence of laboratories with proven expertise in off-site analysis of CWC-related samples provides a high degree of confidence that chemical analyses yield unambiguous results, thus rendering a significant element of deterrence against breaches of the CWC.

The tests are part of a mechanism that ensures laboratory competence in the analysis of chemicals related to the Convention. Unlike many other proficiency test schemes, the OPCW PT is qualitative, i.e. the laboratories must determine if out of a very large set – essentially infinite – of chemicals relevant to the CWC, any are present in the samples.

¹ VA = Verification Annex of the CWC; CA = Confidentiality Annex of the CWC; CVA.II.52 is to be read as: Verification Annex, part II, paragraph 52

3.2 Conduct of OPCW Proficiency Tests

The OPCW PT scheme is accredited by the Dutch Accreditation Council (Raad voor Accreditatie; RvA) since 2001. The OPCW Official PTs are conducted in compliance with ILAC-G13/ISO 17043:2010,[7] with the assistance of two designated laboratories that respectively prepare the samples and assist in the evaluation of results based on OPCW requirements.

When a laboratory wants to participate for the first time, it has to be nominated and notified by its National Authority. Then, the Secretariat informs the participants of the test plan and the scenario before a sample is dispatched. The sample preparation laboratory dispatches the samples to the participating laboratories at no cost and the participating laboratories have to report the identified chemicals relevant to the test within 15 days after receipt. Any laboratory that fails to submit the PT report will fail the test unless it informs the Secretariat of its intention to withdraw before the end of the test and provides a satisfactory explanation. Once all participants complete the test, or the time period ends, the OPCW Laboratory and the assisting evaluating laboratory evaluate the results independently and concurrently in accordance with the relevant work instruction for the evaluation of results of OPCW PTs.[8] The draft report is then discussed by the evaluating laboratory and the OPCW Laboratory. Prior to drawing any conclusions on the performance of individual laboratories, the Secretariat prepares:

- (a) an assessment of the stability, quality and validity of samples to determine the applicability of the test; and
- (b) an assessment of the evaluation of the test results.

The Secretariat prepares a preliminary evaluation report based on all above-mentioned data, and thereafter informs the participants of their performance and requests comments. The Secretariat organises a one-day meeting of all participants at the OPCW headquarters to discuss the preliminary results of the PT. After a general session, the participants are given the opportunity to meet individually with the representatives of the OPCW Laboratory, the evaluating laboratory and the sample preparation laboratory. The participants are given two weeks to inform the Secretariat whether they accept the performance evaluation and to submit written comments on their evaluation. The Secretariat informs the evaluating laboratory of the participants' comments to allow it to take corrective measures if needed, and to finalise its report. The Secretariat then finalises its test report and publishes the final evaluation report.

3.3 Samples and reporting

Unlike most PTs aiming for quantitative analyses of known compounds in pre-determined matrices, the OPCW PT is primarily based on the qualitative analysis of samples of unknown sample composition in order to determine the presence of an essentially infinite number of possible chemicals relevant to the CWC. The participants receive two sets of three samples each (test sample, control, blank).² There is no indication for the type of the sample; thus each sample has to be analysed like a test sample. The laboratory creates its

² The test sample simulates a real sample collected during an inspection. The control sample simulates the control sample prepared at the OPCW Laboratory by spiking known chemicals to matrix similar to the real sample. The blank sample simulates the control sample prepared at the OPCW Laboratory and consists only of matrix similar to the real sample.

own solvent blank to prove the absence of the reported chemicals in the instrument before introducing the sample. The salient features of the test are as follows:

- **Diverse nature of chemicals:** The OPCW PT is primarily based on qualitative analysis and aims to confirm the presence of CWC-related chemicals, which are scheduled chemicals and/or degradation products of scheduled chemicals spiked at 1–10 ppm level. The nature of chemicals may vary from highly volatile early-eluting chemicals (e.g. pinacolyl alcohol: 3,3-dimethylbutan-2-ol) to non-volatile late-eluting chemicals (e.g. BZ: 3-quinuclidinyl benzilate).
- **Blind testing for an almost infinite number of chemicals in complex matrices:** Participants receive samples of unknown composition to be analysed for the presence of a large number of possible chemicals in various matrices (e.g. soil, water, organic waste, wipe), often with a high background comprised of polyethylene glycol (PEG), diesel, etc.
- **No prescription of methods:** Laboratories have no restrictions regarding sample preparation or analytical methods. Sample preparation procedures often require clean-up, extraction, derivatisation and concentration steps. All identifications must be confirmed by at least two different analytical techniques and give consistent results; one technique must be spectrometric.
- **Detailed reporting of results:** Participants' reports must contain names, structures and CAS numbers (if available) of the identified chemicals, a detailed description of the employed sample preparation methods, a description of the employed analytical methods, the analytical data (chromatographic and/or spectrometric), and the analytical conditions. There should be an unbroken chain of evidence linking each sample to the corresponding reported chemicals throughout the entire report.
- **Short timeline:** Participants have only 15 calendar days after receipt of the samples to perform the analyses and dispatch the report.
- **Zero tolerance for false positive:** Any reported chemical that is not contained in the sample matrix or that cannot be formed in the sample matrix constitutes a false positive result. Only chemicals relevant to the aim of the test should be reported. In addition, reporting a chemical that is present in the sample, even as a minor constituent of the spiking chemical, based on erroneous or misinterpreted analytical data will also constitute a false positive result. Reporting any false positive result will constitute a failure of the proficiency test, even if the identification is not supported by sufficient data. Chapter “*False positive identification of Chemical Weapons Convention-related compounds*” (1-B-IV) discusses the issues related to false positive identification in more detail.

3.4 Chemicals

Over the course of the past 29 OPCW PTs, all major classes of scheduled chemicals have appeared as spiking chemicals, with the exception of chemicals in schedules 1.A.07 and 1.A.08 toxins (saxitoxin and ricin, respectively), 2.A.02 (perfluoroisobutylene; PFIB), 1.B.09–12 and some highly reactive and/or volatile Schedule 3 chemicals. In the initial phase of PTs, the concept behind sample composition included at least one spiking chemical that was either completely unknown, or whose analytical data, chemical properties or analytical behaviour were different from already known chemicals. Participating laboratories had to make a significant effort to identify the structure of spiking chemicals, with particular consideration of assignment of certain atoms or functional groups. Over time and partially as a response to new requirements of OPCW inspections, the focus of OPCW PTs has shifted from the testing of participants'

capabilities to identify unknown chemicals to the improvement of sample preparation procedures and reporting skills. As a result, there has been an increase in the complexity of sample matrices, including more challenging sets of background chemicals and a more challenging selection of chemicals to be reported, and a decrease of complexity of the structures of the spiking chemicals, which caused less chemical identification ambiguity.

An overview of the classes of chemicals used as spiking chemicals in the 29 PTs is shown in Fig. 1. The most frequent schedule is 2.B.04, as this is the group with the largest number of chemicals covered by the CWC. Another reason for the predominance of 2.B.04 chemicals is that even though some of them seem to be simple in structure (e.g. methylphosphonic acid), they present an analytical challenge for both sample preparation and instrumental analysis. The vast majority of 2.B.04 chemicals need to be carefully extracted from samples and then derivatised with the most popular technique used by PT participants, gas chromatography–electron ionisation mass spectrometry (GC–EI/MS), before they are analysed. Both processes have an impact on the overall performance of analysis and the correctness of reporting. Furthermore, many 2.B.04 chemicals are direct degradation products or precursors of schedule 1.A chemicals, and thus their presence is almost always expected in samples containing schedule 1.A chemicals.

The second most frequent group of chemicals are non-scheduled chemicals (N.S.), of which those related to the sulphur mustard family (1.A.04), including thiodiglycol (2.B.13; the only scheduled degradation product for sulphur mustard) were most often used as spiking chemicals (13 times). Next are the VX family (1.A.03) related chemicals (5 times) and the tabun and triethanolamine-related chemicals (3 times, respectively).

The reason for including such chemicals in PT samples is that they are known degradation/decontamination products or synthesis impurities of schedule 1 chemicals, and therefore, are expected to be present in authentic samples taken by OPCW inspection teams.

3.5 Criteria for performance evaluation

The performance scoring is done as follows:[9,–10]

- (a) Each correct identification is positively scored (+1 point).
- (b) Identification of a degradation product(s) instead of the spiking chemical is positively scored (+1 point) if the original spiking chemical is no longer present in the sample.
- (c) Identification of nerve agents in Schedule 1A and nerve agent precursors and degradation products in Schedules 1.B and 2.B is considered correct (+1 point) even without specific identification of the locations of alkyl groups in the O-alkyl or O-cycloalkyl side chains. However, the side chain of the P–C bond must be fully identified.
- (d) Identification of minor impurities or degradation products of a spiking chemical(s) while the spiking chemical is still present is considered correct, but does not add to the score.
- (e) False negative results arising from not finding a spiking chemical or its degradation product is scored negatively (–1 point).
- (f) False positive results and reporting of an irrelevant chemical constitutes failure of the proficiency test.

The performance of laboratories in an individual test is rated as shown in Table 1 to allow for a simple and unambiguous comparison in different PTs.

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When a laboratory performs unsatisfactorily at any degree in the tests, it must be prepared to take remedial actions. A laboratory must report on the follow-up actions it takes before it is allowed to participate in the next test. If the remedial actions taken by the laboratory prove to be ineffective, the Director-General may wish to reconsider the certification of the laboratory for this type of analysis.

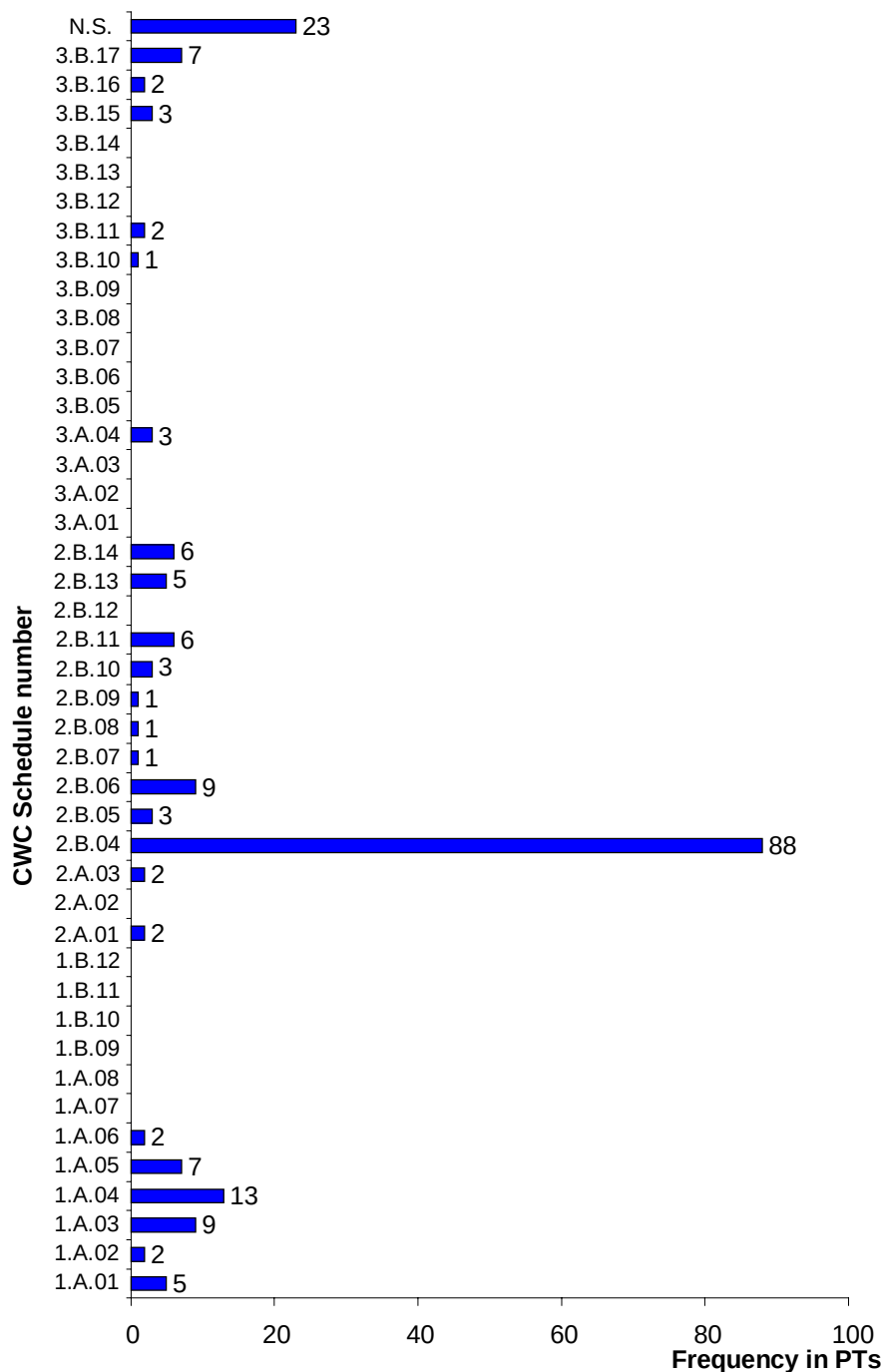


Figure 1. Number of times chemicals from the different schedule numbers have been spiked in samples in OPCW PT.

Table 1. Method for evaluating laboratory performance

Performance criteria fulfilled	Identification of chemicals	Performance scoring	Performance rating
Yes	Lab identifies all chemicals	Maximum score	A
Yes	Lab identifies all chemicals except one	Maximum score minus two	B
Yes	Lab identifies more than half of the chemicals	Score between zero and maximum minus two	C
Yes	Lab misses more chemicals than it identifies	Negative score	D
No	–	No score; Failure	F

3.6 Designation of laboratories

In order to achieve designation, the laboratories must participate in at least one PT per year and have performed successfully in the last three consecutive tests with a minimum rating of three As, or two As and one B. This effectively means that only one spiking chemical can be missed in three consecutive tests. Additionally, the laboratory must be accredited for the analysis of CWC-related chemicals by an internationally recognised accreditation body. The most common international standard DLs are following is the ISO/IEC 17025:2005.[11]

DL that perform unsuccessfully, (e.g. scores C or D), in the OPCW PT scheme are temporarily suspended or may lose the designation status (e.g. scores F). Temporarily suspended laboratories will not be selected for off-site analysis of samples until they regain the full designation status. However, these laboratories can assist in sample preparation or evaluation of results.

3.6 Common errors in laboratories' reports

Reporting of irrelevant chemicals, mistakes in identification of spiking chemicals and breaches in the chain of evidence are the most common errors observed in PT reports. A compound has to be confirmed by at least two analytical techniques, one of which has to be spectrometric. Incomplete confirmation of identification leads to non-scoring.

Only scheduled chemicals that are relevant to the aim of the test should be reported. An understanding of the definition of scheduled chemicals is essential to avoid reporting irrelevant chemicals. Care should be taken to report name, structure, CAS number of the identified chemicals, as all of these must be correct to qualify the chemical for scoring. The chain of custody with respect to the sample identity must be maintained throughout the report. For example, incorrect aliquot name in sample tracking leads to breach of the chain of evidence and, as a consequence, the corresponding chemical is not scored.

Reasons for not reporting a chemical that has been spiked are inadequate sample clean-up, poor extraction and missed derivatisation step during sample preparation. Due to the high background added to the test samples the compounds of interest may be masked and easily missed. Therefore, the sample should be cleaned using suitable ion exchange procedures. Alternative methods, such as solid phase matrix extraction, single drop extraction may also be adopted if necessary. Many of the scheduled chemicals are not GC-amenable and

require appropriate derivatisation, such as silylation and/or methylation. For the detection of arsenical compounds (e.g. Lewisite), derivatisation with reagents such as propane-1-thiol is necessary.

Incorrect analytical conditions for early and late eluting compounds (pinacolyl alcohol, chloropicrin, BZ) may lead to missing these compounds. Other typical errors made during analysis include unexplained peak(s) in the blank (retention time of GC peak within a window of ± 0.1 min of retention time of reference chemical) or that the EI mass range was not presented to at least 10 m/z -units above molecular weight.

4. Global strategy for an OPCW Proficiency Test in a participating laboratory

A PT report must be sent to the OPCW within 15 calendar days after sample arrival at the laboratory. Thus, it is critical to plan the general conduct of sample preparation and analysis (see Fig. 2). Due to the large amount of relevant chemicals of different natures, sample preparation procedures cannot be very specific. It is therefore necessary to design general methods that allow recovery of the spiking chemicals from different matrices. These methods may then be modified to help identification or to improve reporting data depending on the specific characteristics of a sample.

Another aspect to take into account before sample arrival is the clean-up of lab equipment and the set-up of analytical instruments. The supplies used for sample preparation should be new or pre-cleaned ensuring deactivation of glassware active sites. All analytical instrumentation should be calibrated and their performance tested in order to avoid false positives due to equipment contamination or other operational problems.

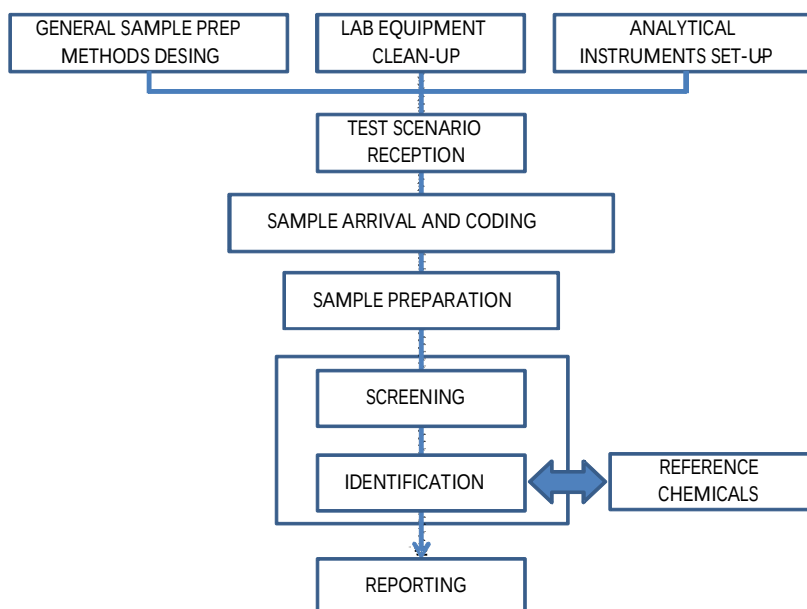


Figure 2. Global strategy for an OPCW Proficiency Test.

The current PT scheme establishes the analysis of two sets of three samples (blank, control and test samples) for the identification of spiking chemicals. Each set of samples can be prepared in a different matrix, with organic solvent, water and soil as the most frequent ones. Fig. 3 depicts all matrices used in PTs.

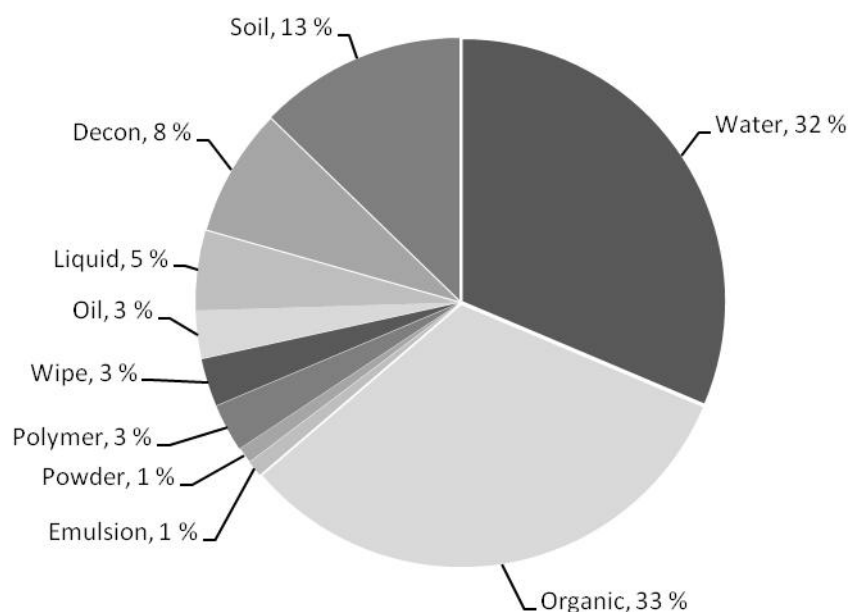


Figure 3. Distribution of matrices used throughout history of Official PT.

A few days before sample dispatch, test scenario – including matrix information – is released by the sample preparation laboratory allowing the sample preparation to be planned in advance. A number of important aspects concerning the strategy for following after arrival of the samples are detailed below.

- During the process of unpacking, the participants should make sure that samples are in good condition. If sample containers are damaged or leaking, degradation and contamination may have occurred, and therefore the analytical results for that sample may be unreliable. The incidence should be reported to the OPCW laboratory and the participating lab must decide whether to continue with the analysis of that sample.
- However, damaged samples are rarely received and sample preparation is carried out normally. The participants should pay special attention to avoid cross-contamination. Some laboratories even handle each set of samples in separate fume hoods and are using different laboratory equipment.
- Usually, each sample is divided in different homogenous subsamples and different sample preparation procedures are carried out on each portion. At least one portion is reserved in case further analyses are needed. Sample division will also minimize cross-contamination during the process.
- When possible, samples are analysed before any preparation procedure is carried out to prevent spiking loss or degradation due to sample handling. If a spiking chemical is detected using direct analysis, any loss or transformation of the molecule during later sample preparation can be recognised. Solid matrices such as polymer, soil, and wipe cannot be analysed without previous sample manipulation. In such cases, matrices are extracted with an organic or aqueous solvent and the extract is analysed using a screening technique. Should sample clean-up, derivatisation or any other process be required to improve analysis data, the risk of reporting a false positive or negative result is minimised, if the original sample was analysed directly.
- The most time-consuming sample preparation procedure should be identified in order to avoid timing problems. In case both sets of samples require intensive preparation, parallel sample preparation is recommended.

- During the preparation procedure, specific blanks of all resulting aliquots must be prepared. The blanks contain the same solvent or mixture of solvent as the analysed aliquot of the sample, including the derivatising agent. All generated aliquots are assigned a unique code in order to assure their identification and to allow a proper documentation of the chain of evidence.
- A first screening of aqueous aliquots is done using analytical techniques compatible with aqueous solvents, for example liquid chromatography. In organic aliquots, a first screening may be performed by gas chromatography (GC)-based techniques.
- In order to support the identification of the reported test chemicals, reference chemicals or reference data are to be used by the analyst. The reference chemicals can be commercial or synthesised in the lab.

More information on the sample preparation can be found in chapters under Section 2 “Sample preparation”. The screening of the samples is discussed in Chapter “Screening” (1-B-I).

4.1 Most frequently used sample preparation techniques and protocols: Effectiveness and limitations

The most reliable sample preparation procedure is one involving no sample preparation. Unfortunately, this cannot always be achieved due to aspects such as composition of the matrix, background chemicals or the nature of the spiking chemicals.

In case of liquid matrices, depending on the analytical technique to be used for analysis, solvent exchange may be required and highly volatile spiking chemicals might be lost during the evaporation process.

When the matrix is a solid (polymer, charcoal, soil, etc.), the spiking chemicals are absorbed in the substrate and liquid extractions need to be done in order to extract them from the matrix. In these cases, the spiking chemicals can be lost due to strong absorbance to the original matrix.

Background chemicals are also spiked in the samples along with the spiking chemicals to provide additional analytical challenges. The background chemicals may interfere with the analysis, detection and identification of spiking chemicals, or make it difficult to obtain good quality analytical data. In those cases, samples require additional preparatory clean-up procedures such as liquid-liquid extraction (LLE), solid phase extraction (SPE) and solid phase microextraction (SPME). For further information please see Section 2 “Sample preparation”.

4.2 Selection of the analytical techniques

The choice of the screening technique will depend on the specific properties of the aliquot matrix. The most typical screening techniques are GC-coupled to selective detectors, GC-MS, LC-MS and NMR [See Chapter “Screening” (1-B-I) for more information]. GC-based techniques coupled to selective detectors allow the detection of specific heteroatoms (nitrogen, phosphorus and sulphur). In conjunction with the retention times of each chemical, this allows the detection of relevant chemicals containing those specific heteroatoms. However, background chemicals spiked in the samples might also contain those heteroatoms.

In order to identify test chemicals, GC–MS, LC–MS, GC–FTIR or NMR spectroscopy are used. The results obtained using these techniques are compared to library data and if the test chemical is not in the database, the identification will be carried out by means of spectral interpretation (see Section 3-A for further information). GC–MS and LC–MS can also be used as screening techniques by selecting characteristic ions from scheduled chemical families. Phosphorous nuclear magnetic resonance ($^{31}\text{P}\{^1\text{H}\}$ NMR) can be used to screen for phosphorous containing compounds. These techniques additionally provide the structural information of the detected compounds.

5. Reporting

5.1 Reporting criteria for sample preparation procedures

The OPCW Technical Secretariat releases revised work instructions (WI) – including the reporting criteria – that need to be followed by all participants before every proficiency test.[8, 12, 13] Not complying with the reporting criteria may lead to failure of the test, even if all spiking chemicals have been identified and reported. Participants must therefore adhere to the reporting criteria fully.

A description for the used sample preparation methods is included in the PT report. The provision of flow charts of sample preparation for each sample is strongly recommended. The OPCW provides a sample preparation description template that should be completed for all samples, even if no chemicals were detected in a sample.

The sample and aliquots generated during the sample preparation must be coded unambiguously in the sample preparation description templates and flow charts in order to guarantee an unbroken chain of evidence throughout the entire report. The flow charts should indicate the aliquots from which results are presented in the report. If an aliquot yielded no results it should be stated in the flow chart.[8, §14.13].

In case a chemical is identified in a derivatised aliquot, the absence of the n-functional chemical³ in the original sample must be proven by showing its absence in the aliquot in which the n-functional chemical is expected.[8, §14.4a].

5.2 Reporting criteria for analytical techniques

The following points are from the current reporting criteria concerning analytical techniques:

- a) Due to the importance of confidentiality of OPCW inspections, only the minimum required data of any test chemical will be reported to prove its identity [8, *Attachment 1*, §1.3].
- b) The reporting templates for ‘method and analysis description’ provided by the OPCW are to be filled with sufficient detail. Samples and aliquots generated must be linked those in sample preparation templates in order to guarantee an unbroken chain of evidence throughout the report.
- c) The identification of each test chemical must be based on at least two different analytical techniques, where the primary technique must be a spectrometric

³ An n-functional chemical is a chemical with one or more functional group that is derivatised in a derivatisation reaction.

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technique and the secondary must be different from the primary. Techniques are not considered as different in the following cases:

- Spectrometric techniques acquired in the same mode with different mass analysers
 - MS and MS/MS when the same separation technique has been used
 - NMR data acquired from different nuclei of the same test chemical
 - GC-based screening techniques using different detectors, e.g. GC–FPD and GC–NPD
- d) It is important to take into account that if more than one set of data obtained with the same analytical technique is reported, any contradicting information in the different set of data shall be considered insufficient for identification purposes.
- e) In GC, CE and LC techniques, the chromatograms/electropherograms of solvent blank, sample and reference chemical must be submitted. They must be presented with the retention parameters of the signal clearly visible and printed by the instrument. Criteria regarding signal/noise ratio shall be follows. The time of the beginning of data acquisition of the solvent blank and sample must be indicated.
- f) Blanks of all aliquots must be shown. The blank shall be run just before the corresponding aliquot. The requirement of the blank chromatogram stipulated by the WI must be carefully met for all presented techniques.
- g) The reported chemicals have to be compared to an identification parameter, reference chemical or data [8, *Attachment 1*, §2.1]. In chromatographic techniques, when compared to a reference chemical, criteria regarding signal/noise ratio, retention/migration parameters, and intensity of the peaks must be fulfilled. When retention indices (RI) are used as an identification parameter the RIs must be compared to OCAD values.[14]
- h) **Mass spectrometry-specific criteria.** When mass spectrometry is presented the total ion chromatogram (TIC) of blank, sample and reference chemical with the retention index printed by the instrument have to be submitted, as well as the test and reference chemical spectra. Extracted ion chromatogram (EIC) of the above mentioned TICs can also be submitted (EIC of test and reference chemical must have the same m/z).
A representative mass spectrum of each test chemical identified must be reported with similar and mass range and isotopic ion ratios [8, *Attachment 2*, §1.5].
If the identification of the test chemicals is not carried out by comparison to a reference chemical but by spectral interpretation, the major fragmentation pathways must be presented and the intense ions must be explained by spectral interpretation.
- i) In mass spectrometric techniques, the extracted ion chromatogram (EIC) of reference chemical and test chemical will be reported using the same m/z . In addition, for EI spectra, a match factor between identified spectrum and reference spectrum as calculated by either the AMDIS,⁴ NIST library or manufacture instrument search routines is required.
- j) **GC–MS/MS-specific criteria.** The data to be presented include:
- TIC or EIC of a precursor/product ion of solvent blank, sample and reference with the retention parameters clearly visible and printed by the instrument.

⁴ The Automated Mass Spectral Deconvolution and Identification System (AMDIS) is a program that extracts spectra for individual components in a GC–MS data file and identifies target compounds by matching these spectra against a reference library. It is freely available from NIST website: <http://chemdata.nist.gov/mass-spc/amdis/>.

- MS/MS spectrum of sample with a precursor ion indicated showing the mass required range.[8]
 - MS/MS spectrum of a reference chemical with a precursor ion indicated (in case a reference chemical is presented) or MS/MS spectrum of a closely related chemical for spectral interpretation.
- k) **LC–MS-specific criteria.** Blank, sample and reference chemical TIC or EIC of a quasi-molecular ion/adduct ion showing the retention times and the time of the beginning of data acquisition. Sample and reference spectra with a quasi-molecular ion visible and indicated and accomplishing the mass range requirements stated in PT03.[8]
- l) **LC–MS/MS-specific criteria.** The data to be presented include:
- TIC or EIC of a precursor/product ion of blank, sample and reference with the retention parameters clearly visible and printed by the instrument.
 - MS/MS spectrum of sample with a precursor ion indicated showing the mass range required by PT03.[8]
 - MS/MS spectrum of a reference chemical with a precursor ion indicated (in case a reference chemical is presented) or MS/MS spectrum of a closely related chemical for spectral interpretation.

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Chapter III. Chain of custody

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1. Introduction

Chain of custody is a term that refers to the maintenance of an unbroken record of the possession of a sample from the time of its collection through analytical or testing procedures to reporting to the customer and finally to the destruction of the samples and also possibly to a court proceedings. The purpose of the procedure is to ensure that the sample has been in possession of, or secured by, a responsible person at all times. It should remove any doubt about sample identification or that the sample has been tampered with. In practice, the laboratory must be able to prove and demonstrate that the obtained results actually correspond to the right samples.

If a laboratory wishes to prove that it has reliable results, it must ensure that chain of custody procedures have been followed strictly. Chain of custody is used in most chemical sampling and analysis situations to maintain the integrity of the sample by providing documentation of the sampling, control, transfer and analysis of samples. In general terms, chain of custody should always be followed in accredited laboratories.

2. Significance of chain of custody

For some laboratories, especially those dealing with forensic, pathological and environmental samples, the establishment of chain of custody procedures is of high importance, because the results of sample testing or analysis might eventually be held as

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evidence in a trial or hearing. Such laboratories should design their sample handling documentation systems so that, during each step of sample collection, delivery, receipt, storage, analysis, disposition, etc., some individual is responsible for the integrity and identification of the sample and its accompanying documentation. The case may occur that a laboratory is obliged to demonstrate an adequate chain of custody. This brings up the important question: What is an “adequate” chain of custody? There is no definite answer to this question but the topics discussed in part 3 of this chapter should be considered.

3. Chain of custody stages

As stated above, there are several stages to be controlled since the sample is collected in order to ensure the chain of custody. This procedure shall concern not only the collected samples and their subsamples but also the sampling and analysis data generated along.

Chain of custody steps can be tracked in Figure 1.

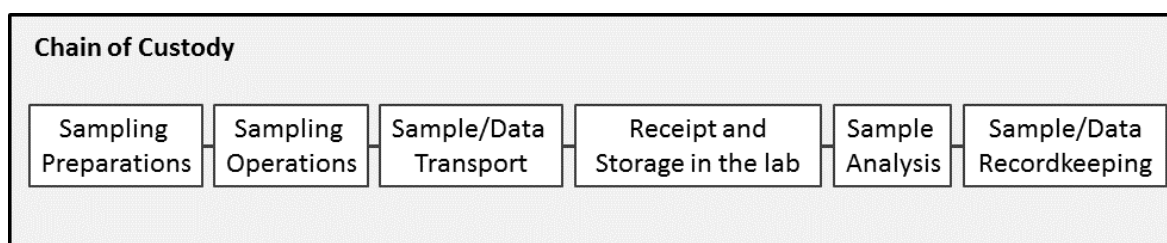


Figure 1. Chain of custody steps of sampling and sample data.

3.1. Chain of custody of samples

3.1.1. Sampling preparations

Previous to sampling operations, the material to be used, such as first response detectors, sample collection kits, individual protection equipment, sampling data sheets, etc., is prepared. Records concerning material preparation date, name of the preparer, and location of the material from preparation through use should be kept in a material preparation form.

Material supplies and records are to be stored in a limited-access locked area until they are used. All exchanges of supplies must be documented. Preferably, the original records stay in locked project files and a copy accompanies the supplies to the field.

3.1.2. Sampling operations

During sampling operations, the desired samples are collected using the proper sampling techniques and methods, and then placed in packaging containers.

In order to ensure the integrity of samples, tamperproof legal custody seals, such as frangible or thermolabile seals, are applied over the top of the lid down the side of the container.

All samples must be assigned a unique and permanent identification code. This code should be written on the sample containers. It will follow the sample from sampling through all the analysis to the final report. There are several ways to create a sample code. Good sample codes are unique but yet simple. For example: FI-Recce-111116-01, where FI is the country code, Recce is the name of the sampling team, 111116 is the date

(yyymmdd) and 01 is the sample number. Whatever way is chosen to develop sample codes, it must be consistent and unequivocal.

A tag or label must be attached to every sample container with at least the following information written in waterproof ink: sample code, name of the team that has taken the samples, the date and time of collection, and initials of the responsible person for that sample. Labeling should not contaminate the sample and should be sufficiently resistant to degradation, fading, or tampering with.¹

Each sample must be accompanied by a sampling data sheet. These records may be used as evidence at a later stage, and therefore, they must give as much information as possible. The sampling team would retain a copy and another copy should go along with the package. The desirable information to be shown in the sampling data sheet includes:

- Sampling data sheet code number;
- Sample code and the legal custody seal number;
- Date and time of sampling;
- Location of sampling;
- Physical and meteorological conditions at time of sampling;
- Sampled medium;
- Sampling method and equipment used;
- First response equipment results;
- Identification of sampling team;
- Signatures of operators.

Other information such as names of people related to the sample, content of conversation held in the sampling field, processes running at the time of sampling (in case the sample was taken in a chemical company), eye witness reports and anything that happened during the sampling event should be documented and must accompany each sample when forwarded for analysis.

All the information regarding sampling procedures (temperature, humidity and location of the scenario, sampling team, result of the first response detectors, sample coding, etc.) is recorded either manually or electronically before leaving the sampling site. Electronic records should be copied, verified, sealed, transported to the storage site and protected from electromagnetic fields that could destroy or corrupt the data.

3.1.3. Sample transport

Shipment of samples is time critical to ensure their integrity. Using custody seal tape on the package is advisable so any tampering with the package can be detected.

In the case of transferring the samples from the on-site inspection site through the OPCW to the designated laboratory, the samples are escorted by a representative of the OPCW and also possible by the representative of the inspected State Party. If an external service provider is used for shipment of the samples, the designated OPCW escort delivers and collects the samples directly to and from (the airport) cargo office.

Transfers of possession must be recorded on a chain of custody form. The chain of custody form is filled during sample collection and travels with the package until it reaches the laboratory. A copy of this form should be kept by the shipper. This form must be updated

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at arrival of the package and when any further transfer occurs. In the latter case, the person handing over the sample and the person receiving it must sign and date/time the chain of custody form. Once the sample is delivered to the laboratory, the escort or the courier and the laboratory representative shall sign the form.

The following information shall be register in the chain of custody form:

- Date and time of collection, shipping and transfers;
- Identification of the items inside the package (sample codes, type of samples, containers, etc.);
- Legal custody seal number of package;
- Seal and package condition (damaged/intact);
- Air bill or package tracking information;
- Names and signatures of all transfer responsible persons (sampler, sender, carrier(s), receiver).
- Transfer from a person to another

3.1.4. Receipt and storage in the lab

When samples are received, the laboratory responsible shall maintain the chain of custody procedures mentioned in paragraph 3.3. The number of samples, their identification and their integrity shall be verified to make sure no unwanted manipulation or damage has occurred. Samples are also photographed or videotaped to demonstrate the integrity of samples during transport. Any discrepancy in the condition of the seal codes and/or the samples, such as a broken container or leaks, should be noted in the transfer form. When not being analysed, the samples should be stored in a locked and conditioned area, according to their needs (temperature, humidity, etc.). If possible, a representative subsample will be stored in case future examinations are required. Samples are destroyed when the permission is granted.

The laboratory may assign its own sample code and register every sample in a sample record form following proper and uniform procedures in order to assure the tracking of samples. This document may be updated every time subsamples are taken for preparation or analysis. Thus, the sample location and disposition is ensured at all times. The lab responsible retains a copy of this document.

A sample record form shall include at least:

- Name of the responsible person;
- Date of sample arrival;
- Sample codes: sampling team code correlated to the laboratory code;
- Sample description: container, matrix, volume, color, etc.;
- Sample condition at arrival.

3.1.5. Analysis

Samples normally go through different stages of sample preparation before they are analysed. After every sample preparation procedure, a new and unique aliquot code is given to the subsample. Each aliquot must be given a unique code which will follow it through the entire testing process so that every sample can be traced down whenever

necessary. Samples and intermediary solutions generated in the analysis processes must be in limited-access areas at all times.

The analyst is responsible for maintaining the chain of custody procedures during this step. In order to determine the appropriate analyses to be performed, the analyst may need access to the information given in the sampling data sheet. The analyst may register all subsamples taken for analysis in the sample record form. Additionally, for every sample a sample preparation sheet shall be filled in including, at least, the following information:

- Sample code;
- Subsample volume;
- Preparation procedure and date;
- Aliquot code;
- Analyst name.

The analyses performed to the aliquots with the different instrumental techniques generate data files which must be coded and registered so that they are unequivocally linked to the corresponding aliquot. An analytical data sheet shall contain:

- Sample code;
- Aliquot code;
- Instrumental technique;
- Analytical method;
- Date of analysis;
- Analyst name.

3.1.6. Sample data recordkeeping

This step involves gathering the sampling and analysis documentation and the chain of custody records generated during the process in limited-access storage. These items, together with the remaining samples, are stored as long as required in order to verify the experimental results by tracing the information back to the raw data.

In order to setup and maintain quality assurance in the lab it is necessary to establish a quality assurance system and name personnel responsible for reviewing and auditing the records maintained in laboratories.

4. Summary

The importance of the chain of custody is remarkable and cannot be stressed too much. It is the key tool, and in fact, the only way to prove that the results produced in a laboratory correspond to the right samples. If a laboratory follows an adequate chain of custody, it will be able to demonstrate the identity of the samples and that they have not been tampered with at any stage of the procedure from sampling to final reporting of the results.

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